

Nullexit: Unified Project Document

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Contents

1	Project Directory Tree	3
2	README.md	5
3	agent.md	11
4	devref.md	18
5	diagrams.md	66
6	toggle.sh	71
7	recover.sh	99
8	setup.sh	108
9	scripts/setup-linux.sh	109
10	docker-compose.yml	110
11	scripts/common.sh	113
12	scripts/watcher.sh	126
13	scripts/crypto.sh	130
14	.aiignore	131
15	.claude/settings.local.json	131
16	.gitignore	131
17	Recover-Gateway.applescript	132
18	Toggle-Gateway.applescript	133
19	benchmarks/benchmark.sh	133
20	benchmarks/test_load.py	134
21	black_list.txt	135
22	cloud-init.yaml	136

23	docker/tor/Dockerfile	137
24	docker/tor/entrypoint.sh	137
25	launchd/com.nullexit.wake-recovery.plist	138
26	post-rules.txt	139
27	scripts/Dockerfile.routing-fix	139
28	scripts/Dockerfile.rule-compiler	139
29	scripts/Toggle-Gateway.bat	139
30	scripts/check_circuits.py	142
31	scripts/diagnose-host-leak.sh	143
32	scripts/dns-proxy.py	152
33	scripts/fix-docker-bridge-collision.sh	154
34	scripts/fix-ssh-delay.sh	158
35	scripts/generate_tex.py	159
36	scripts/logger/go.mod	162
37	scripts/logger/main.go	162
38	scripts/pf.conf	165
39	scripts/reinstall-host-tailscale.sh	166
40	scripts/routing-fix.sh	175
41	scripts/rule-compiler/go.mod	179
42	scripts/rule-compiler/main.go	179
43	scripts/setup-common.sh	190
44	scripts/sweep.sh	195
45	scripts/unlock-files.sh	199
46	tor-bridges.txt	200
47	white_list.txt	200

1 Project Directory Tree

```
nullexit/
|-- .aiignore
|-- .claude
|   |-- settings.local.json
|-- .gitignore
|-- LICENSE
|-- README.md
|-- Recover-Gateway.applescript
|-- Toggle-Gateway.applescript
|-- agent.md
|-- benchmarks
|   |-- benchmark.sh
|   |-- test_load.py
|-- black_list.txt
|-- cloud-init.yaml
|-- devref.md
|-- diagrams.md
|-- docker
|   |-- tor
|       |-- Dockerfile
|       |-- entrypoint.sh
|-- docker-compose.yml
|-- launchd
|   |-- com.nullexit.wake-recovery.plist
|-- post-rules.txt
|-- recover.sh
|-- scripts
|   |-- Dockerfile.routing-fix
|   |-- Dockerfile.rule-compiler
|   |-- Toggle-Gateway.bat
|   |-- check_circuits.py
|   |-- common.sh
|   |-- crypto.sh
|   |-- diagnose-host-leak.sh
|   |-- dns-proxy.py
|   |-- fix-docker-bridge-collision.sh
|   |-- fix-ssh-delay.sh
|   |-- generate_tex.py
|   |-- logger
|       |-- go.mod
|       |-- main.go
|   |-- pf.conf
|   |-- reinstall-host-tailscale.sh
|   |-- routing-fix.sh
|   |-- rule-compiler
|       |-- go.mod
|       |-- main.go
|   |-- setup-common.sh
|   |-- setup-linux.sh
|   |-- sweep.sh
|   |-- unlock-files.sh
|   |-- watcher.sh
```

```
|-- setup.sh  
|-- toggle.sh  
|-- tor-bridges.txt  
`-- white_list.txt
```

2 README.md

```
1 # nullexit: Tailscale + Cloudflare WARP Docker Gateway
2
3 > **Last updated:** July 12, 2026 [Architecture & Flow Diagrams ](./diagrams.md) [Full Dev Reference ](./
  devref.md)
4
5 **nullexit** is a chained network gateway that routes all Tailscale exit-node traffic through a Cloudflare WARP
  VPN tunnel double-encrypting every packet, hiding your ISP metadata, and providing network-wide DNS ad-
  blocking (AdGuard Home) and kernel-level IP threat blocking (`ipset`/`iptables`) for every device on your
  mesh.
6
7 ---
8
9 ## 1. Prerequisites
10 - Docker and Docker Compose installed.
11 - A Tailscale account.
12 - **macOS:** Install the standalone Tailscale CLI via Homebrew (`brew install tailscale`) and register `
  tailscaled` as a system service (`brew services start tailscale`). No `.app` GUI install required.
13 - **OS & Python:** Developed and tested against **macOS 26.5.2**. The project has zero external Python
  dependencies (no `requirements.txt`) and explicitly targets the default Apple-provided **Python 3.9.6** (`/
  usr/bin/python3`). While the scripts could function on newer versions, `nullexit` explicitly avoids
  experimenting with or modifying the built-in system Python environment to prevent unexpected OS-level side
  effects.
14
15 ## 2. Installation & Setup
16
17 Run the interactive setup script for your OS. It downloads `wgcf`, generates fresh Cloudflare WARP WireGuard
  keys, prompts for a Tailscale Auth Key, and writes your `.env`.
18
19 ```bash
20 ./setup.sh # macOS
21 ./scripts/setup-linux.sh # Linux
22 ```
23
24 ### Reference `.env`
25 This is a quick reference of the common variables. For the **complete, authoritative list** of every `.env`
  variable (including all optional/advanced settings), see `devref.md 7`.
26
27 ```env
28 WIREGUARD_PRIVATE_KEY=<Generated>
29 WIREGUARD_PUBLIC_KEY=<Generated>
30 WIREGUARD_ADDRESSES=<Generated>
31 TS_AUTHKEY=tskey-auth-...
32 GATEWAY_RULE_PROFILE=medium # light | medium | heavy
33 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
  have a healthy path.
34 GATEWAY_MSS=1120
35 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
  internet is NOT hijacked).
36 GATEWAY_HIJACK_HOST=true
37 GATEWAY_USE_EXIT_NODE=true
38 WARP_FAIL_THRESHOLD=6 # consecutive warp=off polls before auto-shutdown (default 6 = 30s)
39 KILL_SWITCH=false # enforce strict PF lock that breaks SSH if VPN fails
40 # On campus/enterprise networks (WPA2-Enterprise / 802.1x), AP Client Isolation blocks direct
41 # LAN traffic between devices. Tailscale attempts a local P2P upgrade anyway, which causes
42 # macOS gvproxy to SNAT-mangle the reply's source port poisoning the phone's WireGuard
43 # endpoint and blackholing 100% of its traffic. Setting this to false drops those local
44 # hole-punch packets so the phone safely falls back to Tailscale DERP relays.
45 # watcher.sh auto-detects 802.1x and AP isolation on every Wi-Fi change and overrides
46 # this setting automatically you only need to set this manually if auto-detection fails.
47 TAILSCALE_ALLOW_LAN_P2P=false # auto-managed; true only on trusted hotspots/home networks
48 ADGUARD_USER=admin # Username for AdGuard Home (defaults to admin if not set)
49 ADGUARD_PASSWORD=nullexit # Shared password for AdGuard Home and Tor ControlPort (defaults to nullexit
  if not set)
50 BLOCKED_COUNTRIES="kp il" # dynamically block country IP ranges via ipdeny.com
51 TOR_EXCLUDE_EXIT_NODES="" # comma-separated country codes to exclude as Tor exit nodes (e.g. {us},{gb
  })
52 DEBUG_TRACE=false # true = mirror a full command-by-command trace of the toggle to output.log
  (see 9)
53
54 ```
55
56 ## 3. Deploy the Gateway
57
58 **macOS (Recommended):**
59 ```bash
60 ./toggle.sh
61 ```
62 Running it again stops the gateway and restores your network. Handles the full lifecycle: Colima VM, containers
  , DNS hijacking, Tailscale exit-node routing, rule compilation, and sleep prevention.
```

```

62 **Linux / Manual:**
63 ```bash
64 docker compose up -d
65 ```
66
67 ## 4. Post-Deployment
68
69 ### Approve the Exit Node
70 1. Go to [Tailscale Admin Machines](https://login.tailscale.com/admin/machines).
71 2. Find the `tailscale` device `...` **Edit route settings** enable as Exit Node.
72
73 ### Verify Traffic Flow
74 On any Tailscale client device, select this gateway as the exit node, then visit [api.ipify.org](https://api.ipify.org) or run:
75 ```bash
76 curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
77 # expect: warp=on
78 ```
79
80 ### AdGuard DNS Setup
81
82 To ensure client devices correctly route DNS through the AdGuard container (and do not bypass it via `tailscaled`'s internal resolver), you **must** configure your Tailscale Admin Console:
83 1. Go to the **DNS** tab in the [Tailscale Admin Console](https://login.tailscale.com/admin/dns).
84 2. Enable **MagicDNS**.
85 3. Under **Nameservers**, add a **Custom** nameserver and enter the gateway's Tailscale IPv4 address (run `tailscale ip -4` on the gateway machine to find it, e.g., `100.x.x.x`).
86 4. Click the **three dots (...)** next to the nameserver you just added and enable **"Use with exit node"**.
87 5. Delete any other Global Nameservers (like Google or Cloudflare).
88 6. Toggle ON **Override local DNS**.
89
90 Traffic will now be correctly forced into the AdGuard sinkhole, and mobile devices will be blocked from bypassing it via encrypted DNS (DoH/DoT).
91
92 > [!WARNING]
93 > **Disable "Private DNS" on mobile devices.** Android's Private DNS (DoT/DoH) bypasses port 53 interception entirely. Go to `Settings Connections More connection settings Private DNS Off`.
94
95 > [!WARNING]
96 > **Bandwidth doubling:** Streaming 1GB of video on a remote device consumes 1GB download + 1GB upload on the host's network. Be mindful on metered connections.
97
98 ### AdGuard Dashboard
99 Navigate to `http://<gateway-tailscale-ip>:3000` credentials: **admin` / `nullexit`**.
100
101 ### Access Any Mesh Device From Anywhere (SFTP/SSH)
102
103 Once the gateway exit node is online, **any device on your Tailscale mesh can reach any other mesh device, anywhere in the world** as long as both devices are on the mesh and the target device has an SSH server running. No port forwarding, no public IPs, no firewall holes. You can SSH into any device using its MagicDNS hostname (e.g. `ssh username@my-macbook`) or its assigned Tailscale IP (`100.x.x.x`).
104
105 > **Security Tip (Zero Local Attack Surface):** It is highly recommended to disable macOS's native "Remote Login" and "Screen Sharing" in System Settings to prevent exposing those ports (22, 5900) to physical Wi-Fi networks (which scanners can fingerprint). The gateway enforces `tailscale up --ssh=true` and the `pf` Kill-Switch automatically blocks local Wi-Fi access to these ports. Since both MagicDNS and Tailscale IPs route through the exact same encrypted tunnel, they are equally secure but **MagicDNS is recommended simply for ease of use.**
106
107 ### Extreme OPSEC: The Secret Tor-over-WARP Proxy
108 `nullexit` includes a completely isolated, headless Tor infrastructure proxy designed for terminals and hacking tools (e.g., `curl`, `sqlmap`, `nmap`).
109 To protect against local malware fingerprinting:
110 1. **Procedural Ports:** The proxy ports are randomized into the ephemeral range (10000-65000) on every single boot using a strict kernel socket collision-avoidance check (`netstat`).
111 2. **Honey-Port Tripwire:** The gateway spawns a fake trap port. If any malware attempts a sequential port-scan on `localhost` to find the proxy, it trips the Honey-Port, instantly triggering a macOS desktop notification and a critical log alert.
112
113 > [!NOTE]
114 > **Threat Model limitation:** Port randomization and the Honey-Port only defeat blind, generic port scanners. They do not protect against targeted local malware that runs `lsof -i` or reads the `/tmp/nullexit-ports.env` file directly. To actually prevent malicious local processes from reaching the proxy, you must run a host-level outbound firewall (like LuLu 4.3.0+ or Little Snitch) with **localhost/loopback filtering explicitly enabled** to gate access by process identity.
115
116 To use the Tor proxy, run `cat /tmp/nullexit-ports.env` to find your `$TOR SOCKS_PORT` for the current session, and point your hardened browser (LibreWolf/Tor Browser) or CLI tool to `127.0.0.1:$TOR SOCKS_PORT`.
117
118 > [!WARNING]
119 > **Preventing DNS Leaks (SOCKS5 vs. SOCKS5-Hostname):**

```

```

120 > When routing command-line tools through Tor, you must force the tool to delegate DNS resolution to the
    Tor proxy. Using plain SOCKS5 (e.g., `socks5://` or curl's `--socks5` flag) will resolve hostnames locally
    * using the host's DNS settings (AdGuard/WARP) before establishing the connection. While this DNS traffic
    is encrypted inside the WARP tunnel, the destination domain name is still leaked to AdGuard and Cloudflare,
    undermining your anonymity.
121 >
122 > Use these correct protocols and configurations:
123 > - curl: Use `--socks5-hostname` or the `socks5h://` scheme (e.g., `curl -x socks5h://127.0.0.1:
    $TOR SOCKS_PORT https://check.torproject.org`). Do not use `--socks5` or `socks5://`.
124 > - sqlmap: Pass the proxy URL using the `socks5h://` scheme to ensure remote resolution: `--proxy="
    socks5h://127.0.0.1:$TOR SOCKS_PORT"`.
125 > - nmmap: Nmap's built-in `--proxies` option resolves hostnames locally. To safely scan targets without
    DNS leaks, tunnel it using `proxychains-ng` configured with `proxy_dns` enabled (add `socks5 127.0.0.1 <
    PORT>` to `/etc/proxychains.conf` or a local config, then run `proxychains4 nmap <args>`).
126
127
128 ##### Transparent .onion Browsing & Dark Web Access
129 In addition to the randomized SOCKS5 proxy port, `nullexit` provides seamless, transparent `.onion` domain
    browsing for all devices on your Tailscale mesh:
130 - Zero-Config DNS: AdGuard Home automatically intercepts queries for `.onion` domains and resolves them to
    Tor's virtual mapping subnet.
131 - Tailscale Auto-Routing: The virtual subnet is mapped to the `198.18.0.0/15` benchmark range (RFC 2544).
    Since this range is not a private IP subnet, Tailscale's Exit Node mode automatically routes it to the
    gateway (bypassing local LAN exclusions).
132 - Transparent NAT Proxying: Kernel firewall rules in the gateway redirect all TCP traffic destined for
    `198.18.0.0/15` directly into Tor's transparent port (`9040`).
133 - Exit Node Exclusion: Supports the ability to exclude specific countries from being used as Tor exit nodes
    via the `TOR_EXCLUDE_EXIT_NODES` environment variable in `.env`.
134
135 ##### Enable in Web Browsers
136 To prevent accidental DNS leaks, web browsers block `.onion` lookups by default. To browse dark web sites
    natively:
137 - Firefox: Go to `about:config`, search for `network.dns.blockDotOnion`, and set it to `false`. Type any
    onion address (e.g. `duckduckgogg42xjoc72x3sjasowarfbgcmvfimafitt6twagswzczad.onion`) directly into the URL
    bar and it will load.
138 - Mobile Browsers (iOS/Android): Connect to your Tailscale exit node, open Firefox (with blockDotOnion
    disabled), and browse onion sites natively on the go.
139
140 ##### Hardening: Using obfs4 Tor Bridges
141 If you want to disguise your Tor traffic from the WARP exit node provider, you can optionally enable Tor
    bridges:
142 1. Obtain private `obfs4` bridge lines from [bridges.torproject.org](https://bridges.torproject.org) or the
    official Telegram bot.
143 2. Create a file named `tor-bridges.txt` in the root of the `nullexit` folder and paste your bridge lines
    inside it (one per line).
144 3. Set `TOR_USE_BRIDGES=true` in your `.env` file and restart the gateway. The proxy will refuse to start if
    the bridges file is empty or missing.
145
146 ##### Example: Browse your Android phone's files from your Mac
147
148
149 1. On your Android phone, install [Termux](https://termux.dev/) and [Termux:Boot](https://f-droid.org/
    packages/com.termux.boot/) from F-Droid.
150 2. In Termux, install and start the SSH server:
151 ```bash
152 pkg install openssh
153 sshd
154 ```
155 Termux defaults to port 8022 (ports below 1024 require root on Android).
156 3. Find your phone's Tailscale IP:
157 ```bash
158 tailscale ip -4 # e.g. 100.87.42.87
159 ```
160 4. Stop Android from killing Termux do ALL of these (Samsung One UI is especially aggressive):
161 - Settings Apps Termux Battery Unrestricted
162 - Device Care Battery Background usage limits Sleeping apps / Deep sleeping apps remove Termux if
    listed
163 - In Termux, run `termux-wake-lock` to hold a persistent wakelock
164 5. On your Mac/PC, open [Cyberduck](https://cyberduck.io/) (or any SFTP client: WinSCP, FileZilla, FE File
    Explorer on iOS) and connect to:
165 - Server: `100.87.42.87` (your phone's Tailscale IP)
166 - Port: `8022`
167 - Protocol: SFTP (SSH File Transfer Protocol)
168 - Username: (your Termux username, usually `u0_a123` run `whoami` in Termux to check)
169 - Password: (your Termux password set with `passwd` in Termux if you haven't already)
170
171 That's it you can now browse, download, and upload files to your phone from any device on the mesh, no matter
    where either device is physically located. This works identically for any mesh device: Mac Mac, Windows
    Linux, phone NAS, etc.
172

```

```

173 > **Troubleshooting:** If the connection hangs for ~30 seconds before failing, your phone's SSH server isn't
      running. Open Termux and run `sshd` again. If Android keeps killing it despite Unrestricted battery, the
      termux-wake-lock` command is your fix.
174
175 ---
176
177 ## 5. Multi-Layer Firewall & Kill-Switch
178
179 ### Layer 1 DNS Sinkhole (AdGuard Home)
180 Blocks ads and tracking domains before they are downloaded. In automated Lighthouse tests on ad-heavy sites:
181 - **Time to Interactive:** ~22% faster (51.0s 41.8s)
182 - **Speed Index:** ~20% faster (15.3s 12.8s)
183
184 Customize blocking via `black_list.txt` and `white_list.txt`. Rule profiles (`light` / `medium` / `heavy`) are
      set in `.env`.
185
186 ### Layer 2 Kernel IP Firewall (ipset`/iptables`)
187 On every startup, the `rule-compiler` fetches threat-intelligence feeds (Spamhaus DROP, Feodo Tracker, Emerging
      Threats, CINS) and compiles **~16,700 unique malicious IPs/CIDRs** into the kernel `FORWARD` chain
      blocking both outbound C2 connections and inbound attack traffic. Zero configuration required.
188
189 ### Layer 3 Geo-IP Blocking
190 Dynamically blocks all traffic to and from specific countries using live IP ranges from `ipdeny.com`.
191 To add countries to your blocklist, open your `.env` file and set the `BLOCKED_COUNTRIES` variable with 2-
      letter ISO country codes (e.g. `BLOCKED_COUNTRIES="kp il cn ru"`), then restart the gateway.
192
193 ### Layer 4 macOS PF Kill-Switch
194 A native macOS Packet Filter (`pf`) kill-switch that enforces a strict default-deny policy on your physical Wi-
      Fi interface (`en0`). When `KILL_SWITCH=true` is set in your `.env`, it drops all outgoing traffic except
      the Cloudflare WARP endpoints and Tailscale DERP relays.
195
196 - **Failsafe Design:** If the VPN tunnel crashes or Docker dies, your Mac completely loses internet access
      rather than leaking your real IP.
197 - **Remote-Access Safe:** Carefully designed to whitelist Tailscale's control plane (`192.200.0.0/24`), `utun*`
      tunnel traffic, and local LAN subnets. This guarantees that your remote SSH sessions and local AirDrop
      will survive even when the kill switch engages.
198 - **Setup Requirement:** The toggle scripts need permission to manipulate the network and firewall in the
      background without prompting you for a password. You must configure your passwordless sudo config by
      running exactly:
199 ```bash
200 echo "$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dscacheutil, /usr/bin/killall,
      /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew, /usr/local/
      bin/brew, /usr/bin/python3 -I $PWD/scripts/dns-proxy.py, /opt/homebrew/bin/python3 -I $PWD/scripts/dns-
      proxy.py, /usr/local/bin/python3 -I $PWD/scripts/dns-proxy.py" | sudo tee /etc/sudoers.d/nullexit
201 ```
202
203 ---
204
205 ## 6. Toggle Scripts
206
207 ### macOS `toggle.sh`
208 Fully automated lifecycle script. Also supports a native `.app` launcher:
209 ```bash
210 osacompile -o "Toggle Gateway.app" Toggle-Gateway.applescript
211 ```
212
213 Optional `.env` settings (common subset; the full canonical table lives in `devref.md`):
214 - `GATEWAY_BYPASS_PING=true` proceed even if pre-flight connectivity checks fail.
215 - `GATEWAY_USE_EXIT_NODE=false` DNS-only mode, skip exit-node routing.
216 - `GATEWAY_HIJACK_HOST=false` skip DNS hijacking on the host (provides VPN/adblocking only to external
      Tailscale peers).
217 - `GATEWAY_MSS=1180` override the TCP MSS clamp for the double-tunnel (default `1120`; raise to `1180` for
      more speed on a healthy path).
218 - `HOST_LEAK_PROBE=false` disable background host-egress leak prober (default: true).
219 - `HOST_MTU=1200` override the host Tailscale interface MTU. Defaults to 1200 to fit inside the 1280-byte WARP
      tunnel.
220 - `KILL_SWITCH=true` enforce strict network lock that breaks SSH if the VPN fails.
221 - `STOP_COLIMA_ON_EXIT=true` fully shut down the Colima VM on toggle-off (saves battery on dedicated hosts).
222 - `TAILSCALE_ALLOW_LAN_P2P=true` allow direct Tailscale LAN P2P connections between devices on the same
      network.
223 - **Default: `false`** On campus or enterprise Wi-Fi (WPA2-Enterprise / 802.1x), AP Client Isolation blocks
      local device-to-device traffic. When Tailscale still attempts a local P2P upgrade, macOS's `gvproxy` layer
      SNAT-mangles the reply's source port. WireGuard's endpoint roaming treats this as a legitimate address
      change and updates the phone's outbound path to the newly-mangled port which has no listener. This
      blackholes 100% of the phone's traffic while the DERP relay path is abandoned. Setting this to `false`
      preemptively drops all RFC1918-destined Tailscale UDP from the gateway so the phone receives a clean
      failure and safely falls back to DERP.
224 - **Auto-managed:** `watcher.sh` automatically detects WPA2-Enterprise via `system_profiler SPAirPortDataType`
      (the `airport` binary was removed in macOS 14.4) and probes for AP isolation by pinging the default
      gateway (`route -n get 1.1.1.1`) on every network change. It writes the detected value to
      `lan_p2p_detected` in the repo root, which `routing-fix.sh` re-reads every 30 seconds **no restart required

```

```

    .** Only set this manually if auto-detection fails. See `devref.md 15.7.1` for the full SNAT endpoint
    poisoning analysis and `devref.md 15.11.6` for the `airport` removal background. *(Note: Direct P2P between
    the gateway container and the host Mac itself is always permitted via the Docker bridge to bypass DERP
    relays, regardless of this setting).*
225 - Set to `true` on trusted home networks or Windows/mobile hotspots (no AP isolation) for a faster, lower-
    latency direct path.
226 - `WARP_FAIL_THRESHOLD=3` number of consecutive failed checks before forcing a gateway teardown (default: 6
    checks = 30s).
227 - `WARP_ENDPOINT_1` / `WARP_ENDPOINT_2` override the Cloudflare WARP WireGuard endpoints.
228
229 ### Linux `toggle.sh` / `recover.sh`
230 ```bash
231 ./toggle.sh # toggle ON/OFF (cross-platform, dispatches on $OSTYPE)
232 ./recover.sh # recovery tool (cross-platform)
233 ```
234
235 ### Sleep Prevention (macOS)
236 When the gateway is ON, `caffeinate -i` runs in the background to prevent idle system sleep, keeping Docker and
    all services alive even on battery. The display can still sleep normally.
237
238 ### Auto-Recovery (macOS post-wake / post-roam)
239 Install the launchd LaunchAgent so the gateway self-heals after sleep/wake and Wi-Fi roams:
240 ```bash
241 cp ./launchd/com.nullexit.wake-recovery.plist ~/Library/LaunchAgents/
242 sed -i '' "s|__NULLEXIT_HOME__|$(pwd)|" ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
243 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
244 ```
245 See `devref.md 15.5.1` for the full deep dive.
246
247 ---
248
249 ## 7. Networking Notes
250
251 - **Port conflicts:** None. Tailscale uses dynamic UDP ports; WARP uses outbound UDP:2408. The gateway binds
    `0.0.0.0:41642/udp` on the host for Tailscale's WireGuard listener this is intentional and required to
    receive inbound hole-punch packets from peers on the internet. All other internal ports (AdGuard :5335,
    SOCKS5 :1080) are bound to `127.0.0.1` only and are not reachable from the LAN.
252 - **Tailscale P2P vs DERP:** On the same Wi-Fi, Tailscale establishes a fast P2P connection. On cellular (CGNAT
    ), it always falls back to a DERP relay (~80-200ms). See `devref.md 5.1`.
253 - **AirDrop / AirPlay:** Unaffected the gateway only touches standard Wi-Fi/Ethernet interfaces, not `awdl0`.
254 - **VPN coexistence:** Do not run a local VPN client (WARP, Mullvad, NordVPN) simultaneously with the exit
    node. Both fight for the default route and will blackhole your connection.
255 - **Banking & financial sites:** May intermittently block logins through WARP. Banks use anti-fraud databases
    that flag datacenter IP ranges (like Cloudflare's `104.28.x.x`) as VPN/proxy traffic. Success fluctuates
    depending on which WARP IP you land on. If a site blocks you, either disable the exit node temporarily for
    that session or use the bank's mobile app (which often uses different fraud detection).
256 - **MSS clamping (Bufferbloat):** Double-tunnelling adds ~120-160 bytes of overhead per packet. This is now
    automatically handled natively via the macOS `pf` firewall, which universally applies TCP MSS Clamping to
    all outbound packets to strictly prevent MTU fragmentation and bufferbloat. No manual `.env` configuration
    is required. Additionally, the host's Tailscale interface MTU is explicitly lowered to 1200 (configurable
    via `HOST_MTU` in `.env`) to guarantee packets fit inside the container's WARP tunnel without fragmenting.
257
258 ---
259
260 ## 8. Privacy & Security Hardening
261
262 The stack shifts trust across four layers: WARP hides traffic from your ISP, AdGuard blocks tracking at DNS,
    Tailscale encrypts device-to-device, and `ipset` blocks known malicious IPs at the kernel.
263
264 For higher security, the gateway supports these drop-in upgrades:
265
266 | Layer | Default | Hardened |
267 |---|---|---|
268 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, no-logs, anonymous payment) |
269 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH |
270 | Coordination | Tailscale (US) | Headscale (self-hosted) |
271 | Auth | Persistent OAuth keys | Ephemeral auth keys |
272 | Content | WireGuard asymmetric | WireGuard + Pre-Shared Keys (PSK) |
273
274 #### Python Runtime Airgap (Isolated Mode)
275 Never install third-party `pip` packages (like `requests` or `pyyaml`) into the Python environment that `
    nullexit` uses. All python scripts in this project (e.g., the `dns-proxy.py` daemon) are strictly
    engineered to run on the zero-dependency Python Standard Library. To strictly isolate the environment from
    supply-chain attacks, `nullexit` executes these scripts using Python's Isolated Mode (`python3 -I`).
    This deliberately blinds the execution environment to any malicious user-installed `pip` packages on the
    host, permanently air-gapping the gateway from PyPI.
276
277 See `devref.md 9` for the full threat model, Snowden programme analysis, and trust assumptions.
278
279 ---
280

```

```

281 ## 9. Debugging
282
283 - **`output.log`** All stderr from `toggle.sh`, `recover.sh`, `setup.sh`, and the rule-compiler is written
    here. The WARP Watcher (`WARP_FAIL_THRESHOLD`) and the Host Leak Probe (`HOST_LEAK_PROBE`) also write all
    their events here `LEAK`, `ROTATE`, `HOST-PROBE failed`, and WARP shutdown notices. Check this first.
284 - Every run now records lifecycle breadcrumbs regardless of `DEBUG_TRACE`: an `EXEC:` / `EXIT <code>:` line
    brackets each heavyweight command (`colima start`, `docker compose up -d --build`, `docker compose down`),
    and an `EXIT: code=<N> phase=<init|stop|start>` line is written when the script exits **for any reason**
    including a killed terminal, `kill`, or OOM. This closes a prior blind spot where an interrupted **START**
    (e.g. a `--restart` race while Wi-Fi was re-associating) left the log ending abruptly after the teardown
    with no indication of what failed. If a toggle "does nothing" or the gateway ends up half-up, `grep -E '
    EXEC:|EXIT ' output.log | tail` shows the last command run and its result.
285 - **`DEBUG_TRACE=true`** (in `.env`) Full command-by-command execution trace of `toggle.sh`, mirrored to `
    output.log` for deep post-mortems of intermittent failures. Each traced line is timestamped and tagged with
    `file:line:function` (via a custom `PS4`), so you can follow the *exact* code path a run took including
    subshells, Docker/Colima calls, and where it aborted. Off by default because it is verbose (the log grows
    quickly); enable it only while reproducing a specific problem, then set it back to `false`. Implementation
    note: it prefers bash's `BASH_XTRACEFD` (bash 4.1, e.g. most Linux) to send the trace to the log while
    keeping your terminal clean; on macOS's stock `/bin/bash` 3.2 it transparently falls back to tee'ing stderr
    to the log (trace also appears in the terminal). It never uses the dynamic-fd (`exec {var}>>`) form, so it
    is safe on 3.2. To read a trace: `grep -nE '\^+ ' output.log`.
286 - **`bash scripts/diagnose-host-leak.sh`** One-shot host-routing diagnostic. Runs 8 checks and classifies your
    state into one of four known scenarios (System Extension conflict, SOCKS5 fallback, IPv6 leak, route-table
    freeze) or OK, and prints the exact fix command. Check 5 uses `route -n get 1.1.1.1` to ask the kernel
    which interface real traffic actually resolves through (more accurate than `netstat -rn | grep default` on
    macOS). Pass `--fix` to apply the remediation automatically. Pass `--watch` (or `--watch 30`) to run a full
    baseline diagnostic then continuously monitor warp/IPv6/default-route for leaks, alerting on any state
    change.
287 - **`bash scripts/host-leak-probe.sh`** Sub-second host-egress prober (enabled automatically via `
    HOST_LEAK_PROBE=true` in `.env`). Polls `cdn-cgi/trace` every 300ms directly from the host NIC not via `
    docker exec` so it catches flash-leaks invisible to the in-container WARP Watcher. All state changes, curl
    errors, and probe timeouts land in `output.log`. Grep with `grep LEAK output.log`.
288 - **`bash scripts/fix-docker-bridge-collision.sh`** One-shot fix for Docker bridge IP conflicts. If your local
    Wi-Fi router assigns you an IP in the `172.17.0.0/16` range, it will violently collide with Docker's
    default internal bridge, permanently killing your internet. This script detects the collision and auto-
    patches your Colima VM to use a safe subnet (e.g. `10.200.0.1/24`).
289 - **`devref.md`** Complete architecture reference, routing deep-dives, and full resolved-issues log. Read this
    before modifying any routing or firewall logic.
290
291 > [!CAUTION]
292 > **Two infinite routing loops are possible.** (1) If a hotspot device providing internet to the gateway host
    is *also* using the gateway as its exit node, packets bounce forever between the two. (2) When the host's
    default route is redirected to `utun`, WARP endpoint packets can loop back into the tunnel. Both are
    documented in `devref.md 15.2.1` and `15.2.2` and are automatically handled by `toggle.sh`.
293
294 ---
295
296 ## 10. Unified Project Document (Quine)
297
298 The project includes a `generate_tex.py` script that compiles the entire source code, configurations, and
    documentation into a single, unified LaTeX document (`nullexit_unified.tex` / `.pdf`).
299
300 Funnily enough, this document acts as a "project-level quine." It encapsulates the entirety of its own source
    code, its documentation, and the exact Python code required to generate itself. It serves as a self-
    contained, long-term archiving strategy providing everything needed to reconstruct and understand the `
    nullexit` system from a single file.
301
302 ---
303
304 ## 11. Acknowledgements
305 - **[SyameimaruKoa](https://github.com/SyameimaruKoa):** Dual-stack TCP MSS clamping, `SIGHUP` state-tracking
    in the routing sidecar, and `TS_AUTH_ONCE` integration.
306
307 ## 12. License
308 GNU Affero General Public License v3. See [LICENSE](./LICENSE).
309
310 ## 13. Changelog
311
312 - **July 12, 2026** Tor ControlPort dynamic password authentication (unified with `ADGUARD_PASSWORD`),
    enabling live circuit inspection via `/sweep`. See `devref.md 15.10.2`.
313 - **July 11, 2026** Tor container hardening: `obfs4proxy` restored (base image `debian:bookworm-slim`),
    robust bridge-file validation, and image pinning. See `devref.md 15.10.1`.
314 - **July 10, 2026** Roaming/startup stabilization suite: fixed the Wi-Fi-roam host-IP leak (raw ISP `132.x`)
    caused by the WARP Watcher racing post-wake recovery, plus a batch of pre-flight/concurrency/kill-switch/
    Tailscale-flag bugs and the macOS SNAT endpoint-poisoning P2P blackhole. Full post-mortems in the Incident
    Log see `devref.md 15.2.7`, `15.5.3` `15.5.5`, `15.2.8` `15.2.9`, `15.7.1`, `15.7.3` `15.7.5`, `15.12.18`.
315 - **July 6, 2026** Edge-case security and stability hardening: DNS Watcher interface independence, robust lock
    -file PID verification, fail-closed crypto integrity check, strict `iptables` pinning for tailscale0tun0,
    Docker port binding to `127.0.0.1` to prevent LAN exposure, routing verification via `netstat -nr`. See `
    devref.md 17`.

```

```

316 - **July 4, 2026** Colima bridged networking, SOCKS5 proxy migrated natively to Tailscale, headless prompt
      crashes fixed, proxy bypass domains added to fix LAN P2P. Python dependencies completely eliminated: `
      logger` and `rule-compiler` ported to blazing-fast Go multi-stage Docker builds. Added `crypto.sh` to
      enforce cryptographic signature validation on all core bash scripts. Massively refactored `toggle.sh` and `
      recover.sh` to eliminate ~200 lines of duplicated code by migrating network and process logic to `scripts/
      common.sh`. Added a concurrency mutex to `toggle.sh`, resolved unbound `TS_AUTHKEY` check crashes on setup,
      and integrated Go validation to the CI pipeline.
317 - **July 3, 2026** **Critical Security Updates:** Addressed the overnight silent IP leak and improved geo-
      blocking. Introduced `scripts/host-leak-probe.sh` (a sub-second host-egress leak detector) and a
      statistical threshold auto-shutdown watcher. Added `unlock-files.sh` to safely reset file permissions.
      Resolved numerous startup regressions and system bugs.
318 - **July 2, 2026** Two infinite routing loops resolved (`devref.md 15.2.2`): host-VM tunnel loop (WARP bypass
      routes now via gateway IP + manual `utun` redirection) and Docker Compose subnet takeover (`10.200.1.0/24`
      lock). `--accept-routes=true` now explicit on all `tailscale up --reset` calls.
319 - **July 2, 2026** Auto-recovery daemon documented in 6 above (`scripts/watcher.sh` + launchd plist). See `
      devref.md 15.5.1`.
320 - **July 2, 2026** Linux scripts (`scripts/toggle-linux.sh`, `recover-linux.sh`, `setup-linux.sh`) documented
      in 6.
321 - **July 1, 2026** `recover.sh --post-wake` ships; wired to watcher daemon. Triggered on every sleep/wake or
      network-state change.
322 - **June 28, 2026** 8 internal scripts moved to `scripts/`. User-facing files (`toggle.sh`, `recover.sh`, `
      setup.sh`, `docker-compose.yml`) stay at repo root.
323 - **June 28, 2026** All hardcoded install paths removed. `SCRIPT_DIR`-relative resolution and `
      _NULLEXIT_HOME` placeholder used throughout.
324
325 ### Cryptographic Integrity Verification
326
327 nullexit uses a built-in cryptographic integrity checker designed to provide tamper-*evidence* against
      accidental edits, logic bugs, or non-privileged malware. During setup, a 256-bit `NULLEXIT_SEED` is
      injected into your `.env` file. The core entrypoint scripts (`toggle.sh`, `recover.sh`, etc.) are hashed
      using HMAC-SHA256, and their signatures are stored in `signatures`.
328
329 *(Note: This is not a strict boundary against an attacker who already has write access to the repo, as they
      could simply read the seed and re-sign the scripts. It is designed as a strict footgun-catcher).*
330
331 If any script is modified without authorization, the gateway will loudly fail to boot. If you make intentional
      edits to the bash scripts, you must re-sign them by running:
332 ```bash
333 ./scripts/crypto.sh --sign
334 ```

```

3 agent.md

```

1 # nullexit Detailed Agent Analysis
2
3 > Complete per-file breakdown of every function, constant, duplication, and bug found.
4 > **Note (July 2026):** `sync-rules.py` and `logger.py` have been fully ported to Go (`scripts/rule-compiler/
      main.go` and `scripts/logger/main.go`) to dramatically improve performance and reduce container footprint
      via multi-stage Alpine builds. The analysis below reflects their previous Python states.
5
6 ## Important Agent Instruction
7 - **NEVER use `cat << EOF` or terminal redirections to write or modify files.** You must always use your native
      file-editing tools (`replace_file_content` or `multi_replace_file_content`). Using `cat EOF` leads to
      duplicated text, broken markdown headers, and structural corruption (which previously destroyed the section
      numbering in `devref.md`).
8 - **Always run `bash scripts/crypto.sh --sign` after making any modifications to the core bash scripts.** This
      is required to update the cryptographic HMAC-SHA256 signatures; otherwise, the startup integrity checks
      will fail and block execution.
9 - **Always run `git diff` and print the changes to the user *before* requesting or executing any git commit
      command.** The user must be shown the diff so they can review and understand the changes. Based on this
      diff, formulate a highly precise, detailed, and descriptive commit message (explaining exactly what was
      changed and why) rather than a generic summary, and present it to the user alongside the diff.
10 - **Always run `git status` before `git diff` to identify all modified and untracked files, ensuring no files (
      such as new scripts or configuration assets) are missed before formulating commit diffs and staging.**
11 - **Always use the `--restart` flag when asked to restart the toggle (e.g., run `./toggle.sh --restart`).**
12 - **NEVER execute any `sudo` commands (especially `sudo route`, `sudo dscacheutil`, or any network-modifying
      commands) without explicitly explaining to the user what the command does and why it is needed FIRST. Do
      not assume implicit permission. Wait for the user's approval before running it.**
13 - **When fixing an error and using logs to debug, always show the user the reasoning and the specific logs that
      support this theory.**
14 - **When the user tells you to "push", this means read git diffs and prepare commit messages that correspond to
      these changes (do not ignore any changes). If the changes can be atomized into multiple commits for easier
      understanding, do so.**
15 - **When the user says "latex this", it means you should run `python3 scripts/generate_tex.py` to regenerate
      the unified LaTeX document.**
16 - **NEVER apply changes to running gateway containers directly via `docker compose up` or `docker compose
      restart`.** Recreating or restarting containers (especially the `warp` container which hosts the network

```

```

namespace) breaks the shared network stack for all other services (`adguardhome`, `tailscale`, `routing-fix`), severing host connectivity and freezing macOS DNS. If you need to apply changes or rebuild containers, cleanly stop the gateway first using `./toggle.sh`, or trigger `recover.sh --post-wake` to rebuild and re-verify the network stack safely in the correct dependency order. Do NOT run `./toggle.sh --restart` or any network-rebuilding commands yourself if the execution could sever host network access; instead, explain the changes and ask the USER to run `./toggle.sh --restart` (or `./toggle.sh` stop and start) themselves to safely apply the modifications.
17
18 ## How to Verify Gateway is Working
19 Whenever modifications are made to the gateway scripts, routing, or containers, execute these verification checks in order:
20 1. **Container Health:** Run `docker compose ps` to ensure all containers (`warp`, `adguardhome`, `routing-fix`, `tailscale`) are `running` and `healthy`.
21 2. **DNS Hijacking Status:** Run `networksetup -getdnsservers "Wi-Fi"` (or appropriate network service) and ensure it is set exclusively to the gateway's Tailscale static IP (typically `100.100.21.8`).
22 3. **DNS Query Interception:** Run `dig @100.100.21.8 google.com` (using the gateway static IP from `gateway_ip`) to ensure AdGuard Home is active, intercepting, and resolving queries.
23 4. **Double-Tunnel Egress:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp` to verify the egress is routed through Cloudflare WARP (`warp=on`).
24 5. **Host Leak Monitoring:** Run `tail -n 30 output.log` and verify the background watchers (DNS + WARP) report healthy status without `LEAK` warnings. For a deeper on-demand audit, run `bash scripts/diagnose-host-leak.sh` (add `--watch` to poll, `--fix` to remediate).
25
26 ---
27
28 ## Part 1: macOS Main Scripts
29
30 ### toggle.sh (2191 lines, unified macOS + Linux)
31
32 > **Refreshed 2026-07-15.** Post-unification (`$OSTYPE` switch at L20) + the July-2026 `common.sh` extraction, the tables below reflect the *current* file. Most former "toggle.sh functions" now live in `scripts/common.sh`; the host-leak-probe pair was removed. For the ordered execution flow see **"toggle.sh Execution Map & Critical Path"** above.
33
34 **Functions still defined in `toggle.sh`** several exist in *both* halves (Linux L63848 / macOS L8782191):
35
36 | Function | Linux L | macOS L | Purpose |
37 |-----|-----|-----|-----|
38 | `log_exit_breadcrumb` | 30 | 885 | Write lifecycle-phase breadcrumb on exit (post-mortem trail) |
39 | `get_free_port` | 67 | 944 | Pick a random unused port (self- + kernel-collision checked) |
40 | `run_gui_cmd` | 1033 | | Run a command as the logged-in console GUI user |
41 | `write_gateway_active_marker` | 1053 | | Write `/tmp/nullexit-gateway-active.marker` (macOS only) |
42 | `clear_gateway_active_marker` | 1059 | | Remove active marker + `TUNNEL_FAILED_CLOSED.marker` |
43 | `start_sleep_prevention` | 145 | 1091 | Start `caffeinate` w/ revert trap |
44 | `stop_sleep_prevention` | 188 | | Stop `caffeinate` via PID file |
45 | `stop_dns_watcher` | 202 | | Stop DNS watcher (macOS uses `stop_pidfile_daemon`) |
46 | `start_warp_watcher` | 1164 | | Background WARP liveness monitor `recover.sh` after N fails |
47 | `stop_warp_watcher` | 1241 | | Stop WARP watcher via PID file |
48 | `cleanup_handler` | 215 | 1258 | Fail-closed teardown trap (guarded by `SUCCESS_RUN`) |
49 | `restart_tailscaled_daemon` | 327 | *(common)* | Restart tailscaled (Linux-half local copy; macOS uses common.sh) |
50 | `disconnect_tailscale_host` | 337 | *(common)* | `tailscale down` w/ retry (Linux-half local copy; macOS uses common.sh) |
51 | `setup_exit_node_routing` | 1430 | | Override default route through the Tailscale utun interface |
52 | `cleanup_network_state` | 354 | 1464 | Nuclear cleanup: proxies, DNS cache, routes, Wi-Fi, sharing services |
53
54 **Delegated to `scripts/common.sh`** (moved out of toggle.sh in the July-2026 refactor; line = common.sh):
55
56 | Function | common.sh L | Function | common.sh L |
57 |-----|-----|-----|-----|
58 | `run_with_timeout` | 58 | `enable_socks_proxy` | 769 |
59 | `is_gateway_active` | 168 | `disable_socks_proxy` | 803 |
60 | `gateway_state` | 232 | `reset_dns` | 824 |
61 | `get_active_service` | 284 | `stop_local_dns_proxy` | 898 |
62 | `restart_tailscaled_daemon` | 338 | `start_local_dns_proxy` | 909 |
63 | `disconnect_tailscale_host` | 352 | `force_dns_to_gateway` | 988 |
64 | `stop_pidfile_daemon` | 393 | `add_warp_bypass_routes` | 484 |
65 | `start_dns_watcher` | 715 | `remove_warp_bypass_routes` | 551 |
66
67 **Removed:** `start_host_leak_probe` / `stop_host_leak_probe` and the whole `HOST_LEAK_PROBE` prober subsystem no longer defined or called anywhere in `toggle.sh`, `recover.sh`, or `common.sh`, and `HOST_LEAK_PROBE` is no longer an env var (all verified). The only surviving host-leak tool is the standalone, on-demand `scripts/diagnose-host-leak.sh` (there is **no** `scripts/host-leak-probe.sh`).
68
69 **Constants / anchors (current):**
70
71 | Constant | Value | Line |
72 |-----|-----|-----|
73 | `LOG_FILE` | `$PWD/output.log` | L22 (Linux) / L863 (macOS) |
74 | `LOCK_FILE` | `/tmp/nullexit-toggle.lock` | L1020 |

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```

75 | `PID_FILE` (caffeinate) | `/tmp/nullexit-caffeinate.pid` | L142 |
76 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L200 |
77 | `WARP_WATCHER_PID_FILE` | `/tmp/nullexit-warp-watcher.pid` | L1144 |
78 | Gateway active marker | `/tmp/nullexit-gateway-active.marker` | L1054 (write) |
79 | TUNNEL_FAILED_CLOSED marker | `two halves selected by `
    $OSTYPE` at L20 (if [[ "$OSTYPE" != "darwin" ]] runs the self-contained Linux path L63848 then
    exit 0 at L845; darwin falls through to the macOS path L8782191). Line numbers below are the 
    macOS path (the deployment target). The halves are similar in intent but not identical see "Linux path
    where it diverges" below. Line = current, approximate.
91
92 ##### Entry & dispatch
93
94 ...
95 ./toggle.sh [ (no-arg) | --restart | restart ]
96
97 | -- verify HMAC signatures (crypto.sh --verify) --fail--> exit 1 (L857)
98 | -- rotate output.log if > 1GB
99 | -- arm teardown trap: ERR/INT/TERM/HUP -> cleanup_handler (L1317-1320)
100 | -- generate random ports (Tor / SOCKS / DNS) (L940)
101 |
102 | -- --restart --> gateway_state? running -> run STOP, then START (L999)
103 |
104 | -- no-arg --> gateway_state? (L1535)
105 |     running -> STOP branch (L1538-1584)
106 |     stopped -> START branch (L1590-2191)
107
108 ...
109 `gateway_state` is driven by `/tmp/nullexit-gateway-active.marker` (persisted intent), falling back to a live
    container probe.
110
111 ##### START synchronous critical path (all must complete)
112
113 | # | Step | -Line | Effect if it fails / is interrupted |
114 | ---|---|-----|-----|
115 | 1 | `disconnect_tailscale_host` (clean slate) | 1595 | host left mesh |
116 | 2 | `colima_start_until_ready` (readiness poll, 15.8.7) | 1631 | no Docker VM abort |
117 | 3 | honey-port reap (`kill`) + arm (`nc -l`, disowned) | 1695 | tripwire missing (non-fatal) |
118 | 4 | rule-compile (hash-gated, OFF critical path 15.8.8) | 1708 | reuse prior compiled rules |
119 | 5 | build-gate routing-fix / tor images | 1743 | rebuild only on change |
120 | 6 | `docker compose up -d` warp, tailscale, routing-fix, adguard, tor | 1758 | no gateway |
121 | 7 | wait tailscaled reachable | 1907 | host tailscale unusable |
122 | 8 | host tailscale set --exit-node=$TS_IP | 2046 | host not routed via gateway REAL-IP LEAK |
123 | 9 | add_warp_bypass_routes + setup_exit_node_routing | 2048 | WARP endpoint unreachable / wrong route |
124 | 10 | force_dns_to_gateway + start_local_dns_proxy | 2078 | host DNS not hijacked |
125 | 11 | enable_socks_proxy + verify SOCKSWARP | 2100 | SOCKS off |
126 | 12 | print "STARTED in Ns" (timing marker) | 2127 | cosmetic only NOT the commit point |
127 | 13 | start_sleep_prevention (caffeinate) | 2130 | Mac may sleep |
128 | 14 | start_dns_watcher (re-hijack DNS every 30s) | 2134 | no roam DNS recovery |
129 | 15 | start_warp_watcher (warp liveness recover.sh) | 2138 | no WARP self-heal |
130 | 16 | write_gateway_active_marker | 2146 | watcher believes gateway is DOWN |
131 | 17 | SUCCESS_RUN=true disarms the teardown trap | 2164 | (true commit point) |
132 | 18 | print "You can close this terminal window now." | 2191 | |
133
134 ##### STOP critical path
135
136 `disconnect_tailscale_host` (1543) `docker compose down --remove-orphans` (1552) `colima stop` (1558) `
    reset_dns` (1568) `stop_local_dns_proxy` (1571) stop caffeinate (1574) stop dns watcher (1575) `
    stop_warp_watcher` (1576) `clear_gateway_active_marker` (1577) "STOPPED in Ns" (1584).
137
138 ##### Long-lived background processes spawned by START
139
140 | Process | Started | PID / reap | Role |
141 | ---|---|-----|-----|
142 | honey-port `nc -l localhost $PORT` (disowned) | L1695 | `kill -f "nc -l localhost"` | port-scan tripwire |
143 | `caffeinate` | L2130 | `/tmp/nullexit-caffeinate.pid` | prevent system sleep |
144 | DNS watcher loop | L2134 | `/tmp/nullexit-dns-watcher.pid` | re-hijack DNS every 30s (Wi-Fi roam) |
145 | WARP watcher loop | L2138 | `/tmp/nullexit-warp-watcher.pid` | monitor warp; fire recover.sh after N fails |
146 | local DNS proxy `dns-proxy.py` | L2089 | (killed by `stop_local_dns_proxy`) | UDP:53 gateway DNS fallback |
147

```

```

148 Separately, the launchd `scripts/watcher.sh` daemon (post-wake / post-roam recovery) runs independently of
149 toggle and invokes `recover.sh --post-wake`; toggle does not spawn it.
150 ##### `cleanup_handler` the fail-closed teardown trap
151
152 Armed on `ERR/INT/TERM/HUP` (L1317-1320), guarded by `SUCCESS_RUN`. While `SUCCESS_RUN=false` it runs: `
153   reset_dns`, `disconnect_tailscale_host`, `stop_local_dns_proxy`, stop watchers, `
154   clear_gateway_active_marker`, `docker compose down` (+ `colima stop` on `ERR`). This is the fail-closed
155   guarantee: a half-built gateway is torn down, not left leaking.
156 ##### Correctness findings (verified against source)
157
158 1. "STARTED in Ns" (L2127) is NOT the commit point `SUCCESS_RUN=true` (L2164) is. The marker write (L2146)
159   and watcher starts (L2130-2138) sit between them. Any TERM/INT including a wrapping tool's timeout
160   in the L2127L2164 window fires `cleanup_handler` a FULL teardown (containers + colima + host tailscale),
161   leaving the marker ABSENT and the host on its raw ISP link, despite the printed success. This is exactly
162   the 2026-07-15 incident. (The Linux half has the same shape but a much narrower window: STARTED L814 `
163   SUCCESS_RUN=true` L820, and it maintains no marker.)
164 2. Never pipe `toggle.sh` through a reader (`| tail`, `| grep`). The disowned children (honey-port,
165   watchers, dns-proxy, caffeinate) inherit stdout, so the pipe never EOFs and the reader hangs long after
166   toggle has exited which then trips the wrapper's timeout SIGTERM finding #1. Run detached with a file
167   redirect instead: `nohup ./toggle.sh --restart > /tmp/t.log 2>&1 &`, then poll the file.
168 3. Host wiring is correctly on the synchronous critical path (steps 8 & 10 precede "STARTED"), so a 
169   completed START cannot leave the host unrouted/leaking the only way to that state is an interruption per
170   finding #1.
171 4. Namespace ordering: step 6's `docker compose up` creates warp's network namespace that `tailscale`/`
172   routing-fix`/`adguard` share (compose `depends_on` warp healthy enforces order). Never `docker compose
173   restart warp` on a live gateway it detaches the shared netns (see Agent Instruction, L16).
174 ##### Linux path where it diverges (L63848)
175
176 The Linux half is intent-similar but not a line-for-line mirror of macOS. Verified differences:
177
178 | Aspect | macOS (L8782191) | Linux (L63848) |
179 |-----|-----|-----|
180 | VM layer | Colima (`colima_start_until_ready`, L1631) | none native Docker |
181 | Host exit-node assert | `$TS_BIN set --exit-node="$TS_IP"` (L2046) | `$TS_BIN up --reset --exit-node="
182   $TS_IP"` (L735) |
183 | gateway-active marker | written (L2146) | not maintained (L119) `gateway_state` degrades to a live
184   container probe |
185 | Commit window | STARTED L2127 `SUCCESS_RUN` L2164 | STARTED L814 `SUCCESS_RUN` L820 |
186 | Port gen tooling | `jot` / `netstat` | `shuf`/`$RANDOM` / `ss` |
187
188 The `up --reset` vs `set` distinction matters: `--reset` re-applies all prefs and forces a route re-eval,
189 whereas `set` mutates only the exit-node pref the same asymmetry `recover.sh --post-wake` relies on (
190 devref 15.12.21).
191
192 ---
193
194 ##### recover.sh (566 lines, 23KB)
195
196 Functions (7 total):
197
198 | # | Function | Lines | Purpose |
199 |---|-----|-----|-----|
200 | 1 | `step()` | L40 | Print bold step header |
201 | 2 | `ok()` | L41 | Print green success message |
202 | 3 | `warn()` | L42 | Print yellow warning message |
203 | 4 | `fail()` | L43 | Print red failure message |
204 | 5 | `die()` | L44 | Print red error and exit |
205 | 6 | `run_with_timeout()` | L5674 | Run command with timeout (bash-native, no GNU coreutils) |
206 | 7 | `restart_tailscaled_daemon()` | L118129 | Restart tailscaled daemon via brew services |
207
208 Constants:
209
210 | Constant | Value | Line |
211 |-----|-----|-----|
212 | Color codes | `RED`, `GREEN`, `YELLOW`, `BOLD`, `NC` | L37-38 |
213 | `SCRIPT_DIR` | `${dirname "${BASH_SOURCE[0]}"}` | L50 |
214 | `PATH` export | `/opt/homebrew/bin:/usr/local/bin:$PATH` | L77 |
215 | `PID_FILE` | `/tmp/nullexit-caffeinate.pid` | L387 |
216 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L407 |
217 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L329-336 |
218 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L51 |
219
220 Duplicated logic from toggle.sh:
221
222 *(Note: As of July 2026, ALL of the below duplicated logic has been successfully extracted to `scripts/
223   common.sh`!)*
224
225 - `run_with_timeout()` simpler subshell version vs toggle's PID-tracking version
226 - `restart_tailscaled_daemon()` near-identical (uses warn()/ok() vs echo)
227 - Active network service detection inline at L217-233 (toggle has `get_active_service()` function)

```

```

208 - ENO service resolution inline at L230-233 (same as toggle L515-516)
209 - WARP bypass routes inline at L329-337 (toggle has `add_warp_bypass_routes()`)
210 - Exit node routing setup inline at L340-345 (toggle has `setup_exit_node_routing()`)
211 - DNS cache flushing L296-302 (same as toggle L611-616)
212 - Wi-Fi power-cycling L442-461 (similar to toggle L626-637)
213 - Proxy disabling L282-288 (same as toggle L604-608)
214 - Sharing services reset L586 (same as toggle L642)
215 - PID file stop pattern L387-399, L407-419, L423-435 (same as toggle L157-167, L193-203, L312-322)
216 - KILL_SWITCH check L194 (same as toggle L141, L469)
217 - .gateway_ip reading L138-142, L209-213 (same pattern as toggle L1080)
218
219 ---
220
221 ### setup.sh (410 lines, 18KB)
222
223 **Functions (4 total):**
224
225 | # | Function | Lines | Purpose |
226 |---|-----|-----|-----|
227 | 1 | `step()` | L10 | Print bold blue step header |
228 | 2 | `ok()` | L11 | Print green success message |
229 | 3 | `warn()` | L12 | Print yellow warning message |
230 | 4 | `die()` | L13 | Print red error and exit |
231
232 **Constants:**
233
234 | Constant | Value | Line |
235 |-----|-----|-----|
236 | Color codes | `RED`, `GREEN`, `YELLOW`, `BLUE`, `BOLD`, `NC` | L7-8 |
237 | `RECOMMENDED_TS_VERSION` | `1.98.5` | L71 |
238 | `ADGUARD_PASSWORD` | `nullexit` | L215 |
239 | Default .env values | `GATEWAY_RULE_PROFILE=medium`, `GATEWAY_MSS=1120`, etc. | L240-248 |
240
241 ---
242
243 ## Part 2: Linux Scripts
244
245 ### toggle.sh Linux path (unified into the root cross-platform script)
246
247 The Linux toggle is no longer a separate file. `scripts/toggle-linux.sh` was
248 merged into the repo-root `toggle.sh`, which dispatches on `OSTYPE`: an early
249 `if [[ "$OSTYPE" != darwin* ]]` block runs the self-contained Linux toggle
250 (shuf / ss / nmcli / ip / systemd) and exits before the macOS body. The shared
251 DNS/proxy/daemon primitives it used were already moved to `common.sh` in the
252 `313dc32` unification, so both branches call them by the same names.
253
254 ---
255
256 ### recover.sh Linux path (unified into the root cross-platform script)
257
258 Linux recovery is no longer a separate file. `scripts/recover-linux.sh` was
259 merged into the repo-root `recover.sh`, which dispatches on `OSTYPE`: the Linux
260 branch runs a self-contained teardown (resolvectl / nmcli / ip) and exits before
261 the macOS dual-mode body. All formatting/timeout helpers come from `common.sh`.
262
263 ---
264
265 ### scripts/setup-linux.sh (416 lines, 18KB)
266
267 **Functions (4 total):**
268
269 | # | Function | Line | Purpose |
270 |---|-----|-----|-----|
271 | 1 | `step()` | L10 | Print formatted step header |
272 | 2 | `ok()` | L11 | Print green success message |
273 | 3 | `warn()` | L12 | Print yellow warning message |
274 | 4 | `die()` | L13 | Print red error + exit 1 |
275
276 ***~95% identical to macOS setup.sh.*** Only differences:
277 1. Desktop shortcuts (L366-391 Linux `.desktop` files vs macOS `.app` via `osacompile`)
278 2. Final instructions text
279 3. .env template: macOS adds `GATEWAY_USE_EXIT_NODE`, `WARP_FAIL_THRESHOLD`, `KILL_SWITCH` Linux omits these
280
281 ---
282
283 ## Part 3: Utility Scripts
284
285 ### scripts/diagnose-host-leak.sh (722 lines, 38KB)
286
287 **Purpose:** Comprehensive host-routing diagnostic. 8 checks classify host IP leaks. Supports `--fix` and `--
watch` modes.

```

```

288
289 **Duplicated logic:**
290 - `resolve_active_service()` identical pattern as toggle.sh, recover.sh, fix-docker-bridge-collision.sh
291 - `run_with_timeout()` reimplemented timeout
292 - Color codes RED/GREEN/YELLOW/BOLD/NC with tty detection
293 - WARP check via `cloudflare.com/cdn-cgi/trace`
294 - `export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"`
295 - .gateway_ip reading pattern
296
297 ### scripts/fix-docker-bridge-collision.sh (402 lines, 18KB)
298
299 **Purpose:** Fixes Docker bridge subnet collisions with `10.200.1.0/24`.
300
301 **Duplicated logic:**
302 - `resolve_active_iface()` same interface detection as diagnose-host-leak.sh
303 - Color codes, step()/ok()/warn()/fail()/die() formatting helpers
304
305 ### scripts/fix-ssh-delay.sh (63 lines, 2.7KB)
306
307 **Purpose:** One-shot fix for ~20s SSH delay. Standalone, minimal duplication (uses hardcoded / instead of
    color functions).
308
309 ### scripts/reinstall-host-tailscale.sh (740 lines, 31KB)
310
311 **Purpose:** Complete uninstall + reinstall of host Tailscale.
312
313 **Duplicated logic:**
314 - `with_timeout()` yet another bash timeout implementation (polling-based, different from diagnose and toggle
    versions)
315 - Color codes + logging functions
316 - System Extension detection logic (similar to diagnose-host-leak.sh)
317
318 ### scripts/routing-fix.sh (266 lines, 10.5KB)
319
320 **Purpose:** Runs INSIDE warp container. Maintains routing table 200, FORWARD rules, country blocking, IP
    blocklists, tunnel health checks.
321
322 **Duplicated logic:**
323 - (Fixed) WARP_ENDPOINT defaults are now centralized in common.sh
324 - WARP health check via cdn-cgi/trace (another instance)
325 - iptables dual-backend pattern (for-loop over `iptables iptables-legacy`)
326
327 ### scripts/watcher.sh (269 lines, 13KB)
328
329 **Purpose:** Long-running launchd daemon for post-wake/post-roam recovery.
330
331 **Key function `detect_lan_p2p_mode()`:**
332 Added July 10, 2026. Called on startup and on every network state change (Listener 2 NET: events). Determines
    whether the current Wi-Fi network safely allows direct Tailscale LAN P2P connections:
333 1. **WPA2-Enterprise check:** `system_profiler SPAirPortDataType` reads the `Security:` field for the current
    association. Enterprise = 802.1x = always AP-isolated sets `allow=false`.
334 2. **AP isolation probe:** `route -n get 1.1.1.1` finds default gateway IP, then `ping -c 3 -t 1 -q "$gw"` if
    0 responses on a non-empty network, AP isolation is active sets `allow=false`.
335 3. **Output:** Writes `true` or `false` to `lan_p2p_detected` in the repo root.
336 4. **Consumer:** `routing-fix.sh` reads this file every 30 seconds and enforces/relaxes the RFC1918 DROP rule
    accordingly no restart required.
337
338 **Duplicated logic:**
339 - PATH export (similar to toggle.sh/recover.sh)
340 - Marker file pattern (`/tmp/nullexit-gateway-active.marker`)
341
342 ### scripts/dns-proxy.py (92 lines, 2.9KB) Clean, standalone
343
344 ### scripts/logger.py (144 lines, 5.9KB) Clean, standalone
345
346 ---
347
348 ## Part 4: Cross-Cutting Issues
349
350 ### Functions Identical Across macOS Linux
351
352 | Function | macOS toggle | macOS recover | macOS setup | Linux toggle | Linux recover | Linux setup |
353 |-----|-----|-----|-----|-----|-----|-----|
354 | `run_with_timeout()` | | | identical | identical | | |
355 | `stop_local_dns_proxy()` | | | identical | | | |
356 | `start_local_dns_proxy()` | | | ~95% similar | | | |
357 | `is_gateway_active()` | | | ~80% similar | | | |
358 | `step/ok/warn/die` | | | identical | identical | | |
359
360 ### Platform-Adapted Functions (same logic, different OS commands)
361

```

```

362 | Function | Linux Command | macOS Command |
363 |-----|-----|-----|
364 | `get_active_service()` | `ip route get 1.1.1.1 \| awk '{print $5}'` | `route get default` + `networksetup -
listnetworkserviceorder` |
365 | `reset_dns()` | `resolvectl revert` | `networksetup -setdnsservers` |
366 | `cleanup_network_state()` | `ip link set dev down/up` | `networksetup -setairportpower off/on` + proxy
disable |
367 | `force_dns_to_gateway()` | `resolvectl dns` + `resolvectl domain -ts.net` | `networksetup -setdnsservers` +
`-setsearchdomains ts.net` |
368 | `start_sleep_prevention()` | `systemd-inhibit --what=sleep` | `caffeinate -i` |
369 | `enable/disable_socks_proxy()` | Stub (no-op on Linux) | `networksetup -setsocksfirewallproxy` |
370 | DNS cache flush | `resolvectl flush-caches` | `dscacheutil -flushcache` + `killall -HUP mDNSResponder` |
371
372 ### Features Missing From Linux Scripts
373
374 These exist in `toggle.sh`'s macOS branch but NOT its Linux branch:
375
376 1. `write_gateway_active_marker()` / `clear_gateway_active_marker()`
377 2. `restart_tailscaled_daemon()`
378 3. `start_warp_watcher()` / `stop_warp_watcher()` WARP liveness monitor
379 4. `add_warp_bypass_routes()` / `remove_warp_bypass_routes()` / `setup_exit_node_routing()`
380 5. `LOG_FILE` structured logging (timestamps to output.log)
381 6. MSS clamping injection (step 4c)
382 7. Rule count reporting
383 8. `--post-wake` mode in recover
384 9. Container teardown on failure in cleanup_handler
385
386 ---
387
388
389
390 ## Part 5: Constant Duplication Map
391
392 | Value | Files |
393 |-----|-----|
394 | `162.159.192.1` (WARP endpoint) | docker-compose.yml, routing-fix.sh (now dynamically loaded via .env/common.
sh!) |
395 | `5335` (AdGuard DNS port) | docker-compose.yml, AdGuardHome.yaml, post-rules.txt |
396 | `5354` (mapped DNS port) | docker-compose.yml, dns-proxy.py |
397 | `41642` (Tailscale WireGuard) | docker-compose.yml, routing-fix.sh, post-rules.txt |
398 | `1080` (SOCKS5 port) | docker-compose.yml, toggle.sh |
399 | `1120` (MSS clamp) | post-rules.txt, .env |
400 | `admin:nullexit` (AdGuard creds) | AdGuardHome.yaml (bcrypt) |
401 | `10.200.1.0/24` (Docker subnet) | (Fixed) no longer duplicated |
402 | `America/New_York` (timezone) | docker-compose.yml (4 services) |
403 | `/tmp/nullexit-caffeinate.pid` | (Fixed) centralized in common.sh |
404 | `/tmp/nullexit-dns-watcher.pid` | (Fixed) centralized in common.sh |
405 | `/opt/homebrew/bin:...` (PATH) | (Fixed) centralized in common.sh |
406
407 ---
408
409 ## Part 6: Refactoring Outcome (Implemented July 2026)
410
411 **Decision: "Scripts-only" architecture**
412 Instead of introducing a new `lib/` directory, all shared code was moved into the existing `scripts/` directory
to keep the project hierarchy flat and simple.
413
414 ### 1. `scripts/common.sh` (288 lines, 10KB)
415 Extracts the fundamental primitives and networking logic used across orchestrator scripts:
416 - **Formatters:** `step()`, `ok()`, `warn()`, `fail()`, `die()`
417 - **Timeout Wrapper:** `run_with_timeout()` (Canonical version with PID tracking, graceful SIGTERM, and SIGKILL
fallback)
418 - **Daemon Lifecycle:** `restart_tailscaled_daemon()`, `stop_pidfile_daemon()` (collapsed caffeinate, DNS
watcher, and WARP watcher teardown logic)
419 - **Network Interface/Routing:** `get_active_service()`, `get_en0_service()`, `bounce_wifi_interfaces()`,
`add_warp_bypass_routes()`, `remove_warp_bypass_routes()`
420 - **Proxy/DNS Cleanup:** `disable_all_proxies()`, `flush_dns_cache()`, `reset_sharing_services()`
421 - **Configuration Helpers:** `read_adguard_ip()`, `is_kill_switch_enabled()`, `get_warp_endpoint_1()`,
`get_warp_endpoint_2()` (which centralizes the hardcoded 162.159.192.1 / .193.1 into .env logic)
422
423 ### 2. `scripts/setup-common.sh` (~350 lines)
424 Extracts the heavy, duplicated installation logic shared between `setup.sh` (macOS) and `scripts/setup-linux.sh`
:
425 - Docker and Colima prerequisite checks
426 - `wgcf` binary download and WARP key generation
427 - `.env` configuration file generation
428 - Interactive Tailscale Auth Key and AdGuard Rule Profile prompts
429
430 ### Sourcing Pattern
431 - macOS root scripts (`toggle.sh`, `recover.sh`, `setup.sh`) source via:
432 `source "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/common.sh"`

```

```

433 - Scripts under `scripts/` (`watcher.sh`, `diagnose-host-leak.sh`, etc.) source via:
434 `source "${dirname "${BASH_SOURCE[0]}"}"/common.sh`
435
436 ## Agent Verbs / Protocols
437
438 ### `/sweep` (or `sweep the gateway`)
439 When the user invokes this verb, the agent MUST perform a full end-to-end connectivity, health, security, and
    performance audit of the gateway:
440 1. **WARP Validation:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp=` (must equal `on`).
441 2. **Tor Proxy & Control Validation:**
442    - Source the dynamically generated ports from `/tmp/nullexit-ports.env` (or identify them using `docker
      compose ps`).
443    - **SOCKS5 Check:** Run `curl -s --socks5-hostname 127.0.0.1:$TOR SOCKS_PORT https://check.torproject.org/
      api/ip` to confirm the SOCKS5 proxy successfully connects and routes through the Tor network.
444    - **Control Port Check:** Query the Tor Control Port using netcat: `echo "PROTOCOLINFO" | nc 127.0.0.1:
      $TOR_CONTROL_PORT`. Verify it returns a standard `250-PROTOCOLINFO` response, confirming the Control Port
      is active and communicating.
445    - **Circuit & Node Lookup:** Query the Tor Control Port for the active path (`GETINFO circuit-status`). For
      each active built circuit, parse the relay fingerprints, lookup their IP addresses (`GETINFO ns/id/<
      fingerprint>`), and resolve their country codes using Tor's local GeoIP database (`GETINFO ip-to-country/<
      IP>`). Include the list of active circuits, showing the role (Guard/Middle/Exit), nickname, IP, and country
      of each hop in the final sweep report.
446    - **Transparent Onion Validation:** Run `nslookup duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.
      onion` to verify it resolves to an IP within `198.18.0.0/15` benchmark range. Verify transparent dark web
      routing by running `curl -sI -H "Host: duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion" http
      ://<resolved-ip>/` (should return HTTP 301/200).
447 3. **DNS Leak Validation:**
448    - Audit the upstream resolver currently utilized by the host. Run `curl -s https://edns.ip-api.com/json` to
      fetch details of the DNS resolver processing the client's requests.
449    - Verify that the resolver IP and organization returned belong to Cloudflare (e.g., AS13335) or match the
      WARP tunnel's exit range, and that **no** DNS traffic leaks to the user's raw physical ISP DNS.
450 4. **Performance Benchmark Audit:**
451    - Run the python benchmark script `python3 benchmarks/test_load.py`.
452    - Report the overall load speed and the ratio of successfully fetched vs. blocked/failed subresources for
      the targeted domains to ensure ad-blocking and MTU encapsulation are performing correctly.
453 5. **Tailscale Mesh Validation:**
454    - Run `ping -c 1 100.100.21.8` to ensure the gateway is reachable.
455    - Run `tailscale status` to identify all active/online nodes in the mesh.
456    - For every online peer returned in the status output, execute a ping check (`ping -c 1 <peer-ip>`) to
      verify bidirectional connectivity and confirm healthy routing to all active mesh devices.
457 6. **Log Audit:** Run `tail -n 100 output.log | grep -iE "(error|fail|warn)"` to catch any underlying component
      failures or warnings.
458
459 ### `/latex` (or `generate latex`)
460 When the user invokes this verb, the agent MUST run `python3 -I scripts/generate_tex.py` to regenerate the
    documentation source code (`nullexit_unified.tex`). The agent MUST NOT attempt to compile the `.tex` file
    into a PDF (the user handles PDF compilation locally).

```

4 devref.md

```

1 # nullexit Development Reference & Resolved Issues
2
3 > **Last updated:** July 12, 2026
4 > **Purpose:** Provide any LLM or developer with complete project understanding, debugging history, and
    resolved issues so they can make informed changes without re-reading every file.
5 > **Diagrams:** See [`diagrams.md`](./diagrams.md) for system architecture, toggle.sh flowcharts, monitoring
    layer, traffic sequence, recover.sh decision tree, and the full failureself-healing map.
6
7 This reference is organized into four parts:
8 > - **Part I Architecture & Reference** (18)
9 > - **Part II Design Decisions & Threat Model** (914)
10 > - **Part III Incident Log & Resolved Issues** (15)
11 > - **Part IV Observations, Changelog & TODO** (1618)
12
13 ---
14
15 # Part I Architecture & Reference
16
17 ## 1. What Is nullexit?
18
19 nullexit is a **Tunnel-in-Tunnel VPN gateway** that chains **Tailscale** (mesh VPN) through **Cloudflare WARP**
    (exit VPN). It runs on a macOS host inside Docker containers managed by **Colima** (lightweight Linux VM).
    The goal: any device on the user's Tailscale mesh can use this gateway as an exit node, and all traffic
    exits through Cloudflare WARP achieving double encryption, ISP invisibility, and network-wide ad-blocking
    via **AdGuard Home**.
20

```

```

21 ### Traffic Flow
22 ---
23 Phone/Laptop  Tailscale mesh (WireGuard)  Mac host  Docker container
24   Tailscale decrypts  Gluetun re-encrypts  WARP tun0  Cloudflare  Internet
25 ---
26
27 ### SOCKS5 Failover Path (if exit node is unavailable)
28 ---
29 DNS:  Browser  host 127.0.0.1:53  Python proxy  TCP:5354  AdGuard  WARP  DNS
30 TCP:  Apps  macOS SOCKS5 proxy (127.0.0.1:1080)  container  tun0  WARP  Internet
31 ---
32
33 ---
34
35 ## 2. Architecture
36
37 ### Containers (all share `warp`'s network namespace via `network_mode: service:warp`)
38 | Container | Image | Role |
39 |-----|-----|-----|
40 | `warp` | `qmcgaw/gluetun:v3.41.1` | Gluetun WireGuard client  Cloudflare WARP. Owns the network namespace.
   | Strict firewall. |
41 | `tailscale` | `tailscale/tailscale:v1.98.4` | Advertises as exit node on the mesh. Also provides the built-in
   | SOCKS5 proxy fallback via `TS_SOCKS5_SERVER=:1080`. |
42 | `routing-fix` | `alpine:3.20` | Sidecar that maintains routing tables + iptables rules every 30 seconds. |
43 | `rule-compiler` | `golang:1.22-alpine` / `alpine:3.20` | One-shot startup container that compiles 500k+ DNS
   | block rules in <2s and immediately exits. |
44 | `adguardhome` | `adguard/adguardhome:v0.107.77` | DNS sinkhole for ads trackers. Listens on port 5335.
   | Upstream DNS: `127.0.0.1:53` (through WARP). |
45
46 ### Port Mappings (on host via Colima)
47 | Host Port | Container Port | Protocol | Purpose |
48 |-----|-----|-----|-----|
49 | 5354 | 5335 | TCP/UDP | AdGuard DNS |
50 | (Tailscale IP):3000 | 3000 | TCP | AdGuard web UI (Internal to Mesh) |
51 | 41641 | 41641 | UDP | Tailscale WireGuard direct |
52 | 1080 | 1080 | TCP | SOCKS5 proxy |
53
54 ### Host Components
55 - **Colima VM** Linux VM running Docker (600MB RAM + 400MB swap)
56 - **tailscaled** Standalone Tailscale daemon via `brew install tailscale` + `brew services start tailscale` (
   NOT the GUI `.app`)
57 - **macOS network settings** DNS, SOCKS proxy, routing all manipulated by `toggle.sh`
58
59 ---
60
61 ## 3. Key Files
62
63 | File | Lines | Purpose |
64 |-----|-----|-----|
65 | `toggle.sh` | ~2109 | **Main script (macOS + Linux).** Dispatches on `$OSTYPE` an early Linux block (shuf/ss
   /nmcli/ip, systemd) runs and exits before the macOS body (jot/netstat/networksetup, launchd/pf). Detects
   state, toggles gateway ON/OFF. Handles DNS hijacking, Tailscale exit node, SOCKS proxy, sleep prevention,
   timeouts, cleanup. |
66 | `setup.sh` | 406 | **One-time setup.** Installs deps (Docker, Tailscale, wgcf), generates WARP keys, writes
   `.env`, configures AdGuard via API, starts containers. |
67 | `docker-compose.yml` | 194 | Service definitions for all 5 containers. |
68 | `scripts/routing-fix.sh` | 201 | Maintains routing tables (table 200 for SOCKS5, table 52 for CGNAT) and
   FORWARD iptables rules in both nftables and legacy backends. Loads and enforces the IP blocklist via `ipset`
   . Runs in a 30-second loop. |
69 | `post-rules.txt` | 18 | Gluetun iptables rules loaded at container start. FORWARD accept for tailscale0, NAT
   MASQUERADE on tun0, DNS redirect to port 5335, IPv6 drop, TCP MSS clamping. |
70
71 | `scripts/dns-proxy.py` | 91 | Local DNS proxy. UDP:53 TCP:5354 with proper 2-byte length prefix. Fallback
   when exit node is unavailable. |
72 | `scripts/rule-compiler/` | ~800 | Unified threat compiler (ported to Go from Python). **DNS pipeline:**
   fetches remote blocklists, deduplicates against AdGuard native filters, optimizes subdomains (~50%
   reduction), compiles to AdGuard syntax. **IP pipeline:** fetches threat-intel feeds (Spamhaus, Feodo, ET,
   CINS), strips private/Tailscale ranges, outputs atomic `ipset` restore file. Built dynamically in Docker
   multi-stage build. |
73 | **`adguard/work/`** | | **Bind-mounted container data folder. Off-limits from outside Docker READ-ONLY-
   EXCEPT-VIA-`sync-rules`. See README 6.** |
74 | `recover.sh` | ~742 | Nuclear recovery script (**macOS + Linux** dispatches on `$OSTYPE`; the Linux branch
   runs a self-contained teardown via `resolvectl`/`nmcli`/`ip` and exits, macOS falls through to the dual-
   mode body). Gain: use `--post-wake` (macOS) for non-destructive refresh after sleep/wake or Wi-Fi roam
   keeps the gateway live while still re-hijacking DNS + refreshing Tailscale exit-node + force-recreating
   warp if unhealthy. Resets DNS on ALL services, disables proxies, flushes routes, stops containers, kills
   sleep prevention, power-cycles Wi-Fi. Cryptographically verified. |
75 | `scripts/diagnose-host-leak.sh` | 580+ | **One-shot host-routing diagnostic.** Runs 8 checks (Tailscale,
   SOCKS5, CDN-cgi/trace, IPv6 leak, default route, output.log hints, gateway IP, Tailscale System Extension
   conflict), classifies into ONE of three known leak scenarios (A=SOCKS5 fallback, B=IPv6 leak, C=route
   freeze) or OK, writes a timestamped report to `host-leak-diagnostic-<UTC>.txt`, and prints a ready-to-run

```

```

fix on stdout. Pass `--fix` to apply the matched remediation and re-verify egress. Pass `--watch` (or `--
watch 30`) to run a full baseline then continuously monitor warp/IPv6/default-route every N seconds,
alerting on any state change. See 15.6.1. |
76 | `scripts/host-leak-probe.sh` | | **REMOVED (2026-07-15, verified deleted/untracked).** Was a continuous sub-
second host-egress prober auto-launched via `HOST_LEAK_PROBE`. Its fail-closed role is now covered by the
**PF kill-switch** (`enable_killswitch` in `common.sh`) + the in-container **WARP Watcher** (15.6.2); on-
demand auditing is `scripts/diagnose-host-leak.sh`. See 15.6.1 (retained as historical rationale). |
77 | `scripts/fix-docker-bridge-collision.sh` | ~290 | **One-shot fix for 15.2.3 (Docker bridge subnet collision)
** Detects Docker's `172.17.0.0/16` bridge colliding with the host Wi-Fi subnet (the symptom that produces
`default 172.17.0.1 UGScg en0` in `netstat -rn`), picks a non-colliding `docker.bip` from a small
candidate set, idempotently edits `~/colima/default/colima.yaml` with a timestamped backup, restarts
Colima, rebinds Wi-Fi DHCP, verifies the new default route is no longer Docker's bridge, and re-runs `
toggle.sh` + `scripts/diagnose-host-leak.sh`. Supports `--dry` and `--skip-toggle`. |
78 | `scripts/watcher.sh` | 194 | **Long-running post-wake + post-roam daemon.** Launched by launchd LaunchAgent;
subscribes to `com.apple.powermanagement` events via `log stream` and to network-state changes via `scutil
n.watch`. Both fire `recover.sh --post-wake`. Single-instance lock + SCRIPT_DIR-relative resolve of `
RECOVER`. See 15.5.1. |
79 | `scripts/common.sh` | ~40 | **Shared script primitives.** Centralizes formatted echo statements (`step`, `ok
`, `warn`, `die`) and the safe background `run_with_timeout` execution wrapper. Sourced by all orchestrator
scripts across macOS and Linux. |
80 | `scripts/setup-common.sh` | ~350 | **Shared installation logic.** Centralizes logic for verifying Docker,
installing Tailscale/wgcf, generating `.env`, and prompting for auth keys. Sourced by `setup.sh` and `setup
-linux.sh`. |
81 | `scripts/setup-linux.sh` | ~80 | **Linux sibling of `setup.sh`. Sources `setup-common.sh`. Distro detection
(apt/dnf/pacman), installs Linux-specific dependencies, creates `.desktop` shortcuts. |
82 | `scripts/Toggle-Gateway.bat` | ~300 | **Windows Toggle helper.** Moved from root to `scripts/`. Helps invoke
the framework on Windows if applicable. |
83 | `scripts/reinstall-host-tailscale.sh` | ~740 | **Nuke-and-reinstall tool for host Tailscale.** Completely
purges Tailscale, its settings, macOS Background Items (`.btm` database), and stale System Extensions.
Wipes ALL Login Items via `sfltool reset-login-items` (requires sudo). Useful for crisis recovery but
destructive to other login items. |
84 | `scripts/fix-ssh-delay.sh` | ~100 | **Fixes SSH connection delays.** One-shot script to address SSH delays
caused by DNS or routing issues, useful in crisis situations. |
85 | `scripts/logger/` | ~100 | **Internal logger piped from `scripts/routing-fix.sh`** inside the `warp`
container. Ported to Go from Python. Built via Docker multi-stage build. |
86 | `black_list.txt` | 71 | Custom domains to block (ads, trackers, telemetry). Supports `$important` modifier. |
87 | `white_list.txt` | 222 | Domains to force-allow (YouTube, Apple services, etc.). Always wins over blocks. |
88 | `.env` | ~14 | WARP WireGuard keys, Tailscale auth key, rule profile. **Contains secrets & NULLEXIT_SEED.** |
89 | `gateway_ip` | 1 | Static gateway Tailscale IP (fallback for dynamic resolution). |
90 | `scripts/unlock-files.sh` | ~20 | **One-shot stale-permission fix.** Uses atomic rename (`cp` + `mv`) to
replace locked inodes (mode `000`/`0444` from old `chmod` decisions) with fresh writable ones no `chmod`
called. |
91 | `scripts/crypto.sh` | ~50 | **Cryptographic integrity enforcement.** Uses `NULLEXIT_SEED` from `.env` to sign
core bash scripts (HMAC-SHA256) and strictly verify them on start to prevent manipulation. Pass `--sign`
or `--verify`. |
92 | `scripts/pf.conf` | 35 | **macOS native firewall ruleset.** Enforces the default-deny kill-switch, allows
local LAN / Tailscale traffic, and applies TCP MSS Clamping (`max-mss 1160`) to prevent WireGuard MTU
fragmentation. |
93
94 ---
95
96 ## 4. toggle.sh Flow
97
98 ### State Detection
99 `is_gateway_active()` returns true if EITHER containers are running OR host DNS is not `1.1.1.1`. This dual
check prevents the script from starting when it should stop (e.g., containers crashed but DNS is still
hijacked).
100
101 ### START Path (gateway OFF ON)
102 1. **Reset DNS to 1.1.1.1** Prevents deadlocks during startup
103 2. **Disconnect host Tailscale** `tailscale down` (prevents exit-node routing during container boot)
104 3. **Compile DNS rules** `docker compose run --rm rule-compiler` (Go binary inside multi-stage Docker build)
105 4. **Boot Colima VM** (if not running) via `colima_start_until_ready()`, which returns as soon as Docker is
reachable (~12s) instead of blocking on Colima's ~2-min post-provision hang (15.8.7). The networking mode
is **gated on the LAN-P2P signal** (`.lan_p2p_detected`, falling back to `TAILSCALE_ALLOW_LAN_P2P` in `.env
`): a trusted home network requests `--network-address --network-mode bridged` (or `shared` on WPA2-
Enterprise), while the common AP-isolated / untrusted case boots plain fast user-mode networking (`colima
start --memory 0.6 --vm-type vz`).
106 - **Why Bridged Networking (when enabled)?** The `--network-mode bridged` and `--network-address` flags
force the VM to receive its own IP directly from your local router's DHCP server, placing it on the same
physical subnet as the host. This bypasses macOS network isolation and is what lets **external LAN devices
** (phones, laptops on the same LAN), mDNS, and local service discovery reach across the container boundary
. It requires the `socket_vmnet` helper and is only worthwhile on a trusted LAN, which is why it is gated
rather than always requested. **It is *not* required for direct hostcontainer P2P** that path (the gateway
its own Mac) works even in user-mode networking via the routing-fix `host_ips` RETURN-rule carve-out;
see 15.3.5 and 15.8.7.
107 - **Enterprise Wi-Fi Fallback & Dynamic Roaming:** `toggle.sh` dynamically queries `system_profiler
SPAirPortDataType` on boot. If it detects a **WPA2-Enterprise** connection (802.1X), it forces Colima to
fall back to `--network-mode shared`. This prevents aggressive network intrusion systems on corporate/
university Wi-Fi switches from terminating the host's Wi-Fi connection due to "MAC spoofing" (detecting
multiple MAC addresses originating from a single authenticated port). P2P connections are impossible on

```

```

these networks anyway due to AP Client Isolation.
108 - **Dynamic Roam Downgrading:** `toggle.sh` writes its selected network mode to `/tmp/nullexit_colima_mode
.txt`. When roaming between networks (e.g., Home to University), `recover.sh` actively checks this state
against the new network's security requirement (`.lan_p2p_detected`). If a mismatch is detected (e.g.,
bridged Colima on an Enterprise network), `recover.sh` automatically forces a full `toggle.sh --restart` in
the background to safely transition the Virtual Machine's network mode and protect the host from de-
authentication.
109 5. **Configure VM swap** 400MB swap file inside the VM to prevent OOM
110 6. **Clean corrupted AdGuard config** Remove empty `AdGuardHome.yaml`
111 7. **Start containers** `docker compose up -d`
112 8. **Wait for gateway Tailscale** Poll `tailscale status` for "offers exit node" (up to 60s, abort at 40
consecutive NoState)
113 9. **Resolve gateway IP** From `.gateway_ip` or `docker compose exec tailscale tailscale ip -4`
114 10. **Connect host to mesh** Verify `tailscaled` is running (auto-start if needed), then `tailscale up --reset
--ssh=true --accept-dns=false --accept-routes=true --exit-node=` (---accept-routes=true must be explicit
--reset reverts it to `false` otherwise, silently preventing the default route from switching to `utun
*`)
115 11. **Pre-flight checks** [1/3] `tailscale ping` gateway, [2/3] `dig +tcp` AdGuard DNS, [3/3] WARP container
internet
116 12. **If all pass** Set exit node + hijack DNS to gateway IP (single server, no fallback)
117 13. **If any fail** Enable SOCKS5 proxy + local DNS proxy as fallback
118 14. **Final DNS re-force** `force_dns_to_gateway` called again after exit-node transition (tailscaled clobbers
DNS briefly)
119 15. **Enable sleep prevention** `caffeinate -i` in background to prevent idle sleep
120
121 ### STOP Path (gateway ON OFF)
122 1. Disconnect host Tailscale (`tailscale down`)
123 2. `docker compose down -t 5`
124 3. Reset DNS to 1.1.1.1
125 4. Stop local DNS proxy
126 5. **Stop sleep prevention** Kill caffeinate process
127 6. Full network state cleanup (proxies off, DNS cache flush, route flush, Wi-Fi power cycle)
128
129 ### Safety Mechanisms
130 - `run_with_timeout()` Pure-bash watchdog for every system command (15s/120s)
131 - `cleanup_handler()` Trap on ERR/INT/TERM/HUP restores DNS to 1.1.1.1, kills caffeinate, and tears down
Tailscale
132 - `force_dns_to_gateway()` Sets DNS and VERIFIES via `networksetup -getdnsservers` (3 attempts with backoff)
133 - `SUCCESS_RUN` flag Cleanup only runs if script didn't complete successfully
134 - `start_sleep_prevention()` / `stop_sleep_prevention()` Manages `caffeinate -i` via PID file at `/tmp/
nullexit-caffeinate.pid`
135
136 ---
137
138 ## 5. Routing & Firewall Architecture (Inside Container)
139
140 ### IP Rules (`ip rule show`)
141 | Priority | Match | Table | Purpose |
142 |-----|-----|-----|-----|
143 | 99 | `to 100.64.0.0/10` | 52 | CGNAT range Tailscale routes (injected by routing-fix) |
144 | 100 | `from 172.18.0.2` | 200 | Container's own traffic (SOCKS5 proxy) tun0 |
145 | 101 | all | 51820 | Gluetun's WireGuard fwmark tun0 |
146
147 ### Table 200 (SOCKS5 proxy traffic)
148 - `162.159.192.1 via $DOCKER_GW dev eth0` WARP endpoint exception (prevents tunnel loop)
149 - `default dev tun0` Everything else through WARP
150
151 ### iptables (TWO separate stacks!)
152 - **nftables backend** (`iptables`) Used by Gluetun's `post-rules.txt`. FORWARD policy DROP.
153 - **legacy backend** (`iptables-legacy`) Used by Tailscale's `ts-forward`. FORWARD policy ACCEPT.
154 - Both stacks apply to every packet. FORWARD RELATED, ESTABLISHED must exist in BOTH.
155
156 ### post-rules.txt (loaded by Gluetun)
157 ...
158 FORWARD: ACCEPT for tailscale0 in/out
159 INPUT: ACCEPT for tailscale0
160 NAT POSTROUTING: MASQUERADE on tun0
161 MANGLE: TCP MSS clamp to 1180
162 NAT PREROUTING: Redirect DNS (53) 5335 on tailscale0 and eth0
163 ip6tables: DROP FORWARD on tailscale0
164 ...
165
166 ---
167
168 ### 5.1 Tailscale Local Network Discovery (DERP Bypass)
169 By default, Gluetun's strict NAT swallows all of Tailscale's UDP hole-punching packets, forcing Tailscale to
fall back to incredibly slow DERP relay servers (often adding 500ms+ latency).
170
171 To fix this, we use `FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8` in `docker-compose.yml`
(on the `warp` container). This creates `ip rule` bypasses (Priority 99, Table 199) that route all packets
destined for private IP ranges *outside* the WARP tunnel directly to the host's `eth0`.

```

```

172 - **172.16.0.0/12** is especially critical because many modern Wi-Fi routers (and Docker itself) assign IPs in
    this block (e.g., 172.17.x.x).
173 - This bypass allows Tailscale's UDP packets to reach devices on the same Wi-Fi directly, establishing a fast
    peer-to-peer connection while the actual web traffic inside the tunnel remains fully routed through WARP.
174
175 ---
176
177 ### 5.2 Firewalling & Per-Device Access Control
178 Because every mesh device's traffic passes through the warp container's network namespace, this namespace
    serves as the ultimate choke point for network-wide firewall rules.
179
180 **Where to Put Rules**
181 The right place to add firewall rules is scripts/routing-fix.sh. It runs a 30-second loop inside the
    namespace and already handles re-injecting rules that Gluetun resets on reconnect. **Do not use post-rules
    .txt for dynamic rules**, as Gluetun flushes and resets it upon VPN reconnects.
182
183 **The Dual iptables Backend Constraint (CRITICAL)**
184 The container runs two simultaneous iptables backends: iptables (for the nftables backend used by Gluetun)
    and iptables-legacy (for Tailscale). Every rule **must** be added to both stacks, or it will silently
    fail for certain traffic.
185
186 **Avoiding Duplication**
187 Because scripts/routing-fix.sh runs in a loop, you must always check for a rule's existence with -C before
    appending with -A. Otherwise, your rules will duplicate every 30 seconds.
188
189 **Per-Device Identification & Filtering**
190 Each mesh device has a stable 100.x.x.x Tailscale IP, which is visible as the src IP in the FORWARD chain
    . This is how you identify devices for filtering.
191 * **Domain-level:** AdGuard Home (port 3000, credentials admin/nullexit) provides a REST API that supports
    per-client blocking rules based on their Tailscale IP.
192 * **IP-level:** Use iptables in scripts/routing-fix.sh (e.g., iptables -I FORWARD -s 100.x.x.x -d <
    blocked_ip> -j DROP).
193 * **Country-level:** Add ipset to the routing-fix apk install step in docker-compose.yml, and load CIDR
    ranges from ipdeny.com into an ipset, then block that set in the FORWARD chain. See 5.3 (Geo-IP
    Blocking) for the full configuration flow.
194
195 ### 5.3 Geo-IP Blocking
196 To block traffic to and from specific countries, the system dynamically downloads country-specific IP ranges
    from [ipdeny.com](http://www.ipdeny.com/ipblocks/data/countries/) and drops them via iptables FORWARD
    rules.
197
198 To add a new country to the blocklist:
199 1. Find the 2-letter ISO country code (e.g., cn for China, ru for Russia, il for Israel, kp for North
    Korea).
200 2. Open your .env file.
201 3. Update the BLOCKED_COUNTRIES variable: e.g., BLOCKED_COUNTRIES="kp il cn ru".
202 4. Run toggle.sh to restart the gateway and apply the new environment variables to the routing-fix container.
203
204 > [!NOTE]
205 > **Limitations of Geo-IP Blocking:** IP-based blocking is highly effective at blocking servers, direct apps,
    and raw infrastructure physically located and registered in a specific country (e.g., www.gov.il).
    However, it will **not** block high-profile websites (e.g., jpost.com) that route their traffic through
    global Content Delivery Networks (CDNs) like Google Cloud, Cloudflare, or Fastly. Because the CDN's IP
    addresses are registered in the US or globally, the network traffic physically never routes to the blocked
    country, allowing it to bypass the Geo-IP firewall.
206
207 ## 6. macOS Quirks to Remember
208
209 1. **Colima SSH tunnel is TCP-only** UDP ports (like 5354/udp) are NOT accessible from host. Use dig +tcp or
    the Python DNS proxy (UDPTCP converter).
210 2. **docker compose exec -T injects \r Always tr -d \r when capturing output for comparisons or networksetup
    commands.
211 3. **brew services can get stuck** Fix with sudo brew services restart tailscale. Common state: brew
    services list shows started but tailscale status shows "Tailscale is stopped."
212 4. **No timeout command** macOS lacks GNU timeout. Use the run_with_timeout() bash function.
213 5. **sudo -v prompts even with NOPASSWD** Use sudo -n (non-interactive) everywhere.
214 6. **DNS is scoped to en0** macOS scutil DNS resolver is scoped to en0 (Wi-Fi). DNS changes on other
    interfaces (like USB Ethernet) are ignored unless en0's service is also updated. The script always sets DNS
    on BOTH $ACTIVE_SERVICE and $ENO_SERVICE.
215 7. **tailscaled clobbers DNS during exit-node transition** force_dns_to_gateway is called a second time at
    the end of the ENABLE branch to counteract this.
216 8. **tailscale up --reset** resets all unspecified flags to defaults Every --reset call MUST also pass --
    accept-dns=false explicitly or tailscaled re-enables DNS management.
217 9. **docker compose ps CWD pitfall** docker compose commands require a docker-compose.yml in the current
    working directory. Use docker ps for raw container listing from any CWD; docker compose ps requires
    the project directory.
218
219 ---
220
221 ## 7. Environment Variables
222

```

```

223 ### `.env` File
224 | Variable | Purpose |
225 |-----|-----|
226 | `WIREGUARD_PRIVATE_KEY` | WARP WireGuard private key (from `wgcf generate`) |
227 | `WIREGUARD_PUBLIC_KEY` | WARP WireGuard public key |
228 | `WIREGUARD_ADDRESSES` | WARP WireGuard address (e.g., `172.16.0.2/32`) |
229 | `TS_AUTHKEY` | Tailscale auth key for the container |
230 | `NULLEXIT_SEED` | 256-bit seed for HMAC-SHA256 script integrity signing (generated by `setup.sh`) |
231 | `GATEWAY_RULE_PROFILE` | Rule compilation tier: `light`, `medium`, `heavy` |
232 | `GATEWAY_BYPASS_PING` | (Optional) `true` to proceed even if pre-flight checks fail |
233 | `GATEWAY_USE_EXIT_NODE` | (Optional) `false` to skip exit node, DNS-only mode |
234 | `GATEWAY_HIJACK_HOST` | (Optional) `false` to skip DNS hijacking on the host (VPN/adblocking for Tailscale peers only) |
235 | `GATEWAY_MSS` | (Optional) TCP MSS clamp value (default 1120); 1180 for speed on healthy paths |
236 | `WARP_FAIL_THRESHOLD` | (Optional) Consecutive `warp=off` polls before auto-shutdown (default 6 = 30s; 3 = 15s) |
237 | `WARP_ENDPOINT_1` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 1 (default `162.159.192.1`) |
238 | `WARP_ENDPOINT_2` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 2 (default `162.159.193.1`) |
239 | `KILL_SWITCH` | (Optional) `true` to enforce strict PF lock drops all host traffic if VPN fails. Breaks SSH if VPN dies. |
240 | `HOST_LEAK_PROBE` | (Optional) `false` to disable the 300ms host-egress leak prober (default: `true`) |
241 | `STOP_COLIMA_ON_EXIT` | (Optional) `true` to fully shut down the Colima VM on toggle-off (saves battery on dedicated hosts) |
242 | `ADGUARD_USER` | (Optional) AdGuard Home web UI username (default: `admin`) |
243 | `ADGUARD_PASSWORD` | (Optional) Shared password for AdGuard Home and Tor ControlPort (default: `nullexit`) |
244 | `BLOCKED_COUNTRIES` | Space-separated 2-letter ISO codes to block via ipdeny.com CIDR ranges (e.g. `"kp il cn ru"`) |
245
246 ---
247
248 ## 8. Diagnostic Commands
249
250 ### Container Health
251 ```bash
252 docker ps --format '{{.Names}} {{.Status}}'
253 docker compose exec -T tailscale tailscale status
254 docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace
255 ```
256
257 ### SOCKS5 Proxy
258 ```bash
259 curl --socks5-hostname 127.0.0.1:1080 --max-time 5 -s https://www.cloudflare.com/cdn-cgi/trace
260 ```
261
262 ### AdGuard DNS
263 ```bash
264 dig +tcp @127.0.0.1 -p 5354 google.com +short +timeout=5
265 ```
266
267 ### Firewall & Routing
268 ```bash
269 # nftables FORWARD chain
270 docker compose exec -T warp iptables -L FORWARD -v --line-numbers
271 # Legacy FORWARD chain
272 docker compose exec -T warp iptables-legacy -L FORWARD -v --line-numbers
273 # IP rules + routing tables
274 docker compose exec -T warp ip rule show
275 docker compose exec -T warp ip route show table 199
276 docker compose exec -T warp ip route show table 200
277 ```
278
279 ### Host macOS
280 ```bash
281 tailscale status
282 brew services list | grep tailscale
283 networksetup -getdnsservers Wi-Fi
284 networksetup -getsocksfirewallproxy Wi-Fi
285 sysctl net.inet.ip.forwarding
286 ```
287
288 ---
289
290 # Part II Design Decisions & Threat Model
291
292 ## 9. Privacy Architecture & Threat Model
293
294 ### Layered Defense
295 The gateway stacks four layers targeting different attack surfaces:
296
297 - **Layer 1 ISP-Level Surveillance (WARP):** Your ISP sees only an encrypted WireGuard tunnel to a Cloudflare IP. In the US, ISPs can legally sell your browsing metadata and are subject to secret NSLs (National

```

Security Letters). WARP removes their visibility entirely.

298 - ****Layer 2 DNS Tracking (AdGuard Home):**** DNS is the phone book of the internet. By default, queries go to your ISP's resolver, giving them a complete log of your traffic. AdGuard intercepts all DNS from mesh devices, blocks trackers, and resolves queries through the WARP tunnel.

299 - ****Layer 3 Device Identity & IP Exposure (Tailscale):**** On untrusted networks (coffee shops, public Wi-Fi), your device's IP is exposed. Tailscale creates an encrypted WireGuard mesh between your devices, routing all traffic to your gateway exit node.

300 - ****Layer 4 Kernel-Level IP Blocking (ipset/iptables):**** Sophisticated malware bypasses DNS entirely with hardcoded IPs. The `rule-compiler` fetches ~16,700 threat-intelligence IPs/CIDRs (Spamhaus, Feodo Tracker, Emerging Threats, CINS) and enforces them as `DROP` rules in the kernel `FORWARD` chain blocking both outbound C2 connections and inbound attack traffic.

301

302 **### The Documented Threat: Why This Matters**

303 This architecture is calibrated against documented mass-surveillance programs revealed in the Snowden disclosures:

304 - ****PRISM:**** Bulk collection of communication content from major tech providers.

305 - ****UPSTREAM:**** Tapping physical backbone fiber, collecting metadata and content in bulk.

306 - ****XKeyscore:**** Retroactive search of unencrypted traffic.

307 - ****MUSCULAR:**** Infiltration of unencrypted internal datacenter interconnect links.

308

309 Passive bulk collection is not paranoia it is a documented reality. By double-encrypting and routing traffic, nullexit prevents your ISP from harvesting your metadata.

310

311 **### Trust Assumptions & Shifting**

312 Every component shifts trust rather than eliminating it:

313 1. ****Physical Device Integrity:**** You trust your physical devices are not compromised.

314 2. ****Cloudflare Integrity:**** You trust Cloudflare not to correlate your WARP tunnel with exit traffic.

315 3. ****Tailscale Integrity:**** You trust Tailscale's coordination server not to MITM WireGuard keys.

316

317 The goal: no single provider has a complete picture. Your ISP sees an encrypted tunnel. WARP sees traffic with no account identity. Tailscale sees node topology but not content.

318

319 **### 9.1 Exit Node IP Rotation (Design Decision)**

320

321 A common question for privacy architectures is whether the system should aggressively rotate its public exit IP (e.g., every 5 minutes, like a Tor circuit or a proxy rotator).

322

323 `nullexit` uses Cloudflare WARP via Gluetun. WARP provides ****anonymity in a crowd**** rather than aggressive rotation. Thousands of users share the same Cloudflare Edge IP simultaneously, making your traffic indistinguishable from the herd. The IP may occasionally rotate on its own (historically surfaced as a `ROTATE` event in `output.log`), but it remains generally stable.

324

325 ****Why aggressive IP rotation was intentionally excluded:****

326 1. ****Broken TCP Connections:**** Every rotation instantly drops all long-lived TCP sessions (SSH, large downloads, WebSockets, gaming, Zoom).

327 2. ****CAPTCHA & 2FA Hell:**** Modern web security (e.g., Cloudflare, Akamai) aggressively flags IP-hopping as bot behavior. Aggressive rotation guarantees the user will be bombarded with "Verify you are human" CAPTCHAs and "New Login Detected" 2FA emails constantly.

328 3. ****WireGuard Architecture:**** WireGuard is stateless and does not natively support IP hopping on the fly. To rotate an IP, the VPN client must actively drop the tunnel, request a new endpoint, and re-handshake. This introduces latency gaps where the kill-switch would repeatedly trigger and blackhole the user's internet.

329

330 ****Verdict:**** For a daily-driver secure gateway, shared-IP crowding (WARP) is superior to aggressive rotation. As an added benefit, Cloudflare WARP IPs are actually highly static (tied to the persistent WireGuard public key generated for your device identity). Because your IP stays static and is part of Cloudflare's own trusted network, it builds a high "trust score," virtually eliminating CAPTCHA loops while still hiding you within the massive crowd of other users in that data center. All mesh devices routing through your gateway will reliably share this exact same trusted IP.

331

332 ***(Note: The only exception where aggressive IP rotation is genuinely required is for specialized offensive security tasks like high-volume web scraping or dark web research where avoiding IP bans or active tracking overrides the need for TCP stability.)***

333

334 **### 9.2 Cloudflare WARP Privacy & Logging**

335

336 Because `nullexit` uses Cloudflare WARP (via Gluetun) as its upstream exit node, the system inherits Cloudflare's consumer privacy policy.

337

338 ****What Cloudflare DOES NOT log (Strict No-Logs Policy):****

339 * ****Browsing History:**** They do not log the domains or URLs you visit.

340 * ****Traffic Destinations:**** They do not log the IP addresses of the servers you connect to.

341 * ****Content:**** They do not inspect or log the payload/data of your traffic.

342 * ****Data Brokering:**** They explicitly guarantee they will never sell or rent your data to advertisers.

343

344 ****What Cloudflare DOES log (Minimal Telemetry):****

345 * ****Data Transfer Volume:**** Total bandwidth used (required to enforce 10GB/mo quotas if you are not using a paid WARP+ key).

346 * ****Performance Metrics:**** Anonymized connection speeds and latency to optimize their network routing.

347 * ****Aggregate Volume:**** Total traffic volumes to popular destinations in aggregate (e.g., "datacenter X sent 50 TB to Netflix"), stripped of all user IDs.

```

348 * Device ID (Public Key): They store the persistent WireGuard public key generated by your container to
    authorize the connection.
349
350 **Privacy Verdict:**
351 Cloudflare WARP provides excellent **historical privacy**. Because they do not log your destinations, they
    cannot comply with historical subpoenas asking what websites you visited yesterday. However, it is **not
    absolute anonymity. They still know your real ISP IP address while you are actively connected. For daily-
    driver privacy against ISPs, hackers, and corporate trackers, it is top-tier. For nation-state evasion, use
    Tor (see 11).
352
353 ### 9.3 OPSEC: Python Runtime Airgap (Isolated Mode)
354 `nullexit` explicitly relies on the host's native system Python 3 runtime for critical components like `scripts
    /dns-proxy.py`. Because this script runs with elevated privileges (`sudo -n`) to bind to the privileged UDP
    /53 socket, modifying or polluting this Python environment is strictly forbidden.
355
356 **The Zero-Dependency Rule & Isolated Mode (-I)**
357 You must **NEVER** install third-party `pip` packages (e.g., `requests`, `pyyaml`) into the system Python
    environment that `nullexit` uses.
358 - All Python scripts in this repository must be written using *only* the Python Standard Library (`socket`, `
    urllib`, `json`, `ssl`).
359 - Third-party packages introduce a massive supply-chain attack vector. A compromised `pip` package installed
    globally could allow local malware to hook into the Python runtime and exploit the `sudo` privilege of the
    DNS proxy to gain full root access.
360 - To enforce strict immunity against user-installed malicious packages, `toggle.sh` explicitly invokes the
    proxy using **Python's Isolated Mode** (`python3 -I`). This flag forces the interpreter to completely
    ignore `PYTHONPATH`, `PYTHONHOME`, and the user's `site-packages` directory, executing exclusively from the
    read-only, Apple-signed system framework.
361
362 ### 9.4 Recommended Upgrades
363
364 | Layer | Default | Upgraded |
365 |---|---|---|
366 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, proven no-logs, anonymous payment) |
367 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH (Cloudflare-blind) |
368 | Coordination | Tailscale (US, OAuth) | Headscale (self-hosted, no third-party) |
369 | Auth | OAuth / persistent keys | Ephemeral auth keys (no identity persistence) |
370 | Content | WireGuard asymmetric | WireGuard + PSKs (coordination server blind) |
371
372 #### Mullvad (Replacing WARP)
373 Swedish-based. Police raided their offices in 2023 and left empty-handed no logs exist to seize. Accounts are
    random numbers. Payment by cash or Monero. Replace `VPN_SERVICE_PROVIDER=custom` with `VPN_SERVICE_PROVIDER
    =mullvad` in `docker-compose.yml`.
374
375 #### Mullvad DoH Upstreams
376 Point AdGuard's upstream resolvers to `https://adblock.dns.mullvad.net/dns-query`. Cloudflare WARP only sees
    encrypted HTTPS to Mullvad's IPs completely blind to your DNS queries.
377
378 #### Headscale
379 Self-hosted Tailscale coordination server. Eliminates Tailscale the company, keeping all node topology under
    your control.
380
381 #### Ephemeral Auth Keys
382 Generate in the Tailscale admin panel. Used once to authenticate; node disappears from the admin panel after
    disconnect. Breaks persistent identity linkage.
383
384 #### Pre-Shared Keys (PSKs)
385 Add a WireGuard PSK between peer nodes. Adds a symmetric encryption layer that Tailscale's coordination server
    has no knowledge of, blinding it from reading transit content.
386
387 ### 9.5 Structural Limitations
388 - **Traffic Analysis / Metadata:** Passive fiber tapping (UPSTREAM) can observe timestamps, data volumes, and
    IP ranges without reading content. Defeating traffic analysis requires mixing networks (Tor) or continuous
    dummy packet padding both heavily degrade performance.
389 - **Targeted State-Level Adversaries:** Consumer-grade VPNs are not a complete shield against a nation-state
    actively targeting you. This stack is designed to defeat bulk passive surveillance and corporate dragnet
    collection.
390
391 ---
392
393 ## 10. Per-Device Access Control
394
395 Because every mesh device routes internet-bound traffic through the `warp` container's `FORWARD` chain, they
    all appear with their stable `100.x.x.x` Tailscale IP as the `src` address. This gives two powerful
    enforcement surfaces:
396
397 ### IP & Port Level (`scripts/routing-fix.sh`)
398 Write raw `iptables` rules targeting specific devices. The 30-second idempotent loop auto-survives Gluetun
    reconnects:
399
400 ```bash
401 # Block a specific device from a target subnet

```

```

402 iptables -I FORWARD -s 100.x.x.x -d 203.0.113.0/24 -j DROP
403
404 # Restrict a device to only HTTP/HTTPS
405 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 3000 -j DROP
406 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 443 -j DROP
407
408 # Block a device from internet egress entirely (mesh only)
409 iptables -I FORWARD -s 100.x.x.x -j DROP
410
411 # Time-based (e.g., cut off 10 PM 7 AM)
412 iptables -I FORWARD -s 100.x.x.x -m time --timestart 22:00 --timestop 07:00 -j DROP
413 ```
414
415 ### DNS Level (AdGuard REST API)
416 AdGuard Home supports per-client rules keyed by Tailscale IP:
417
418 ```bash
419 curl -X POST http://localhost:80/control/clients/add \
420   -H "Content-Type: application/json" \
421   -d '{
422     "name": "kids-ipad",
423     "ids": ["100.x.x.x"],
424     "blocked_services": ["youtube", "tiktok", "instagram"],
425     "safesearch_enabled": true
426   }'
427 ```
428
429 ---
430
431 ## 11. Tor & OPSEC Architecture
432
433 ### 11.1 Tor-over-WARP Topology
434
435 When extreme anonymity is required, integrating the Tor network into `nullexit` provides a powerful Tor-over-VPN topology:
436 * The University/ISP sees only an encrypted Cloudflare WireGuard tunnel, rendering them completely blind to the fact that you are using Tor (effectively bypassing Enterprise firewall Tor blocks).
437 * Cloudflare sees an encrypted TCP stream heading to a Tor Guard Node. They cannot see your DNS lookups, the destination website, or the content of your traffic.
438 * The Tor Exit Node sees your traffic, but not your real IP (only the Cloudflare IP).
439
440 ##### The "Transparent Proxy" Trap (What NOT to do)
441 It is tempting to build a system-wide "Transparent Tor Proxy" that forces all host traffic (e.g., standard Chrome/Safari browsers, background iCloud syncs) through Tor automatically. This is a massive OPSEC trap and destroys anonymity.
442 Modern operating systems and browsers will instantly leak your identity through background syncing (e.g., Apple Mail, Google Drive) and browser fingerprinting (canvas fingerprints, WebRTC, screen resolution). The Tor network protects your *IP*, but if your payload contains identifying cookies or unique fingerprints, the Tor Exit Node (which could be run by malicious actors) will instantly tie your session to your real identity.
443
444 ##### The nullexit Implementation: Isolated Procedural Proxy
445 To safely integrate Tor without triggering the transparent proxy trap, `nullexit` implements an Isolated Tor SOCKS5 Proxy with Procedurally Generated Ports:
446 1. Isolated Container: A lightweight `tor` Docker container runs securely inside the `warp` network namespace. It does not hijack system traffic.
447 2. Opt-In Usage: You must consciously configure a highly-hardened browser (like the official Tor Browser or a dedicated Firefox profile) to use the proxy. This prevents accidental background sync leaks.
448 3. Procedurally Generated Ports: Rather than hardcoding the standard proxy ports (like `127.0.0.1:9050` or `1080`), `toggle.sh` randomizes all internal proxy ports on every boot into the ephemeral range (10000-65000) and silently writes them to `/tmp/nullexit-ports.env`. This completely defeats static fingerprinting by malware.
449 4. Kernel-Level Collision Avoidance: To prevent the randomizer from picking a port already in use by a root daemon or Apple service (which would crash the gateway), the script directly queries the macOS kernel socket table (`netstat -an -p tcp`) to verify the port is absolutely free before assigning it.
450 5. The Honey-Port Tripwire: While procedural ports defeat static fingerprinting, advanced malware might attempt a full sequential port scan of `localhost` to find the hidden proxy. To counter this, `toggle.sh` spawns a "Honey-Port" (a fake random port with a netcat listener attached). If any local application attempts to connect to the Honey-Port, it instantly logs a critical security alert to `output.log` and fires a macOS desktop notification, serving as an early-warning Intrusion Detection System (IDS) for local malware.
451
452 > Threat Model note: The Tor SOCKS proxy and Honey-Port Tripwire only bind to `127.0.0.1` (loopback). Port randomization and the Honey-Port only defeat blind, generic port-scanners they do NOT protect against a targeted local process that reads `/tmp/nullexit-ports.env`, greps process memory, or runs `lsof -i`. The proper mitigation is a host-level loopback-filtering firewall (LuLu/Little Snitch). See 15.9 (Threat Model: Local Proxy Discovery) for the full analysis and the rogue-binary verification test.
453
454 ### 11.2 Hardening: Tor Bridges and obfs4 Pluggable Transports
455 For users operating under severe Deep Packet Inspection (DPI), `nullexit` supports configuring Tor to connect exclusively through private bridges using the `obfs4` pluggable transport.

```

```

456 When `TOR_USE_BRIDGES=true` is set:
457 - What it does: It hides the entry IP from being matched against the public Tor consensus list, and
458   disguises the traffic's protocol fingerprint from the WARP exit provider (Cloudflare).
459 - What it does NOT do: It does not add any extra layers of anonymity beyond what Tor already provides. It
460   is purely a censorship-circumvention and obfuscation mechanism.
461 - Limitations: `obfs4` is highly effective against passive DPI, but it is not guaranteed to defeat active-
462   probing-resistant detection by a highly sophisticated state-level adversary.
463
464 > Architecture Rule: The `nullexit` gateway deliberately does NOT bundle, hardcode, or automatically fetch
465   bridge lines. Bridge lines are highly sensitive and expire. Users must obtain their own bridge lines from
466   https://bridges.torproject.org` and manually populate the `tor-bridges.txt` file. If bridges are enabled
467   but the file is missing or empty, the Tor container will intentionally crash and fail to start rather than
468   silently falling back to public guard relays.
469
470 The Bootstrapping Paradox (Out-of-Band Key Exchange)
471 Automating the bridge acquisition process (e.g., scripting the gateway to log into the Telegram @GetBridgesBot`
472   via MTPROTO API) introduces catastrophic OPSEC flaws:
473 1. Identity Leaks: Baking a Telegram session into the container explicitly links the "anonymous" gateway to
474   a real-world physical phone number.
475 2. The Chicken & Egg Deadlock: In heavily censored regimes, Telegram itself is usually blocked. If the
476   container requires Telegram to fetch bridges to bypass the firewall, it will fail because it cannot bypass
477   the firewall to reach Telegram.
478 3. Bridge Exhaustion & CAPTCHAs: Automated scraping behaves identically to state-sponsored scrapers,
479   triggering Tor's anti-bot CAPTCHAs which leads to silent proxy failures and exhausts the limited public
480   bridge pool.
481
482 Instead, users should rely on a less secure layer to manually reach Telegram, obtain the bridges, and populate
483   `tor-bridges.txt`. This can be done via the standard `nullexit` WARP tunnel, a mobile phone on cellular
484   data, Mullvad VPN, or a secure remote VPS. (Note: We plan on fully integrating Mullvad and VPS
485   architectures directly into `nullexit` in the future, but have not had the time to build it yet).* This
486   intentionally "air-gaps" the cryptographic key exchange from the proxy infrastructure, leaving zero API
487   tokens and zero network traces for an adversary to exploit.
488
489 11.3 Tor Container Kill-Switch Inheritance & Fail-Closed Egress Verification
490
491 Context
492 Unlike the host's SOCKS5 fallback path (whose kill-switch interaction is documented in 15.7), the `tor`
493   container's connection status and fail-closed behavior under VPN tunnel drops is not governed by macOS `pf`
494   rules. Instead, it relies on network namespace inheritance inside Docker Compose.
495
496 Mechanics
497 1. Shared Network Namespace: The `tor` service is configured with `network_mode: service:warp` in `docker-
498   compose.yml`. This shares the network namespace of the `warp` (GlueTun) container.
499 2. Inherited Egress Rules: All socket communication within the namespace is bound to the same networking
500   stack. GlueTun manages this stack by redirecting traffic through `tun0` (the WireGuard/WARP tunnel) and
501   applying `iptables` rules that explicitly drop outbound traffic originating on `eth0` (the physical/bridge
502   interface), except for permitted local subnets or the VPN server's endpoint.
503 3. Tunnel Fail-Closed: If the WARP connection drops or the `tun0` interface is brought down, the routing
504   table entries for `tun0` become invalid or disappear. Because the `iptables` rules in the shared namespace
505   continue to drop any outgoing internet traffic attempting to exit via `eth0`, the `tor` container cannot
506   leak raw traffic to the host's underlying networks. Egress fails closed.
507
508 Verification / Manual Test Case
509 To manually verify that the Tor container fails closed and does not leak traffic when the tunnel dies:
510
511 1. Verify active path: Ensure the gateway is active (`toggle.sh start`) and fetch the Tor port (`
512   $TOR SOCKS PORT` in `/tmp/nullexit-ports.env`). Verify Tor traffic routes correctly:
513   ```bash
514   curl --socks5-hostname 127.0.0.1:$TOR SOCKS PORT https://check.torproject.org/api/ip
515   ```
516   *(This should output a Tor exit node IP).*
517
518 2. Simulate tunnel drop: Bring down the virtual interface inside the shared network namespace:
519   ```bash
520   docker compose exec warp ip link set dev tun0 down
521   ```
522
523 3. Re-run the egress check:
524   ```bash
525   curl --socks5-hostname 127.0.0.1:$TOR SOCKS PORT https://check.torproject.org/api/ip
526   ```
527
528 Expected Result: The request must hang and eventually time out, or immediately fail with a proxy error
529   (e.g., `curl: (97) SOCKS5: connection failed` or `curl: (7) Failed to connect`).
530 Leaked Egress Check: Check the host's Wi-Fi router or firewall logs. No outbound packets from the Tor
531   container should route directly over the bridge network to the internet.
532
533 11.4 Transparent .onion Routing via RFC 2544 Subnet (198.18.0.0/15)
534
535 Context

```

```

506 To enable seamless dark web browsing across the entire mesh, the gateway requires a mechanism to dynamically
    map `.onion` domains to IP addresses that the host operating system and network layers can route.
507
508 ##### IP Address Conflict & Solution
509 1. **Initial Conflict:** Initially, the Tor virtual address network was configured to map `.onion` sites within
    the `10.192.0.0/10` range. However, the Colima Docker daemon dynamically allocated `10.200.1.0/24` to the
    containers. Because this Docker subnet mathematically falls within the `10.192.0.0/10` block, a routing
    loop occurred: all host packets destined for the Docker containers on ports like `9050` or `9051` were
    intercepted by the transparent proxy NAT rules and dropped, causing proxy failures.
510 2. **Tailscale Private IP Exclusion:** Shifting the subnet to another private range like `10.123.0.0/16`
    resolved the port conflict, but led to connection timeouts. On macOS, Tailscale's Exit Node routing
    actively excludes RFC 1918 private subnets (e.g. `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`) to
    maintain accessibility to local LAN resources (like printers/routers). Thus, packets targeting `10.123.x.x`
    were dropped locally by Tailscale before reaching the gateway tunnel.
511 3. **The Benchmarking Subnet Solution:** To bypass these private IP exclusions without manual static routing
    tables, Tor's virtual network was changed to **198.18.0.0/15** (reserved by RFC 2544 for benchmarking).
    Since this is not a private RFC 1918 range, Tailscale exit-node routing automatically encapsulates and
    routes all traffic destined for `198.18.0.0/15` through the tunnel to the gateway.
512 4. **Transparent NAT Redirection:** The `routing-fix` sidecar applies NAT rules in the shared network namespace
    :
513     ````bash
514     iptables -t nat -I PREROUTING -d 198.18.0.0/15 -p tcp -j REDIRECT --to-ports 9040
515     ````
516     This intercepts all incoming TCP traffic targeting the mapped `.onion` range and transparently proxies it
    through Tor's transparent port (`9040`).
517
518 ##### Exit Node Exclusions
519 The gateway supports the ability to exclude specific countries from being used as exit nodes (e.g. to avoid
    certain jurisdictions or nodes with poor network latency). This is configured globally via the `
    TOR_EXCLUDE_EXIT_NODES` environment variable in `.env`, which gets dynamically appended to Tor's `
    ExcludeExitNodes` and strictly enforced with `StrictNodes 1`.
520
521 ---
522
523 ## 12. Censorship-Resistant Transport & Egress Compartmentalization
524
525 ### 12.1 Why this is needed
526 WARP can be blocked from deployment countries with strict internet controls. The most likely cause isn't that "
    encrypted traffic is blocked" in general it's that WireGuard has a recognizable handshake signature, and `
    WIREGUARD_ENDPOINT_IP=162.159.192.1` on `WIREGUARD_ENDPOINT_PORT=2408` is a fixed, easily-enumerable target
    (Cloudflare's known WARP range). That combination is exactly what IP/ASN-based and DPI-based blocking is
    good at catching.
527
528 ### 12.2 Important correction: Gluetun's `SHADOWSOCKS=on` is the wrong direction
529 Gluetun's built-in Shadowsocks option runs a Shadowsocks **server** inside the already-connected VPN tunnel, so
    other LAN devices can reach out through Gluetun's **existing** connection via the SS protocol. It is not a
    way to make Gluetun itself connect **outbound** through a Shadowsocks server Gluetun only speaks OpenVPN
    or WireGuard as its actual transport. There is no `VPN_TYPE=shadowsocks`. Confirmed against Gluetun's own
    maintainer/community discussion: people have asked for a "Shadowsocks-client" mode specifically to chain
    through Gluetun, and the standing answer is "run a separate Shadowsocks client container."
530
531 ### 12.3 Diagnose before building anything
532 The right fix depends entirely on what's actually being blocked:
533 - Point Gluetun at a **different provider/IP/ASN** (any other Gluetun-supported WireGuard/OpenVPN provider, or
    a self-hosted WireGuard server) and test. If that gets through, the block is Cloudflare/WARP-specific, not
    a general WireGuard block **no new component needed**, just swap `VPN_SERVICE_PROVIDER` / the endpoint.
534 - If WireGuard fails regardless of endpoint, the block is protocol-level (DPI fingerprinting the handshake)
    proceed to obfuscation.
535
536 ### 12.4 Path A: Swap WARP for a different Gluetun-native transport
537 Simplest fix, only applies if 12.3 shows the block is Cloudflare-specific. Change `VPN_SERVICE_PROVIDER` / `
    VPN_TYPE` / endpoint in `docker-compose.yml` and in the now-centralized `WARP_ENDPOINT_*` values in `.env`.
    Nothing else in the stack needs to change.
538
539 ### 12.5 Path B: Add an obfuscation hop
540 Needed if WireGuard itself is being fingerprinted/blocked. Two ways to slot it in:
541
542 **B1 Wrap (recommended):** Run a Shadowsocks (or obfs4/v2ray) client as a new sidecar container. Point Gluetun
    's `WIREGUARD_ENDPOINT_IP` at `127.0.0.1:<local-port>` instead of Cloudflare directly; the sidecar relays
    that traffic to a Shadowsocks server you run yourself abroad. Keeps Gluetun's kill-switch, healthchecks,
    and the entire rest of the stack (Tailscale, AdGuard, ipset, `routing-fix.sh`) untouched, since none of
    them know the transport changed underneath `warp`.
543
544 **B2 Replace:** Swap the `warp` service entirely for a Shadowsocks-client + `tun2socks` container that owns
    the network namespace instead of Gluetun. More invasive loses Gluetun's built-in kill-switch/healthcheck,
    which would need to be rebuilt by hand (see Section 12.9 for the pluggable architecture to support this
    natively).
545
546 **Recommendation: B1.** Smaller surface area, preserves the self-healing architecture already built and
    documented.
547

```

```

548 ### 12.6 Fingerprinting caveat
549 Plain Shadowsocks can still be caught by active-probing DPI in more aggressive censorship environments. If
    plain SS gets blocked too, the next step up is an obfuscation plugin (`v2ray-plugin`, `Cloak`) or a newer
    protocol designed against active probing (VLESS+Reality, Hysteria2). Test plain SS first don't over-build
    before confirming it's needed.
550
551 ### 12.7 AmneziaWG (The High-Speed Alternative)
552 If you want to maintain the bare-metal speeds of WireGuard while bypassing DPI (e.g. in Egypt or Russia), the
    best alternative to Shadowsocks/V2Ray is AmneziaWG. AmneziaWG is a fork of WireGuard that pads the
    handshake packets with randomized "junk" data, entirely destroying the 148-byte DPI signature that
    firewalls look for.
553 - **Pros:** Because it runs entirely as UDP without TCP-over-TCP overhead, it completely avoids "TCP meltdown"
    latency spikes associated with Shadowsocks/V2Ray wrappers.
554 - **Cons:** You cannot use Cloudflare WARP. Furthermore, it requires completely ripping Gluetun out of the `
    nullexit` stack and building a custom Docker container, meaning you lose Gluetun's built-in kill-switches
    and health-check logic.
555
556 ### 12.8 Infrastructure Costs & Privacy
557 To use either Shadowsocks (Path B) or AmneziaWG, you cannot use the free Cloudflare WARP infrastructure,
    because Cloudflare servers only speak standard WireGuard. The software for both custom protocols is 100%
    free and open-source, but you must host the remote server infrastructure yourself.
558 - **Free Tier (Oracle Cloud):** You can host your remote server on Oracle Cloud's "Always Free" tier for $0.
    However, this routes your highly sensitive, censorship-evading traffic directly through Oracle's massive
    corporate data broker. If privacy is the core objective, handing all your metadata to Oracle defeats the
    purpose.
559 - **Paid Tier (Hetzner / Mullvad):** The recommended path is to rent a standard ~$4/month Virtual Private
    Server (VPS) in a free country (e.g., Hetzner in Germany, subject to strict EU GDPR privacy laws) and self-
    host the server. Alternatively, Mullvad VPN (5/month) provides native v2ray/Shadowsocks bridges built into
    their network, allowing you to bypass censorship with a strict zero-logs policy. It is highly recommended
    to pay with untrackable payment methods to maintain complete anonymity, which these providers typically
    support without requiring local residency.
560
561 ### 12.9 The Future: Egress Compartmentalization (Pluggable Architecture)
562 To natively support invasive replacement paths like **Path B2** or **AmneziaWG** (Section 12.7) without
    breaking the core `nullexit` routing and monitoring logic, the architecture must be refactored into a
    modular, "pluggable" design.
563
564 1. **The Egress Plugin System:** Currently, WARP-specific logic (like `cdn-cgi/trace` health checks and
    hardcoded interface names) is scattered across `toggle.sh` and `scripts/routing-fix.sh`. This should be
    abstracted into a `scripts/plugins/` directory, where each egress method (`warp.sh`, `v2ray.sh`) implements
    standardized functions: `get_egress_interface()`, `check_health()`, and `get_bypass_ips()`.
565 2. **Dynamic Docker Compose:** Based on an `EGRESS_TYPE` variable in `.env`, dynamic compose file generation
    would spin up only the required egress containers (e.g., skipping the `warp` container when `EGRESS_TYPE=
    v2ray` is set).
566 3. **Agnostic Routing & Watchers:** `routing-fix.sh` would dynamically read the `EGRESS_INTERFACE` from the
    active plugin to apply `iptables` rules, and the background watchers in `toggle.sh` would become an
    agnostic `Egress Watcher` that invokes `check_health()` periodically.
567
568 This compartmentalization transforms `nullexit` into a universal, censorship-resistant gateway where the entire
    egress layer can be swapped by changing a single `.env` variable and running `./toggle.sh --restart`.
569
570 ---
571
572 ## 13. Roaming & Recovery Design: How nullexit Mimics Commercial VPNs
573
574 The `nullexit` roaming/network recovery subsystem replicates the exact engineering mechanics used by commercial
    , premium VPN clients (like Mullvad or NordVPN) to transition across Wi-Fi networks safely and seamlessly.
    This is the design overview; the specific bugs encountered and fixed along the way live in the Incident Log
    (15.5 Roaming/Sleep-Wake, 15.7 Tailscale P2P/SNAT).
575
576 ### The 4-Step Transition Cycle
577 When your Mac disconnects from one Wi-Fi and associates with another, `nullexit` coordinates the following
    events:
578
579 1. **Firewall Lock (Kill-Switch):** The macOS Packet Filter (PF) anchor `com.apple/nullexit` remains locked on
    the physical interface (`en0`), blocking all direct outbound IPv4 and IPv6 traffic. This guarantees that
    your real ISP IP or unencrypted DNS requests **never leak** onto the new network while the connection is
    rebuilding.
580 2. **System Event Interception:** A launchd LaunchAgent (`watcher.sh`) listens to the macOS System
    Configuration framework for the `State:/Network/Global/IPv4` key changes, spawning `recover.sh --post-wake`
    in under 500ms when a change is detected.
581 3. **Bypass Routing:** The script dynamically resolves the new physical network's IP gateway (e.g.
    `192.168.137.1`), cleans up stale bypass routes from the previous network, and adds static bypass routes
    pointing to Cloudflare's WARP endpoints and the Tailscale control plane directly via the new gateway. This
    allows the VPN client to negotiate its handshake outside of the tunnel.
582 4. **Hole-Punching / NAT Re-negotiation:** The script runs a target check on the `warp` container's tunnel. If
    the WireGuard connection is wedged by the network switch, it force-recreates the container to establish a
    fresh NAT binding to Cloudflare, restoring connection in ~15 seconds.
583
584 ---
585

```

```

586 ## 14. Deployment, Packaging & Known Limitations
587
588 ### 14.1 Packaging nullexit as a macOS Application (.dmg)
589
590 nullexit can be packaged into a native `.app` bundle, but this introduces nested virtualization requirements.
591
592 nullexit uses **Colima** (Apple Virtualization Framework `vz`) to run a Linux VM hosting Docker. Installing the
593   `.app` inside a virtualized macOS environment (Parallels, cloud Mac) means running a Linux VM inside a
594   macOS VM requiring hardware-level **nested virtualization**.
595
596 ##### Hardware Support
597 - **Supported:** Apple Silicon M3/M4 (macOS Sequoia 15+), Intel Macs with VMX passthrough.
598 - **Unsupported:** M1, M2, A18 Pro. Colima crashes silently; the terminal (`setup.sh`) surfaces crash logs.
599
600 ##### Build Steps
601 ```bash
602 # 1. Verify nested virtualization support
603 sysctl -a | grep hv_nested_virt_supported # expect: 1
604
605 # 2. Compile the AppleScript launcher
606 osacompile -o "Nullexit.app" "Toggle Gateway.applescript"
607
608 # 3. Embed scripts into app resources
609 mkdir -p "Nullexit.app/Contents/Resources/scripts"
610 cp toggle.sh setup.sh recover.sh docker-compose.yml "Nullexit.app/Contents/Resources/"
611
612 # 4. Package into DMG
613 hdiutil create -volname "Nullexit Installer" -srcfolder "./Nullexit.app" -ov -format UDZO Nullexit.dmg
614
615 # 5. Bypass Gatekeeper (unsigned app)
616 xattr -cr /Applications/Nullexit.app
617 ```
618
619 Without a paid Apple Developer certificate ($99/yr), users see an "App is damaged" Gatekeeper warning and need
620 to run step 5.
621
622 ### 14.2 Moonlight/Sunshine + Tailscale Race Condition
623
624 When streaming from a remote Windows host running Sunshine over a Tailscale mesh (e.g., while the client is on
625 a cellular hotspot), a race condition can occur on boot:
626
627 1. Both Tailscale and Sunshine are set to start automatically on boot.
628 2. Tailscale takes a few seconds to fully initialize its `100.x.x.x` virtual adapter and establish a connection
629 3. **Sunshine starts too fast.** It launches before Tailscale is fully ready. Since Sunshine only binds to
630   network interfaces that are active at the exact moment of startup, it misses the Tailscale adapter entirely
631 4. Moonlight clients (even when configured with the correct Tailscale IP) will fail to connect because Sunshine
632   isn't listening on that interface.
633
634 **The "Magic Toggle" Symptom:**
635 If the user manually enables or disables Wi-Fi on the host PC, it triggers a global Windows network change
636 event. Sunshine listens for these events, re-enumerates all network adapters, and says, "Oh, there's a
637 Tailscale adapter here!" and finally binds to it. Suddenly, Moonlight can connect.
638
639 **The Permanent Fix:**
640 On the Windows host, open `services.msc`, locate the **Sunshine Service**, and change its Startup type from **Automatic** to **Automatic (Delayed Start)**. This forces Windows to wait a minute or two after booting
641 before launching Sunshine, ensuring Tailscale's virtual adapter is fully initialized first.
642
643 ### 14.3 Chrome Remote Desktop Performance (UDP NAT)
644
645 ##### Observation (July 11, 2026)
646 When the user connects to their host Mac using Chrome Remote Desktop (CRD) while `nullexit` is active, the
647 connection suffers from massive latency, lag, and degraded video quality. AdGuard Home's `blocked.log`
648 shows zero blocked DNS requests for Google, WebRTC, or `chromoting` endpoints, indicating this is not a DNS
649 sinkhole issue. *(See also 15.11 for the separate CRD-on-remote-device connection-failure quirk.)*
650
651 ##### Root Cause
652 This is a fundamental limitation of tunneling all host traffic through a strict commercial NAT (Cloudflare WARP
653 ).
654 1. Chrome Remote Desktop attempts to use **STUN (UDP hole-punching)** to establish a lightning-fast, direct
655   peer-to-peer WebRTC connection between the client device and the host Mac.
656 2. Because all non-Tailscale traffic is forcefully routed through the `warp` container, the STUN packets exit
657   the network from a Cloudflare IP.
658 3. Cloudflare's enterprise NAT infrastructure strictly drops unsolicited inbound UDP packets. The UDP hole-
659   punch fails.
660 4. Because a direct P2P connection cannot be established, CRD falls back to **Relay Mode** (TURN), bouncing all
661   video data back and forth through Google's cloud servers instead of sending it directly over the local or
662   mesh network. This relaying introduces massive latency.

```

```

646 ##### Workaround / Fix
647 This lag is the accepted "tax" for maintaining absolute cryptographic anonymity; the only way to restore CRD
648 speed would be to leak its UDP traffic outside the tunnel (violating zero-trust).
649 To achieve low-latency remote access, users should bypass CRD entirely and use macOS's built-in **Screen
650 Sharing (VNC)** directly over the Tailscale IP. Tailscale's sophisticated custom DERP/WireGuard
651 architecture successfully UDP hole-punches through strict NATs, providing a direct, encrypted, high-speed
652 connection without relying on external cloud relays.
653
654 ### 14.4 Dynamic IP Fetch vs. MagicDNS
655 ##### Observation
656 The `nullexit` gateway uses a dynamically fetched IP address (cached in `.gateway_ip`) to configure host
657 routing and the `pf` kill-switch, rather than utilizing Tailscale's user-friendly MagicDNS hostname (e.g.,
658 `nullexit-gateway`).
659
660 ##### Why MagicDNS is Not Used
661 1. **The "Chicken and Egg" Bootstrapping Problem:**
662 macOS's Packet Filter (`pf`) and low-level routing commands (`route add`) operate strictly at Layer 3 and
663 require raw IP addresses. If `toggle.sh` or `recover.sh` attempted to use `nullexit-gateway`, the host Mac
664 would need to perform a DNS lookup to resolve it. However, the very first step of the gateway's
665 initialization is to completely hijack the host's DNS and point it *at the gateway itself*. The Mac cannot
666 resolve the gateway's MagicDNS name if it needs the gateway's IP to know where to send the DNS query!
667 2. **Dynamic Self-Healing (Avoiding Stale Cache):**
668 To solve the bootstrapping problem without causing bugs if the node is deleted and re-authenticated on the
669 Tailscale Admin Console, `toggle.sh` explicitly runs `docker compose exec ... tailscale ip -4` during boot
670 to fetch the absolute freshest IP dynamically. It saves this to `.gateway_ip` so that fast-roaming scripts
671 like `recover.sh` can instantly retrieve the routing target without incurring the latency of querying
672 Docker.
673
674 ### 14.5 Battery & Power Measurement Quirks
675
676 When trying to optimize this framework (or any daemon) for battery life, developers often look for a way to
677 programmatically measure the exact Wattage consumed by a specific process (e.g., "How many Watts is `toggle
678 .sh` using?").
679
680 **This is a hardware impossibility on macOS and Linux.**
681
682 ##### Why Activity Monitor is Lying to You
683 Your machine's logic board only has physical power sensors for the *entire* CPU package, the GPU, and the RAM.
684 It knows the CPU is pulling 5W total, but it has no physical hardware capability to know whether Docker is
685 using 3W and Chrome is using 2W.
686
687 The "Energy Impact" score you see in macOS Activity Monitor is a **synthetic, heuristic score** calculated by `
688 powerd`. It penalizes apps for waking up the CPU (idle wakes), doing disk I/O, or keeping the screen awake.
689 It is a relative score, not actual Wattage.
690
691 ##### The Proper A/B Testing Methodology
692 If you want to truly know how many Watt-hours or mAh your background daemons (`routing-fix.sh`, the DNS/WARP
693 watchers, etc.) are costing you, you must perform a hardware-level A/B drain test:
694
695 1. Unplug the laptop, run the gateway, and note your exact battery mAh using:
696 ```bash
697 ioreg -l | grep "AppleRawCurrentCapacity"
698 ```
699 2. Leave the laptop idle for exactly 1 hour.
700 3. Run the command again to see exactly how many mAh were drained.
701 4. Turn the gateway off and repeat the test for another hour.
702
703 The delta in mAh drained between the two tests is the true, exact cost of running the software. When optimizing
704 polling loops (e.g., changing `sleep 5` to `sleep 30`), this is the only reliable way to measure the
705 actual battery life returned to the user.
706
707 ### 14.6 Host Lockdown Mode (Why It's Impossible on macOS)
708
709 A common privacy feature request is a "Host Lockdown Mode" (Mode 3): A mode where the Docker exit node remains
710 perfectly functional for remote devices, but the host machine itself is completely blocked from accessing
711 the internet (to prevent even encrypted host traffic from leaking metadata).
712
713 **Why this is impossible on macOS:**
714 On Linux (e.g., a Raspberry Pi), Docker runs natively. The host's traffic traverses the `OUTPUT` iptables chain
715 , while Docker traffic traverses the `FORWARD` chain. We could easily drop all `OUTPUT` traffic to kill the
716 host's internet, and Docker would continue routing traffic perfectly.
717
718 However, Docker on macOS runs inside a Linux Virtual Machine (via `qemu` or Apple `Virtualization.framework`).
719 This VM runs as a standard macOS user-space application. If you configure the macOS firewall (`pf`) to drop
720 all host outbound traffic, it will indiscriminately drop the VM application's traffic as well, instantly
721 killing the Cloudflare WARP tunnel and the exit node.
722
723 Writing complex macOS `pf` rules to "allow the VM app, allow Cloudflare WARP IPs, allow Tailscale DERP IPs, but
724 block everything else" is highly fragile and heavily discouraged. Since the current `toggle.sh` default
725 mode already fully encrypts the host's traffic via the WARP tunnel, the host is already protected from ISP

```

```

leaks.
695
696 ---
697
698 # Part III Incident Log & Resolved Issues
699
700 This is the single, deduplicated log of every incident, bug, and resolved issue. Entries are organized into
    subsystem groups (15.115.12). Each entry follows a **Symptom / Root Cause / Fix** shape where applicable;
    packet-level analyses are preserved as clearly-labeled **Deep dive** blocks. The terse dated changelog
    lives in 17; this section holds the full post-mortems.
701
702 ## 15. Incident Log
703
704 ### 15.1 Exit-Node Return Path & SOCKS5 Failover
705
706 ##### 15.1.1 Exit Node Return Path (`rx 0`) & The SOCKS5 Failover RESOLVED
707 **Symptom:** Gateway showed `tx 936 rx 0` transmitted but never received return traffic. For days, Tailscale's
    exit node refused to return traffic back to the client, leading to the assumption that Tailscale's
    userspace forwarding was bypassing `iptables` and ignoring the WARP `tun0` interface. In a desperate
    attempt to fix this, a **SOCKS5 proxy** (`scripts/socks5-proxy.py`) was introduced as a replacement.
708
709 **Root Cause:** Tailscale's userspace forwarding was not the problem. `docker-compose.yml` included `
    FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`. Gluetun created a strict routing rule (`ip rule add to
    100.64.0.0/10 lookup 199`, priority 99) that forced all Tailscale CGNAT traffic to bypass the VPN via the
    unencrypted Docker bridge (`eth0`). Return packets were sent to the macOS host instead of back through `
    tailscale0`, blackholing all return packets back to the client.
710
711 **Fix:** Removed `FIREWALL_OUTBOUND_SUBNETS` entirely, immediately restoring the Tailscale exit node. `scripts/
    routing-fix.sh` now injects `lookup 52`, and return packets correctly flow back into `tailscale0`. Instead
    of removing the SOCKS5 proxy, we repurposed it into a **bulletproof failover** for when restrictive Wi-Fi
    networks block Tailscale UDP ports.
712
713 Architecture after the fix:
714 ```
715 PRIMARY (Exit Node):
716 DNS/TCP/UDP: Apps Tailscale Exit Node gateway container tun0 WARP internet
717
718 FAILOVER (SOCKS5 - if Tailscale is blocked by local Wi-Fi):
719 DNS: Browser 127.0.0.1:53 Python proxy TCP:5354 AdGuard WARP
720 TCP: Apps SOCKS5:1080 gateway container tun0 WARP internet
721 Mesh: SSH/ping 100.x.x.x utun5 WireGuard peers (independent of proxy)
722 ```
723
724 **Deep dive: Exit Node Return Path Analysis (June 26, 2026).**
725 This preserves the detailed packet-level analysis that led to identifying and fixing the `rx 0` bug.
726
727 *The exit node was probably never going to work in this container topology.* Here's the packet path traced by
    reading `ip rule show` and the routing tables live:
728
729 **Outbound (Phone-1 internet):**
730 1. Phone-1 sends packet to gateway via Tailscale mesh
731 2. Gateway's tailscale0 (kernel TUN, `TS_USERSPACE=false`) decrypts packet enters FORWARD chain
732 3. Packet has src=`100.87.42.87` (Phone-1), dst=some internet IP
733 4. FORWARD chain (nftables): Rule 1 matches (`-i tailscale0 -j ACCEPT`)
734 5. Routing decision for the forwarded packet. Which ip rule matches?
735 - Rule 98: `to 172.18.0.0/16` no (dst is internet)
736 - Rule 99: `to 100.64.0.0/10` no (dst is internet)
737 - Rule 100: `from 172.18.0.2` **NO** (src is `100.87.42.87`, not the container IP!)
738 - Rule 101: `not fwmark 0xca6c lookup 51820` **YES** (no fwmark on forwarded packets)
739 6. Table 51820: `default dev tun0` packet goes out through WARP
740 7. NAT POSTROUTING: `MASQUERADE on tun0` src becomes WARP IP
741 8. Packet exits through WARP to internet (in theory)
742
743 **Return (internet Phone-1):**
744 1. Return packet arrives on tun0, conntrack de-MASQUERADES dst back to `100.87.42.87`
745 2. Routing decision for dst=`100.87.42.87`:
746 - Rule 99: `to 100.64.0.0/10 lookup 199` **YES**
747 3. Table 199: `100.64.0.0/10 via 172.18.0.1 dev eth0` **sends it to the Docker gateway!**
748 4. **This is wrong.** The packet should go to tailscale0 (to be re-encrypted and sent to Phone-1), but table
    199 sends it out eth0 to the Docker bridge host.
749
750 **Table 199 is the smoking gun.** It has one route: `100.64.0.0/10 via 172.18.0.1 dev eth0`. This sends ALL
    Tailscale CGNAT return traffic out the Docker bridge to the host not back through tailscale0. The return
    packet leaves the container's network namespace entirely, hits the host's routing stack, and gets dropped
    because the host has no idea what to do with a raw packet for `100.87.42.87`.
751
752 **Why table 199 exists and why it's wrong for exit node traffic.* Table 199 + the CGNAT rule (pref 99) was
    designed so that the **container itself** can reach other Tailscale peers (e.g., for mesh management, DERP
    relay). For that use case, routing CGNAT traffic via the Docker bridge to the host (which has its own
    tailscaled) makes sense. But for **forwarded exit node traffic**, the return packet needs to go back
    through tailscale0 inside the container not out to the host. The CGNAT rule intercepts the return packet

```

```

before it can be routed back to tailscale0 through the main table or Tailscale's own routing (table 52,
which is at pref 5270).
753
754 *The SOCKS5 path works because it dodges all of this.* SOCKS5 proxy creates connections **from the container's
own IP** (`172.18.0.2`). So:
755 - ip rule 100 (`from 172.18.0.2 lookup 200`) matches
756 - Table 200 has `default dev tun0`
757 - Return traffic comes back to `172.18.0.2` on tun0, conntrack de-NATs, proxy sends response to client
758
759 No FORWARD chain involved. No CGNAT rule involved. No table 199. It just works.
760
761 *Summary of findings:*
762
763 | Finding | Status | Evidence |
764 |-----|-----|-----|
765 | Table 199 CGNAT route hijacks exit node return traffic | **Fixed** | Changed `lookup 199` to `lookup 52` in `
scripts/routing-fix.sh`. Return packets now route via tailscale0. |
766 | FORWARD RELATED, ESTABLISHED missing | **Fixed** | Installed `iptables` via apk in `docker-compose.yml`. Rule
now injects successfully. |
767 | DOCKER_GW variable parsing error | **Fixed** | Added `head -1` in `scripts/routing-fix.sh`. |
768 | `FIREWALL_OUTBOUND_SUBNETS` was the root cause | **Fixed** | Removed from `docker-compose.yml`. Gluetun no
longer hijacks CGNAT routing. |
769 | Host tailscaled error state from aggressive `tailscale down` | **Mitigated** | Toggle script now detects and
auto-restarts. |
770
771 ##### 15.1.2 Container Default Route Bypasses WARP
772 **Problem:** Inside the WARP container's network namespace, the main routing table's default route was via `
eth0` (Docker bridge), not `tun0` (WARP tunnel). While gluetun uses policy routing (rule 101 table 51820
`tun0`) for most traffic, traffic originating from the container's own IP (`172.18.0.2`) hits rule 100 (`
from 172.18.0.2 lookup 200`), whose default was also via `eth0`. This caused the SOCKS5 proxy's outbound
connections to bypass WARP.
773
774 **Fix:** Updated the `routing-fix` container to:
775 1. Change the main table's default route to `default dev tun0`
776 2. Change table 200's default route to `default dev tun0`
777 3. Add a specific route for the WARP WireGuard endpoint (`162.159.192.1`) through `eth0` via the Docker gateway
in table 200, preventing a tunnel loop
778 4. Re-assert all routes every 30 seconds (gluetun may reset them on health checks)
779 5. Dynamically detect the Docker subnet from `ip route show dev eth0` instead of hardcoding `172.18.0.0/16`
780
781 ##### 15.1.3 Hardcoded Docker Subnet in Routing Fix
782 **Problem:** The `routing-fix` container hardcoded `172.18.0.0/16` and `172.18.0.1` for the Docker bridge
subnet and gateway. If Docker's bridge network changed (after `docker network prune`, Colima restart, or on
a different machine), these routes would be wrong.
783
784 **Fix:** Detection is now dynamic using `ip route show`:
785 - `DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
786 - `DOCKER_GW=$(ip route show default | awk '{print $3}')
787
788 This extracts the actual subnet and gateway directly from the routing table, working with any subnet size
(`/16`, `/24`, `/20`, etc.).
789
790 ### 15.2 Routing Loops & Subnet Collisions
791
792 ##### 15.2.1 The Infinite Recursive Routing Loop (Hotspot Paradox)
793 **Problem:** A critical network topology crash occurs if the host Mac connects to a Mobile Hotspot (e.g., a
Windows PC or Phone) that is *also* connected to the Tailscale mesh and has "Use Exit Node" enabled.
Because the Mac is hosting the Exit Node, it routes its internet traffic to the Hotspot. The Hotspot
intercepts the Mac's traffic on its way to the cellular tower and routes it *back* to the Exit Node (the
Mac) via Tailscale. The Mac receives it, tries to send it out again, and the Hotspot intercepts it again.
This creates an infinite, recursive encryption loop that instantly destroys network bandwidth and drops all
connections.
794
795 **Fix:** This is a physical routing paradox that cannot be resolved in software. The upstream physical router
providing internet to the Mac *must* be allowed to route its traffic directly to the physical WAN
interface. The user must manually disable "Use Exit Node" on the upstream device providing the hotspot.
796
797 **Advanced Workaround:** If the upstream hotspot is a Windows machine and the user explicitly wants it to
remain connected to the Exit Node, a permanent static route can be injected into the Windows routing kernel
to forcefully bypass the Tailscale WFP interception for WARP packets. By binding the route to the physical
adapter's Interface Index (IF), it becomes a permanent, roaming-aware bypass.
798
799 On Windows (Admin Command Prompt):
800 ```cmd
801 route print (find Interface List, note the IF number for the active Wi-Fi adapter, e.g. 15)
802 route -p add 162.159.192.1 mask 255.255.255.255 0.0.0.0 IF 15
803 route -p add 162.159.193.1 mask 255.255.255.255 0.0.0.0 IF 15
804 ```
805
806 On Linux (Root Terminal):
807 ```bash

```

```

808 ip route add 162.159.192.1 dev wlan0
809 ip route add 162.159.193.1 dev wlan0
810 ```
811
812 On macOS (Root Terminal):
813 ```bash
814 route add -host 162.159.192.1 -interface en0
815 route add -host 162.159.193.1 -interface en0
816 ```
817
818 *(Note: If the upstream router is an iPhone or Android device, you cannot inject static routes without
    jailbreak/root. You must simply disable "Use Exit Node" in the mobile Tailscale app).*
819
820 This punches a tiny hole straight through the paradox, letting the Mac's WARP packets escape to the physical
    internet while keeping the rest of the upstream router's traffic securely trapped in the Tailscale Exit
    Node.
821
822 ##### 15.2.2 Infinite Egress Routing Loops & Subnet Takeover Paradoxes
823 **Problem:** Two distinct network loops can break host egress connectivity entirely when the exit node is
    active:
824
825 1. **The Recursive Host-VM Tunnel Loop:**
826 - **Mechanism:** The host Mac's default route points to the Tailscale exit node (`utun*`). The exit node
    container sends its encrypted tunnel packets (destined for the WARP endpoint `162.159.192.1`) back to the
    host via the VM NAT. Without a static route exception on the host Mac, the host Mac routes those packets
    right back into the exit node (`utun*`), creating an infinite loop that blackouts all host network egress.
827 - **ARP-Scoping Pitfall:** Simply binding the bypass route to the interface (`route add -host 162.159.192.1
    -interface en0`) fails when the default route is deleted or redirected to `utun*`. Since `162.159.192.1` is
    not local to the physical link, macOS fails to resolve it via ARP, dropping all tunnel packets.
828 - **Fix:** Dynamically resolve the active physical gateway IP (e.g. `172.17.0.1`) on startup, and add the
    static host routes for the WARP endpoints via the gateway IP directly (`route add -host 162.159.192.1 <
    gateway_ip>`). Additionally, because standalone `tailscaled` on macOS does not automatically modify the
    system default route via the CLI, we manually redirect the default gateway to `utun*` using `
    setup_exit_node_routing`.
829
830 2. **The Docker Compose Subnet Takeover Collision:**
831 - **Mechanism:** To avoid collisions with the host's campus network, Colima's default bridge (`docker.bip`)
    is moved (e.g. to `10.200.0.1/24`). However, because `172.17.0.0/16` is now free inside the VM, Docker
    Compose's IPAM dynamically takes it over for the project's custom network (`nullexit_default`). Because the
    containers receive IPs on `172.17.0.0/16`, the host Mac cannot send local peer discovery packets (e.g.,
    Tailscale `magicsock` packets on `172.17.0.2:41641`) directly. The host routing table sends them out `en0`
    to the physical Wi-Fi, returning `host is down` or `no route to host`.
832 - **Fix:** Explicitly configure a custom default network subnet `10.200.1.0/24` in `docker-compose.yml` to
    prevent Docker Compose from ever reclaiming the conflicting `172.17/172.18` ranges.
833
834 > See also 15.2.1 (The Hotspot Paradox) for the related infinite loop that occurs when the physical upstream
    router is also a Tailscale exit-node client.
835
836 ##### 15.2.3 Docker Default Bridge vs Host Wi-Fi Subnet Collision (The 172.17.0.0/16 Subnet Overlap)
837 **Context:** When attempting to ping a local Windows client (`172.17.22.187`) or the default gateway
    (`172.17.0.1`) directly over Wi-Fi, the Mac host returns `No route to host` (displaying a `REJECT` / `!`
    flag in the routing table). Tailscale on the client also drops offline shortly after starting when using
    the exit node, failing to establish direct P2P connections and falling back to slow DERP relays.
838
839 **Root Cause & Subnet Overlap Mechanism:**
840 1. **Docker Default Subnet:** By default, the Docker daemon initializes its default bridge network (`docker0`)
    on the `172.17.0.0/16` IP range.
841 2. **Subnet Conflict:** The host's physical Wi-Fi network happens to be assigned the exact same
    `172.17.0.0/16` range by the local network router.
842 3. **Routing Clash:** Colima launches a Linux VM on the Mac host to run the Docker engine. Because Docker
    inside that VM creates `docker0` and claims the `172.17.0.0/16` subnet, any packet sent to `172.17.x.x`
    from within the containers (such as Tailscale local discovery) is captured by the VM's internal virtual
    bridge interface (which is down/empty) and is blackholed. On the Mac host, macOS dynamically marks routes
    as `REJECT` (!) when local ARP resolution fails, causing local pings to immediately abort with `No route
    to host`.
843 4. **Tailscale Relay Saturation:** Because local peer-to-peer discovery is blackholed, Tailscale falls back to
    public DERP relay servers (e.g. `relay "tor"` in Toronto). When exit-node routing is enabled, all client
    web traffic is routed through the relay. Public relays impose strict rate limits and throttle high-volume
    traffic, resulting in packet drops. This saturation drops Tailscale's control packets, causing the
    connection to time out and showing the client as offline.
844
845 **Fix:**
846 The canonical script for this is `scripts/fix-docker-bridge-collision.sh`. **One command applies the entire fix
    :**
847
848 ```bash
849 bash scripts/fix-docker-bridge-collision.sh # apply
850 bash scripts/fix-docker-bridge-collision.sh --dry # preview changes without applying
851 ```
852

```

```

853 It auto-detects the host's actual Wi-Fi subnet (no more guessing whether the campus DHCP handed out `172.x`,
    `10.x`, or `192.168.x`), picks a non-conflicting `docker.bip` from a small candidate set (defaulting to
    `172.26.0.1/24` if no overlap is matched), backs up `~/colima/default/colima.yaml` with an ISO-8601
    timestamp, edits the file **in place** (idempotent replaces an existing `docker.bip` if present, appends
    under an existing `docker:` block, or creates a fresh block), restarts Colima, rebinds Wi-Fi DHCP, verifies
    the new default route is no longer Docker's bridge, and re-runs both `./toggle.sh` and `scripts/diagnose-
    host-leak.sh` to print the post-fix verdict.
854
855 For reference, the underlying manual fix is: edit the Colima configuration file (`~/colima/default/colima.yaml
    `) on your Mac to define:
856 ```yaml
857 docker:
858   bip: 172.26.0.1/24   # any /24 that does NOT overlap your Wi-Fi subnet
859 ```
860 Then, restart Colima to apply the new subnet inside the VM:
861 ```bash
862 colima restart
863 ```
864 This forces Docker to use `172.26.x.x` for its default bridge, freeing the physical `172.17.x.x` range and
    letting pings and direct Tailscale peer-to-peer connections flow normally.
865
866 ##### 15.2.4 July 4, 2026: The "Network Unreachable" Host Leak & Post-Wake Container Destructions
867 **Symptom:**
868 1. Running `docker compose config` with dummy keys corrupted the user's `.env` file by appending invalid
    WireGuard keys. This caused the `warp` container to become unhealthy on boot, which triggered a full
    teardown of the gateway by `toggle.sh`.
869 2. After fixing `.env`, `toggle.sh` successfully booted the gateway, but the host's internet started bypassing
    the tunnel entirely (a massive host leak). The Cloudflare trace returned `warp=off` and exposed the host's
    physical ISP IP.
870 3. Running `recover.sh --post-wake` would violently force-recreate all gateway containers and drop the host's
    internet connection.
871
872 **Root Cause:**
873 - **Bug 1 (The Host Leak):** In `toggle.sh`, the logic to find the Tailscale `utun` interface used `ifconfig |
    grep -B4 "inet 100."`. Due to the 4 lines of before-context, this regex was accidentally capturing the
    interface *above* the actual Tailscale interface (e.g. grabbing `utun4` instead of `utun5`). Because `utun4`
    had no IP address, the subsequent `sudo route add -net 0.0.0.0/1 -interface utun4` silently failed with "
    Network is unreachable", leaving the host's default route exposed.
874 - **Bug 2 (The Post-Wake Destructions):** The health check in `recover.sh --post-wake` relied on `docker
    compose exec -T warp curl -sf https://www.cloudflare.com/cdn-cgi/trace`. However, the newer `qmcgaw/glutun`
    Alpine-based Docker image no longer ships with `curl` pre-installed. The health check failed with `
    executable file not found in $PATH`, tricking `recover.sh` into thinking the tunnel was completely dead and
    needed to be forcefully recreated via `docker compose up -d --force-recreate`.
875 - **Bug 3 (Restart Flags):** Previous refactors incorrectly permitted `toggle.sh restart` instead of strictly
    enforcing `toggle.sh --restart`.
876
877 **Resolution:**
878 - Cleaned the `.env` file of duplicate dummy entries.
879 - Replaced the brittle `grep -B4` interface parsing in `toggle.sh` with a strict AWK parser: `awk '/^[a-z0
    -9]+/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ' '`. This mathematically isolates the exact
    interface possessing the Tailscale `100.x.x.x` IP address, preventing the `0.0.0.0/1` route from silently
    failing.
880 - Changed the health check in `recover.sh` to use `wget -qO- --timeout=3` which is natively installed in the
    Glutun image. This stopped the unnecessary destructive recreations of the containers during post-wake
    events.
881 - Reverted the `restart` alias in `toggle.sh` to strictly require `--restart`.
882
883 ##### 15.2.5 July 4-5, 2026: Tailscale Data-Plane Loop (The DERP Relay Deadlock)
884 **Symptom:**
885 After bypassing the Tailscale control plane (`192.200.0.0/16`) to fix the "offline" mesh status, the Mac
    appeared online. However, external peer devices (like the user's phone on cellular) could not successfully
    ping or SSH into the Mac. `tailscale netcheck` reported `Nearest DERP: unknown (no response to latency
    probes)`.
886
887 **Root Cause:**
888 While bypassing the control plane kept the daemon connected to the coordination servers, Tailscale's actual
    peer-to-peer data plane relies on STUN (for NAT traversal) and DERP relays. These relays span ~80
    dynamically allocated IP addresses across multiple cloud providers. Because we were forcing `0.0.0.0/1`
    through the `utun` exit node, the Mac's own outbound connection attempts to DERP relays were being routed
    back into the tunnel. To prevent an infinite loop, Tailscale's daemon dropped its own packets when it saw
    them returning on the TUN interface. This effectively left the daemon deaf and blind to peer connections.
889
890 **Resolution:**
891 - Modified `add_warp_bypass_routes` in `scripts/common.sh` to execute a live API fetch (`curl -s https://login.
    tailscale.com/derpmap/default`) during startup.
892 - The script parses out all active DERP relay IPv4 addresses (~80 IPs) and adds a physical host bypass route
    for every single one of them.
893 - These IPs are written to a temporary file (`/tmp/nullexit-derp-ips.txt`) so they can be cleanly un-routed by
    `remove_warp_bypass_routes` when the gateway shuts down. Peer connectivity (ping, SSH, SFTP) was fully
    restored.
894

```

```

895 ##### 15.2.6 July 5, 2026: Hardcoded WARP Endpoints in docker-compose
896 **Symptom:**
897 The bash scripts allowed overriding the default Cloudflare WARP IP endpoints via `.env` variables (`
WARP_ENDPOINT_1` and `WARP_ENDPOINT_2`) to establish bypass routes. However, `docker-compose.yml`
statically hardcoded `162.159.192.1` for the `warp` container's `WIREGUARD_ENDPOINT_IP`.
898
899 **Root Cause & Risk:**
900 If a user set a custom endpoint in `.env`, the script would successfully bypass the custom IP on the host level
. However, Gluetun would still attempt to connect to the hardcoded `162.159.192.1`. Since `162.159.192.1`
was no longer explicitly bypassed, its packets would be sucked into the `utun*` exit node, causing an
infinite WireGuard routing loop that instantly breaks the gateway.
901
902 **Resolution:**
903 Modified `docker-compose.yml` to dynamically read the environment variable via `${WARP_ENDPOINT_1
:-162.159.192.1}`, ensuring both the host routing scripts and the container use the exact same endpoint.
904
905 ##### 15.2.7 Incident Post-Mortem: WARP Watcher Race vs Post-Wake (July 10, 2026)
906 **Symptom:** After switching Wi-Fi networks, the host IP appeared as the raw ISP IP (`132.x.x.x`) instead of a
Cloudflare/WARP IP. The gateway appeared to complete post-wake recovery successfully in the logs, but
nuclear shutdown fired immediately after and killed everything.
907
908 **Root Cause:** A fatal race condition between two concurrent processes:
909
910 1. **Wi-Fi roam** `watcher.sh` fires `recover.sh --post-wake`
911 2. **Post-wake** finds the `warp` container missing/dead calls `docker compose up -d --force-recreate` (~35
seconds)
912 3. **In parallel**, the WARP Watcher (running as a child of `toggle.sh`) polls the warp container every 5s
during failures
913 4. While the container is being recreated it returns `warp=off` for those 35 seconds 7 consecutive failures
**WARP SHUTDOWN fires**, calling nuclear `recover.sh`
914 5. Nuclear `recover.sh` tears down Tailscale, resets DNS to 1.1.1.1, stops all containers **gateway dead, IP
exposed**
915 6. The post-wake `docker compose up` finishes at ~35s mark, container healthy but the gateway is already gone
916
917 The evidence in `output.log`:
918 ```
919 [2026-07-10T06:07:27Z] NET: ... bash recover.sh --post-wake
920 [2026-07-10T06:07:47Z] WARP DOWN cdn-cgi/trace reports warp=off watcher starts counting
921 ...
922 add net 0.0.0.0: gateway utun5 post-wake successfully re-routes at 02:08:22
923 ...
924 [2026-07-10T06:09:04Z] WARP SHUTDOWN 6 consecutive failures watcher fires nuclear
925 ```
926
927 **Fix (July 10, 2026):** Added a **WARP Watcher inhibit marker** (`/tmp/nullexit-warp-inhibit.marker`):
928 - `recover.sh --post-wake` writes the marker at startup **before** any container operations
929 - The WARP Watcher loop checks for the marker at the top of every iteration; if present it resets `consec_off
=0` and sleeps 5s (skips the poll entirely)
930 - `recover.sh` removes the marker at the very end (on both success and failure via `trap cleanup_inhibit EXIT`)
931 - This guarantees the watcher never counts failures during the intentional container-down window of a post-wake
force-recreate
932
933 The marker file path is hardcoded to `/tmp/nullexit-warp-inhibit.marker` in both `toggle.sh` and `recover.sh`.
934
935 ##### 15.2.8 Incident Post-Mortem: Concurrency Collision between Nuclear Teardown and Post-Wake (July 10, 2026)
936 **Symptom:** When the background WARP Watcher triggers a nuclear shutdown (due to a real or simulated tunnel
failure), the gateway is completely torn down. However, the network configuration changes made during the
teardown trigger a `State:/Network/Global/IPv4` network change event. The LaunchAgent-based network watcher
(`watcher.sh`) intercepts this event and concurrently spawns `recover.sh --post-wake` to heal/restore the
gateway. This causes `docker compose down` (from nuclear teardown) and `docker compose up` (from post-wake)
to run in parallel, resulting in container errors like `dependency failed to start: container warp exited
(0)` and the gateway becoming wedged.
937
938 **Root Cause:** A lack of coordination/locking between the lifecycle control scripts. While `toggle.sh` used `/
tmp/nullexit-toggle.lock` to prevent concurrent `toggle.sh` runs, `recover.sh` (both nuclear and `--post-
wake` modes) did not utilize or check any lock files.
939
940 **Fix (July 10, 2026):** Implemented a shared lifecycle lock file (`/tmp/nullexit-toggle.lock`) across both
scripts:
941 - **`recover.sh` (all modes):** Checks `/tmp/nullexit-toggle.lock` at startup. If the lock is held by another
running lifecycle script (`toggle.sh` or `recover.sh`):
942 - In `--post-wake` mode, it logs a skip message to `output.log` and exits cleanly (`exit 0`). This safely
prevents the auto-recovery agent from recreating containers during a teardown.
943 - In nuclear recovery mode, it exits with an error.
944 - **`recover.sh` Lock Cleanup:** Cleans up the lock file upon completion using `trap cleanup_recover EXIT` (
which checks if the lock file belongs to the current PID).
945 - **`toggle.sh` update:** The lock check in `toggle.sh` was updated to look for both `toggle.sh` and `recover.
sh` in the running processes to enforce proper exclusion.
946
947 ##### 15.2.9 Incident Post-Mortem: Infinite Auto-Recovery Loop after WARP Watcher Nuclear Shutdown (July 10,
2026)

```

948 ****Symptom:**** When the background WARP Watcher triggered a nuclear shutdown (due to a tunnel outage), it successfully executed `recover.sh` (nuclear). However, immediately after the teardown completed, the system started spawning `recover.sh --post-wake` and rebuilding the containers again. This created an infinite loop of tearing down and auto-restarting the gateway, keeping the host's internet route permanently wedged.

949

950 ****Root Cause:**** An active-state marker file leak. The LaunchAgent-based network watcher (`watcher.sh`) only fires `recover.sh --post-wake` when `/tmp/nullexit-gateway-active.marker` is present on disk. While `toggle.sh` (STOP path) correctly deleted this marker, the nuclear `recover.sh` script did not. When the WARP Watcher ran `recover.sh` (nuclear), the containers were stopped but the active-state marker was left on disk. The network changes from the teardown then triggered `watcher.sh`, which saw the marker, believed the gateway was still supposed to be active, and triggered `--post-wake` to start it back up.

951

952 ****Fix (July 10, 2026):**** Modified the start of `recover.sh` to explicitly delete `/tmp/nullexit-gateway-active.marker` if `POST_WAKE` is `false` (nuclear mode). This ensures that any subsequent network configuration changes made during the teardown are correctly ignored by `watcher.sh` because the marker file is gone.

953

954 **### 15.3 MTU, Fragmentation & Throughput**

955

956 **#### 15.3.1 Network Bufferbloat & MTU Optimizations (Double-VPN UDP Crash)**

957 ****Symptom:**** When running aggressive UDP bandwidth tests (such as macOS `networkQuality`, which uses QUIC/HTTP3), the `nullexit` tunnel completely crashes or exhibits severe lag (6,000ms+ bufferbloat).

958

959 ****Root Cause:**** The bug is a cascading MTU mismatch. Tailscale generates a UDP packet. Cloudflare WARP (also WireGuard) receives the packet, wraps it in another layer of encryption, pushing the packet size beyond the standard 1500 byte limit. Unlike TCP, UDP packets cannot be resized dynamically via MSS negotiation, resulting in massive IP packet fragmentation. The Apple Virtualization framework (`vz`), which Colima uses for bridged networking, silently drops heavily fragmented UDP packets under high load. This blackholes the entire UDP upload stream, instantly dropping the connection.

960

961 ****Fix (Partial):**** To prevent MTU fragmentation for standard web traffic, ****TCP MSS Clamping**** is strictly enforced via the macOS `pf` firewall. By adding `scrub out all max-mss 1160` to `scripts/pf.conf`, macOS unilaterally shrinks all outgoing TCP headers to fit comfortably inside the double-encrypted tunnel, bypassing Path MTU Discovery blackholes. `TS_DEBUG_MTU=1200` was also added to `docker-compose.yml` to minimize internal fragmentation.

962 ****Unresolved:**** The only way to solve the UDP crash bug natively is to artificially cap the maximum tunnel bandwidth using SQM (`fq_codel`). Because a static bandwidth cap would artificially throttle high-speed networks, `nullexit` leaves the bandwidth uncapped, optimizing perfectly for TCP while accepting UDP fragmentation under extreme edge-case load tests.

963 ****Why not Dynamic SQM (Aurorate)?**** Implementing an EMA/PID-controlled aurorate daemon (like `sqm-aurorate`) requires a constant 100ms polling loop to measure the kernel packet queue, which severely degrades laptop battery life by preventing deep C-state CPU idling. Furthermore, macOS's native firewall (`dummynet`) only supports `fq_codel` and does not support the newer Linux `CAKE` algorithm, which is the only algorithm that offers kernel-native, event-driven `aurorate-ingress` (zero polling penalty). Therefore, dynamic SQM is computationally hostile to battery-powered macOS hosts.

964

965 **#### 15.3.2 Double-Tunneling MSS Clamping (Stalling Bug)**

966 ****Problem:**** Traffic double-wrapped through Tailscale (WireGuard) and WARP (WireGuard) suffered 120 bytes of MTU overhead (60 + 60). The default MSS clamp of 1180 was still slightly too large, causing large packets to fragment or drop, which resulted in mysterious web page stalls on strict-MSS endpoints.

967

968 ****Fix:**** Lowered the TCP MSS clamp in `post-rules.txt` to `1120` to guarantee double-tunneled payloads fit safely within standard internet MTUs.

969

970 **#### 15.3.3 Wi-Fi Edge Packet Loss via QUIC (UDP) Fragmentation**

971 ****Problem:**** Applications that heavily utilize HTTP/3 (QUIC) over UDP such as Facebook, Instagram, and Google services experience massive stuttering and stalled image loading when the client device is far away from the router (weak Wi-Fi signal). The issue did not occur when close to the router.

972

973 ****Root Cause:**** The `nullexit` gateway uses a double-encrypted tunnel (Tailscale + WARP). To prevent packets from fragmenting when entering these tunnels, a firewall rule in `post-rules.txt` uses `--set-mss 1120` to safely clamp TCP packets. However, because QUIC runs entirely over UDP, it bypasses the TCP MSS handshake completely. QUIC blasts maximum-MTU (1280 byte) UDP packets. The inner `tailscale0` interface (MTU 1200) forces these large UDP packets to fragment into two pieces. When the client is on the edge of Wi-Fi range (where 5-10% packet loss is common), losing *either* UDP fragment destroys the *entire* image frame. This exponential amplification of packet loss stalled QUIC streams.

974

975 ****Fix & Trade-offs:**** Appended an `iptables` rule to `post-rules.txt` that explicitly blocks `UDP --dport 443` (QUIC). When modern web apps detect QUIC is blocked, they instantly fall back to HTTP/2 (TCP 443). Because TCP is routed through the MSS clamp, the packets are resized *before* they leave, entirely eliminating IP fragmentation and restoring high-speed loading over weak Wi-Fi.

976 ***Consequences:** This adds a minor latency penalty (TCP 3-way handshake) compared to QUIC's zero-RTT connection. It may also force WebRTC video calls (Google Meet/Discord) to fall back to TCP TURN servers, slightly inflating real-time video latency. However, in a constrained double-tunnel mesh, the absolute stability gained from eliminating fragmentation vastly outweighs these trade-offs.

977

978 **#### 15.3.4 Cellular Networks & PMTUD Blackholes (The 1280 Byte Limit)**

979 ****Experiment:**** Conducted a live packet sweep from the PC-1 host to Phone-1 connected via a cellular network + Tailscale DERP relay using `ping -D -s <size>`.

980

981 ****Result:****

```

982 - Packets with a payload of `1252` bytes (+ 28 bytes IP/ICMP headers = **1280 bytes total**) succeeded
    perfectly with ~60ms latency.
983 - Packets with a payload of `1253` bytes (**1281 bytes total**) and above resulted in a silent timeout.
984
985 **Analysis:** The cellular network (or the Tailscale DERP relay) enforces a strict MTU of 1280 bytes.
    Critically, it acts as a "PMTUD Blackhole" it silently drops packets larger than 1280 bytes instead of
    returning an ICMP "Fragmentation Needed" warning. If the exit node tries to send a standard 1500-byte
    internet packet to the phone over this link, it vanishes, causing web pages to stall permanently.
986
987 **Conclusion:** This empirically validates the absolute necessity of the TCP MSS clamping rule in `post-rules.
    txt`. By artificially clamping the MSS to `1120` (or `1180`), we ensure the TCP payload + headers + double-
    WireGuard overhead never exceeds the strict 1280-byte ceiling of the cellular mesh link.
988
989 #### 15.3.5 Low Gateway Throughput (DERP Relay & MTU Fragmentation) Fixed
990 **Problem:** The host Mac's internet throughput through the gateway dropped significantly (e.g., from 34Mbps
    raw to 2.1Mbps via gateway). This was caused by a combination of two bottlenecks:
991
992 **1. Tailscale DERP Relay Bottleneck:**
993 Because the host Mac and the Docker container operate behind the same public IP, standard hole-punching for
    Tailscale P2P failed, forcing traffic through Tailscale's NYC DERP relay. Normally, `routing-fix.sh`
    explicitly drops container Tailscale UDP packets destined for local subnets (when `TAILSCALE_ALLOW_LAN_P2P=
    false` in `.env` or auto-detected). We nuke these P2P connections on purpose to prevent the severe macOS `
    gvproxy` SNAT endpoint poisoning issue we faced earlier (see 15.7 SNAT Endpoint Poisoning). However, this
    strict policy unintentionally blocked direct communication between the container and the Mac host itself
    via the Docker bridge network.
994
995 **Fix:** Tailscale's control plane registers the Mac host using its physical IP (e.g., `172.17.52.223`), not
    the Docker bridge IP. If this physical IP falls within the dropped local subnets (like `172.16.0.0/12`),
    the P2P handshake is dropped. To solve this natively, `scripts/common.sh` now features a `write_host_ips()`
    function that actively grabs all physical IPv4 addresses on the Mac and writes them to `.host_ips`. `
    routing-fix.sh` dynamically reads this file every 30 seconds and injects priority `RETURN` rules for those
    exact physical IPs. This guarantees direct container-to-host P2P over the local bridge (bypassing DERP)
    while continuing to block external LAN devices (like phones) to prevent the 15.7 SNAT poisoning bug.
996
997 **2. MTU Double-Encapsulation Fragmentation:**
998 The host's Tailscale interface (`utun5`) defaulted to an MTU of 1280. The WARP tunnel inside the container also
    uses an MTU of 1280. When a full 1280-byte Tailscale packet from the host entered the container and was
    encapsulated by WARP (adding ~60 bytes of overhead), the resulting packet exceeded 1280 bytes, leading to
    severe fragmentation and performance degradation.
999
1000 **Fix:** In `toggle.sh`'s `setup_exit_node_routing` function, the host's Tailscale `utun` interface MTU is
    explicitly lowered to `1200` (configurable via `HOST_MTU` in `.env`). This ensures that host-originated
    packets can comfortably fit inside the container's WARP tunnel encapsulation without fragmenting.
1001
1002 > **Design Note Is direct hostcontainer P2P a Docker bug, or are we "exploiting a VM/host relationship"?
1003 > Neither. It is expected, well-defined behavior, and the RETURN-rule carve-out is a *defensive de-restriction
    *, not an exploit. The reasoning:
1004 > - **On native Linux**, the container and host share one kernel; the Docker bridge is a first-class local
    interface and hostcontainer is *trivially* direct. Nobody calls that an exploit. Our RETURN rules simply
    reproduce that same local reachability on macOS.
1005 > - **The RETURN rules add nothing new; they *stop blocking* something legitimate.** `routing-fix.sh` broadly
    DROPS containerLAN Tailscale UDP to prevent the real 15.7 `gvproxy` SNAT endpoint-poisoning bug caused by *
    external* LAN peers (e.g. a phone on another AP). Tailscale's control plane registers the Mac host by its *
    physical* LAN IP (e.g. `172.17.52.223`), which falls inside those DROPPed ranges so the host, the one peer
    that is literally the same physical machine, became collateral damage. The `.host_ips` RETURN rules are
    the minimal, precise exception that re-permits exactly that same-machine path and nothing else.
1006 > - **The genuinely VM-specific wart is `gvproxy` SNAT poisoning (15.7), and we route *around* it, not through
    a hole in it.** Keeping hostcontainer traffic on the Docker bridge (direct) avoids letting it get NATed
    through the userspace `gvproxy` endpoint rewriter that confuses Tailscale's peer-endpoint learning. That is
    the opposite of exploiting the VM boundary it is respecting it.
1007 > - **What genuinely *doesn't* work on a VM host is UDP hole-punching to arbitrary *external* peers** (see
    15.13). That is a real limitation of userspace VM networking, and it is why remote peers fall back to DERP
    unless bridged mode is enabled on a trusted LAN. Direct hostcontainer P2P is the one case that works
    regardless because both endpoints live on the same physical machine, no hole-punching is needed at all. It
    was verified `direct 172.17.52.223` even in plain user-mode networking (no `--network-address`).
1008 >
1009 > In short: the DROP is the deliberate policy, the RETURN is a surgical exception for same-machine traffic, and
    the whole thing is standard bridge networking not a Docker bug and not an exploit.
1010
1011 ### 15.4 DNS Interception & Proxy
1012
1013 #### 15.4.1 DNS Proxy: TCP Wire Format Mismatch (socat / dnsmasq)
1014 **Problem:** When the Tailscale data plane is unavailable (DERP relay only), the script falls back to a local
    DNS proxy that forwards queries from host UDP:53 to Docker's TCP:5354 (AdGuard). The `socat` and `dnsmasq`
    approaches both failed.
1015
1016 - **socat** forwards raw bytes it does not prepend the 2-byte length prefix that DNS-over-TCP requires.
    AdGuard parses the stream incorrectly and returns garbage.
1017 - **dnsmasq** tries UDP first to reach the upstream (`127.0.0.1#5354`), but Colima's SSH tunnel only forwards
    TCP. With a single upstream server, dnsmasq doesn't fall back to TCP even with `--timeout=1`.
1018

```

1019 ****Solution:**** Replaced both with a ****Python DNS proxy**** (``scripts/dns-proxy.py``, ~25 lines) that properly
handles the DNS-over-TCP wire format: reads a UDP query from the host, prepends the 2-byte length prefix,
sends it over TCP to AdGuard, strips the prefix from the response, and sends it back over UDP.

1020

1021 ***Architectural Note:** Why is the DNS proxy in Python while the SOCKS5 proxy was moved to a compiled Go Docker
container?*

1022 The SOCKS5 proxy handles heavy data streaming (e.g., video, downloads). Python introduces massive CPU/battery
overhead for raw socket streaming, which is why we rely on the compiled Go implementation built directly
into the ``tailscaled`` daemon via ``TS_SOCKS5_SERVER=:1080``.

1023 Conversely, the DNS proxy only handles intermittent UDP queries (a few bytes per minute) generated solely by
the local host machine. The Python overhead for this is effectively 0.001 Watts. We deliberately kept ``dns-
proxy.py`` in Python because it runs natively on the macOS/Linux host rewriting it in Go would force users to
install a Go compiler (``brew install go``) on their host machine for zero noticeable battery gain. (Note:
If this architecture is ever adapted to serve as a public DNS resolver for thousands of users or for
massive automated web-scraping clusters, ``dns-proxy.py`` must be rewritten in Go to survive the concurrent
packet flood).

1024

1025 **#### 15.4.2 Tailscale MagicDNS Bypassing the AdGuard PREROUTING Intercept**

1026 ****Context:**** The ``routing-fix.sh`` script applies an iptables NAT rule (``PREROUTING -i tailscale0 -p udp --dport
53 -j REDIRECT``) to intercept all incoming DNS queries from connected Tailscale peers and force them into
the AdGuard container on port 5335.

1027 ****Problem:**** When remote clients (like Android or iOS) have "Use Tailscale DNS settings" turned on, they query
Tailscale's MagicDNS IP (``100.100.100.100``). This packet enters the exit node, but the ``tailscaled`` daemon
running ``*inside*`` the gateway container intercepts the ``100.100.100.100`` packet internally. ``tailscaled``
then generates a brand new, local DNS request to the upstream DNS server to resolve the query. Because this
new request is generated ``*locally*`` by the daemon (not arriving externally via the ``tailscale0`` interface),
it bypasses the ``PREROUTING`` chain completely. The request glides straight out the WARP tunnel to
Cloudflare/Google, entirely bypassing the AdGuard sinkhole.

1028 ****Fix:**** The only way to fix this blindspot is to force ``tailscaled`` to use AdGuard as its upstream. In the
Tailscale Admin Console, the gateway's Tailscale IPv4 address (e.g., ``100.x.x.x``) must be added as the sole
Custom Global Nameserver, and "Override local DNS" must be enabled. Additionally, to block Android devices
from bypassing this via Private DNS (DNS-over-TLS), outbound Port 853 traffic is explicitly REJECTED in
both ``routing-fix.sh`` and ``post-rules.txt``.

1029

1030 **#### 15.4.3 Seamless Wi-Fi Roaming & The macOS DNS Wipeout Loophole**

1031 ****Problem:**** The Tailscale, WireGuard, and Colima NAT stacks are natively designed for connectionless roaming
and will seamlessly survive when the macOS host switches Wi-Fi networks (e.g., roaming from home Wi-Fi to a
cellular hotspot). However, the internet completely breaks upon network transition. This occurs because
macOS intentionally wipes out all custom DNS settings (``networksetup -setdnsservers``) and reverts to the
new network's default DHCP nameserver whenever the active Wi-Fi BSSID changes. Since the Mac is locked to
the exit node, its DNS requests are swallowed by WARP and dumped onto the public internet, which cannot
route private DHCP IPs. This results in a total DNS failure.

1032

1033 ****Solution:**** Implementing a silent background "DNS Watcher" daemon (``nullexit-dns-watcher``) in ``toggle.sh``.
When the gateway starts, it spawns a background polling loop that checks the active Wi-Fi DNS every 30
seconds. If macOS resets the DNS during a network change, the watcher instantly detects the drift and
forcefully re-injects ``100.100.21.8`` via ``networksetup``. When the gateway stops, the watcher process is
cleanly killed. This allows true, seamless roaming across physical networks without ever dropping the
gateway state.

1034

1035 **#### 15.4.4 docker compose exec ``\r`` Injection**

1036 ****Fix:**** All ``docker compose exec`` output piped through ``tr -d '\r'``.

1037

1038 **#### 15.4.5 Stderr Invisibly Swallowed**

1039 ****Symptom:**** Failures were silent due to stderr redirected to ``output.log``.

1040

1041 ****Fix:**** Tailscale wait loop now shows output. Critical commands show errors inline.

1042

1043 **### 15.5 Roaming, Sleep/Wake & Auto-Recovery**

1044

1045 **#### 15.5.1 Gateway Breakage on Lid Close / Wi-Fi Roam Post-Wake + Post-Roam Auto-Recovery**

1046 ****Context / Symptom:**** Closing the lid (sleep/wake) OR losing + reconnecting Wi-Fi (e.g. in an elevator, caf,
or roaming between APs) leaves the gateway in a partly-dead state. Symptoms range from a ~30s window of
dead DNS to a permanent outage that only full ``recover.sh`` or ``toggle.sh`` fixes. The root cause is that
nullexit has ``*no daemon watching macOS sleep/wake or network-state-change events*`` the existing DNS
Watcher inside ``toggle.sh`` only runs in-process and is suspended along with the rest of the user shell on
lid close, so the gateway cannot self-heal after either event.

1047

1048 ****Diagnosis:**** Two distinct failure modes share the same root gap.

1049

1050 *** ** (A) Lid close sleep wake. **** ``caffeinate -i`` (used by ``toggle.sh``) blocks ``*idle*`` sleep but ``**does not``
block forced sleep from lid close. Closing the lid hard-suspends Colima VM + every container + every
Tailscale-quit-shells-except-tailscaled. On wake the VM clock needs ~5-30s to resync via NTP, during which
TLS cert validation fails: Cloudflare WARP ``162.159.192.1:2408`` rejects the handshake gluetun's
healthcheck (``interval: 2s, retries: 15``) eventually flips unhealthy Docker ``unless-stopped`` restarts the
``warp`` container (~30s gap). ``mDNSResponder``'s in-memory cache still holds pre-sleep entries (stale A
records for tailnet hosts) so the first dozen DNS queries after wake time out. ``tailscaled`` on the host
often keeps its stale DERP relay preference and routes packets through a dead relay for another 30-60s.

1051 *** ** (B) Wi-Fi roam / loss in elevators, captive portals. **** WARP is a UDP tunnel to Cloudflare's anycast edge.
Carrier NATs and captive-portal networks silently invalidate the UDP binding on roam and on hotspot-bound
re-association. macOS ``networksetup`` ``*should*`` preserve DNS settings on the same Wi-Fi service across a roam

, but most captive-portal networks DHCP-replace DNS to the captive-portal resolver ****during the captive-portal dance itself****, before the user has authenticated so even DNS hijack survives a clean roam and quietly breaks on captive-portal handoff.

1052

1053 ****Resolution.**** A single ****launchd LaunchAgent**** (``com.nullexit.wake-recovery``) runs a long-running shell daemon (``scripts/watcher.sh``) that listens for both event sources and, on each fire, executes ``bash recover .sh --post-wake`` a `*lighter-touch*` recovery mode that's the opposite of the existing ``--nuclear`` semantics : it keeps the gateway live while still refreshing every stale subsystem.

1054

1055 ****Deep dive: Apple Power Management Primer (for the unfamiliar).****

1056 macOS exposes several relevant surfaces; the right primitive depends on what you actually need to detect. None of these are obvious and none of them have a standard splash screen in the docs.

1057

1058 * ****caffeinate <flags>**** creates I/O Kit power assertions via ``IOPMAssertionCreateWithName``. It does NOT block forced sleep (lid close, low-battery shutdown, sleep timer firing on a per-set Energy Settings policy). Flags:

1059 * ``-i`` PreventIdleSleep (the only one ``toggle.sh`` uses; protects against the OS going idle while the user is active but a clamshell close still hard-puts it to sleep)

1060 * ``-s`` PreventSystemSleep (only honored on AC power; the user may be on battery)

1061 * ``-u`` DeclareUserActive (resets the idle timer every 30s by default; equivalent to keeping the cursor moving)

1062 * ``-d`` PreventDisplaySleep

1063 * ``-m`` PreventDiskIdleSleep

1064 * There is NO ``-l`` (lid-block) flag in any modern macOS. Lid close is a kernel-firmware-level event that ignores user-space assertions.

1065 * To prove any of this on live hardware: ``pmset -g assertions`` lists every active assertion with type / process / reason / timeout.

1066 * ****`pmset`**** is the read-side of the same system:

1067 * ``pmset -g log | grep -E 'Sleep|Wake|DarkWake`` shows the recent timeline.

1068 * ``pmset -g history`` is the per-day summary.

1069 * ``pmset -g assertions`` is the live assertion list (matches ``-i -s -d -u`` to processes).

1070 * ****`launchd`** does NOT have a native "on-wake" hook.** Not in any version of macOS as of Sonoma. The canonical recipe is ``StartCalendarInterval`` with a sub-minute cadence and let `launchd` queue missed intervals for replay-on-wake, or use a third-party daemon (Sleepwatcher). For our use case a long-running shell daemon listening to the unified log is more responsive than a polling agent.

1071 * ****`com.apple.system.powermanagement.\*`** Darwin notifications** (``willSleep``, ``didWake``, ``didDim``, ``didUndim``) are the C-level signal. From a shell, they are best reached through the unified log with ``log stream --predicate`` see the watcher implementation.

1072 * ****`scutil n.watch`**** is the canonical CLI for live-following network-state changes. Pairs with ``n.add State:/Network/Global/IPv4`` to get a single tap that fires on ****any**** IPv4 link/route/DNS/address change across ****all**** interfaces. This is what ``scripts/watcher.sh``'s network-change listener uses.

1073 * ****`com.apple.networkChange`** Darwin notification** is the C-level equivalent but has no shell-friendly listener; use ``scutil`` instead.

1074 * ****`SCNetworkReachability`**** is the Objective-C API used by SOCKS5/HTTP clients to detect route changes per target. Never a fit for shell scripts.

1075

1076 ****The Fix (component-by-component).****

1077

1078 1. ****`recover.sh --post-wake`**** (new flag, ****non-destructive**** by design). Adding ``--post-wake`` to the existing nuclear recovery script inverts the semantics. The default mode still tears down Tailscale, resets DNS to empty, stops caffeinate, runs ``docker compose down``, power-cycles Wi-Fi. The new mode does:

1079 * Skip Tailscale disconnect (we WANT it to keep the mesh connection)

1080 * ****Re-hijack**** DNS to the gateway Tailscale IP read from ``gateway_ip`` (instead of resetting to empty) applies to both ``ACTIVE_SERVICE`` and ``ENO_SERVICE`` because the system resolver is scoped to `en0`

1081 * ****Refresh**** the exit-node preference with ``tailscale up --reset --ssh=true --accept-dns=false --exit-node=$TS_IP`` `--exit-node-allow-lan-access=true`` (the exact flags ``toggle.sh`` START uses). This re-asserts DERP mapping without dropping the mesh

1082 * Inspect ``docker inspect --format '{{.State.Health.Status}}`` `nullexit-warp-1`` and **\*\*only\*\*** ``docker compose up -d --force-recreate warp`` if `gluetun` is unhealthy this single targeted recreate nudges the UDP NAT binding back to Cloudflare without disturbing Tailscale or AdGuard

1083 * Run the `sharingd-reset` step (existing fix from ``toggle.sh`` START path; see 15.11) so AirDrop doesn't freeze on the new IP lease

1084 * Verify the gateway with ``dig +tcp @127.0.0.1 -p 5354 google.com`` (AdGuard via TCP through Colima's SSH tunnel) and a 4-second ``tailscale ping`` to the gateway Tailscale IP, instead of the default mode's direct-to-1.1.1.1 check

1085 * Crucially: ****skip `sudo -v`**** because the LaunchAgent has no TTY to prompt on; all privileged calls use ``sudo -n`` and lean on ``/etc/sudoers.d/nullexit`` `NOPASSWD` entries (``networksetup``, ``dnsmasq``, ``dscacheutil``, ``killall``, ``pkill``).

1086 2. ****`toggle.sh`**** writes and clears ``/tmp/nullexit-gateway-active.marker`` (an ISO-8601 UTC timestamp) at the bottom of the START path and the top of the STOP path / ``cleanup_handler``. Two new helpers: ``write_gateway_active_marker`` and ``clear_gateway_active_marker``. The watcher uses this file as its ****only signal**** that the gateway is live: if ``toggle.sh`` was stopped (or never started), the watcher no-ops. This prevents waking-up-only-on-counterfactual events from churning DNS.

1087 3. ****`scripts/watcher.sh`**** (long-running daemon, sibling-style source file). Two backgrounded pipelines:

1088 * ****Wake listener****: ``log stream --predicate 'composedMessage CONTAINS[c] "didWake" OR composedMessage CONTAINS[c] "Wake from Sleep" OR composedMessage CONTAINS[c] "Waking from" OR composedMessage CONTAINS[c] "DarkWake" --style compact --info`` piped through ``while read; do run_recover "WAKE: ..."; done``. ``[c]`` makes the predicate case-insensitive (the unified log writes phrases in mixed case across versions).

1089 * ****Network listener****: ``(echo "n.add State:/Network/Global/IPv4"; echo "n.watch"; while true; do sleep 86400; done) | scutil 2>/dev/null`` piped through a ``case`` match on ``n.state*`` / ``SCEventUpdate*``.

```

1090 * **run_recover** checks the marker, debounces via `/tmp/nullexit-watcher.last-recovery` (default 10s,
    configurable via `NULLEXIT_DEBOUNCE_SECONDS`), and shells out to `bash recover.sh --post-wake`. The 10s
    debounce prevents a single wake event (which typically emits 2-3 log lines) from triggering multiple
    recoveries.
1091 * `trap ... TERM INT` `kill 0` cleanly tears down the process group on launchctl bootout` so the `sleep
    86400` inside the scutil pipe also dies.
1092 * `wait` blocks indefinitely while both pipelines feed it.
1093 4. **launchd/com.nullexit.wake-recovery.plist** (LaunchAgent in the user domain runs as the console user at
    every login without needing sudo).
1094 * `Label: com.nullexit.wake-recovery`
1095 * `ProgramArguments: ["/bin/bash", "<absolute-path-to-your-nullexit-install>/scripts/watcher.sh"]`
1096 * `RunAtLoad: true` starts immediately on `launchctl load` (no need to log out and back in)
1097 * `KeepAlive: { SuccessfulExit: false, Crashed: false }` **don't** restart on clean SIGTERM exit (so `
    launchctl bootout` doesn't get stuck in a relaunch loop). Also **don't** restart on hard crash: we observed
    that macOS Sonoma's sleep/wake suspend-resume path accumulates orphan watcher.sh PIDs because SIGCONT
    does not reliably clean up the prior instance's listener grandchildren, and a `KeepAlive.Crashed=true`-
    driven relaunch actively races with the still-live prior process. The script enforces single-instance on
    its own via a `flock`-style PID-file lock, so a relaunch-on-crash would be skipped by the lock and produce
    0 listener coverage until the live instance happened to die. Trade-off: a real crash leaves the user
    without auto-recovery until they run `launchctl load -w` manually acceptable because (a) gateway still
    works without auto-recovery, (b) a relaunch wouldn't fix whatever caused the crash, and (c) the gateway-
    recovery itself is observably broken (no DNS, no exit-node) if it does go down.
1098 * `ProcessType: Background` hint to App Nap not to suspend us.
1099 * `StandardOutPath/StandardErrorPath: /tmp/nullexit-watcher.{out,err}.log` separates launchd-level
    diagnostics from the script's own `/tmp/nullexit-watcher.log` (which `exec >>` redirects).
1100
1101 **Install (one-time)** The plist lives in the repo at **launchd/com.nullexit.wake-recovery.plist** for
    version control. The installed (live) copy must land in `~/Library/LaunchAgents/` so launchd picks it up on
    the user session.
1102
1103 ```bash
1104 # Copy the plist to the user-launchd location (run from repo root)
1105 cp ./launchd/com.nullexit.wake-recovery.plist \
1106 ~/Library/LaunchAgents/
1107
1108 # Substitute __NULLEXIT_HOME__ with your local install absolute path.
1109 # The plist uses WorkingDirectory + a relative program arg, so this
1110 # single substitution is the only place it references your machine.
1111 sed -i ' "s|__NULLEXIT_HOME__|$(pwd)|" \
1112 ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1113
1114 # Validate the XML
1115 plutil -lint ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1116
1117 # Load + persist this user session + every future login
1118 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1119
1120 # Confirm it actually started
1121 launchctl list | grep nullexit
1122 ```
1123
1124 To **stop** the watcher (without uninstalling):
1125
1126 ```bash
1127 launchctl bootout gui/$UID/com.nullexit.wake-recovery
1128 # or:
1129 launchctl unload -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1130 ```
1131
1132 To **uninstall** completely:
1133
1134 ```bash
1135 launchctl bootout gui/$UID/com.nullexit.wake-recovery 2>/dev/null || true
1136 rm ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1137 ```
1138
1139 **Why this fix was preferred over alternatives.**
1140
1141 * **Polling with `StartCalendarInterval`** would have given ~30-60s detection latency per wake. The unified-log
    stream is sub-second.
1142 * **Sleepwatcher** would have covered wake events but not network changes. We'd still need a separate `scutil`
    listener, and introducing a third-party dependency for what amounts to ~150 lines of shell is unjustified.
1143 * **Forcing `caffeinate -l`** doesn't exist and the closest equivalent (`caffeinate -s`) only works on AC power
    . The whole point of the gateway is to stay usable on battery while traveling lid close must NOT silently
    kill the gateway.
1144 * **A kernel extension** is impossible (no entitlement) and out of scope.
1145
1146 **Reproduction recipe.** Test both events in isolation to confirm the fix actually closes the gap.
1147
1148 * ** (A) Lid-only repro**: with the gateway up, run in a terminal: `pmset displaysleepnow && sleep 30 && pmset
    wake` (without physically closing the lid). Watch the wake wake-fires the watcher post-wake routine runs

```

login should still work in <5s after wake. Compare to baseline pre-fix: pre-fix the gateway would be dead for 30-90s.

1149 * **(B) Roam-only repro****: with the gateway up: ``networksetup -setairportpower off && sleep 30 && networksetup -setairportpower on`` (or simply walk out of Wi-Fi range and back). The captive-portal WILL repave DHCP DNS for a few seconds; the watcher fires on the second ``State:/Network/Global/IPv4`` change (after the Wi-Fi re-associates) and the post-wake routine re-hijacks DNS within 2-3s.

1150

1151 ****Operative files.****

1152

1153 * ``recover.sh`` added arg parsing; wrapped destructive sections behind ``POST_WAKE``; added re-hijack DNS / refresh exit-node / force-recreate-if-unhealthy path.

1154 * ``toggle.sh`` added ``write_gateway_active_marker`` / ``clear_gateway_active_marker`` called at the END of START and TOP of STOP / cleanup_handler.

1155 * ``scripts/watcher.sh`` new long-running daemon.

1156 * ``launchd/com.nullexit.wake-recovery.plist`` launchd LaunchAgent descriptor.

1157 * ``~/Library/LaunchAgents/com.nullexit.wake-recovery.plist`` the installed copy (one-time copy).

1158

1159 ****Honest assessment (read before testing).**** This fix has two asymmetric halves:

1160

1161 * ****Network / roam / captive-portal path RELIABLE.**** ``scutil n.watch`` on ``State:/Network/Global/IPv4{6}`` catches every Wi-Fi roam, captive-portal DHCP-rebind, hotspot handoff, and VPN change with zero missed events. To validate, drop Wi-Fi for 30 s and re-add it; ``~/tmp/nullexit-watcher.log`` will record a ``NET:`` trigger and ``recover.sh --post-wake`` will run within ~10 s of the network coming back.

1162 * ****lid close / wake path BEST-EFFORT on macOS Sonoma.**** Apple ``*suspends*`` LaunchAgents process-state during forced sleep and ``*resumes*`` them on wake; it does NOT re-launch them every wake. As a result: (a) ``log stream`` is a live subscription events emitted during the agent's suppressed window are DROPPED, not buffered, so the wake event that prompted the resume is invisible to ``log stream`` on resume; (b) the "fire on startup" guard runs once on ``launchctl load -w`` and after a ``KeepAlive`` crash-recovery, NOT on every successive wake; (c) ``subsystem == "com.apple.powermanagement"`` will catch FUTURE emissions (display dim/restore, manual sleep/wake) but not the lid-close/open wake directly.

1163

1164 In practice the lid-wake gap is short, because ``toggle.sh``'s existing DNS Watcher already polls every 5 s and re-hijacks DNS (so DNS recovers within 5 s of any roam or wake), ``tailscaled`` self-heals via DERP fallback within 3060 s, and ``gluetun`` reconnects via its healthcheck retry within 30 s. The user-visible pain on lid open is ~30 s of wonky DNS, not a permanent outage.

1165

1166 ****Test the ROAM path FIRST.**** If ROAM works in your testing, the lid-wake gap is acceptable. If lid-wake handling is critical, the two clean follow-ups are:

1167 1. ****Sleepwatcher**** (one canonical install): ``brew install sleepwatcher``; drop ``recover.sh --post-wake`` into ``~/sleep`` and ``~/wake`` so Sleepwatcher invokes it directly on every wake without the launchd propagation problem.

1168 2. ****Move the watcher to a LaunchDaemon**** (root) and use ``IOPMSchedulePowerEvent`` via a tiny C wrapper not portable to shell.

1169

1170 A truly reliable shell-only solution to "wake events from inside a LaunchAgent" does not exist on macOS Sonoma +.

1171

1172 ****Empirical lessons from deployment.**** Two findings came up while deploying this fix end-to-end; recording so the next operator doesn't lose an evening to each.

1173

1174 1. ****Bash function-call-before-definition gotcha.**** While introducing the gateway-active marker helpers, the defensive ``clear_gateway_active_marker`` call was inserted at the very top of ``toggle.sh`` (right after ``set -e``), but the function definition lived near ``PID_FILE="/tmp/nullexit-cafeinate.pid"`` ~90 lines down. Bash parses top-to-bottom and resolves names at first invocation, so the call site exited with code 127 (``clear_gateway_active_marker: command not found``). The gateway stayed unreachable from the START path even though ``bash -n`` passed (the parser doesn't catch order-of-resolution errors). Rule: define functions BEFORE any block that calls them. Verify with ``grep -n '^name{\s*{'`` followed by ``grep -n '^name\s*$`` the latter must appear on a line number AFTER the former.

1175 2. ****`networksetup -setairportpower off/on` is not a faithful roam reproduction.**** Synthetic Wi-Fi radio-cycling (``sudo networksetup -setairportpower Wi-Fi off`` for 30 s, then back on) is expected to produce a ``NET:`` trigger in ``scutil n.watch`` but produced ZERO during testing because ``setairportpower`` powers the radio at the driver level while leaving the network SERVICE in the "up" state from ``scutil``'s perspective, so no ``n.state State:/Network/Global/IPv4`` event is generated. A real roam (macOS briefly loses the AP and re-associates into a different one) DOES fire the event because the link actually changes.

1176 Better reproduction recipes in priority order:

1177 * Walk between two real SSIDs via ``networksetup -switchtolocation`` (or actual physical station movement) guaranteed to fire the ``scutil`` event.

1178 * Force a DHCP rebind on the same SSID with ``sudo ipconfig set en0 DHCP`` triggers ``State:/Network/Global/IPv4`` to update.

1179 * Trust the existing 30-second DNS Watcher inside ``toggle.sh`` for the DNS path: it re-hijacks DNS automatically. The post-wake watcher adds clear value mostly for tailscale exit-node re-assertion and warp gluetun force-recreate, both of which self-heal within ~30 s anyway.

1180

1181 **#### 15.5.2 July 8, 2026: Network Watcher Event Mismatch, Sudo-in-Background Crash, and Loop-Prevention**

1182 ****Symptom:****

1183 1. Switching Wi-Fi or waking the Mac from sleep did not trigger a post-wake recovery (``recover.sh --post-wake``) automatically.

1184 2. The network connection would eventually get completely sinkholed/blocked. If the user tried to restart or wait, it did not self-heal.

1185 3. During the recovery attempt, checking the public IP on the host would return the unencrypted campus/ISP IP (starting with ``132.``) instead of the encrypted tunnel IP.

```

1186
1187 **Root Cause:**
1188 - **Bug 1 (Network Watcher Mismatch):** In `scripts/watcher.sh`, the network change listener watched `State:/
Network/Global/IPv4` changes via `scutil n.watch` but matched them against `*.state*|*SCEventUpdate*`. On
macOS, the actual output is `changedKey [0] = State:/Network/Global/IPv4`. Because the pattern did not
match, all network switches were silently ignored.
1189 - **Bug 2 (Sudo Caching Background Crash):** When the background WARP Watcher eventually detected the broken
tunnel (after 6 failures/30s), it triggered the default nuclear `recover.sh` (with `POST_WAKE=false`).
Because there was no active TTY in the background context, the `sudo -v` caching check aborted instantly
with a terminal required error. Due to `set -e`, `recover.sh` died immediately before resetting DNS,
disabling the packet-filter killswitch, or stopping containers, leaving the host network completely
sinkholed.
1190 - **Bug 3 (Post-Wake Infinite Loop):** Once the watcher patterns were fixed, `recover.sh --post-wake` triggered
successfully on network changes. However, because the script deleted the default routing entries at the
start, spent ~15 seconds rebuilding 88 DERP relay bypass routes, and re-added the default routes at the end
, this execution time exceeded the watcher's 10-second debounce threshold. Furthermore, the watcher
recorded the `start` timestamp in the debounce file. Consequently, the route changes made by the script
itself triggered the watcher again immediately after completion, trapping the system in an infinite loop of
recovery runs. During this loop, the VPN routes were constantly deleted, resulting in the host leaking
traffic through the real ISP interface (an IP starting with `132.`) for 15s out of every cycle.
1191
1192 **Resolution:**
1193 - **Fixed Watcher Matching:** Updated `scripts/watcher.sh` to match `*changedKey*` and `*State:/Network/Global/
IPv4` to correctly catch network changes.
1194 - **Bypassed Background Sudo:** Added `--auto`/`--non-interactive` flags to `recover.sh` and added a TTY check
`[ -t 0 ]` to skip interactive `sudo -v` authentication when running headlessly in the background. Updated
the WARP Watcher call in `toggle.sh` to pass `--auto`.
1195 - **Prevented Infinite Loops:** Modified `scripts/watcher.sh` to write the **completion** timestamp of `recover
.sh` to `DEBOUNCE_FILE` (instead of only the start timestamp). This ensures all buffered network config
events generated during route reconstruction are evaluated against the end-of-run timestamp and
successfully debounced, breaking the infinite loop.
1196 - **Centralized & Timestamped Logs:** Redirected `watcher.sh` stdout/stderr from `/tmp/nullexit-watcher.log` to
the repository's main `output.log` and updated `scripts/common.sh`'s `step`, `ok`, `warn`, `fail`, and `
die` helpers to prefix logs with a local `[YYYY-MM-DD HH:MM:SS]` timestamp.
1197
1198 ##### 15.5.3 Incident Post-Mortem: Startup Pre-Flight Check Tailscale Race (July 10, 2026)
1199 **Symptom:** Immediately after startup via `toggle.sh`, the pre-flight connectivity check `[1/3] Gateway
reachable via Tailscale` returned `FAIL`, forcing `toggle.sh` to skip the full exit-node activation and
fall back to SOCKS5/local DNS proxy mode.
1200
1201 **Root Cause:** A timing race condition during startup. In `toggle.sh`, the host joins the mesh using `
tailscale up --reset` (Phase A) right before performing the pre-flight check. Because `tailscale up` resets
and reconfigures the host's `tailscaled` daemon, the local daemon must fetch the latest netmap
asynchronously from the coordination servers. This network map propagation and route establishment takes a
few seconds. Running `tailscale ping` immediately after `tailscale up` returns (with no delay) causes the
check to fail because the local daemon has not yet discovered the gateway node `100.100.21.8`.
1202
1203 **Fix (July 10, 2026):** Upgraded pre-flight Check 1 in both `toggle.sh` and `scripts/toggle-linux.sh` to use a
5-attempt retry loop with a 1-second delay between checks. This allows up to 5 seconds for local tailnet
routing paths to settle, ensuring the gateway is correctly reached and a pong is received before deciding
whether to enable exit-node routing.
1204
1205 ##### 15.5.4 Incident Post-Mortem: Pre-Flight Check False Negatives due to VPN Firewall STUN Blocks (July 10,
2026)
1206 **Symptom:** Immediately after a cold boot or full restart of the gateway, the pre-flight connectivity check
`[1/3] Gateway reachable via Tailscale` failed, causing `toggle.sh` to skip the full exit-node
configuration and fall back to SOCKS5/local DNS proxy mode. However, manually running `tailscale ping
100.100.21.8` immediately after startup succeeded in 1ms.
1207
1208 **Root Cause:** An asynchronous Tailscale handshake delay caused by firewall restrictions. Inside the container
network namespace, the `warp` container (gluetun) implements a strict firewall that blocks all outbound
UDP traffic to the public internet except to the designated WireGuard endpoint. Because of this, the `
tailscale` container's STUN requests to public STUN servers are blocked, causing the local daemon to report
`UDP is blocked, trying HTTPS`. Without UDP STUN capability, Tailscale is forced to fall back to a slower
DERP-assisted handshake (relayed via HTTPS/TCP). This negotiation and direct-path punch-through takes
roughly 30 to 35 seconds to establish on cold boots (after a full `tailscale up --reset`). Since the pre-
flight check only waited 15 seconds (15 attempts with 1-second sleeps), the check timed out and aborted
before the path completed.
1209
1210 **Fix (July 10, 2026):** Increased the pre-flight ping retry limit from 15 to 45 attempts (45 seconds) in both
`toggle.sh` and `scripts/toggle-linux.sh`. This provides sufficient time for the DERP-assisted handshake to
complete on cold boots, while still passing immediately (in 1-2 seconds) on warm starts or once the path
is established.
1211
1212 ##### 15.5.5 Incident Post-Mortem: Docker Image Build DNS Failures on Immediate Restart (July 10, 2026)
1213 **Symptom:** When executing `toggle.sh --restart`, the script successfully stops the gateway but then
immediately fails during startup during `docker compose build` with DNS resolution errors:
1214 `failed to do request: Head "https://registry-1.docker.io/...": dial tcp: lookup registry-1.docker.io on
192.168.5.1:53: no such host`.
1215

```

```

1216 **Root Cause:** A race condition between physical link negotiation and virtual machine startup. The STOP path
    of the restart command invokes `cleanup_network_state` which power-cycles the host's physical Wi-Fi
    interface (`en0`) to flush stale routing states. Immediately after, `toggle.sh` starts the gateway boot
    sequence. Because `toggle.sh` did not verify the status of the physical interface, it proceeded to boot
    Colima and build Docker containers while `en0` was still negotiating a DHCP lease. Consequently, the host (
    and by extension, the VM bridge) had no active internet gateway/DNS path, causing Docker's registry check
    to fail.
1217
1218 **Fix (July 10, 2026):**
1219 1. Created a unified `wait_for_dhcp_settle` helper function in `scripts/common.sh` that polls `ipconfig
    getpacket` and `ifconfig` until a valid IP and router are assigned. To handle typical macOS Wi-Fi
    reassociation and DHCP negotiation latency (which commonly takes 15-20 seconds after a power-cycle), this
    loop was configured to poll up to 60 times (30 seconds total).
1220 2. Injected a call to `wait_for_dhcp_settle` at the beginning of the `START` action in `toggle.sh` (right
    before checking Colima status) to block container builds until the host's physical network is online.
1221 3. Refactored `recover.sh` to reuse the unified `wait_for_dhcp_settle` function.
1222
1223 #### 15.5.6 Watcher.sh Suppressed Exit Code Bug (Fixed)
1224 **Observation:** The `scripts/watcher.sh` script is a long-running daemon that invokes `recover.sh --post-wake`
    when it detects system wake or network roam events. Previously, it contained the following bug:
1225 ```bash
1226 bash "$RECOVER" --post-wake
1227 local exit_code=?
1228 ...
1229 return $exit_code || true
1230 ```
1231 The `|| true` mistakenly swallowed the actual `$exit_code` returned by `recover.sh`, causing `run_recover()` to
    always return `0` (success), masking any underlying failures in the wake recovery process.
1232
1233 **Fix:** The `|| true` statement was removed from the return line:
1234 ```bash
1235 return $exit_code
1236 ```
1237 Since `watcher.sh` explicitly omits `set -e` (to prevent pipe breakages from killing the daemon), removing `||
    true` safely allows the underlying exit code to propagate without risking daemon termination. The watcher
    daemon was then reloaded (`launchctl kickstart -k`) to pick up the script changes in memory.
1238
1239 #### 15.5.7 Network Readiness Race Condition on `--restart` (Fixed)
1240 **Problem:** When running `./toggle.sh --restart`, the script tears down the gateway and immediately begins the
    startup sequence. On systems with Wi-Fi (or any wireless interface), the OS takes a moment to re-associate
    with the network after the teardown's DNS/routing changes. If the startup sequence ran before the OS had a
    default route, `get_active_service` would return nothing, routing bypass targets would be wrong, and DNS/
    WARP setup would silently fail resulting in a partially configured gateway that self-heals only once the
    `watcher.sh` re-runs.
1241
1242 **Root cause:** The original code had no gate before the first network-dependent call (`ACTIVE_SERVICE=$(
    get_active_service)` at the top of the start branch). This was a pure timing race.
1243
1244 **First fix (Wi-Fi specific):** A polling loop on `ipconfig getifaddr en0` was added to wait for an IPv4
    address on `en0` before proceeding. This worked but was brittle it only handled Wi-Fi and would fail for
    Ethernet, USB-C adapters, or iPhone USB tethering.
1245
1246 **Generalized fix:** Replaced the `en0` check with `route get default | grep 'gateway:'`. This detects a valid
    default gateway on `any` interface, covering all connection types. The loop polls every 2 seconds with a
    60-second hard timeout (then proceeds with a warning rather than hanging forever). The output prints the
    detected gateway IP so failures are easy to diagnose in logs.
1247
1248 ```bash
1249 # In toggle.sh, before ACTIVE_SERVICE=$(get_active_service)
1250 while true; do
1251     if route get default 2>/dev/null | grep -q 'gateway: '; then
1252         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
1253         echo " Network ready (default gateway: $_gw)."
1254         break
1255     fi
1256     ...
1257 done
1258 ```
1259
1260 **Why not ping?** Pinging an IP (e.g. `1.1.1.1`) during restart is unreliable because DNS may still be hijacked
    from the previous session, and the exit node routing may not yet be cleared, causing the ping to loop back
    through the tunnel. `route get default` is a pure local kernel query no packets sent.
1261
1262 ### 15.6 WARP Tunnel Liveness & Host Leaks
1263
1264 #### 15.6.1 Host-Side Traffic Leaking Past the Exit Node (with Remote Clients Routing Correctly)
1265 **Symptom:** `toggle.sh` reports success. Remote tailnet clients (e.g. an S24 phone) correctly egress through
    Cloudflare WARP and see a Cloudflare IP on `whatismyip.com`. But on the **HOST Mac running the gateway,
    `whatismyip.com` reports the underlying physical ISP's ASN (e.g. `campus_isp` for a university campus Wi-Fi)
    . The gateway container itself is healthy; the leak is specifically on the host.
1266

```

```

1267 **Root causes.** Three distinct mechanisms can each produce this exact symptom on the host alone, while leaving
      remote clients unaffected. They are mutually rankable so the diagnostic picks the active one
      deterministically.
1268
1269 * *(A) SOCKS5 fallback lane active.** `toggle.sh`'s three Phase-B pre-flight checks (`tailscale ping` AdGuard
      via `dig +tcp` WARP internet, toggle.sh -lines 750-820) sometimes flake during the first minute. When ANY
      of the three fails, the script sets `SKIP_EXIT_NODE=true` (toggle.sh:849) and instead activates the SOCKS5
      fallback path (toggle.sh:894). The SOCKS5 fallback hijacks DNS to `127.0.0.1` (so AdGuard filtering still
      works) but **modern browsers on macOS do NOT honor the system SOCKS5 proxy** they ignore `networksetup -
      setsocksfirewallproxy`. Result: DNS gets ad-filtered but actual HTTP/S traffic exits direct through the
      campus ISP. This branch is invisible from a remote tailnet client because the client's tunnel is unaffected
      .
1270
1271 * *(B) IPv6 leak over campus Wi-Fi.** The Tailscale exit node and WARP tunnel are both IPv4-only (gluetun + `
      post-rules.txt` drop all IPv6 forwarded traffic; see 15.12 IPv6 Exit-Node Leak). But many university and
      enterprise APs are dual-stack the host happily accepts AAAA DNS responses and sends IPv6 traffic straight
      out `en0`. macOS happily encrypts that traffic through Tailscale ONLY if Tailscale has IPv6 advertised;
      with an IPv4-only exit node, IPv6 traffic skips Tailscale entirely. This branch is invisible from a phone
      because iOS/Android Tailscale apps actively sinkhole IPv6 in their VPN profile, but the macOS host's `
      tailscaled` does not.
1272
1273 * *(C) Route-table freeze and `--accept-routes` reset after `tailscaled` wake/toggle.** Running `tailscale up
      --reset` clears the exit-node preference and reverts `--accept-routes` back to its default (`false`).
      Without explicitly passing `--accept-routes=true`, `tailscaled` will not attempt to install the default
      route through the `utun*` tunnel interface even if the exit node preference is reconciled. Additionally,
      macOS occasionally freezes and fails to apply route-table changes (see 15.11 Standalone Tailscale Daemon
      Freeze). In both cases, `netstat -rn` shows the default route still pointing to the physical Wi-Fi gateway
      (e.g. `172.17.0.1`) instead of the Tailscale `utun*` interface, causing host traffic to exit direct while
      remote clients route correctly.
1274
1275 **Why remote clients don't see this.** Each remote phone/laptop uses its OWN Tailscale tunnel to reach the
      container's `100.x.x.x:443` over the mesh they're end-to-end encrypted regardless of how the host Mac
      itself is configured. Only the HOST's own egress sockets (HTTP requests from the host's browser, curl, etc
      .) are affected.
1276
1277 **Resolution.** A single script: `scripts/diagnose-host-leak.sh`. It runs the 8 checks, classifies into one of
      A / B / C / OK, prints a verdict + a ready-to-run fix command on stdout, AND writes the full report to `
      host-leak-diagnostic-<UTC>.txt` so the user can paste the file back instead of scrolling-and-copying from
      the terminal. With `--fix`, the matched remediation is applied and the two checks that could change (warp +
      ipv6) are re-verified. With `--watch` (or `--watch 30`), it runs the full baseline diagnostic then enters
      a continuous loop checking warp/IPv6/default-route every N seconds, alerting only on state changes logs to
      `host-leak-watch.log`.
1278
1279 ```bash
1280 # Diagnose only:
1281 bash scripts/diagnose-host-leak.sh
1282
1283 # Diagnose + apply matched fix + re-verify:
1284 bash scripts/diagnose-host-leak.sh --fix
1285 ```
1286
1287 **What it prints.** A 7-section report (`tailscale status`, SOCKS5 state, `cdn-cgi/trace` warp=, IPv6 leak
      probe, host default route, last toggle.sh output.log hints, resolved gateway Tailscale IP), followed by a
      classification block (`Scenario: A | B | C | OK`) with the exact command(s) the user needs to paste into
      their terminal they don't have to figure out which `tailscale up` flags match their environment.
1288
1289 **What it does NOT do.** It does not touch `toggle.sh` or `toggle.sh`'s pre-flight logic, does not modify `.env
      ` (except `--fix` mode appends `GATEWAY_BYPASS_PING=true` when fixing Scenario A), and does not restart
      Docker. It is a read-only diagnostic with an opt-in remediation flag. Safe to run on a healthy gateway the
      worst case is a 30-line report that says "OK".
1290
1291 **Why cdn-cgi/trace over whatismyip.com for the verdict.** `whatismyip.com`'s ISP database routinely mislabels
      campus IPv6 ranges as residential ISPs (that's why the local ISP's ASN appears even when you're routing
      through Cloudflare). `cdn-cgi/trace` is hosted by Cloudflare ITSELF and reports `warp=on|off` plus the
      exact edge colo serving the request. It is the only public endpoint that can truthfully confirm whether the
      WARP tunnel is in use.
1292
1293 **Common false alarms.**
1294
1295 * **`warp=on` but whatismyip.com still shows the local ISP.** `whatismyip.com`'s ISP-detection database is
      unreliable on academic networks. Trust `cdn-cgi/trace`'s `warp=on` over `whatismyip.com`'s ISP string. Use
      `https://api.ipify.org` (returns just the IP, no ISP DB lookup) as a third confirmatory check.
1296 * **`tailscale status` shows the host online but exit node preference is missing.** `tailscale up --reset --
      exit-node=` (empty value) clears any stale preference; `tailscale up --reset --exit-node=<100.x.x.x>` re-
      establishes it. The diagnostic prints the exact command with the resolved TS_IP filled in.
1297 * **Both IPv6 leak AND route-freeze flags set simultaneously.** Scenario B outranks Scenario C fixing IPv6
      makes IPv4 the only viable egress, after which the route-freeze usually self-resolves within ~30s (or one
      more `toggle.sh` cycle).
1298
1299 **Deep dive: Host Leak Probe Continuous Sub-Second Egress Monitor (`scripts/host-leak-probe.sh`).**
1300

```

```

1301 > ** STATUS: REMOVED (historical). ** `scripts/host-leak-probe.sh` and the `HOST_LEAK_PROBE` env flag no longer
    exist the file is deleted/untracked and nothing launches it (verified 2026-07-15). Its fail-closed role
    was superseded by the **PF kill-switch** (`enable_killswitch` in `common.sh`) plus the in-container **WARP
    Watcher** (15.6.2); the only surviving host-leak tool is the on-demand `scripts/diagnose-host-leak.sh`. The
    text below is kept for design rationale only treat every present-tense claim (auto-launch, PID file, `
    HOST_LEAK_PROBE=false` toggle) as past tense.

1302
1303 *What it is.* `scripts/host-leak-probe.sh` is a lightweight background daemon (~1.4 MB RSS, ~01% CPU) that
    polls `https://www.cloudflare.com/cdn-cgi/trace` every 300ms directly from the host via `curl` not via `
    docker compose exec`. This is a fundamentally different vantage point from the WARP Watcher (15.6.2): the
    WARP Watcher asks the WARP *container* whether it sees `warp=on`; the Host Leak Probe asks what a real
    browser request leaving the host's physical NIC looks like. The two blind spots it covers that the WARP
    Watcher cannot:

1304
1305 1. **Host-side flash-leaks** a Cloudflare anycast edge re-route, a browser reusing a stale pooled connection
    across a tunnel state change, or a split-second routing gap during a Gluetun healthcheck restart (see
    15.12.13 Routing Fix 30-second polling acknowledged 14s window).
1306 2. **Host egress while container is healthy** the container can report `warp=on` while the host's own traffic
    exits direct (the three-scenario leak class documented above in 15.6.1).

1307
1308 *Lifecycle.* Started by `toggle.sh`'s START path (`start_host_leak_probe`) immediately after `
    start_warp_watcher`. Controlled by:

1309
1310 | Signal path | What happens |
1311 |---|---|
1312 | `toggle.sh` STOP | `stop_host_leak_probe()` SIGTERM + PID file cleanup |
1313 | `toggle.sh` `cleanup_handler` (error/INT/TERM/HUP) | Same |
1314 | `recover.sh` (default, non `--post-wake`) | 6d block kill by PID file |
1315 | `recover.sh --post-wake` | Left running the gateway is still live |
1316
1317 PID file: `/tmp/nullexit-host-leak-probe.pid`. Toggle with `HOST_LEAK_PROBE=false` in `.env` to disable at next
    gateway start.

1318
1319 *Output format.* All events are written to `output.log` (repo root) with UTC timestamps. Only state *changes*
    are logged the probe is silent during normal healthy operation.

1320
1321 ...
1322 [09:10:26] LEAK warp=off ip=45.23.1.1 prev_warp=on prev_ip=104.28.246.50
1323 [09:10:26] ROTATE warp=on ip=104.28.248.11 prev_ip=104.28.246.50
1324 [09:10:26] HOST-PROBE failed/timeout (count=1)
1325 ...
1326
1327 curl transport errors (`TLS handshake failed`, `Connection refused`, etc.) are also appended to `output.log`
    via `2>>output.log` nothing is silenced to `/dev/null`.

1328
1329 *Grep cheatsheet.*
1330
1331 ```bash
1332 grep LEAK output.log          # real warp=off events from the host
1333 grep ROTATE output.log        # Cloudflare anycast edge rotations (harmless)
1334 grep 'HOST-PROBE' output.log  # curl failures / timeouts
1335 ```
1336
1337 *Resource budget.* ~1.4 MB RSS, ~0% CPU at rest, ~3 HTTPS requests/second to Cloudflare. Safe to leave running
    for hours. Back off the polling interval (`bash scripts/host-leak-probe.sh 1.0`) if you observe probe-
    failure pile-ups indicating Cloudflare rate-limiting.

1338
1339 *Detection vs prevention.* This script detects leaks; it does not prevent them. A sub-300ms flash-leak can
    still slip between two polls undetected. If `grep LEAK output.log` shows confirmed leak events, the next
    step is a kill-switch (prevention) an `iptables` rule on the host's `en0` that drops non-Tailscale, non-
    local egress whenever the gateway is active. That is a separate engineering task not yet implemented.

1340
1341 ##### 15.6.2 In-Flight WARP Tunnel Liveness Monitor & Statistical Auto-Shutdown
1342 **Why.** nullexit's core promise is that your IP always appears as Cloudflare. If the WARP tunnel silently dies
    due to a UDP NAT timeout, a Cloudflare edge rotation, a Colima VM clock skew causing TLS cert rejection,
    or any of a dozen other failure modes the host silently falls back to egressing through the physical ISP.
    The user sees the gateway as "up" (containers running, Tailscale connected, DNS hijacked) but their real IP
    is exposed. This is the exact class of failure that Ingo Blechschmidt documented in his chilling 2024
    Linux kernel post-mortem: for over two years, LUKS disk encryption on suspend was silently broken because a
    refactored kernel syscall returned success while doing nothing ([source](https://mathstodon.xyz/@iblech
    /116769502749142438)). The lesson: **a security mechanism that fails silently is worse than no mechanism at
    all** it creates a false sense of safety that prevents the user from taking corrective action.

1343
1344 **Design principle.** The WARP Watcher (`start_warp_watcher` in `toggle.sh`) is a background daemon that
    polls `cdn-cgi/trace` every 30 seconds and applies a **statistically calibrated threshold** before
    triggering any safety response. This threshold distinguishes transient network blips (which self-heal
    within seconds) from genuine outages (which require intervention). A single `warp=off` reading is
    meaningless Docker might be slow, the network might be congested, a container healthcheck might be
    restarting. But 6 consecutive off readings (30 continuous seconds of downtime) has vanishingly low
    probability of being a false positive: even at an unrealistically high 10% false-positive rate per poll, P
    (6 consecutive) = 0.1 0.0001%. In practice, the per-poll false-positive rate is far lower than 10%, making

```

```

1345         the real probability astronomically small.
1346 **What it does.**
1347
1348 1. **Always logs.** Every state transition (onoff, offon) is timestamped to `output.log`. The user can `grep
1349     WARP output.log` to audit the tunnel's entire lifetime. This is the "never fail silently" mandate.
1350
1351 2. **Tracks consecutive failures.** A counter increments on each `off` reading and resets to 0 on any `on`
1352     reading. This ensures the watcher never fires on intermittent flapping.
1353
1354 3. **Auto-shuts down at threshold.** When the counter reaches `WARP_FAIL_THRESHOLD` (default 6, configurable in
1355     `.env`), the watcher declares a statistically significant outage and runs `recover.sh` the nuclear
1356     recovery script that disconnects Tailscale, stops Docker containers, resets DNS to `1.1.1.1`, flushes
1357     routes, and power-cycles Wi-Fi. This instantly reverts the host to normal (un-encrypted) internet,
1358     guaranteeing no traffic leaks through a dead tunnel while the user thinks they're protected. A trigger file
1359     at `/tmp/nullexit-warp-shutdown-triggered` prevents double-firing.
1360
1361 4. **Exits cleanly.** After shutdown, the watcher exits (the gateway is dead, there's nothing left to monitor).
1362     `stop_warp_watcher()` is also called by `toggle.sh`'s STOP path and `cleanup_handler` to terminate the
1363     daemon cleanly during normal teardown.
1364
1365 **Configuration.** Set `WARP_FAIL_THRESHOLD=3` in `.env` for a more aggressive 15s timeout, or `
1366     WARP_FAIL_THRESHOLD=12` for a conservative 60s window. The variable is parsed from `.env` using the same `
1367     grep` pattern as `GATEWAY_MSS` and `GATEWAY_BYPASS_PING`.
1368
1369 **Log sample.** After a real outage (or a forced test via `docker compose pause warp`):
1370
1371     ...
1372     [2026-07-03T21:15:17Z] WARP DOWN    cdn-cgi/trace reports warp=off
1373     [2026-07-03T21:15:47Z] WARP SHUTDOWN 6 consecutive failures (threshold=6). Running recover.sh to kill the
1374     gateway and restore normal internet. Your IP is NO LONGER Cloudflare.
1375     [2026-07-03T21:15:52Z] WARP SHUTDOWN recover.sh completed, watcher exiting. Your IP is now your real ISP IP.
1376     ...
1377
1378 Or, if WARP self-recovers before the threshold:
1379
1380     ...
1381     [2026-07-03T21:15:17Z] WARP DOWN    cdn-cgi/trace reports warp=off
1382     [2026-07-03T21:15:32Z] WARP RECOVERED warp=on after 3 consecutive off readings (threshold was 6)
1383     ...
1384
1385 **Relationship to other monitors.** The WARP Watcher is the **innermost safety layer** it runs as a child of `
1386     toggle.sh` and only while the gateway is active. It complements (does not replace) the `scripts/watcher.sh`
1387     daemon (15.5.1, which handles sleep/wake and Wi-Fi roam) and `scripts/diagnose-host-leak.sh` (15.6.1,
1388     which is a manual diagnostic tool). The WARP Watcher is the only component that actively verifies end-to-
1389     end tunnel liveness and takes autonomous action when it fails.
1390
1391 ##### 15.6.3 Incident: The Overnight Silent IP Leak (July 2026)
1392 **The Problem:** The host leaked traffic over its raw, unencrypted `eth0` IP continuously for roughly 8 hours,
1393     completely bypassing the VPN. The `toggle.sh` state and the container liveness checks all reported the
1394     gateway as healthy and active.
1395
1396 **The Investigation:**
1397 1. **Sleep/Wake Checks:** We initially suspected macOS sleep cycles broke the routing table. A check of `pmset
1398     -g log` confirmed the laptop literally never slept (0 sleep events recorded since boot), ruling out all
1399     sleep/wake code paths.
1400 2. **Keepalive Expiry:** We discovered `docker-compose.yml` was missing `
1401     WIREGUARD_PERSISTENT_KEEPLIVE_INTERVAL`. Without a keepalive, standard router NAT tables (which typically
1402     garbage collect UDP after 30 to 120 seconds of silence) quietly expired the WireGuard port mapping
1403     overnight.
1404 3. **Container State Masking:** Because WireGuard is stateless, the `warp` container didn't immediately crash.
1405     It stayed "Up", which meant Docker's healthchecks and our `toggle.sh` liveness checks never noticed the
1406     tunnel inside it was totally dead.
1407
1408 **The Solution:**
1409 1. Added `WIREGUARD_PERSISTENT_KEEPLIVE_INTERVAL=25s` (the WireGuard standard to beat standard 30s NAT
1410     timeouts) to keep the UDP tunnel permanently pinned open through the router.
1411 2. Injected a **Fail-Closed Kill-Switch** into `routing-fix.sh`: A 30-second loop now actively verifies tunnel
1412     health by running `curl` over `tun0` to Cloudflare's trace endpoint. If it fails 3 consecutive times, it
1413     immediately rewrites the default route to `blackhole`, severing all traffic instead of letting it leak
1414     through `eth0`.
1415 3. Added a listener in `watcher.sh` that detects this fail-closed event and instantly pushes a native macOS UI
1416     notification to the desktop, alerting the user to manually intervene.
1417
1418 > [!NOTE]
1419 > **Why dual failure systems?**
1420 > The system now relies on two independent failure handlers that cover each other's blind spots:
1421 > 1. **Automated Self-Healing (WARP Watcher + `recover.sh`):** Triggered when the `warp` container logs `warp=
1422     off`. This happens when the server actively kicks the client. Because the container knows it is
1423     disconnected, it triggers `recover.sh` to safely reboot Docker and reconnect.

```

```

1392 > 2. **Fail-Closed Kill-Switch (`routing-fix.sh` + `watcher.sh`):** Triggered when a silent network failure
1393 occurs (e.g., NAT timeouts). The container incorrectly believes it is connected (`warp=on`), so auto-
1394 recovery never fires. Instead of auto-recovering (which risks falling back to raw Wi-Fi mid-process while
1395 the user isn't looking), this kill-switch actively blackholes the internet and forces a manual intervention
1396 via a desktop notification.
1397
1398 ### 15.7 Tailscale P2P, SNAT & Control Plane
1399
1400 #### 15.7.1 macOS SNAT Endpoint Poisoning & The P2P Blackhole (July 10, 2026)
1401 **Symptom:** When a Tailscale client (like an S24) connected to the exact same Wi-Fi network as the macOS
1402 gateway, it completely lost internet connectivity. However, if the S24 connected to a Windows PC's hotspot
1403 on the same network, the tunnel worked flawlessly.
1404
1405 **Root Cause:** Claude's critique correctly dismantled the port collision theory: the gateway daemon wasn't
1406 failing to bind, and macOS wasn't load-balancing ports. The real culprit was **macOS SNAT Source Port
Mangling poisoning Tailscale's path discovery**.
1407
1408 Here is the exact step-by-step of the "infinite loop" blackhole:
1409 1. **The P2P Handshake:** The S24 and the Mac are on the same Wi-Fi (`192.168.1.x`). The S24 discovers the Mac's
1410 local IP and sends a UDP hole-punch packet to `192.168.1.5:41641`. Colima successfully forwards this to
1411 the gateway container.
1412 2. **The Bypass Rule:** The gateway replies. Because `routing-fix.sh` forces all Tailscale UDP traffic out `
eth0` to bypass WARP, the reply goes to the macOS host.
1413 3. **macOS SNAT Mangling:** macOS routes the reply out `en0` to the S24. Because the packet crosses from the
1414 Colima VM bridge to the Wi-Fi interface, macOS applies Source NAT (SNAT). Critically, macOS **randomizes
the source port** (e.g., changes it from `41641` to `58291`).
1415 4. **Endpoint Poisoning:** The S24 receives the authenticated Tailscale packet from `192.168.1.5:58291`.
1416 Tailscale's `magicsock` is designed to handle NAT dynamically, so it assumes the Mac has roamed to port
1417 `58291`! The S24 updates its endpoint and starts sending all its encrypted internet traffic to
1418 `192.168.1.5:58291`.
1419 5. **The Blackhole:** macOS has no port forward for `58291`. It drops 100% of the S24's outbound data packets.
1420 The connection is completely dead.
1421
1422 **Why did the Hotspot work?** When the S24 connected to the PC hotspot, it was behind the PC's NAT. The S24
1423 sent the packet, PC NATted it, and the Mac replied (mangled to `58291`). When the reply hit the PC, the PC's
1424 strict NAT dropped it because the source port (`58291`) didn't match the destination port it originally
1425 sent to (`41641`). Because the mangled packet was dropped, the S24 never poisoned its endpoint! It safely
1426 gave up on P2P, fell back to the 100% reliable TCP DERP relay, and the internet worked perfectly.
1427
1428 **Fix & Resolution (July 10, 2026):**
1429
1430 **Phase 1 Port disambiguation:** Changed the gateway container's Tailscale listen port from the default
1431 `41641` to `41642` across `docker-compose.yml`, `post-rules.txt`, and `routing-fix.sh`. This prevents any
1432 bind conflict with the macOS host's native Tailscale daemon.
1433
1434 **Phase 2 RFC1918 DROP rules:** Updated `routing-fix.sh`'s `add_tailscale_p2p_bypass()` to explicitly drop all
1435 outgoing Tailscale UDP packets destined for private subnets (`192.168.0.0/16`, `10.0.0.0/8`,
1436 `172.16.0.0/12`) when `TAILSCALE_ALLOW_LAN_P2P=false`. This surgically kills the local P2P hole-punch
1437 attempt *before* it can trigger the macOS gvproxy SNAT mangling. The S24 receives a clean failure and
1438 immediately locks onto the public DERP relay with 0% packet loss.
1439
1440 **Phase 3 Automatic network detection:** Because `TAILSCALE_ALLOW_LAN_P2P=false` breaks P2P on trusted
1441 hotspots that genuinely don't have AP isolation, a `.env` toggle and a full auto-detection system were
1442 implemented:
1443
1444 - **`.env` toggle:** `TAILSCALE_ALLOW_LAN_P2P=true/false` static fallback, used only if auto-detection cannot
1445 run.
1446 - **`watcher.sh` auto-detection (`detect_lan_p2p_mode`):** On every network change, `watcher.sh` runs two
1447 checks:
1448 1. **WPA2-Enterprise detection:** `airport -I` is used to read the current Wi-Fi `link auth` field. If the
1449 auth type does not contain `-psk` (e.g., it is plain `wpa2` = 802.1x), the network is classified as
1450 enterprise and P2P is forced off. *(Note: the `airport` binary was removed in macOS Sonoma 14.4 the
1451 detection now uses `system_profiler SPAirPortDataType`; see 15.11 for the full migration.)*
1452 2. **AP Isolation probe:** For WPA2-Personal networks, a short `ping` is sent to the default gateway. If the
1453 gateway responds (it always should on a real home/hotspot network), P2P is allowed. If not, AP isolation is
1454 inferred and P2P is forced off.
1455 - The result (`true` or `false`) is written to `.lan_p2p_detected` in the repo root.
1456 - **`routing-fix.sh` dynamic reading:** `add_tailscale_p2p_bypass()` re-reads `.lan_p2p_detected` on every 30-
1457 second loop iteration and atomically adds or removes the RFC1918 DROP rules. **No container restart is
needed when switching networks.**
1458
1459 **Known Unknowns (Pending Verification):**
1460 - The gvproxy SNAT source port mangling theory is mechanistically sound and backed by observed behavior (
gvproxy's UDP proxy changelog has documented edge cases in this exact code path), but has not yet been
1461 confirmed via `tcpdump -i en0 udp portrange 40000-60000` while triggering a P2P handshake from the phone. A
1462 packet capture cross-referenced with `tailscale ping` endpoint reports on the phone side would confirm the
1463 theory definitively.
1464 - Claude's recommendation: run `tcpdump -i en0 udp port 41642 or portrange 40000-60000` on the Mac while
1465 triggering the handshake from the phone to see whether the reply leaves with a different source port than
1466 `41642`.

```

1428 - ****Client-Side VPN Cycling (Roaming Recovery):**** An observation (July 10, 2026) suggests that when a hung DNS probe state is triggered by roaming between Access Points (APs) on a WPA2-Enterprise network, simply toggling the VPN connection (Tailscale) off and on locally on the client phone immediately resolves the issue. You do not need to cycle the phone's physical Wi-Fi. This reinforces the theory that roaming on Enterprise networks leaves a stale/poisoned P2P endpoint in the phone's Tailscale magicsock state, which gets instantly flushed upon a local VPN restart. (Status: Possible but not formally verified).

1429

1430 **#### 15.7.2 July 6, 2026: Packet Filter (pfctl) Defaulting Local LAN Rules to TCP-Only**

1431 ****Symptom:**** Devices on the exact same local Wi-Fi network as the gateway were experiencing ~500ms latency to the gateway instead of the expected ~80ms.

1432

1433 ****Root Cause:**** When writing the ``pf.conf`` rules to whitelist local LAN traffic (``pass out quick on $ext_if to { 192.168.0.0/16, ... }``), the rule omitted an explicit protocol. The macOS ``pfctl`` compiler implicitly applies state tracking to all ``pass`` rules. However, because state tracking (``keep state``) natively evaluates TCP sessions, ``pfctl`` automatically appended the ``flags S/SA`` modifier to the compiled rule in the kernel. This silently restricted the whitelist to ****TCP only****, causing all local UDP traffic to be dropped by the default-deny kill-switch. Since Tailscale relies on UDP port 41641 for direct P2P connections, the local devices were unable to hole-punch and were forced to fall back to routing traffic through a remote Tailscale DERP relay, massively inflating latency.

1434

1435 ****Resolution:**** Split the single implicit rule into explicit ``proto tcp``, ``proto udp``, and ``proto icmp`` rules in ``scripts/pf.conf`` to guarantee the macOS compiler allows all three core transport protocols on the local subnet without mutating the rule into a TCP-only lock.

1436

1437 **#### 15.7.3 Incident Post-Mortem: PF Kill-Switch Bricks Host Internet in SOCKS5 Fallback Mode (July 10, 2026)**

1438 ****Symptom:**** When the pre-flight checks fail (e.g. because host Tailscale was unauthenticated/logged out), the script falls back to SOCKS5 proxy mode but the host immediately loses all internet connectivity and tailscaled reports ``You are logged out... connect: no route to host``. Even running ``sudo tailscale up`` to log in fails because the connection to the control plane times out.

1439

1440 ****Root Cause:**** An architectural mismatch. The macOS Packet Filter (PF) Kill-Switch works at the IP level: it blocks all outbound traffic on the physical interface (``en0``) except to explicitly whitelisted VPN endpoints, expecting all other traffic to go through the virtual VPN interface (``utun*``). In SOCKS5 fallback mode, the exit-node (``utun*``) is NOT enabled; instead, only specific applications (like browsers) use the localhost SOCKS5 proxy, while all other host traffic (including the ``tailscaled`` daemon's UDP packets) continues to go through ``en0``. Because the script still enabled the PF Kill-Switch during SOCKS5 fallback, it blocked all non-SOCKS5 traffic on ``en0``, completely bricking the host's internet and locking the Tailscale daemon out of the control plane.

1441

1442 ****Fix (July 10, 2026):**** Modified ``toggle.sh`` to remove the ``enable_killswitch`` call from the SOCKS5 fallback path. The firewall kill-switch is now strictly reserved for exit-node VPN mode, ensuring SOCKS5 fallback leaves the host's direct internet route open for system daemons to authenticate.

1443

1444 **#### 15.7.4 Incident Post-Mortem: Host Tailscale Logouts due to Aggressive reset Flags (July 10, 2026)**

1445 ****Symptom:**** Every time the user runs ``toggle.sh`` or ``recover.sh`` which disconnects/reconnects the host client, the host client randomly gets logged out, forcing them to re-run ``sudo tailscale up`` to authenticate.

1446

1447 ****Root Cause:**** An overly aggressive configuration reset. The host-side ``tailscale up`` calls (used in ``disconnect_tailscale_host`` and the startup/enable exit node paths) all passed the ``--reset`` flag. On macOS, ``--reset`` resets the entire configuration state and authentication token cache of the daemon. Since the script does not pass a ``TS_AUTHKEY`` to the host's commands (which are meant for the user's private interactive client), the ``--reset`` flag effectively logged the user out of their tailnet, requiring interactive re-login.

1448

1449 ****Fix (July 10, 2026):**** Removed the ``--reset`` flag from all host-side ``tailscale up`` invocations in ``toggle.sh`` and ``scripts/common.sh``. This ensures that exit-node state transitions (enabling/disabling) are handled safely while preserving the host client's authenticated session.

1450

1451 **#### 15.7.5 Incident Post-Mortem: macOS Tailscale Flag Requirement Abort (July 10, 2026)**

1452 ****Symptom:**** When the ``toggle.sh`` stop trap or the roaming watcher triggered, the gateway failed to shut down or disconnect properly. The ``output.log`` showed the error: ``Error: changing settings via 'tailscale up' requires mentioning all non-default flags. To proceed, either re-run your command with --reset...``

1453

1454 ****Root Cause:**** In a previous attempt to stop host-side logouts (15.7.4), we removed the ``--reset`` flag from all ``tailscale up`` invocations. However, if the Tailscale daemon on macOS is already running with non-default flags (like ``--ssh`` or ``--accept-routes``), invoking ``tailscale up`` with only a subset of flags (like ``--exit-node=``) triggers a strict safety check in the Tailscale CLI that aborts the command completely. Because the command aborted, exit-node routing remained stuck.

1455

1456 ****Fix (July 10, 2026):**** Migrated all specific preference changes in ``toggle.sh`` and ``scripts/common.sh`` from ``tailscale up`` to a two-step sequence:

1457 1. ``tailscale up`` (with no flags) to safely ensure the interface is brought up using the current authenticated state.

1458 2. ``tailscale set --exit-node=...`` to modify the specific target preference cleanly without triggering the missing flags error or requiring ``--reset``.

1459

1460 **### 15.8 Docker / Colima Lifecycle**

1461

1462 **#### 15.8.1 Colima VM Memory / Swap Thrashing Latency**

1463 ****Symptom:**** After running nullexit flawlessly for over 24 hours straight to test stability, the 0.5GB VM RAM configuration began experiencing severe network latency on the Tailscale mesh later in the day.

1464
1465 ****Root Cause:**** Services (like AdGuard, Tailscale, Docker logs) slowly accumulate cache and state over extended periods. Under the tight 512MB physical RAM limit, this slow creep forced the Linux kernel to aggressively swap memory to the 512MB SSD swap file. The constant disk I/O thrashing blocked process execution, causing DNS resolutions and packet forwarding to delay significantly (making the internet feel "very slow").

1466
1467 ****Proof (Empirical Observation):**** While in this OOM-thrashing state, direct DNS queries through the Tailscale mesh (`dig @100.100.21.8 google.com`) would intermittently hang or take several seconds to resolve. After restarting with the 600MB configuration, the exact same query immediately dropped to a lightning-fast **41ms** response time. (Note: Querying the mapped host port via `127.0.0.1:5354` will always time out due to Gluetun's strict leak-prevention firewall blocking non-VPN ingress traffic).

1468
1469 ****Fix:**** Increased VM base physical RAM to 600MB (`--memory 0.6`) and reduced the swap file to 400MB. This provides enough native RAM headroom for long-running state caches without forcing heavy I/O swapping. Subdomain deduplication still reduces blocklists by ~60%. Memory profiles (`light`/`medium`/`heavy`) let users trade off coverage for memory.

1470
1471 **#### 15.8.2 Missing iptables in routing-fix Container**

1472 ****Root Cause:**** `alpine:3.20` does not have iptables installed. Commands failed silently with `2>/dev/null || true`.

1473
1474 ****Fix:**** Updated `docker-compose.yml` to `apk add --no-cache iptables iproute2` before `scripts/routing-fix.sh`.

1475
1476 **#### 15.8.3 DOCKER_GW Variable Parsing Bug**

1477 ****Root Cause:**** `ip route show default` printed two lines; `awk '{print \$3}'` picked up both, causing route commands to fail.

1478
1479 ****Fix:**** Added `head -1` to the parsing chain.

1480
1481 **#### 15.8.4 Hard Reboots Leave Stale Docker Socket in Colima VM**

1482 ****Problem:**** If the macOS host machine is subjected to a hard reboot, sudden power loss, or a kernel panic, the Colima VM running in the background is abruptly killed. Upon next boot, running `toggle.sh` resulted in the contradictory output:
`Colima is already running.` followed by `Cannot connect to the Docker daemon at unix:///./colima/default/docker.sock.`

1483
1484 ****Root Cause:**** The hard crash prevents the inner Docker daemon from executing its graceful shutdown routines. It leaves behind a corrupted `docker.sock` file inside the VM. On boot, the outer VM spins up (hence "already running"), but the inner Docker daemon sees the stale socket, assumes another instance is running, and crashes.

1485
1486 ****Fix:**** Added an Auto-Healing routine directly into `toggle.sh` (Step 4b). The script now explicitly tests the Docker daemon with `docker ps`. If the daemon is unresponsive despite Colima reporting as active, it assumes a stale socket and automatically runs `colima restart` in the background to cleanly rebuild the VM and socket before proceeding.

1487
1488 **#### 15.8.5 Cloudflare WARP 24-Second Startup Delay (UDP Session Rate-Limiting)**

1489 ****Context:**** When running `toggle.sh` to turn the gateway ON, `docker compose up -d` often hangs for exactly 24-25 seconds waiting for the `warp` container to become `healthy`. Users might mistakenly blame the Go `rule-compiler` processing 400k+ rules for this delay, as Docker prints `rule-compiler Exited 24.3s` at the end of the wait.

1490
1491 ****Root Cause:**** The `rule-compiler` actually finishes its work in <2 seconds. The 24-second blockage is entirely due to Cloudflare WARP's edge server rate-limiting. WireGuard is a stateless protocol with no "disconnect" packet. When the toggle is turned OFF, the old UDP session state remains active on Cloudflare's backend for 20-30 seconds. When the toggle is instantly turned back ON, Docker starts a new WireGuard connection from a new ephemeral UDP source port but uses the **same static cryptographic key** (from `.env`). Cloudflare detects this as a potential replay attack or key-sharing abuse, and intentionally drops all packets (rate-limits) for the new connection until the old session's ghost state fully expires (~20-25 seconds).

1492 ****Result:**** Gluetun's healthcheck fails repeatedly every 2 seconds until Cloudflare lifts the penalty box, at which point traffic flows and Docker unblocks the rest of the startup sequence. This is a known, expected behavior of reusing static keys rapidly on Cloudflare's network and cannot be bypassed.

1493
1494 **#### 15.8.6 July 4, 2026: Multi-stage Docker Builds for Go Ports & Teardown Hang Fix**

1495 ****Symptom:**** The Python implementations of `logger.py` and `sync-rules.py` were slow and required a heavy python runtime inside Alpine containers. Also, `routing-fix` container teardown was hanging for 30s during `toggle.sh` shutdown.

1496 ****Root Cause:****

1497 - Python is slower than Go and installing it dynamically via `apk add` added overhead and complexity to the containers.

1498 - Docker containers ignore `SIGTERM` if the init process doesn't handle it, causing a full 30s timeout on shutdown.

1499 ****Resolution:****

1500 - Ported `logger` and `sync-rules` to Go using Goroutines for massive concurrent speedups.

1501 - Implemented ****Multi-stage Docker builds**** (using `golang:1.22-alpine` as builder) to compile the static binaries inside Docker. The final containers are raw Alpine with zero dependencies.

1502 - Added `--build` to `docker compose up -d` in `toggle.sh` to ensure updates are consistently applied on boot.

1503 - Added `stop\_grace\_period: 1s` to `routing-fix` in `docker-compose.yml` which eliminates the 30-second teardown hang instantly.

```

1504 - Implemented a cryptographic integrity checker (`scripts/crypto.sh`) using HMAC-SHA256 and `NULLEXIT_SEED` in
1505   `.env` to prevent tampering of bash scripts. (See also 15.9.4.)
1506 ##### 15.8.7 Colima `start` Post-Provision Hang (~2 min) & the Docker-Readiness Poll (July 14, 2026)
1507 **Symptom:** `toggle.sh` took ~197s of wall-clock time to bring the gateway up, and almost the entire delay
    lived in a single step: `colima start`. Timestamps in `output.log` showed the VM reaching `READY` and
    creating the `colima` Docker context in ~8-12 seconds, after which `colima start` simply *did not return*
    for ~110 more seconds until the `run_with_timeout 120` watchdog `SIGKILL`ed it. The rest of the boot (
    containers, routing) then proceeded normally.
1508
1509 **Root Cause:** A colima-internal hang that occurs *after* `Successfully created context colima` is printed. It
    is **mode-independent**. It was initially misattributed to `--network-address` / `--network-mode bridged`
    needing the `socket_vmnet` helper, but a controlled test with plain user-mode networking (`colima start --
    memory 0.6 --vm-type vz`, no `--network-address`) reproduced the identical ~120s hang (04:40:34 04:42:34,
    watchdog-killed) while Docker was fully usable the entire time. Crucially, killing the hung `colima start`
    process does **not** stop the VM and does **not** corrupt state `colima status` correctly reports `running`
    afterwards and `docker` keeps working via the `colima` context. The ~110s was therefore pure dead time
    waiting on a CLI that had already done its real work.
1510
1511 **Fix:** `scripts/common.sh` now provides `colima_start_until_ready()`. It launches `colima start "$@"` in the
    background and polls `docker --context colima info` every 3s. The moment Docker answers (~12s in practice)
    it stops waiting, reaps the (usually still-hung) starter with `kill`, runs `docker context use colima` to
    guarantee the default context is set, and returns 0. If Docker never answers within a 90s cap it returns
    non-zero and `toggle.sh` continues to the existing Step 4b `docker ps` auto-heal (15.8.4). Both Colima
    start call sites in `toggle.sh` the LAN-P2P bridged/shared path and the user-mode fast path now go
    through this helper instead of `run_with_timeout 120 colima start`.
1512
1513 **Result:** The Colima phase dropped from ~120s to ~12s and total `toggle.sh --restart` wall time from ~197s to
    ~106s, with the double-tunnel and direct hostcontainer P2P confirmed intact after the change. The key
    insight: treat "Docker is reachable" not "the `colima start` CLI returned" as the definition of *started
    *.
1514
1515 ##### 15.8.8 Toggle-On Startup Optimizations: Rule-Compiler Off the Critical Path (July 14, 2026)
1516 **Context:** A genuine cold toggle-on (VM bounced for RAM, WARP down) measured **95s**. Profiled per-phase with
    the now-fixed `DEBUG_TRACE` (15.11.8-A): Colima cold start **19s** (near floor), `docker compose up` **45s
    **, tailscale connect+poll **21s**, final checks **7s**.
1517
1518 **Two wrong hypotheses first (documented so they aren't re-tried):** (1) that BuildKit `--build` was the ~30s
    cost it wasn't; every layer was already `CACHED`, so gating `--build` saved only ~2s. (2) that `docker
    compose exec` was slow in the poll loop it isn't (0.18s steady-state vs 0.06s for `docker exec`); the ~5s/
    iteration was `tailscale status` **blocking during cold connect** and hitting the `run_with_timeout 5` cap.
1519
1520 **Real bottleneck (evidence-backed):** the 45s `compose up` was **not** WARP. On a genuine cold start WARP goes
    healthy fast (tor starts right with it no 15.8.5 same-key penalty; that only bites rapid offon). The long
    pole is the **rule-compiler re-deduping ~340k rules inside the 600MB VM (~28s), every toggle** `
    adguardhome` started 28s after warp because it waited on `rule-compiler: service_completed_successfully`,
    even though all 16 remote lists loaded from cache unchanged.
1521
1522 **Fix hash-gated, off the critical path (kept in-container):**
1523 - `rule-compiler` is now a **`compile`-profiled** service, excluded from `docker compose up`. `adguardhome` and
    `routing-fix` no longer `depends_on` it they boot instantly on the `compiled_rules.txt` / `ip_blocklist.
    ipset` already in `./adguard/work`.
1524 - `toggle.sh` runs it on-demand via `docker compose run --build --rm rule-compiler`, **gated on `
    compute_rules_hash`** `** = sha256 of `black_list.txt` + `white_list.txt` (the lists the user controls).
    Recompile only when that hash changes, when an artifact is missing, or when the compiled file is older than
    `RULES_MAX_AGE_DAYS` (default 7, so the remote blocklists still refresh occasionally). A compile failure
    is **non-fatal** we keep the previous compiled rules rather than bricking the tunnel (ad-block the kill-
    switch).
1525 - Also trimmed the tailscale-connect poll timeouts (status `5s3s`, `ip -4` `10s5s`) to reduce overshoot.
1526
1527 **Why NOT run the compiler natively on the Mac host (the tempting idea):** rejected on purpose. There is no Go
    on the host, so "native" means either `brew install go` (a host toolchain dependency that drifts the
    project from *portable+secure-on-any-device* toward *faster-on-mine*) or committing a prebuilt arch-
    specific **binary blob** you'd then have to trust/sign. Worse, it would make the **host** write into the
    `./adguard/work` bind-mount violating the deliberate *single allowed writer is the rule-compiler
    container* invariant (hostcontainer UID/perms mismatches corrupt AdGuard's BoltDB store). Staying in-
    container preserves both properties; the gate delivers the speed.
1528
1529 **Result:** unchanged toggle `compose up` **45s 9s**, total `--restart` ***95106s 63s**, with AdGuard still
    serving the full 342,519 rules (it reuses the existing compiled file) and the double-tunnel + direct
    hostcontainer P2P intact. Editing `black_list.txt` / `white_list.txt` flips the hash one-off recompile new
    rules take effect. Both lists were added to the `crypto.sh` signed set (they're security inputs), so
    editing them requires a re-sign which is the same moment you'd want the recompile. Local state files `
    build_hash` / `rules_hash` are gitignored.
1530
1531 ### 15.9 Permissions, Crypto & Security Hardening
1532
1533 ##### 15.9.1 Sudo Credential Caching Removed
1534 **Problem:** The script used `sudo -v` at startup to cache sudo credentials, plus a background `SUDO_KEEPER_PID`
    loop refreshing them every 60 seconds. This required interactive password entry even with NOPASSWD
    sudoers configured, because `sudo -v` itself prompts.

```

```

1535
1536 **Fix:** Removed the entire `sudo -v` + `SUDO_KEEPER_PID` section. All privileged commands now use `sudo -n` (
        non-interactive), which works silently because the user has NOPASSWD rules in `/etc/sudoers.d/nullexit` for
        the specific commands needed (`networksetup`, `python3`, `dscacheutil`, `killall`, `pkill`, `route`, `
        ifconfig`).
1537
1538 ##### 15.9.2 Background Sudo Failures
1539 **Context:** The `--post-wake` script is intentionally designed to skip the `sudo -v` interactive prompt
        because it is launched by `watcher.sh` running via `launchd` in the background (which has no TTY and cannot
        accept password input). Thus, it relies entirely on `sudo -n` (non-interactive sudo) for all its internal
        routing changes.
1540 **Problem:** If the user has not configured the `/etc/sudoers.d/nullexit` file properly, any automated
        background wake event will silently fail: `sudo -n` commands drop out, the WARP bypass routes fail to apply
        , and the `warp` container loses internet connectivity.
1541
1542 ##### 15.9.3 Tailscale Hijacking the Default Route (PHYSICAL_GW Bug)
1543 **Problem:** When `--post-wake` runs, it clears the routing table using `sudo route -n flush` to ensure a clean
        slate after the Wi-Fi card roams or wakes up. Because it skips bouncing the Wi-Fi interface (to make the
        reconnect faster), macOS's DHCP client does not automatically rebuild the physical default route. To
        counteract this, the script must manually restore the physical default route. Originally, the script tried
        to capture the physical gateway via `route get default`. However, because the gateway is already running
        and Tailscale is active, the default route is often already pointing to `utunX`. When this happens, `route
        get default` does not output a `gateway:` field (it only outputs the `interface:`), causing the captured `
        PHYSICAL_GW` variable to be empty.
1544 When `PHYSICAL_GW` is empty, the physical default route is never restored, causing the `en0` interface to be
        isolated from the physical network. The WARP bypass routes then fall back to targeting `interface en0` (
        instead of routing via the gateway), which breaks WireGuard's remote connections completely.
1545 **Fix:** The script now bypasses the OS routing table entirely and directly queries the hardware DHCP lease (
        `ipconfig getpacket en0`) to reliably extract the physical router IP, ensuring the physical route is
        correctly restored even when Tailscale is active.
1546
1547 ##### 15.9.4 July 4, 2026: Consolidation of Core Bash Scripts (Refactoring)
1548 **Symptom:** Over 200 lines of bash logic were directly duplicated across `toggle.sh` and `recover.sh` (e.g. `
        restart_tailscaled_daemon`, `stop_sleep_prevention`, WARP bypass routes). This caused drift when one script
        was updated without the other.
1549 **Root Cause:** Historical separation of responsibilities allowed `recover.sh` to grow complex alongside `
        toggle.sh`.
1550 **Resolution:**
1551 - Massively refactored both scripts by extracting all duplicated networking logic, PID lifecycle management,
        and environmental configuration down to `scripts/common.sh`.
1552 - Specifically, the `stop_sleep_prevention`, `stop_dns_watcher`, `stop_host_leak_probe`, and `stop_warp_watcher
        ` functions were collapsed into a single `stop_pidfile_daemon()` abstraction.
1553 - Proxy disable loops were abstracted to `disable_all_proxies()`.
1554 - The `162.159.192.1/193.1` WARP edge IPs are now dynamically resolved from `.env` instead of hardcoded.
1555
1556 ##### 15.9.5 Threat Model: Local Proxy Discovery
1557 The Tor SOCKS proxy and Honey-Port Tripwire only bind to `127.0.0.1` (loopback). The real security boundary
        here is "can an unprivileged local process reach the loopback interface," not "does the malware know the
        port number."
1558
1559 The port-randomization and the Honey-Port trap only catch blind or generic port-scanners that do not already
        know where to look. They **do not protect** against a process that directly reads the `/tmp/nullexit-ports.
        env` file, greps the `toggle.sh` memory space, or runs `lsof -i` to inspect open socket bindings.
1560
1561 Because modifying file ownership on `/tmp/nullexit-ports.env` via `chown`/`chmod` to restrict read-access
        introduces an unavoidable Time-Of-Check to Time-Of-Use (TOCTOU) race condition (the file is created user-
        owned before privilege escalation, meaning file descriptors can be snatched), it is deliberately left as a
        standard user-owned file.
1562
1563 The actual, proper mitigation for preventing a malicious local process from reaching the proxy is not hiding
        the portit is running a host-level outbound firewall with strict localhost/loopback filtering enabled (e.g
        ., LuLu 4.3.0+ or Little Snitch). These firewalls gate access by cryptographic process identity (binary
        signature) at the kernel/network-extension level, rather than by port secrecy.
1564
1565 **Verifying Localhost Filtering (The Rogue Binary Test).** To empirically validate that the host firewall
        properly intercepts local proxy hijacking, you can compile and execute a completely unknown, un-whitelisted
        "rogue" binary to attempt a connection to the proxy port:
1566
1567 ```c
1568 // lulu-test.c
1569 #include <stdio.h>
1570 #include <stdlib.h>
1571 #include <arpa/inet.h>
1572
1573 int main(int argc, char *argv[]) {
1574     int port = atoi(argv[1]);
1575     int sock = socket(AF_INET, SOCK_STREAM, 0);
1576     struct sockaddr_in server = { .sin_family = AF_INET, .sin_port = htons(port) };
1577     server.sin_addr.s_addr = inet_addr("127.0.0.1");
1578
1579     printf("Rogue binary attempting connection to 127.0.0.1:%d...\n", port);

```

```

1580     if (connect(sock, (struct sockaddr *)&server, sizeof(server)) < 0) {
1581         printf("Connection BLOCKED by firewall.\n");
1582         return 1;
1583     }
1584     printf("Connection SUCCESSFUL (Firewall bypassed!).\n");
1585     return 0;
1586 }
1587 ...
1588 Compile with `gcc -o lulu-test lulu-test.c`. When executed against the `$_TOR SOCKS_PORT`, a properly configured
    host firewall (with "Allow Loopback" disabled in LuLu Preferences) will immediately pause the socket
    execution and generate a desktop alert for the unrecognized binary signature. This physically stops
    targeted local malware from exfiltrating data through the Tor tunnel, proving the threat model mitigation.
1589
1590 ##### 15.9.6 Unlocking Mode-Locked Output Files Without `chmod` (`unlock-files.sh`)
1591 **The one-command fix:** `bash scripts/unlock-files.sh` replaces every known locked inode in the project with
    a fresh writable one. No `chmod`. Covers `.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`
    `, `ip_blocklist.ipset`, `cache/*.txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode
    `0444`). Run it once after setup or after pulling an old repo; `toggle.sh` and `sync-rules.py` will then
    work without permission errors.
1592
1593 **Context:** The nullexit project has a strict project-wide policy of `no chmod from scripts` (see README 6).
    This rule exists because every prior `chmod 0444` (post-write tamper-proof lock) and `chmod 000` (post-
    toggle `.env` lock) call from `scripts/sync-rules.py` / `toggle.sh` could leave a file permanently
    unreadable on disk if the script exited early, was killed mid-flight, or simply set a restrictive mode
    without ever being re-flipped. We've stripped every `chmod` from the codebase. But files **written by older
    versions of those scripts** are still on disk in those restrictive modes (`0444` for `compiled_rules.txt`,
    `ip_blocklist.ipset`, `data/filters/<id>.txt`; `000` for `.env` after end-of-toggle lock). Re-running `
    toggle.sh` or `scripts/sync-rules.py` against those leftovers hits `PermissionError: [Errno 13] Permission
    denied` immediately.
1594
1595 **Why the stale permissions also block `mv`/`rm` of the parent folder:** The restrictive modes (`000` on `.env
    `, `0444` on output files) don't just block writes they prevent the kernel from unlinking the file's
    dentry during a `mv` or `rm` of the **containing directory** (`adguard/work/`). macOS enforces this at the
    VFS layer: you own the directory, but you can't delete a directory that contains entries you can't write to
    . This made the entire `adguard/work/` tree unmovable/undeletable until the locked inodes inside it were
    replaced. `unlock-files.sh` resolves this permanently by swapping each locked inode for a fresh `0644` one
    via atomic rename (`cp` + `mv`), which operates on the directory entry rather than the file contents no `
    chmod` called, no permission bypass needed.
1596
1597 **Problem:** You (the owner) cannot `cat`, `cp`, `rm`, `>` redirect, or any normal POSIX write against a `mode
    =000` file even though you own it the basic mode bits block ALL access for owner, group, and other. The
    script does not call `chmod` and you want a permanent fix that never relies on a chmod from any future run
    either.
1598
1599 **Solution:** Replace the locked inode with a fresh readable one. `mv` is atomic; mode bits live in the inode,
    not the directory entry. Two flavors cover every case we hit on a real machine:
1600
1601 **Flavor A locked file is mode `000` (owner can't even READ it):**
1602 NOPASSWD sudoers (`/etc/sudoers.d/nullexit`) already includes `/usr/bin/python3`. So we read via `sudo python3`
    (which has the DAC override root has) and pipe the bytes around as base64 in user space, where the
    redirect happens with the user's normal umask = `0644`:
1603 ```bash
1604 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
    decode())" > /tmp/snap.b64
1605 base64 -d < /tmp/snap.b64 > <FILE>.new
1606 mv -f <FILE>.new <FILE>
1607 ls -la <FILE>          # mode is now 0644
1608 ```
1609 The old `mode=000` inode is unlinked by `mv`; the new `mode=0644` inode (created by base64-decode via the
    calling user's umask) takes the path. No chmod ever called. The parent directory just needs to be writable
    by you (it always is, since you own it).
1610
1611 **Flavor B locked file is mode `0444` (you can READ but cannot WRITE; `scripts/sync-rules.py` needs to re-
    write):**
1612 You own the file and can read it, you just cannot truncate/overwrite it. The simplest path is to rename it
    aside and let the writer create a fresh inode from scratch (which gets the umask-default `0644`):
1613 ```bash
1614 mv <FILE> /tmp/old.<FILE>.$$
1615 python3 scripts/sync-rules.py          # now writes <FILE> cleanly at mode 0644
1616 ```
1617 Or apply Flavor A if you'd prefer to preserve the contents while resetting the mode.
1618
1619 **Why this is the canonical recovery, not a hack:**
1620 - Mode bits live in the inode, not the path. `mv -f` replaces the path's inode reference; the old locked inode
    drops out of the directory entirely (dentry eviction) and loses its last link, so the kernel reclaims it.
    The new inode (whatever it is: a freshly-written one, or a base64-decoded copy) gets the path.
1621 - The only prerequisite to apply Flavor B's `mv` is: you own the parent directory (so you can unlink entries
    from it) AND you have a NOPASSWD-capable binary that can read the byte stream if mode prohibits user read.
1622 - This pattern generalizes. Any time a script ends up producing a "locked file that the next run can't update"
    situation, the same trick applies without a single chmod anywhere.
1623

```

```

1624 **Empirical verification (user machine, July 1, 2026):**
1625 - `.env` was `mode=000` (owner $USER, i.e. you): Flavor A unblocked it in 3 commands, no chmod.
1626 - Before: `----- 1 $USER staff 899 Jun 28 07:07 .env`
1627 - After: `--r--r--@ 1 $USER staff 899 Jul 1 20:39 .env`
1628 - `docker compose config` immediately picked up `TS_AUTHKEY` + `WIREGUARD_PRIVATE_KEY` substitution.
1629 - `compiled_rules.txt`, `ip_blocklist.ipset`, `data/filters/1782645604.txt` were `mode=0444`: Flavor B (`mv`-
    aside) unblocked all three.
1630 - `scripts/sync-rules.py` then ran clean: 279587 block rules + 171 allow rules + 16710 IP entries compiled; new
    files came out at `0644`.
1631 - `toggle.sh` then went end-to-end in 48s (warp / routing-fix / adguardhome / tailscale all healthy, gateway
    mesh-joined, DNS hijack verified single-server).
1632
1633 **When this trick is appropriate vs. inappropriate:**
1634 - You own the parent directory and the file (typical desktop scenario).
1635 - The lock is a leftover from a removed `chmod` call that you want off disk forever.
1636 - NOPASSWD sudo includes at least one interpreter (`python3` on this system) that can be used to read mode-0
    files. If it does not, add it first: `**Reusable cookbook:**
1642
1643 ```bash
1644 # Quick diagnostic: figure out which flavor you need.
1645 WHOAMI=$(whoami)
1646 ls -la <FILE>
1647 stat -f 'mode=%Sp owner=%Su group=%Sg' <FILE>
1648 # mode=----- or mode=----r----- : you cannot even read it Flavor A.
1649 # mode=r--r--r-- but writer fails : you are blocked from write but can read Flavor B.
1650 # mode=rw-r--r-- : not actually locked at all; unrelated write failure.
1651
1652 # Flavor A (mode=000):
1653 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
    decode())" > /tmp/snap.b64
1654 base64 -d < /tmp/snap.b64 > <FILE>.new && mv -f <FILE>.new <FILE> && rm -f /tmp/snap.b64
1655
1656 # Flavor B (mode=0444, file is readable):
1657 mv <FILE> /tmp/old.<FILE>.$(date +%s)
1658 # (let the original writer recreate <FILE>; new file lands at mode 0644)
1659
1660 # Verify after either flavor:
1661 ls -la <FILE> # mode should be 0644
1662 <your normal validation command>
1663 ```
1664
1665 ### 15.10 Tor Container
1666
1667 #### 15.10.1 July 11, 2026: Tor Container Hardening & obfs4proxy Restoration
1668 **Symptom:**
1669 1. The Tor container entered a fatal crash loop upon boot, throwing `exec: "obfs4proxy": executable file not
    found in $PATH` when `TOR_USE_BRIDGES=true`.
1670 2. When the user pasted Tor bridge lines containing comments (`#`) or blank lines into `tor-bridges.txt`, the
    container failed to validate the file properly and crashed.
1671
1672 **Root Cause:**
1673 - **Bug 1 (Missing Pluggable Transports):** The Tor Dockerfile was built on `debian:bullseye-slim`, which had
    silently dropped or failed to resolve the `obfs4proxy` package in its latest apt repositories.
1674 - **Bug 2 (Fragile Validation):** The `entrypoint.sh` bridge validation check merely checked `[ -s tor-bridges.
    txt ]` (file size > 0). It did not verify if the file actually contained valid, uncommented bridge lines.
    Passing empty lines or comments tricked the validation script into appending invalid `Bridge` directives to
    the `torrc`, causing the Tor daemon to instantly exit.
1675
1676 **Resolution:**
1677 - **Upgraded Base Image:** Shifted the Tor Dockerfile base image to `debian:bookworm-slim`, restoring access to
    the `obfs4proxy` pluggable transport package.
1678 - **Robust Bridge Validation:** Replaced the fragile `[ -s ]` check with a strict `grep -vE '\s*(#|$\n)'`
    command that strips comments and empty lines. If no valid lines remain, the container safely falls back to
    standard public guard relays instead of crashing, ensuring Tor remains available even with a misconfigured
    bridge file.
1679 - **Image Pinning:** Explicitly pinned `image: nullexit-tor:v1.0.0` in `docker-compose.yml` to prevent
    arbitrary local builds from drifting to `:latest`.
1680
1681 #### 15.10.2 July 12, 2026: Tor ControlPort Dynamic Password Authentication & User Config
1682 **Symptom:** Querying the Tor Control Port (e.g., via the `/sweep` script or `check_circuits.py`) returned
    empty responses or connection errors. Logs showed that Tor automatically closed its ControlPort:
1683 `You have a ControlPort set to accept unauthenticated connections from a non-local address. ... That's so bad
    that I'm closing your ControlPort for you.`

```

```

1684
1685 **Root Cause:** Docker port mapping (especially under Colima on macOS) requires container sockets to bind to
    `0.0.0.0` to be reachable from the host. However, when Tor's ControlPort is bound to `0.0.0.0` (non-local)
    with no authentication, Tor closes it by default as a safety precaution. Sourcing SOCKS5/ControlPort to
    `127.0.0.1` inside the container was not an option as it broke the Docker bridge routing path.
1686
1687 **Resolution:**
1688 - **Dynamic Password Authentication:** Updated the Tor container's `entrypoint.sh` to read `TOR_PASSWORD` and
    dynamically generate its HMAC hash via `tor --hash-password`, adding `HashedControlPassword <hash>` to `
    torrc`. This secures the ControlPort on `0.0.0.0` so Tor keeps it open.
1689 - **Unified Credentials in .env:** Exposed `ADGUARD_USER=admin` and `ADGUARD_PASSWORD=nullexit` in `.env` to
    serve as the unified gateway credentials. Sourced `TOR_PASSWORD` directly from `ADGUARD_PASSWORD` in `
    docker-compose.yml`.
1690 - **Client Authentication:** Updated `scripts/check_circuits.py` to fetch `TOR_PASSWORD` (defaulting to `
    ADGUARD_PASSWORD` or `nullexit`) and authenticate with the ControlPort using `AUTHENTICATE "<password>"` to
    regain access.
1691
1692 > See also 11.3 (Tor Container Kill-Switch Inheritance & Fail-Closed Egress Verification) for the design-level
    fail-closed guarantee.
1693
1694 ### 15.11 macOS Platform Quirks
1695
1696 #### 15.11.1 macOS Application Firewall Silently Blocking AirDrop & Continuity
1697 **Problem:** Users of complex network extensions (Tailscale, LuLu, Docker, etc.) often run into an issue where
    AirDrop, AirPlay, and Universal Clipboard silently fail, even with the gateway inactive and Wi-Fi/Bluetooth
    correctly configured. The internal Apple Wireless Direct Link (`awdl0`) interface appears healthy, but
    connections never establish.
1698
1699 **Root Cause:** The built-in macOS Application Firewall has a specific list of explicitly blocked applications.
    Apple's own core networking services (e.g., `/usr/libexec/sharingd`, `/usr/libexec/rapportd`, `/usr/
    libexec/avconferenced`) can be silently added to the "Block incoming connections" list either through
    accidental "Deny" clicks on permission prompts, execution of aggressive privacy-hardening scripts, or a
    known macOS bug that retains strict blocks even after toggling the "Block all incoming connections" setting
    off. Since these are background daemons, macOS provides no visible error.
1700
1701 **Fix:** The firewall rules must be cleared via the terminal (or System Settings > Network > Firewall > Options
    ):
1702 ```bash
1703 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/sharingd
1704 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/rapportd
1705 ```
1706 Once unblocked, `sharingd` immediately resumes broadcasting mDNS/Bonjour discovery packets, and AirDrop
    restores instantly without requiring a reboot.
1707
1708 #### 15.11.2 Standalone Tailscale Daemon Freeze after macOS Sleep/Wake
1709 **Problem:** When the macOS host goes to sleep and wakes up, the network interfaces and routing tables are
    rebuilt. The standalone `tailscaled` daemon (installed via Homebrew) occasionally fails to handle this
    transition and becomes completely frozen/unresponsive. In this state, any command like `tailscale down` or
    `tailscale status` hangs indefinitely, and all DNS requests sent to the local Tailscale resolver
    (`100.100.21.8`) time out, causing a total internet blackout on the host.
1710
1711 **Fix:** Enhanced the Tailscale verification and teardown logic in `toggle.sh` and `recover.sh`:
1712 1. **Unresponsive Daemon Detection:** Instead of assuming the daemon is healthy just because the Unix socket `
    var/run/tailscaled.socket` exists, the scripts now actively run a status check with a 5-second timeout (`
    run_with_timeout 5 tailscale status`). If the check times out (exit code 143), the daemon is classified as
    unresponsive.
1713 2. **Auto-Recovery Loop:** If the daemon is unresponsive, the script now automatically attempts to restart the
    service using `brew services restart tailscale` (falling back to `sudo -n brew services restart tailscale`)
    .
1714 3. **Graceful Retry:** Once restarted, the script pauses for 3 seconds for the daemon to initialize before
    proceeding with connection or teardown commands.
1715
1716 #### 15.11.3 macOS Sharing Services (AirDrop/AirPlay) Freeze on Network Transitions
1717 **Problem:** When the gateway is turned on or off, the scripts perform network configuration cleanups which
    include flushing the routing tables (`route -n flush`), restarting the `mDNSResponder` daemon (`killall -
    HUP mDNSResponder`), and power-cycling the Wi-Fi interface (`setairportpower` or `ifconfig en0 down/up`).
    While AirDrop BLE discovery continues to function, the actual file transfers stall at "Waiting..."
    indefinitely.
1718
1719 **Root Cause:** The rapid tear-down and rebuild of the local routing table and Wi-Fi interface causes Apple's
    core sharing and connection daemons (`sharingd` and `rapportd`) to lose their socket bindings to the mDNS/
    Bonjour discovery interface. Instead of reconnecting gracefully, they enter a wedged state and try to route
    peer-to-peer (AWDL) IPv6/TCP transfer traffic into the default gateway (the Tailscale interface `utun`)
    instead of the direct wireless link, causing the TLS verification handshake to time out.
1720
1721 **Fix:** Integrated an automatic sharing services reset into both `toggle.sh` and `recover.sh`:
1722 1. The script now automatically runs `sudo -n killall sharingd rapportd` at the end of both the **START** path
    and the **STOP** path (as well as inside `recover.sh`).
1723 2. Killing these daemons forces macOS to immediately relaunch them, binding them fresh to the newly initialized
    network interfaces and routing tables, allowing AirDrop to work seamlessly while the gateway is active.
1724

```

```

1725 ##### 15.11.4 Chrome Remote Desktop Connection Failures (Tailscale on Remote Device)
1726 **Context:** When attempting to access a remote machine via Chrome Remote Desktop (CRD), the connection fails
or hangs if Tailscale is active **on the remote device**. The connection works fine even if Tailscale (and
the exit node) is active **on the local client/viewer device**, but only if Tailscale is disabled on the
remote machine. *(This is distinct from the CRD-through-WARP latency limitation in 14.3.)*
1727
1728 **Why:** Having Tailscale active on the remote machine creates virtual network interfaces and DNS/routing
modifications that conflict with Chrome Remote Desktop's WebRTC bindings and signaling traffic.
1729
1730 **Workaround:**
1731 - **Use Tailscale Native RDP/RustDesk (Recommended):** Connect directly to the remote device's Tailscale IP
(`100.X.Y.Z`) via an RDP or RustDesk client. This is faster and avoids third-party relay conflicts.
1732 - **Disable Tailscale on the Remote Device:** If you must use CRD, disable Tailscale on the remote machine
before connecting.
1733
1734 ##### 15.11.5 tailscaled Daemon Broken State (brew services)
1735 **Symptom:** `brew services list` shows `started` but `tailscale status` shows "stopped".
1736
1737 **Fix:** `sudo brew services restart tailscale`. Toggle script now detects and auto-restarts.
1738
1739 ##### 15.11.6 Apple Silently Removed the `airport` Binary in macOS Sonoma 14.4 (March 2024)
1740 **What happened:** `watcher.sh`'s `detect_lan_p2p_mode()` was written to use the `airport` CLI tool at:
1741 ...
1742 /System/Library/PrivateFrameworks/Apple80211.framework/Versions/Current/Resources/airport
1743 ...
1744 This path returned `exit 127: no such file or directory`. The function silently fell back to `reason="no-wifi"`
even while the Mac was actively connected to WPA2 Enterprise Wi-Fi.
1745
1746 **Why Apple removed it:** `airport` was never a real public binary it lived inside a **private framework** and
was always technically unsupported. Apple deprecated it quietly years ago and finally removed it in **
macOS Sonoma 14.4 (March 2024)**. The removal is part of a broader privacy clampdown on Wi-Fi metadata:
1747 - Starting in **macOS 14.5**, BSSIDs and other Wi-Fi association details are **redacted** (`<redacted>`) in
most tools, including the official replacement `wdutil`
1748 - Apple's official recommendation is to use the **CoreWLAN framework** (Swift/Objective-C) with proper
entitlements useless for a bash script
1749 - The official CLI replacement `wdutil` requires `sudo` for most useful output also useless for a background
LaunchAgent
1750
1751 **Fix:** Replaced `airport -I | awk '/link auth/'` with:
1752 ```bash
1753 system_profiler SPAirPortDataType | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF
; exit}'
1754 ...
1755 This returns human-readable strings like `WPA2 Enterprise`, `WPA2 Personal`, or `None` for the currently
associated network without `sudo`, and without any removed private binary.
1756
1757 **Confirmed working output on this machine:**
1758 ...
1759 P2P detect: security='Enterprise' allow=false (reason: 802.1x-enterprise)
1760 ...
1761
1762 > ** Risk: `system_profiler` may not survive forever either.** `system_profiler SPAirPortDataType` still works
as of macOS Sequoia (15.x) without sudo and without redaction. However, given Apple's trajectory on Wi-Fi
privacy, this could be locked down in a future release. If `security` comes back empty on a future macOS
version while the Mac is actively on Wi-Fi, the function will silently fall back to `allow=false` (safe
default), but the detection will be broken again. The long-term fix would be a small Swift helper using
CoreWLAN that we compile once and bundle in the repo. *(This forward-looking caveat is also tracked as an
observation in 16.)*
1763
1764 ##### 15.11.7 Changing macOS Local Hostname
1765 **Context:** Users may wish to change their Mac's computer name/hostname in macOS System Settings.
1766 **Effect:** Changing the Mac's hostname will **not** break Nullexit (which relies on internal routing/localhost
). However, Tailscale will automatically detect this change and update the machine name on the Tailnet.
1767 - The underlying Tailscale `100.x.x.x` IPv4 address will remain exactly the same.
1768 - The **MagicDNS name** will change to match the new hostname. Users must remember to update their SFTP/SSH
clients to use the new MagicDNS address.
1769
1770 ##### 15.11.8 Shell-Language Landmines: macOS `/bin/bash` 3.2 + `set -e` (Field Notes)
1771
1772 This project's lifecycle scripts (`toggle.sh`, `recover.sh`, `common.sh`, ) run under `#!/bin/bash`, which on
macOS is **GNU bash 3.2.57 (2007)** Apple has never shipped a newer bash because bash 4+ is GPLv3. (
Homebrew's bash 5 lives at `/opt/homebrew/bin/bash` but is *not* the shebang target, and cannot be assumed
on PATH a bare `bash --version` on a stock Mac reports 3.2.) Every script here also runs `set -e` (exit-on
-error) with an `ERR`/`INT`/`TERM`/`HUP` trap. That combination **ancient bash + `set -e` + traps + `set -
x` xtrace redirection** is a minefield. The traps below were all hit and fixed during development;
document them so they are not re-discovered.
1773
1774 **A. `set -e` + `if then else fi` can abort at the `else` (bash 3.2 heisenbug).**
1775 The nastiest one. A construct as innocent as:
1776 ```bash
1777 if [ -f "$MARKER_FILE" ]; then

```

```

1778 X="yes"
1779 else
1780 X="no"
1781 fi
1782 ```
1783 aborted `toggle.sh` at init under `set -e` whenever the file was absent the false status of the `[ -f ]`
test leaked through the compound and `set -e` killed the script (EXIT: code=1 phase=init), before any
real work ran. Two things made this vicious: (1) it is intermittent/context-dependent it did not
reproduce in isolated `bash -c` snippets of the identical block; it only manifested inside the full
script with the `set -x` xtrace fd-redirect (`exec 2>>(tee)`) active. We only proved causation by 
bisecting the live script (replace the block with a trivial assignment the script sailed past init). (2)
Static tools (`bash -n`, `shellcheck`) cannot see it it is a runtime interaction. Safe pattern:
default-assign, then an `if` with no `else` (an `if` whose false condition has no else-branch yields
status 0):
1784 ```bash
1785 X="no"
1786 if [ -f "$MARKER_FILE" ]; then X="yes"; fi
1787 ```
1788 Confirmed root cause (2026-07-13, reproduced head-to-head). The trigger is specifically set -x plus a
PS4 that contains a command substitution here the `DEBUG_TRACE` prompt `PS4='+ [$(date)] '`. Forcing
the false-condition/else path (`gateway_state` `stopped`, so `if [ "$(gateway_state)" = "running" ]` takes
the else/START branch) on stock `/bin/bash` 3.2.57: with the default `PS4` it survives (3/3, exit 0);
with toggle's $(date) `PS4` the `ERR` trap fires on the else branch (`rc=1` from the false condition,
3/3, exit 42). The $(date) runs a subshell on every traced command; while tracing the if-condition
it perturbs ` $? ` so the condition's non-zero status leaks past the normal if-exemption and trips the `ERR`
trap. This is a true heisenbug enabling tracing (`DEBUG_TRACE=true`) is what causes the abort,
which is exactly why it's invisible with `DEBUG_TRACE=false` (the default) and why every isolated repro on
the default `PS4` passes (this misled an entire debugging session into briefly "disproving" the landmine)
. It aborted the gateway START decision (`toggle.sh:689`) under `DEBUG_TRACE=true`; with `DEBUG_TRACE=false`
the gateway starts cleanly (verified: 192 s, all five containers healthy). Fixed (2026-07-14).
`DEBUG_TRACE`'s `PS4` now uses the subshell-free `${SECONDS}` (elapsed seconds since the shell started)
instead of `$(date +%H:%M:%S)` , removing the per-traced-command subshell that perturbed ` $? ` . bash 3.2
has no subshell-free wall-clock token for `PS4` (`$EPOCHSECONDS` is 5.0+, `printf '%()T'` is 4.2+), so
the per-line stamp is now elapsed-seconds the absolute start time is still logged once in the `DEBUG_TRACE`
enabled header line, so wall-clock is recoverable. Bonus: this also drops a date fork on every
traced line (previously thousands). Verified head-to-head on stock `/bin/bash` 3.2.57: toggle.sh --
restart with `DEBUG_TRACE=true` now completes `code=0` with tracing active straight through the exact
decision that aborted before the trace shows gateway_state stopped (`common.sh:229`) feeding the false
if [ "$(gateway_state)" = "running" ] , then the else branch reaching Action: STARTUP GATEWAY (1436
traced lines, no ERR trap firing, all five containers healthy). DEBUG_TRACE is now safe to use on the
STOP/START decision path.
1789
1790 More generally, any command whose "failure" is normal control flow (`grep -q`, `[ ]`, `diff`, `kill -0`) that
ends up as the last command of a function/branch/script under set -e is a landmine; guard it (`|| true`
) or restructure so a non-zero status can't be the tail value.
1791
1792 B. `BASH_XTRACEFD` and the exec {var}>>file dynamic-fd form are bash 4.1+ (hard-fail on 3.2).
1793 Our `DEBUG_TRACE` feature wants xtrace (`set -x`) sent to output.log while keeping the terminal clean the
clean way is exec {BASH_XTRACEFD}>>"$LOG_FILE" . On 3.2 this is a syntax/runtime error: exec: {
BASH_XTRACEFD}: not found (dynamic-fd allocation `{var}>>` didn't exist until 4.1, and BASH_XTRACEFD
itself is 4.1+; on 3.2 it's just an ordinary variable that set -x ignores). Left unguarded it would abort
the script at startup i.e. the debug feature would break the very thing it's meant to observe. Safe
pattern: branch on `${BASH_VERSINFO[0]}.${BASH_VERSINFO[1]}` on 4.1 use exec 9>>"$LOG_FILE"; export
BASH_XTRACEFD=9; on 3.2 fall back to exec 2>>(tee -a "$LOG_FILE" >&2) (xtrace goes to stderr, which we
duplicate to the log; the terminal also shows it, which is acceptable). Never use the exec {var}>> form
unless you know you're on 4.1.
1794
1795 C. trap ERR + $LINENO misattributes the failing line on 3.2.
1796 When a command fails under set -e, our trap runs clean_up_handler ERR $LINENO . On bash 3.2, $LINENO inside
an ERR trap frequently reports the line of the enclosing fi/closing brace of a compound statement,
not the true failing line. During the STOP/START-decision bug this produced Script failed at line 1202
where 1202 is a bare fi actively misleading. Mitigation: don't trust the trap's line number as
gospel on macOS; corroborate with the EXEC:/EXIT: lifecycle breadcrumbs and the DEBUG_TRACE xtrace (
which prints file:line:function via a custom PS4), which pin the actual last-executed command.
1797
1798 D. set -e is not inherited by command substitutions / subshells the way people expect, and is suppressed
inside if`/`&&`/`||` conditions.
1799 This cuts both ways and is a frequent source of "why didn't it fail?" and "why did it fail?": a non-zero
command used as an if condition (or left/right of &&`/`||`) is exempt from set -e (by design), but
the moment that same call is a bare statement it aborts. is_gateway_active() relies on this (it's always
called as if is_gateway_active; then `), which is precisely why moving the decision to a marker file
instead of a probe that must be if-guarded to be safe is more robust (see 15.12.19). Also note local X=
$(cmd) masks cmd's exit status (the local builtin's status wins) a shellcheck SC2155 we see
repeatedly; harmless when we don't check ` $? ` , but a real footgun if you do.
1800
1801 E. Process substitution (>>(), <()) is a bashism, and its lifecycle is loose.
1802 Our 3.2 xtrace fallback (exec 2>>(tee -a "$LOG_FILE" >&2)) uses process substitution fine under bash, but
(1) it is not POSIX sh, so these scripts must never be invoked as sh script.sh; and (2) the
background tee it spawns is reaped asynchronously, so the very last few lines written just before exit
can occasionally be truncated in the log. Acceptable for a debug trace; do not rely on it for critical last
-gasp logging (use a direct >> "$LOG_FILE" append for must-capture lines, as the EXIT breadcrumb does).

```

```

1803
1804 **F. GNU-vs-BSD userland divergence (not bash itself, but same class of pain).**
1805 macOS ships BSD variants of the core tools, so idioms differ from Linux: `sed -i ''` (BSD requires the empty
    backup-suffix arg) vs `sed -i` (GNU); `stat -f%z` (BSD) vs `stat -c%s` (GNU); `date` format flags; `jot` (
    BSD) vs `shuf` (GNU) for random port generation; `route`/`networksetup`/`dscacheutil` (macOS-only) vs `ip
    `/'`resolvectl` (Linux). This is *why* the codebase branches on `[[ "$OSTYPE" == "darwin"* ]]` everywhere
    `toggle.sh` and `recover.sh` are each single cross-platform scripts that dispatch on `OSTYPE` (an early
    Linux block runs and exits before the macOS body). `timeout(1)` doesn't exist on stock macOS at all hence
    the pure-bash `run_with_timeout` in `common.sh`.
1806
1807 **Working rules distilled from the above:**
1808 - Assume **bash 3.2** semantics for anything under `#!/bin/bash` on macOS; gate any 4.0+ feature (
    BASH_XTRACEFD`, dynamic `{var}` fds, associative arrays, `${var^^}` case-conversion, `mapfile`/`readarray`,
    negative array indices) behind a `BASH_VERSINFO` check or avoid it.
1809 - Under `set -e`, never let a "benign non-zero" command (`[ ]`, `grep -q`, `kill -0`, `diff`) be the tail
    statement of a branch/function/script; use `no`/`else`/`if`s, `|| true`, or explicit `$?` handling.
1810 - Prefer **persisted state (marker files)** over **live probes** for control-flow decisions a probe forces an
    `if`-guard and can hang/abort; a file read cannot (see 15.12.19).
1811 - `bash -n` and `shellcheck` catch syntax and many smells but **cannot** catch `set -e`/trap runtime
    interactions a **live run is mandatory** to validate lifecycle-script changes; the `DEBUG_TRACE` +
    breadcrumb instrumentation exists precisely to make those live runs diagnosable.
1812 - Never run these scripts as `sh` they are bashisms end to end.
1813
1814 ### 15.12 Misc Resolved
1815
1816 #### 15.12.1 Gluetun Resets nftables FORWARD Rules
1817 **Symptom:** FORWARD RELATED,ESTABLISHED rule disappears after Gluetun health check.
1818
1819 **Fix:** `scripts/routing-fix.sh` re-adds to both iptables backends every 30 seconds. Legacy backend (policy
    ACCEPT) acts as safety net.
1820
1821 #### 15.12.2 Native SSH & SFTP Security (Zero Local Attack Surface)
1822 **Context:** macOS "Remote Login" (`sshd`) and "File Sharing" (`smbd`) open ports on the local network (e.g.
    `172.x.x.x`), exposing the host to brute-force attacks on public Wi-Fi.
1823
1824 **Implementation:** The script strictly enforces `tailscale up --ssh=true` whenever the mesh state is reset.
    This empowers the user to completely disable native macOS Remote Login and File Sharing. This provides an
    incredible zero-trust security advantage: even if an attacker steals your Mac username and password, they
    **cannot** access your files remotely. Disabling the native services completely closes the listening ports
    on the local Wi-Fi interface (`172.x.x.x`), rendering the Mac inaccessible to local network logins.
    Meanwhile, Tailscale intercepts port 22 traffic exclusively over the encrypted mesh and authenticates
    cryptographically based on your Tailscale SSO identity. Stolen Mac passwords are mathematically useless
    without first bypassing your Tailscale Two-Factor Authentication and registering a device onto the mesh.
1825
1826 **Cross-Platform File Exchange:** To easily browse and exchange files securely, simply use an SFTP client and
    connect to your Mac's MagicDNS name on port 22. Recommended clients: **WinSCP** or **FileZilla** (Windows),
    **Cyberduck** (macOS), and **FE File Explorer** or **Solid Explorer** (iOS/Android).
1827
1828 #### 15.12.3 Logging Architecture
1829 For debugging, logs are strictly segmented based on the component's lifecycle:
1830 - **`output.log` (Host-side):** Contains all standard error (`stderr`) and verbose output from the host scripts
    (`toggle.sh`, `recover.sh`, `setup.sh`). Since the `rule-compiler` container is ephemeral and deleted
    after running, its logs (and any Python errors from `scripts/sync-rules.py`) are extracted and appended to
    this file before deletion.
1831 - **Log Rotation Policy:** On startup, `toggle.sh` checks if `output.log` exceeds **1GB** (`1,073,741,824`
    bytes). If it does, it performs a metadata-only rename (`mv output.log output.log.old`), preserving history
    while resetting the active log to 0 bytes. This rename-based rotation avoids the heavy disk I/O and CPU
    overhead of rewriting a massive file line-by-line. The threshold is intentionally large because with
    `DEBUG_TRACE=true` the log accumulates a full command-by-command execution trace (effectively the framework's
    entire compilation/warning history), which is valuable to retain for post-mortems; a smaller cap would
    rotate that history away too quickly.
1832 - **Egress Probe Logging:** The background host-egress leak probe checks if stdout is a TTY (`[ -t 1 ]`) and
    will only print live status updates (using carriage return `\r`) or duplicate console warnings in
    interactive mode. It automatically detects macOS/BSD vs. Linux environments to dynamically construct
    timestamps with date and time (omitting milliseconds on macOS to prevent high CPU utilization from spawning
    sub-second processes).
1833 - **`docker logs <container>` (Guest-side):** The persistent containers (`warp`, `tailscale`, `routing-fix`,
    `adguardhome`) use the Docker `json-file` logging driver with a strict `max-size` (1m-10m) to prevent VM
    disk exhaustion. Use standard `docker logs` to view them.
1834
1835 #### 15.12.4 Pre-Flight Check 1: Wrong `tailscale ping` Flag
1836 **Problem:** Check 1 of the pre-flight connectivity checks used `--c 1` (double dash) but `tailscale ping`
    accepts `-c 1` (single dash). The invalid flag caused the command to exit with code 1, marking the check as
    FAIL even though the gateway was reachable.
1837
1838 **Fix:** Changed `--c 1` to `-c 1`.
1839
1840 #### 15.12.5 Pre-Flight Check 1: DERP Relay Exit Code
1841 **Problem:** Even with the correct `-c 1` flag, `tailscale ping` exits with code 1 when the connection goes
    through a DERP relay ('direct connection not established') even though a pong was successfully received
    (28ms via DERP(tor)). The check treated relayed connections as failures.

```

```

1842
1843 **Fix:** Changed the check from relying on exit code to piping stdout through `grep -q "pong"`. This accepts
1844 relayed connections as valid (they work fine for the exit node use case), while still failing on true
1845 timeouts or unreachable peers.
1846
1847 #### 15.12.6 IPv6 Exit-Node Leak
1848 **Problem:** Tailscale supports IPv6, but WARP/gluetun is IPv4-only. IPv6 packets forwarded through the exit
1849 node would leak through the Docker bridge unencrypted.
1850
1851 **Fix:** Added `ip6tables -A FORWARD -i tailscale0 -j DROP` to `post-rules.txt`, blocking all IPv6 forwarded
1852 traffic from the Tailscale interface. *(This is the fix referenced by the IPv6-leak scenario in 15.6.1.)*
1853
1854 #### 15.12.7 SOCKS5 Proxy Threading Race Condition
1855 **Problem:** The `forward` function in `scripts/socks5-proxy.py` called `select.select([src, dst], [], [])`
1856 which raised `ValueError: file descriptor cannot be a negative integer (-1)` when one thread closed a
1857 socket while the other thread was selecting on it.
1858
1859 **Fix:** Added `ValueError` exception handling around `select.select()` and `recv()` calls. When a file
1860 descriptor becomes negative, the thread breaks out of the forwarding loop cleanly instead of crashing.
1861
1862 #### 15.12.8 Docker Healthcheck: Multi-line Python in CMD Array
1863 **Problem:** The socks-proxy healthcheck used a `CMD` array with a multi-line Python string. YAML collapsed the
1864 newlines, causing the `#` comment to comment out the rest of the code and the `assert` statement to be
1865 mangled into `as`.
1866
1867 **Fix:** Changed from `["CMD", "python3", "-c", "..."]` to `["CMD-SHELL", "python3 -c '...']` with a single-
1868 line Python expression. Uses a real SOCKS5 handshake (sends `[5, 1, 0]`, expects `[5, 0]`) instead of a
1869 simple TCP port check.
1870
1871 #### 15.12.9 Graceful Shutdowns Leave DNS Hijacked
1872 **Problem:** The gateway overrides the macOS host's DNS settings (via `networksetup`) to route queries to the
1873 AdGuard container. Because macOS persists `networksetup` changes to disk, shutting down the Mac while the
1874 gateway is running causes the Mac to boot up later with hijacked DNS. Since the Colima VM and Docker
1875 containers don't auto-start on boot, the user would have no internet access until they manually run the
1876 toggle script.
1877
1878 **Root Cause:** The `toggle.sh` script exits after the gateway starts. There was no background daemon listening
1879 for macOS system shutdown signals to revert the network settings.
1880
1881 **Fix:** Wrapped the background `caffeinate` process (used to prevent system sleep) in a bash `trap` command.
1882 The trap listens for `SIGTERM`, `SIGINT`, and `SIGHUP`. When the user initiates a graceful shutdown, macOS
1883 sends `SIGTERM` to the background trap, which instantly executes `recover.sh` to flush the host DNS back to
1884 `1.1.1.1` right before the machine powers off.
1885
1886 *Note:** We explicitly use `recover.sh` instead of `toggle.sh` because during a shutdown, the OS limits process
1887 cleanup time and is already independently sending termination signals to Docker and Colima. Trying to run
1888 `docker compose down` inside a shutdown trap creates a race condition that can hang the script until macOS
1889 forcefully `SIGKILL`s it. `recover.sh` abandons container management and surgically flushes the macOS
1890 network settings in milliseconds, guaranteeing completion before the OS pulls the plug.
1891
1892 #### 15.12.10 Linux Native Wi-Fi Bouncing
1893 **Problem:** The generated `scripts/recover-linux.sh` incorrectly retained the macOS `networksetup` commands,
1894 which would crash and fail to bounce Wi-Fi on Linux hosts.
1895
1896 **Fix:** Swapped the macOS logic for native Linux commands. The script now attempts `nmcli radio wifi off/on`
1897 first, and gracefully falls back to `ip link set wlan0 down/up` if NetworkManager is unavailable.
1898
1899 #### 15.12.11 TS_AUTH_ONCE Cloud Footgun & Ephemeral Keys
1900 **Decision:** The compose file uses `TS_AUTH_ONCE=true` to prevent authentication loops. However, on headless
1901 cloud VPS deployments, if a standard auth key expires (usually 90 days), the container will silently drop
1902 off the mesh. We documented this trade-off explicitly in `docker-compose.yml` and recommended the use of **
1903 Tailscale Ephemeral Keys** for cloud deployments, which automatically clean themselves up and bypass this
1904 issue.
1905
1906 #### 15.12.12 Hardcoded AdGuard Credentials
1907 **Decision:** AdGuard is deployed with the hardcoded credentials `admin / nullexit`. This is perfectly safe
1908 behind a NAT on a local macOS laptop. However, if deployed on a cloud host with exposed ports, this becomes
1909 a severe security risk. We added a stark warning comment in `docker-compose.yml` to ensure cloud users
1910 change this before deployment.
1911
1912 #### 15.12.13 Routing Fix: 30-second polling vs Netlink Socket
1913 **Decision:** Instead of building a complex Netlink socket listener (`ip monitor`) to react instantly to
1914 iptables and routing flushes from Gluetun, we retained the 5-second `sleep` loop in `scripts/routing-fix.sh`
1915 .
1916
1917 - **Reasoning:** Building a netlink listener in pure bash on Alpine is incredibly brittle and complex (
1918 especially for iptables events). The 30-second polling loop is bulletproof. The only trade-off is a
1919 potential 1-4 second stutter if Gluetun reconnects, which was deemed an acceptable edge case for absolute
1920 reliability.
1921
1922 #### 15.12.14 Cache Poisoning and URL Rot in scripts/sync-rules.py
1923 **Problem:** If a remote blocklist URL went permanently offline (404 Not Found), the Python `urllib` request
1924 would return the 404 HTML string. The script would aggressively regex-parse this HTML, find no domains, and

```

overwrite the healthy local cache with a 0-domain file. This effectively disabled ad-blocking until the URL was fixed.

1885
1886 ****Fix:**** Implemented a defensive programming sanity check in `scripts/sync-rules.py`. Before overwriting the cache, the script now verifies that the compiled domain count is greater than 10 (and 1 for IP feeds). If it falls below this threshold (indicating a catastrophic 404 failure across multiple URLs), it aborts the cache overwrite, raises a `ValueError`, and keeps the previous day's healthy cache intact.

1887
1888 **#### 15.12.15 Gluetun nf_tables Parser Crash (Bash Interpolation Failure)**

1889 ****Problem:**** Attempting to make the TCP MSS dynamically configurable by using a standard bash variable inside `post-rules.txt` (`iptables ... --set-mss \${GATEWAY\_MSS:-1120}`) caused the Gluetun container to catastrophically crash during startup. Gluetun's internal parser executes `post-rules.txt` line by line without a shell, meaning the variable was never expanded. It literally attempted to inject the string `\${GATEWAY\_MSS:-1120}` into iptables, resulting in a fatal `bad value for option "--set-mss" error`.

1890
1891 ****Fix:**** Removed the bash variable from `post-rules.txt` and reverted it to a pure hardcoded integer. To retain dynamic `.env` configuration, we moved the interpolation logic to `toggle.sh`. Right before Docker starts, the host script parses `GATEWAY\_MSS` from the `.env` file and uses a cross-platform `sed` command to dynamically overwrite the integer directly inside `post-rules.txt`. This perfectly mimics manual configuration and completely bypasses Gluetun's strict parser.

1892
1893 **#### 15.12.16 AdGuard Filter Redundancy & Memory Optimization**

1894 ****Problem:**** AdGuard Home does not perform cross-list deduplication in memory. When it loads its own native subscription filters (e.g., `AdGuard DNS filter`) alongside our massive `compiled\_rules.txt`, overlapping rules are loaded twice, wasting significant RAM in the Colima VM. The UI would show ~500k rules, representing the sum of all lists rather than unique domains.

1895
1896 ****Fix:**** Updated `scripts/sync-rules.py` to intelligently cross-reference AdGuard's configuration and deduplicate our list against it.

1897 1. The script now reads `adguard/conf/AdGuardHome.yaml` to dynamically fetch the exact URLs of any enabled native AdGuard filters.

1898 2. The parsing engine was upgraded to normalize basic AdGuard syntax (`||domain`) back into raw base domains during the build process.

1899 3. The script subtracts the native AdGuard domains from our custom blocklist *before* compiling, immediately purging ~83,000 completely redundant rules.

1900 4. The normalized base domains then pass through our subdomain optimizer, which squashes an additional ~50% of the remaining rules.

1901
1902 This reduces the final compiled output from ~325k to ~281k rules on the `heavy` profile, saving ~45k rules from being loaded into memory twice and reducing the overall RAM footprint of the `adguardhome` container.

1903
1904 **#### 15.12.17 Kernel-Level IP Blocklist (Threat Intelligence Firewall)**

1905 ****Problem:**** DNS sinkholing cannot stop malware or botnets that bypass DNS entirely by hardcoding direct IP addresses (a common technique for C2 communication). These connections pass straight through AdGuard unseen.

1906
1907 ****Fix:**** Added a second compilation pipeline to `scripts/sync-rules.py` that runs on every startup alongside the DNS pipeline.

1908 1. The compiler concurrently fetches four curated threat-intelligence feeds: ****Feodo Tracker**** (abuse.ch botnet C2 IPs), ****Spamhaus DROP**** (IPs allocated to criminal organizations), ****Emerging Threats**** (Proofpoint active C2 and scanners), and ****CINS**** (active brute-force sources).

1909 2. All entries are normalized via Python's `ipaddress` module. Any IP or CIDR that overlaps with RFC1918 private ranges, loopback, link-local, or the Tailscale CGNAT range (`100.64.0.0/10`) is stripped to prevent accidentally locking users out of their own LAN or mesh.

1910 3. The cleaned list (16,721 unique IPs/CIDRs) is written as an `ipset restore` file using an ****atomic swap pattern****: `create\_new` populate swap with live `destroy\_new`. This guarantees the live `block\_malicious` ipset is never empty during a reload.

1911 4. `scripts/routing-fix.sh` watches the file's mtime every 30 seconds. On change it runs `ipset restore` and idempotently re-injects `FORWARD DROP` rules for both `src` and `dst` into both iptables backends (nftables + legacy).

1912 5. `docker-compose.yml` mounts `adguard/work/userfilters/` into `routing-fix` as `/userfilters:ro` and adds `rule-compiler: service\_completed\_successfully` to its `depends\_on`, eliminating the race condition where routing-fix could start before the file existed.

1913
1914 ****Memory cost:**** The entire 16,721-entry ipset costs only ~1.6 MiB of kernel memory negligible in the 600MB VM budget.

1915
1916 **#### 15.12.18 Incident Post-Mortem: Discarded Docker Exec Errors in logs (July 10, 2026)**

1917 ****Symptom:**** When the `warp` container is stopped, dead, or failing to connect to its endpoints, the `toggle.sh` and `recover.sh` connectivity health checks fail, but no details are printed to `output.log`. The logs only show generic `FAIL` outputs, making it hard to see if the failure was a network timeout, Docker daemon error, or a missing binary.

1918
1919 ****Root Cause:**** Overly broad error silencing. The `docker compose exec` commands in the health checks (Check 3 in `toggle.sh` and the traffic check in `recover.sh`) both redirected stderr to `/dev/null` (`>/dev/null` and `2>/dev/null` respectively), completely throwing away any diagnostic error messages produced by Docker or `wget`.

1920
1921 ****Fix (July 10, 2026):**** Redirected stderr of the `docker compose exec` check commands to `output.log` (`2>> output.log`). This ensures that any command execution failures or connection timeout details are logged.

1922

1923 ##### 15.12.19 Incident Post-Mortem: Toggle Decided STOP/START From a Live Probe, Aborting When Docker Was Down (July 13, 2026)

1924 ****Symptom:**** Running `./toggle.sh` (or `--restart`) while the Colima VM / Docker daemon was ****not**** running caused the script to hang for ~15 seconds and then abort during the START phase without ever booting Colima. With `DEBUG_TRACE` the lifecycle breadcrumb showed `toggle.sh EXIT: code=1 phase=start`, and the trace showed execution jumping straight from the START-branch entry to the `ERR` trap (`cleanup_handler ERR`) with the START body never executing. Historically this silent failure occurred 57 in `output.log`. Spam-clicking the toggle during the resulting window (each retry rejected by the lock guard, then finally succeeding) is what produced the earlier "pile of stacked toggle processes" and Wi-Fi-bounce confusion.

1925

1926 ****Root Cause:**** `toggle.sh` decided whether to STOP or START by calling `is_gateway_active()` (`scripts/common.sh`), which ****live-probes**** the running system its first step is `run_with_timeout 15 docker compose ps --status running`. When `/var/run/docker.sock` is absent (Colima down), that call blocks for the full 15 s timeout and returns non-zero. Combined with the script's global `set -e` and `ERR` trap, a non-zero result evaluated at the `if is_gateway_active`; then else `fi` boundary aborted the whole script ****before**** the `else` (START) branch body ran precisely the branch whose job is to start Colima. Design-wise the deeper flaw is that a ****disable**** should not depend on whether the gateway is ****currently**** healthy; it should act on whether the gateway is ****supposed**** to be running (persisted intent). The correct signal already existed `MARKER_FILE=/tmp/nullexit-gateway-active.marker`, written at the end of a successful START and cleared on STOP but `toggle.sh` only wrote/cleared it and never read it for this decision. ****Note:**** this was a long-standing latent bug, not a regression from the July 2026 `common.sh` platform-function unification or the `DEBUG_TRACE` work; those changes only made the previously-silent abort ****visible**** (via the `EXEC:/EXIT:` breadcrumbs and `xtrace`).

1927

1928 ****Fix (July 13, 2026):**** Introduced `gateway_should_be_running()` in `common.sh` and routed all four decision sites through it (`toggle.sh` main decision + `--restart` pre-check; `scripts/toggle-linux.sh` main decision + `--restart` pre-check). It decides from persisted intent:

1929 1. `toggle.sh` captures the marker's existence into `GATEWAY_MARKER_AT_STARTUP` ****before**** the top-of-script "defensive stale-marker clear" wipes it, and the helper honors that captured value (falling back to the live marker file for callers that run before the clear, e.g. the `--restart` pre-check).

1930 2. Marker present STOP path; marker absent START path. This is instant and cannot hang or abort.

1931 3. Only when the marker is absent ****and**** the Docker socket is actually reachable (`/var/run/docker.sock` or `~/colima/default/docker.sock` on macOS) does it consult `is_gateway_active` as a non-fatal reconciliation fallback (covers a marker lost to a `/tmp` wipe while containers are genuinely up). A dead socket short-circuits to "stopped" with no probe, so `set -e` can never be tripped.

1932 4. The STOP path's `docker compose down` is now `|| true`-guarded so a disable always completes even with Docker down; the START path's `docker compose up` is deliberately left unguarded (a failed start ****should**** trigger cleanup). On Linux the marker isn't maintained, so the helper degrades to the `is_gateway_active` fallback fast there because native Docker has no Colima VM to hang.

1933

1934 ****How we got here (methodology note):**** This one is worth recording because the ****process**** mattered as much as the patch. The failure had been happening silently for a long time the log simply ended after a teardown with no error, so there was nothing to chase. We first attacked the ****observability gap**** rather than guessing at the cause: we wrapped the heavyweight lifecycle commands (`colima start`, `docker compose up/down`) so their output and exit codes always land in `output.log`, added an `EXIT: code= phase=` breadcrumb that fires even when the process is killed, and put a full opt-in `xtrace` behind `DEBUG_TRACE`. Only ****then**** did we reproduce the failure and this time the trace said, unambiguously, `EXIT: code=1 phase=start` with the START body never entered. The instrumentation didn't fix anything, but it converted an invisible abort into a precise, reproducible fact. (Lesson, consistent with 15.12.18 and the LUKS post-mortem referenced in 15.11: ****a mechanism that fails silently is worse than one that fails loudly**** so we make it fail loudly first.)

1935

1936 The root-cause leap, though, came from a ****design-smell intuition**, not the stack trace: the observation that it is nonsensical for a ****disable**** to first check whether the gateway is ****currently online**** a teardown should act on whether the gateway is ****supposed**** to be running, not probe a subsystem to find out. That instinct turned out to name the bug exactly. The smell was a ****conflated responsibility**** with an inverted dependency: to decide whether to ****start**** the gateway, the code first had to ****talk to**** the gateway (via Docker) the very thing START is responsible for bringing up so the check was guaranteed to be unavailable in precisely the cold-start case that mattered. A second tell was ****false symmetry****: `enable` (builds state) and `disable` (tears it down) are not symmetric operations, yet both ran the identical expensive probe; whenever two operations that should differ share suspiciously identical machinery, the code is usually lying about the structure of the problem, and reality eventually calls the bluff. The fix invented nothing the honest signal (`MARKER_FILE`), literally "the gateway is supposed to be up" already existed and was maintained in all the right places; it was simply never consulted for the decision it was made for. We just pointed the decision at the answer the codebase already knew. General principle worth keeping: ****design smells here are load-bearing they tend to be* latent bugs, not merely cosmetic****, and are worth chasing on intuition even before a trace confirms them.

1937

1938 ****Acceptance verified (static + live):**** with marker present STOP (instant, even with Docker down); with marker absent + Docker down START (0 s, no `set -e` abort); a real run now proceeds past init to the gateway-status decision. `bash -n` clean on all three scripts; no new `shellcheck` findings; re-signed signatures (`toggle.sh`/`common.sh`/`toggle-linux.sh` are in the signed set) and `crypto.sh --verify` exits 0.

1939

1940 ****Follow-up (same day)** a `set -e` regression the fix introduced, and why only a live run caught it. ****The first cut of the marker capture wrote the intent as a full `if [ -f "$MARKER_FILE" ]; then GATEWAY_MARKER_AT_STARTUP="yes"; else GATEWAY_MARKER_AT_STARTUP="no"; fi`. On macOS `/bin/bash` 3.2, with `set -e` active and the `DEBUG_TRACE` `xtrace` redirect (`exec 2> >(tee )`) in effect, that construct ****aborted the script at init**** the moment the marker was absent the false `[ -f ]` status leaked through the compound and `set -e` killed the run (`EXIT: code=1 phase=init`), before the gateway logic ran at all. Notably this did ****not**** reproduce in isolated `bash -c` snippets of the same block it only surfaced in**

the script's real runtime context so we had to bisect against the actual `toggle.sh` (swap the block for a trivial assignment the script sailed past init) to prove causation. Fix: write it as a default assignment plus an `if` **without an `else`** (`GATEWAY\_MARKER\_AT\_STARTUP="no"` then `if [ -f "\$MARKER\_FILE" ]; then GATEWAY\_MARKER\_AT\_STARTUP="yes"; fi`), which yields status 0 when the condition is false. Lesson reinforced: `set -e` on legacy bash is genuinely treacherous around conditionals, and the observability work paid for itself again the breadcrumb pinned the abort to init instantly, and a live run (not static checks) was the only thing that exposed a heisenbug invisible to isolated reproduction.

15.12.20 Honey-Port Tripwire Accumulation (July 14, 2026)

Symptom: After many toggle-ons, `ps` showed a pile of idle `bash ./toggle.sh` subshells, each blocked on an `nc -l localhost <port>` listener one leaked per toggle-on, never reaped.

Root Cause: The Honey-Port Tripwire arms a fresh random port each START via a disowned `( nc -l ) &` subshell. `nc -l` blocks until a connection that (by design) almost never comes, so each toggle-on leaks one idle listener; because the port is random, successive listeners never collide and simply stack up over time.

Fix: `toggle.sh` now reaps the previous tripwire before arming a new one `kill -f "nc -l localhost" ( macOS ) / `kill -f "nc -l .127.0.0.1" + "nc -l localhost" (Linux)` immediately before `HONEY_PORT=\$(get_free_port)` at both arming sites. No `sudo` needed (the listeners are user-owned). Net effect: at most one tripwire alive at a time instead of one per toggle. (The same `kill` pattern cleared the backlog that this session's ~15 test toggles produced.)

15.12.21 Post-Wake Self-Feedback Loop: The Watcher Re-Fires On The Route Change It Caused (July 14, 2026)

Symptom: After a container restart (or wake), `output.log` showed `recover.sh --post-wake` launching ~6 times in a row, one launch roughly every ~40s, each exiting `exit=0` recovery *worked* every time yet kept re-firing "until it worked." Every launch was triggered by `NET: changedKey State:/Network/Global/IPv4`.

Root Cause a self-feedback loop, NOT PF. **scripts/watcher.sh** Listener 2 watches `State:/Network/Global/IPv4` via `scutil n.watch` and launches `recover.sh --post-wake` on any change. But post-wake re-asserts the exit node (`recover.sh` ~L367, `tailscale set --exit-node`), and even though `set` is deliberately used instead of `--reset` to minimise churn (recover.sh ~L360-366), **re-asserting the exit node re-writes the host default route and the default route IS `State:/Network/Global/IPv4`**. So each run mutates the exact key the watcher triggers on. The self-triggered change lands ~15-45s after the run finishes; `run\_recover()`'s old 10s debounce (reset to the run's *finish* time at the tail of the function) had long expired by then, so the watcher re-fired. The loop self-terminates only once `tailscale set` becomes a no-op (route already correct) no route change no re-trigger.

Why the debounce couldn't stop it: the loop period (~40s = one full post-wake run + settle) far exceeds the 10s debounce, which therefore only ever caught macOS's rapid *duplicate* scutil notifications for a single change (the ~0.2s doublets), not the self-trigger.

Not PF: post-wake never calls `pfctl` (the kill-switch is armed once, in `scripts/common.sh`, not per cycle); no PF-enable line falls inside the loop window. The "No ALTQ support in kernel" warning is normal macOS pfctl noise (Apple ships pfctl without ALTQ compiled in), and "Could not capture PF reference token" is a separate benign ref-count bookkeeping wart both unrelated to the loop.

Fix: `scripts/watcher.sh` `run\_recover()` now arms a **post-recovery settle cooldown** (`/tmp/nullexit-watcher.settle-until`, `NULLEXIT\_POSTWAKE\_SETTLE\_SECONDS` default 45s) *after* each post-wake run, and skips launching while `now < settle-until`. Because it is armed only after a recovery runs, a genuine first wake/roam is still handled immediately; only the Global/IPv4 change the run caused itself is absorbed. A real new roam during the window is delayed at most `POSTWAKE\_SETTLE\_SECONDS`, an acceptable trade for killing the loop. Watcher-side only `recover.sh` is unchanged. Severity was low (self-correcting, healthy every iteration, no leak/outage; cost was wasted ~28s runs + log noise).

15.12.22 FaceTime (and Real-Time P2P Media) Over the Double-Tunnel: Why It Breaks + the Split-Route Plan (July 15, 2026)

Symptom: A FaceTime call placed while nullexit is up fails to connect.

Diagnosis nullexit is NOT blocking it; every layer passes the traffic (all verified live):

- **DNS:** no Apple/FaceTime domain is sinkholed `dig` via the gateway (`100.100.21.8`) returns real Apple A/CNAME records, none `0.0.0.0`/`127.0.0.1`. The `no answer` domains (`gateway.push.apple.com`, `fmn.apple.com`, `identityservices.apple.com`, `fdr.apple.com`) return empty on `1.1.1.1` too CNAME/non-A, not our block.

- **routing-fix FORWARD:** the exit-path rule `-i tailscale0 -o tun0 -j ACCEPT` carries ~203K pkts / 73M bytes and accepts *everything*.

- **Country / threat blocklists** (`block\_il`, `block\_kp`, `block\_malicious`): **0** packets dropped.

- **WARP (gluetun) egress:** `-A OUTPUT -o tun0 -j ACCEPT` + `-A POSTROUTING -o tun0 -j MASQUERADE` all outbound UDP (any port) leaves and is SNAT'd through WARP.

Root cause double symmetric NAT ending at Cloudflare WARP. **Path:** `host Tailscale (MASQUERADE) WARP (MASQUERADE) Cloudflare shared egress`. WARP is a CGNAT-style shared egress with **no inbound UDP mappings**, and the double-MASQUERADE presents as **symmetric NAT** the exact condition that defeats FaceTime's STUN/ICE UDP hole-punching. No P2P path can form; the Apple-relay fallback is unreliable through the stacked tunnel, and the 12801200 MTU squeeze degrades media. This is inherent to routing real-time P2P media through WARP, **not** a firewall rule. (Zoom/WhatsApp *calls* hit the same wall.)

Latent finding (separate bug, NOT yet fixed): the `FORWARD` chain's `-i tailscale0 -p udp --dport 443` (QUIC) and `--dport 853` (DoT) `REJECT` rules are **shadowed** by the earlier `-i tailscale0 -o tun0 -j ACCEPT` (first-match wins) verified **0 pkts** on each. So the intended QUIC-block / force-through-AdGuard (block DoT) on the exit path is **not enforced**: a mesh client can use DoT (853) to bypass AdGuard

```

filtering entirely. To actually enforce, those port-`REJECT` rules must be inserted before the `
tailscale0 tun0` ACCEPT (or scoped `-o tun0`). TODO.
1972
1973 Fix plan (chosen: Option 2 narrow, always-on, opt-in split-route). DESIGNED / STAGED, not yet implemented in
code:
1974 - Mechanism: add more-specific host routes for FaceTime's Apple `/16`s via the physical gateway
(`172.17.0.1`/`en0`). Longest-prefix match beats Tailscale's exit-node `0.0.0.0/1` + `128.0.0.0/1` routes,
so only those `/16`s leave direct while everything else stays double-tunneled. Mirrors `
add_warp_bypass_routes`.
1975 - Wiring: `add_apple_split_routes()` / `remove_apple_split_routes()` in `common.sh`; called in toggle START
(after exit-node assert) + `recover.sh --post-wake` (survives roam) + removed on STOP; guarded by an opt
-in `env` flag (default OFF) so the privacy posture is unchanged unless enabled.
1976 - Privacy tradeoff: whatever `/16`s go direct see the real ISP IP (not WARP). Apple already ties
traffic to the Apple ID, so Apple-direct is the accepted scope; user chose narrow (FaceTime-infra `/16`
s only) over all-`17.0.0.0/8` to minimise the surface.
1977 - CIDR list to be finalised from an empirical capture (PENDING, deferred at user request): host-side `sudo
tcpdump -n -i <exit-utun, e.g. utun6> net 17.0.0.0/8` during a real call, then extract distinct `/16`s.
Known-likely from DNS today: `17.57/16` (APNs push/ringing), `17.248/16` (iCloud/FaceTime gateway),
`17.253/16` (aapling push CDN), `17.157/16` (GSA auth). The relay/STUN `/16`s the load-bearing media
path are call-time-discovered and MUST come from the capture; routing signaling-only will leave media
broken. Signed scripts are untouched until the capture yields the media ranges. See P2P note 16.4.
1978
1979 ---
1980
1981 # Part IV Observations, Changelog & TODO
1982
1983 ## 16. Unverified Observations
1984
1985 ### 16.1 AdGuard Returns Two Different Sinkhole IPs (Unverified)
1986
1987 > Status: Observed but not formally verified. Needs a deeper audit of the rule-compiler sources to confirm.
1988
1989 Observation (July 10, 2026): When querying AdGuard Home for blocked ad domains, some domains resolve to
`0.0.0.0` and others resolve to `127.0.0.1`. Both are blocked, both cause connection failure, but the
different responses were unexpected.
1990
1991 ...
1992 doubleclick.net 0.0.0.0 (blocked)
1993 ads.google.com 127.0.0.1 (blocked)
1994 pagead2.googlesyndication.com 0.0.0.0 (blocked)
1995 scorecardresearch.com 127.0.0.1 (blocked)
1996 ...
1997
1998 Likely Explanation (Unverified): There are three historical schools of thought on the "correct" DNS
sinkhole response, and AdGuard faithfully preserves whichever the source blocklist used:
1999
2000 | Sinkhole IP | Convention | Origin |
2001 | ---|---|---|
2002 | `0.0.0.0` | AdBlock-style lists (`\|\|domain`) | Modern DNS-level blocking; AdGuard's native format. Fails
fast OS doesn't attempt a TCP connection. Works for both IPv4 and IPv6 without a separate rule. |
2003 | `127.0.0.1` | Hosts-file-style lists (`127.0.0.1 domain`) | 90s-era `/etc/hosts` tradition. Steven Black's
hosts list (500k+ domains) still uses this. |
2004 | `NXDOMAIN` | Purist / Pi-hole default | Returns "domain doesn't exist." Semantically most accurate but
requires browsers to handle NXDOMAIN gracefully. |
2005
2006 The dual-response behavior in this project is because `rule-compiler` pulls from both AdBlock-format sources (
Hagezi, uBlock origin lists) and hosts-format sources (Spamhaus DROP, Steven Black, Feodo Tracker), and
AdGuard returns whatever sinkhole IP the matching rule specifies.
2007
2008 To Verify: Run `docker compose exec tailscale cat /etc/hosts` to confirm no hosts-file rules are being
injected at the container level, and check which specific AdGuard rule matched each domain via the AdGuard
query log at `http://100.100.21.8:3000/#logs`.
2009
2010 ### 16.2 AGY CLI Appears to Generate Native macOS Notification Pop-ups (Unverified)
2011
2012 > Status: Observed but source unverified. macOS security logs ruled out system-level origin.
2013
2014 Observation (July 10, 2026): While running a DNS blocklist verification via the Antigravity CLI (`agy`), a
command was proposed that included `malware.wicar.org` (a known-safe DNS blocklist test domain) in a `dig`
loop. Shortly after, a macOS-style notification popup appeared describing a "malicious" or "blocked" event.
The AGY CLI terminal session then closed.
2015
2016 Investigation:
2017 - macOS XProtect / Gatekeeper logs: Empty. Zero events in the 30-minute window around the incident.
2018 - macOS Endpoint Security / System Policy logs: Empty.
2019 - Broad `log show` search for "malicious", "malware", "blocked": Empty.
2020 - Conclusion: macOS itself did not generate the notification. No system security subsystem fired.
2021
2022 Likely Explanation (Unverified): The notification appears to have originated from AGY CLI's own internal
safety system. AGY CLI is a native macOS application (not just a terminal process) and has access to the
macOS Notification Center API (`UNUserNotificationCenter`). When its guardrails detected a domain

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containing the word "malware" (`malware.wicar.org`) in a proposed shell command, it likely:

- Blocked the command from executing
- Posted a native macOS-style notification via the Notification Center

This is surprising behavior for a CLI tool most terminal programs don't send GUI notifications but AGY CLI appears to bridge both worlds: it runs in a terminal but is packaged as a native app with full system API access. The notification would be indistinguishable from a system security alert at a glance.

****Notes:****

- `malware.wicar.org` is the DNS/AV equivalent of the EICAR test file a deliberately benign domain that triggers blacklist/AV systems to verify they work. It was included in the test set to confirm the AdGuard blacklist was catching it. AGY's safety system is not aware of this distinction.
- For future DNS blacklist testing, use only ad/tracking domains (e.g., `doubleclick.net`, `adnxs.com`) to avoid triggering AGY's guardrails.

16.3 `system\_profiler SPAirPortDataType` Longevity (Wi-Fi Detection)

The Wi-Fi security detection used by `watcher.sh`'s `detect\_lan\_p2p\_mode()` currently relies on `system\_profiler SPAirPortDataType` (after the `airport` binary removal full history in 15.11.6). This works as of macOS Sequoia (15.x) without sudo and without redaction, but given Apple's ongoing Wi-Fi privacy clampdown, it may be locked down in a future release. If `security` comes back empty on a future macOS version while on Wi-Fi, detection silently falls back to `allow=false` (a safe default). The long-term fix would be a small Swift CoreWLAN helper compiled and bundled in the repo.

16.4 Tailscale LAN P2P Succeeds Cross-AP But Fails Same-AP (Client Isolation + No NAT Hairpin)

****Observation (July 15, 2026):**** On the residence Wi-Fi, a Galaxy S24 in the lobby reception 6 floors away, associated to a *different* AP establishes a ****direct, low-ping**** Tailscale P2P session to an iPhone 11. The ****same**** pair on the ****same**** AP / same floor / close range does ****not**** go direct. Counterintuitively, ****farther apart = P2P works****. The effect shows for the iPhone 11 (a plain client) but ****not**** for S24macOS or S24the gateway container.

****Mechanism:****

- ****Same AP:**** the AP enforces ****client isolation**** (L2 device/device block on one radio). Both devices then share one public IP; STUN returns identical external mappings, so reaching each other would need the residence router to ****NAT-hairpin**** which most campus/consumer gear does not do direct P2P fails and falls back to a DERP relay.
- ****Different AP (floors away):**** large residence networks ****segment per AP/area into distinct VLANs/subnets****, so the two devices sit on different L3 networks routed through the campus core, where client isolation does not apply and the STUN path works ****without**** hairpin clean direct P2P.
- ****Why macOS / the container are immune:**** nullexit pins a ****direct WireGuard endpoint**** (Tailscale port `41642`, `direct 172.x`), so those peers already hold a negotiated direct path and do not depend on same-AP L2 discovery. A plain iPhone 11 has no pinned endpoint, so its path is entirely at the mercy of the Wi-Fi topology which is exactly why ****it**** exposes the isolation effect.

****Relevance:**** refines `watcher.sh`'s `detect\_lan\_p2p\_mode()` (which probes same-subnet AP isolation via a broadcast ping). The probe correctly flags same-AP isolation, but this observation adds that isolation is ****per-AP**** and cross-VLAN P2P can succeed even when same-AP fails so a single "LAN P2P allowed?" boolean under-describes a multi-AP network. No code change; documents expected behavior. Related: the FaceTime NAT analysis in 15.12.22 (same STUN/hairpin physics, different symptom).

17. Changelog / Recent Updates

Reflects the current branch's state. Each entry is a one-line summary + commit hash; read the commit message (and the linked Incident Log entry) for full detail. Full post-mortems live in 15.

- ****July 10, 2026 `fix(race): suppress WARP Watcher during post-wake force-recreate\*\*** Fixed a race where switching Wi-Fi caused the host IP to leak as the raw ISP IP (`132.x`) on every roam; `recover.sh --post-wake`'s warp force-recreate tripped the WARP Watcher into a nuclear shutdown. Fixed via `/tmp/nullexit-warp-inhibit.marker`. See 15.2.7.
- ****July 10, 2026 startup/roaming stabilization suite**** Pre-flight Tailscale race (15.5.3), STUN-block cold-boot false negative (15.5.4), restart DNS-build race via `wait\_for\_dhcp\_settle` (15.5.5), nuclear/post-wake concurrency lock (15.2.8), infinite auto-recovery loop marker leak (15.2.9), PF kill-switch bricking SOCKS5 fallback (15.7.3), `--reset` host logouts (15.7.4), missing-flags abort (15.7.5), discarded docker-exec errors (15.12.18), and the SNAT endpoint poisoning fix (15.7.1).
- ****July 6, 2026 `fix: resolve edge-case vulnerabilities and system state leaks**** Security-review hardening suite:
 - **DNS Watcher Interface Independence:**** `start\_dns\_watcher` now detects the active interface via `get\_active\_service()` instead of hardcoding a `grep` for "Wi-Fi".
 - **Robust Lock Verification:**** `toggle.sh` lock-file logic uses `ps -p` verification to prevent permanent lockouts from a recycled PID.
 - **Fail-Closed Crypto:**** `crypto.sh` integrity check switched from `[ -x ]` to `[ -f ]` so it runs even if git strips the `+x` bit.
 - **Strict `iptables` Pinning:**** `post-rules.txt` FORWARD-chain rules explicitly pin `tailscale0` both directions, preventing escape via `eth0` during WARP drops.
 - **Docker Local Network Isolation:**** `docker-compose.yml` binds exposed WARP ports (`1080`, `5354`) to `127.0.0.1` to prevent LAN access.
 - **Routing Verification & Sudoers:**** Added a `netstat -nr` default-route override check and ensured `/usr/bin/python3` is permitted in the sudoers template.
- ****July 4, 2026 `cc789c5`** Concurrency mutex on `toggle.sh` (`/tmp/nullexit-toggle.lock`); defensive `TS\_AUTHKEY` init under `set -u`; Go `go vet`/`go build` steps in CI; AdGuard log capping; macOS BSD `grep` fixes in `setup-common.sh`; `recover-linux.sh` quoting/PID fixes; removed redundant bypass-routes block in

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`recover.sh`.
2060 - **July 4, 2026 `fix: post-wake routing loop & destructive recovery`** (1) `add_warp_bypass_routes` now
accepts explicit interface args so post-wake stops routing WARP traffic back into `utun*` (was an infinite
loop killing internet on every wake). (2) Replaced `sudo route -n flush` with targeted deletions of
`0.0.0.0/1`/`128.0.0.0/1` + WARP bypass routes, so recovery no longer destroys loopback/multicast routes
and wedges `configd`/`mDNSResponder` until reboot.
2061 - **July 4, 2026 `feat: secure execution`** Restricted the blanket `/usr/bin/python3` NOPASSWD sudo to
exactly `/usr/bin/python3 -I ../scripts/dns-proxy.py`; locked `dns-proxy.py` with `chown root:wheel` + `
chmod 755`; added native auth prompts to `Toggle Gateway.app`/`Recover Gateway.app`; `toggle.sh` now
captures `warp` logs on failure before teardown.
2062 - **July 4, 2026 Go ports + refactor`** `logger` and `rule-compiler` ported to Go multi-stage Docker builds
(15.8.6); ~200 lines of duplicated bash extracted to `scripts/common.sh` (15.9.4); `crypto.sh` HMAC-SHA256
script integrity added.
2063 - **July 3, 2026 `fix: resolve numerous system regressions`** Fixed truncated `/etc/sudoers.d/nullexit` line;
temporary `$.tmp`+`os.replace()` atomic writes in `sync-rules.py` (later reverted, see below); `recover.sh`
`ts_args` quoting; ANSI `%b`/`~e` output in `diagnose-host-leak.sh`/`fix-docker-bridge-collision.sh`; `
toggle-linux.sh` using `resolvectl dns` instead of macOS `networksetup`; AdGuard REST refresh POST
targeting port `3000`; restored `START_GATEWAY=true`; purged stale port `80` mapping; verified `output.log`
in `$.gitignore`.
2064 - **July 3, 2026 `unlock-files.sh` + revert atomic writes`** Reverted all 5 `$.tmp`+`os.replace()` sites in `
sync-rules.py` to direct writes; created `scripts/unlock-files.sh` to permanently fix stale `0444`/`000`
file permissions via atomic rename (no `chmod`). Covers `.env`, `compiled_rules.txt`, `ip_blocklist.ipset`,
`cache/*.txt`, `cache/ip/*.txt`, `data/filters/*.txt`. See 15.9.6 + 3.
2065 - **July 3, 2026 Overnight leak + geo-blocking`** Introduced `scripts/host-leak-probe.sh` (sub-second host-
egress detector, 15.6.1) and the statistical-threshold WARP auto-shutdown watcher (15.6.2); fixed the
overnight silent IP leak (15.6.3).
2066 - **July 2, 2026 `8c5fae1` + `181e1e1`** Two infinite routing loops fixed: host-VM tunnel loop (WARP bypass
routes via physical gateway IP + `setup_exit_node_routing` forcing default to `utun*`) and Docker Compose
subnet takeover (`10.200.1.0/24` lock). `--accept-routes=true` added explicitly to all `tailscale up --
reset` calls. See 15.2.2.
2067 - **July 2, 2026 auto-recovery daemon`** `scripts/watcher.sh` + launchd LaunchAgent documented. See 15.5.1.
2068 - **July 1, 2026 `c5b92c0` (refactor: personal-leak cleanup)`** Eliminated every residual personal-username
leak across source + config; launchd plist uses `WorkingDirectory` + relative program arg with the `
__NULLEXIT_HOME__` sentinel (install recipe gains a single `sed -i 's|__NULLEXIT_HOME__|$(pwd)|'` step);
8 devref probe sites genericized to `~/colima/...`, `~/Library/LaunchAgents/...`, `$USER`, ``.
2069 - **July 1, 2026 `a5e2a5a` (refactor: script-relative paths)`** Replaced 6 hardcoded `/Users/<user>/.../
nullexit/...` literals with script-relative resolution. `recover.sh`/`watcher.sh` inject `SCRIPT_DIR=$(cd
$(dirname "${BASH_SOURCE[0]}") && pwd)`; `RECOVER="$SCRIPT_DIR/./recover.sh"` resolves without launch-
path dependency.
2070 - **June 28, 2026 `f8f8e4e` (M1: scripts/ move)`** 8 internal scripts consolidated into `scripts/`; the 4 user
-facing/orchestrator/config files (`toggle.sh`, `recover.sh`, `setup.sh`, `docker-compose.yml`) stayed at
repo root.
2071 - **June 28, 2026 `0875ef3` (ci: glob)`** CI `py_compile` switched to `python3 -m py_compile scripts/*.py`
after M1 left root with zero `.py` files.
2072 - **June 28, 2026 `25b37a1` (docs: scripts/ prefix)`** `devref.md` + `README.md` updated to the `scripts/`
prefix for all 8 moved files (~31 patches).
2073
2074 ### End-to-end verification (commit `c5b92c0`)
2075 Manual START STOP cycle ran clean on macOS after path relativization + personal-leak cleanup:
2076 - `bash toggle.sh` START: exit 0, ~135s (cold Colima boot); marker present (`2026-07-02T03:41:41Z`), all 5
containers healthy, host DNS hijacked to `100.100.21.8` (no fallback), exit node selected, Cloudflare trace
through WARP succeeds.
2077 - `bash toggle.sh` STOP: exit 0, ~12s; marker cleared, all nullexit containers gone, host DNS restored to
`1.1.1.1`, Tailscale disconnected.
2078
2079 ## 18. TODO & Future Work
2080
2081 ### 18.1 TODO
2082 * **FaceTime / real-time P2P split-route (Option 2, narrow):** Implement `add_apple_split_routes()` / `
remove_apple_split_routes()` in `common.sh` (route FaceTime's Apple `/16`'s direct via the physical gateway,
opt-in `.env` flag, default OFF), wired into toggle START/STOP + `recover.sh --post-wake`. **Blocked on**
an empirical `tcpdump` capture of the relay/STUN `/16`'s during a real call (user-triggered). Full design +
diagnosis in 15.12.22.
2083 * **Fix the shadowed exit-path port REJECTs (DoT/QUIC bypass):** The `FORWARD` `-i tailscale0 -p udp --dport
853`/`--dport 443` REJECT rules are shadowed by the earlier `tailscale0 tun0` ACCEPT (0 pkts), so the
intended block-DoT / force-AdGuard and QUIC-block are **not enforced** a mesh client can bypass AdGuard
via DoT (853). Re-order the REJECTs before the ACCEPT (or scope `-o tun0`). See 15.12.22.
2084 * **Decide toggle STOP/START from persisted *intent*, not a live probe:** ~~`toggle.sh` decided STOP-vs-START
by calling `is_gateway_active()`, which live-probes the running system (`docker compose ps` + DNS + SOCKS
checks). When Colima/Docker was down the `docker compose ps` blocked for the full 15 s `run_with_timeout`
window and, under `set -e` + the `ERR` trap, aborted the script *before* the START body ran* so the path
that would boot Colima never executed (`EXIT: code=1 phase=start`, seen 57 historically).~~ **Closed**
added `gateway_should_be_running()` in `common.sh`, which reads persisted intent (`MARKER_FILE`) captured
into `GATEWAY_MARKER_AT_STARTUP` before the defensive stale-marker clear, and only falls back to `
is_gateway_active` when the marker is absent *and* the Docker socket is actually reachable (so a dead-
socket probe can neither hang nor trip `set -e`). All four decision sites (`toggle.sh` main + `--restart`,
`toggle-linux.sh` main + `--restart`) now use it, and the STOP `docker compose down` is `|| true`-guarded
so a disable always completes even with Docker down. See 15.12 Misc Resolved.
2085 * **Verify Wi-Fi Roaming:** Investigate and test whether switching Wi-Fi networks now successfully and smoothly
recovers the gateway end-to-end without any tailscale flag errors or dead tunnels. (Pending user
verification).

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2086 * **Fix Diagnostic Script Check 5/8:** ~~Investigate why `diagnose-host-leak.sh` check 5/8 (`Host default route
    `) sometimes reports "default route goes via physical Wi-Fi Tailscale route assertion failed" on macOS
    even though the `warp=on` egress checks confirm that traffic is successfully tunneling.~~ **Closed** fixed
    by switching check 5 from `netstat -rn | grep default` to `route -n get 1.1.1.1`. macOS Tailscale uses
    interface-scoped routes and does not replace the DHCP default route entry. See 15.7.1 Known Unknowns.
2087 * **Expose Tor Control Port (9051):** ~~Consider mapping the Tor Control port to localhost and adding `nyx` (
    the Tor terminal status monitor) to allow users to inspect their entry, middle, and exit node IPs in real-
    time.~~ **Closed** mapped the Tor Control Port to a procedurally generated, localhost-bound port `
    TOR_CONTROL_PORT`, configured `ControlPort` and disabled cookie authentication in `docker/tor/entrypoint.sh
    `, and integrated it into the `/sweep` verification protocol.
2088 * **Host-Only Tor Mode (Pluggable Egress):** Implement an `EGRESS_TYPE=tor_host_only` mode for users in heavily
    censored (DPI-restricted) environments. This mode would skip booting `warp` and `tailscale`, boot `
    adguardhome` and `tor` (using Telegram `obfs4` bridges), set AdGuard's upstream to Tor's `DNSPort`, and use
    the `pf` kill-switch to force all host Mac traffic strictly through the Tor network. This provides a 100%
    free, DPI-resistant evasion path without requiring an external VPS.
2089 * **The Technical Realities (What You Need to Watch Out For):**
2090 * **The UDP Problem:** Tor only transports TCP traffic. It does not support UDP (aside from DNS requests
    passed directly to its `DNSPort`). macOS is incredibly chatty over UDP (FaceTime, mDNS, QUIC protocols). Your
    pf rules defined in `scripts/pf.conf` must be configured to aggressively and silently DROP all host UDP
    traffic (except for local DNS queries to AdGuard). If you don't drop it cleanly, macOS services might hang
    indefinitely while waiting for timeouts.
2091 * **The "Daily Driver" Friction:** Routing a whole host OS through Tor is brutal on performance. Background
    daemons (iCloud sync, macOS software updates, Spotlight indexing) will blindly attempt to download
    gigabytes of data over Tor, which could saturate the circuits or time out completely.
2092 * **Bridge Rot:** State firewalls actively hunt and block obfs4 bridges. When a user's bridges burn out,
    they will lose all internet access due to the pf kill-switch. You will need to ensure the user has a
    frictionless way to update the `tor-bridges.txt` file and restart the Tor container without exposing their
    IP.
2093 * **Exit Node Discrimination:** Because all traffic will exit from known Tor nodes, the user will face an
    onslaught of Cloudflare CAPTCHAs, and many banking or streaming services will flat-out reject the
    connections.
2094
2095 ### 18.2 Future Work
2096 - **Direct P2P on VM Hosts:** Non-Linux hosts run Docker inside a VM, making true UDP hole-punching **to
    arbitrary external peers **impossible; those peers fall back to the DERP relay unless bridged mode is
    enabled on a trusted LAN. The only fully general solution is a native Linux host (Raspberry Pi, Intel NUC)
    where Docker runs without VM hypervisor translation. *(The one case that works regardless is direct **
    hostcontainer** P2P the gateway and its own Mac, the same physical machine re-permitted via routing-fix
    `host_ips` RETURN rules and confirmed `direct` even in plain user-mode networking; see 15.3.5 and 15.8.7.)
    *
2097 - **Native Linux Deployment:** Benchmark on a Raspberry Pi native container footprint is ~75MB without macOS
    hypervisor overhead.
2098 - **Mesh-Wide Filesystem (SFTP over Tailscale):** Any mesh device passively exposes its filesystem over SFTP.
    Android via Termux + OpenSSH + wakelock; iOS is client-only due to background process limits.
2099 - **Post-Quantum Cryptography (PQC):** Tailscale uses Curve25519, theoretically vulnerable to "harvest now,
    decrypt later" quantum attacks. A future iteration could replace the Tailscale container with raw WireGuard
    + [Rosenpass](https://rosenpass.eu/) to negotiate post-quantum PSKs.
2100 - **Egress Compartmentalization (Pluggable Architecture):** See 12.9 for the full plan to make the entire
    egress layer swappable via a single `.env` variable.

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5 diagrams.md

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1 # nullexit Flow Diagrams & Architecture
2
3 ---
4
5 ## 1. System Architecture What's Running & Where
6
7 ```mermaid
8 graph TB
9     subgraph INTERNET[" Internet / Cloudflare Edge"]
10         CF["Cloudflare WARP\n(Exit Point)\nwarp=on"]
11     end
12
13     subgraph TAILNET[" Tailscale Mesh (100.x.x.x)"]
14         PHONE[" Phone / Laptop\n(Tailnet Client)"]
15     end
16
17     subgraph MACHOST[" macOS Host"]
18         direction TB
19         PFFIREWALL["macOS pf Firewall\n(Kill-Switch & TCP MSS Clamping)"]
20         TSDAEMON["tailscaled\n(brew service)\nhost Tailscale"]
21         HOSTDNS["Host DNS\nGateway IP\n(hijacked by toggle.sh)"]
22         HOSTSOCKS["Host SOCKS5\n127.0.0.1:1080\n(fallback only)"]
23         HOSTTOR["Host Tor SOCKS5\n127.0.0.1:$TOR SOCKS_PORT\n(randomized)"]
24         HOSTTORCTRL["Host Tor Control\n127.0.0.1:$TOR_CONTROL_PORT\n(randomized)"]

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25
26     subgraph MONITORS[" Background Daemons (toggle.sh children)"]
27         WARPWATCH["WARP Watcher\n(every 30s, via docker exec)\n output.log"]
28         LEAKPROBE["Host Leak Probe\n(every 300ms, via curl from HOST)\n output.log"]
29         DNSWATCHER["DNS Watcher\n(re-hijacks if drift detected)\n output.log"]
30         CAFFEINATE["caffeinate -i\n(sleep prevention)"]
31     end
32
33     subgraph COLIMAVM[" Colima VM (Linux, 600MB RAM)"]
34         subgraph NETNS["warp container network namespace (shared by ALL containers)"]
35             GLUETUN["warp / Gluetun\nWireGuard tun0\n(owns the namespace)"]
36             TS["tailscale container\nAdvertises exit node\ntailscale0 interface"]
37             SOCKS["tailscale\nSOCKS5\nport 1080"]
38             ADGUARD["adguardhome\nDNS sinkhole\nport 5335"]
39             ROUTINGFIX["routing-fix sidecar\nGeo-IP Blocking & Go Logger\n(writes blocked.log)"]
40             TOR["tor container\nSOCKS5: port 9050\nControl: port 9051\nTransparent Proxy: port 9040\n
nDNSPort: port 5353"]
41         end
42     end
43
44     subgraph LAUNCHD[" launchd (always-on)"]
45         WATCHER["scripts/watcher.sh\n(sleep/wake + Wi-Fi roam\n+ LAN P2P auto-detection)"]
46     end
47 end
48
49 subgraph RECOVERY[" Recovery"]
50     RECOVER["recover.sh\n(nuclear reset)"]
51     RECOVERPOST["recover.sh --post-wake\n(light refresh)"]
52 end
53
54 %% Traffic flow
55 PHONE -->|"WireGuard / Tailscale mesh"| TSDAEMON
56 TSDAEMON -->|"exit-node route utun*"| TS
57 TS -->|"FORWARD tun0 (MASQUERADE)"| GLUETUN
58 GLUETUN -->|"WireGuard UDP (macOS Host Egress)"| PFFIREWALL
59 PFFIREWALL -->|"TCP MSS Clamped / Kill-Switch check"| CF
60 TS -->|"198.18.0.0/15 PREROUTING redirect"| TOR
61 TOR -->|"internal path / exit nodes via tun0"| GLUETUN
62
63 %% DNS
64 HOSTDNS -->|"UDP:53 TCP:5354"| ADGUARD
65 ADGUARD -->|"DNS upstream via tun0"| CF
66 ADGUARD -->|"onion queries localhost:5353"| TOR
67
68 %% Monitoring
69 WARPWATCH -->|"docker exec warp wget cdn-cgi/trace"| GLUETUN
70 LEAKPROBE -->|"curl cdn-cgi/trace (HOST NIC)"| CF
71 DNSWATCHER -->|"networksetup -getdnsservers"| HOSTDNS
72
73 %% Logging
74 ROUTINGFIX -->|"writes"| BLOCKEDLOG["blocked.log\n(Host-side)"]
75 HOSTTOR -->|"port mapping"| TOR
76 HOSTTORCTRL -->|"port mapping"| TOR
77
78 %% Recovery triggers
79 WARPWATCH -->|"6 consecutive warp=off"| RECOVER
80 WATCHER -->|"sleep/wake / roam"| RECOVERPOST
81 RECOVERPOST -->|"if gateway unhealthy"| RECOVER
82 WATCHER -->|"writes .lan_p2p_detected"| ROUTINGFIX
83
84 style MONITORS fill:#1a1a2e,color:#e0e0ff
85 style COLIMAVM fill:#0d2137,color:#e0e0ff
86 style NETNS fill:#0a1628,color:#e0e0ff
87 style RECOVERY fill:#2d0a0a,color:#ffe0e0
88 style INTERNET fill:#0a2d1a,color:#e0ffe0
89 style TAILNET fill:#1a2d0a,color:#e8ffe0
90 style LAUNCHD fill:#2d1a2d,color:#ffe0ff
91 ...
92
93 ---
94
95 ## 2. toggle.sh Full START Flow
96
97 mermaid
98 flowchart TD
99     START(["./toggle.sh"])
100     CHECK{"is_gateway_active?\n(containers running OR\nDNS 1.1.1.1)"}
101
102     START --> CHECK
103     CHECK -->|"YES already ON"| STOPFLOW[" See STOP Flow"]
104     CHECK -->|"NO start it"| S1

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105 S1["1. Reset DNS 1.1.1.1\n(prevent deadlocks)"]
106 S2["2. tailscale down\n(prevent exit-node routing during boot)"]
107 S3["3. Check Colima VM\n(colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged)"]
108 S4["4. Configure VM swap\n(400MB, prevent OOM)"]
109 S5["5. Clean corrupted AdGuard config\n(prevent container crash loop)"]
110 S6["6. docker compose up -d --build\n(compile Go threat rules & boot containers)"]
111 S7["7. Poll tailscale status\n(wait for 'offers exit node'\nup to 60s)"]
112 TSREADY{"container\nTailscale ready?"}
113 ABORT(["ABORT\n cleanup_handler"])
114 S8["8. Resolve gateway Tailscale IP\n(.gateway_ip or docker exec)"]
115 S9["9. Verify tailscaled running\n(auto-start if needed)"]
116 S10["10. tailscale up --reset\n--exit-node=\n--accept-routes=true\n--ssh=true --accept-dns=false"]
117
118 PF1{"Pre-flight 1/3\ntailscale ping gateway"}
119 PF2{"Pre-flight 2/3\ndig +tcp AdGuard DNS"}
120 PF3{"Pre-flight 3/3\nWARP internet check"}
121 ALLPASS{"All 3 pass?"}
122
123 EXITPATH["EXIT NODE PATH\n tailscale up --exit-node=\n force_dns_to_gateway (3 retry)\n SOCKS5 disabled"]
124 SOCKSPATH["SOCKS5 FALLBACK PATH\n macOS SOCKS5 127.0.0.1:1080\n DNS proxy 127.0.0.1:53\n No exit node (SKIP_EXIT_NODE=true)"]
125
126 DAEMONS["Start background daemons\n caffeinate -i (sleep prevention)\n DNS Watcher\n WARP Watcher\n Host Leak Probe\n(all write output.log)"]
127
128 RULECOMPILE["Rule compiler status\n(log rule/IP counts)"]
129 WRITEMARKER["write_gateway_active_marker\n(/tmp/nullexit-gateway-active.marker)"]
130 DONE(["Gateway UP"])
131
132 S1 --> S2 --> S3 --> S4 --> S5 --> S6 --> S7
133 S7 --> TSREADY
134 TSREADY -->|"NO (40 NoState)"| ABORT
135 TSREADY -->|"YES"| S8
136 S8 --> S9 --> S10
137 S10 --> PF1 --> PF2 --> PF3
138 PF1 & PF2 & PF3 --> ALLPASS
139 ALLPASS -->|"YES"| EXITPATH
140 ALLPASS -->|"NO"| SOCKSPATH
141 EXITPATH --> DAEMONS
142 SOCKSPATH --> DAEMONS
143 DAEMONS --> RULECOMPILE --> WRITEMARKER --> DONE
144
145 style ABORT fill:#5c1010,color:#fff
146 style EXITPATH fill:#0a3d1a,color:#c0ffc0
147 style SOCKSPATH fill:#3d2d00,color:#ffe0a0
148 style DAEMONS fill:#0a1f3d,color:#c0d8ff
149 style DONE fill:#0a3d1a,color:#fff
150
151 ---
152
153 ---
154
155 ## 3. toggle.sh STOP Flow
156
157 ``mermaid
158 flowchart TD
159     STOP(["./toggle.sh\n(gateway already ON)"])
160
161     ST1["1. tailscale down\n(disconnect from mesh)"]
162     ST2["2. docker compose down -t 5\n(stop all containers)"]
163     ST3["3. Reset DNS 1.1.1.1\n(networksetup on all services)"]
164     ST4["4. stop_local_dns_proxy\n(kill Python UDP proxy)"]
165     ST5["5. stop_pidfile_daemon\n(kill caffeinate, rm PID)"]
166     ST6["6. stop_pidfile_daemon\n(kill DNS Watcher, rm PID)"]
167     ST7["7. stop_pidfile_daemon\n(kill WARP Watcher, rm PID)"]
168     ST8["8. stop_pidfile_daemon\n(kill Leak Probe, rm PID)"]
169     ST9["9. clear_gateway_active_marker\n(rm /tmp/nullexit-gateway-active.marker)"]
170     ST10["10. cleanup_network_state\n disable_all_proxies\n Flush DNS cache (dscacheutil)\n Flush stale routes\n Power-cycle Wi-Fi (offon)"]
171
172     DONE(["Gateway DOWN\nInternet restored"])
173
174     STOP --> ST1 --> ST2 --> ST3 --> ST4
175     ST4 --> ST5 --> ST6 --> ST7 --> ST8 --> ST9 --> ST10 --> DONE
176
177     style DONE fill:#0a3d1a,color:#fff
178 ---
179
180 ---
181
182 ## 4. Monitoring Layer The 4 Background Daemons

```

```

183
184 ```mermaid
185 graph LR
186     subgraph DAEMONS["Background Daemons (all start with gateway, stop on teardown)"]
187         direction TB
188
189         subgraph D1[" Sleep Prevention"]
190             CAFF["caffeinate -i\nPID: /tmp/nullexit-caffeinate.pid\n\nKeeps Mac awake while\ngateway is active"]
191         end
192
193         subgraph D2[" DNS Watcher"]
194             DNSW["Polls networksetup every ~30s\nPID: /tmp/nullexit-dns-watcher.pid\n\nIf DNS drifts from gateway IP\nre-hijacks automatically\noutput.log"]
195         end
196
197         subgraph D3[" WARP Watcher"]
198             WARPW["Polls every 30s via:\ndocker compose exec warp wget cdn-cgi/trace\nPID: /tmp/nullexit-warp-watcher.pid\n\nCounts consecutive warp=off\nWARP_FAIL_THRESHOLD (default 6=30s)\nruns recover.sh (nuclear)\noutput.log"]
199         end
200
201         subgraph D4[" Host Leak Probe"]
202             LEAKP["Polls every 300ms via:\ncurl cdn-cgi/trace (HOST NIC directly)\nPID: /tmp/nullexit-host-leak-probe.pid\n\nLogs ONLY on state change:\n LEAK / ROTATE / HOST-PROBE failed\nAll errors output.log\n(HOST_LEAK_PROBE=true in .env)"]
203         end
204     end
205
206     subgraph BLIND["What each monitor can/can't see"]
207         B1["WARP Watcher sees:\n Container WARP tunnel state\n Host NIC traffic\n Flash leaks < 5s"]
208         B2["Host Leak Probe sees:\n Host NIC egress (browser path)\n Sub-second transitions\n Container internals"]
209     end
210
211     D3 -->|"covers"| B1
212     D4 -->|"covers"| B2
213
214     style D1 fill:#1a2d0a
215     style D2 fill:#0a1a2d
216     style D3 fill:#2d0a2d
217     style D4 fill:#2d1a00
218     style BLIND fill:#1a1a1a,color:#aaa
219 ```
220
221 ---
222
223 ## 5. Traffic Flow Data Path (Normal Operation)
224
225 ```mermaid
226 sequenceDiagram
227     participant PHONE as Phone<br/>(Tailnet client)
228     participant TS_HOST as tailscaled<br/>(macOS host)
229     participant TS_CONTAINER as tailscale<br/>(container)
230     participant GLUETUN as Gluetun/warp<br/>(container)
231     participant CF as Cloudflare<br/>WARP Edge
232     participant INTERNET as Internet
233
234     Note over PHONE,INTERNET: Normal EXIT NODE path (all 3 pre-flights passed)
235
236     PHONE->>TS_HOST: WireGuard packet<br/>(encrypted, UDP 41641)
237     TS_HOST->>TS_CONTAINER: route via tun* tailscale0
238     TS_CONTAINER->>GLUETUN: FORWARD chain<br/>(iptables MASQUERADE on tun0)
239     GLUETUN->>TS_HOST: WireGuard tunnel (tun0) egresses macOS Host
240     Note over TS_HOST: macOS pf Firewall applies<br/>Kill-Switch & TCP MSS Clamping
241     TS_HOST->>CF: re-encrypted UDP packet (MSS 1160)
242     CF->>INTERNET: Plain HTTPS from<br/>Cloudflare IP
243
244     Note over PHONE,INTERNET: DNS Resolution path
245     PHONE->>TS_HOST: DNS query
246     TS_HOST->>TS_CONTAINER: UDP:53 TCP:5354 AdGuard :5335
247     TS_CONTAINER->>GLUETUN: AdGuard upstream DNS via tun0
248     GLUETUN->>CF: Encrypted DNS query
249     CF-->>GLUETUN: DNS response
250     GLUETUN-->>TS_CONTAINER: response
251     TS_CONTAINER-->>TS_HOST: response
252     TS_HOST-->>PHONE: DNS answer
253 ```
254
255 ---
256

```

```

257 ## 6. recover.sh Decision Tree
258
259 ```mermaid
260 flowchart TD
261     INVOKE(["recover.sh called"])
262     MODE{"--post-wake\nflag?"}
263
264     subgraph POSTWAKE["--post-wake (light refresh, gateway stays UP)"]
265         PW1["Re-hijack DNS gateway IP"]
266         PW2["Flush DNS cache"]
267         PW3["Refresh tailscale exit-node"]
268         PW4["Force-recreate warp container\n(if Gluetun healthcheck failed)"]
269         PW5["Leave: containers, Wi-Fi,\ncaffeinate, DNS watcher,\nHost Leak Probe untouched"]
270     end
271
272     subgraph NUCLEAR["Default (nuclear gateway torn down)"]
273         N0["Sudo keep-alive loop (background)"]
274         N1["1. tailscale down\n(disconnect from mesh)"]
275         N2["2. Reset DNS 1.1.1.1\n(ALL network services)"]
276         N3["3. disable_all_proxies\n(all interfaces)"]
277         N4["4. Flush DNS cache\n(dscacheutil -flushcache)"]
278         N5["5. Flush stale routes\n(WARP endpoint routes)"]
279         N6a["6a. docker compose down -t 30\n(stop all containers)"]
280         N6b["6b. stop_pidfile_daemon\n(kill caffeinate)"]
281         N6c["6c. stop_pidfile_daemon\n(kill DNS Watcher)"]
282         N6d["6d. stop_pidfile_daemon\n(kill Leak Probe)"]
283         N7["7. Power-cycle Wi-Fi\n(networksetup offslepp 2on)"]
284         N8["8. Verify internet working\n(curl ifconfig.io)"]
285     end
286
287     DONE_PW(["Gateway refreshed\n(still running)"])
288     DONE_NUC(["Internet restored\n(gateway fully stopped)"])
289
290     INVOKE --> MODE
291     MODE -->|"yes"| PW1
292     PW1 --> PW2 --> PW3 --> PW4 --> PW5 --> DONE_PW
293     MODE -->|"no"| N0
294     N0 --> N1 --> N2 --> N3 --> N4 --> N5
295     N5 --> N6a --> N6b --> N6c --> N6d --> N7 --> N8 --> DONE_NUC
296
297     style POSTWAKE fill:#0a2d1a,color:#c0ffc0
298     style NUCLEAR fill:#2d0505,color:#ffc0c0
299     style DONE_PW fill:#0a3d1a,color:#fff
300     style DONE_NUC fill:#3d1a00,color:#ffe0c0
301
302 ---
303 ---
304
305 ## 7. Failure Paths & Self-Healing
306
307 ```mermaid
308 flowchart LR
309     subgraph EVENTS["Failure / Event"]
310         E1["WARP tunnel drops\n(warp=off)"]
311         E2["Mac wakes from sleep\nor Wi-Fi roams"]
312         E3["DNS drifts from\ngateway IP"]
313         E4["toggle.sh crashes /\nCtrl-C / SIGTERM"]
314         E5["Host-side flash leak\ndetected (HOST NIC)"]
315         E6["Silent NAT timeout\n(ping stops working)"]
316     end
317
318     subgraph DETECTORS["Detector"]
319         D1["WARP Watcher\n(30s poll, docker exec)"]
320         D2["scripts/watcher.sh\n(launchd, always on)"]
321         D3["DNS Watcher\n(30s poll, networksetup)"]
322         D4["cleanup_handler\n(ERR/INT/TERM/HUP trap)"]
323         D5["Host Leak Probe\n(300ms poll, HOST curl)"]
324         D6["routing-fix.sh\n(30s curl over tun0)"]
325     end
326
327     subgraph ACTIONS["Response"]
328         A1["threshold consecutive off\nrecover.sh (nuclear)"]
329         A2["recover.sh --post-wake\n(gentle refresh)"]
330         A3["Re-hijack DNS\n(networksetup setdnsservers)"]
331         A4["Stop all daemons\nReset DNS 1.1.1.1\nTear down Tailscale"]
332         A5["Log LEAK to output.log"]
333         A6["Blackhole internet traffic\n+ Drop TUNNEL_FAILED_CLOSED.marker"]
334         A7["Push macOS Desktop Notification\n(via watcher.sh listener)"]
335     end
336
337     E1 --> D1 --> A1

```

```

338     E2 --> D2 --> A2
339     E3 --> D3 --> A3
340     E4 --> D4 --> A4
341     E5 --> D5 --> A5
342     E6 --> D6 --> A6
343     A6 -.->|"marker detected"| D2
344     D2 --> A7
345
346     style EVENTS fill:#2d1010,color:#ffcccc
347     style DETECTORS fill:#0a1a2d,color:#cce0ff
348     style ACTIONS fill:#0a2d0a,color:#ccffcc
349 ---
350
351 ---
352
353 ## 8. output.log Who Writes What
354
355 All events converge into a single `output.log` at the repo root.
356
357 | Writer | Event Types | Format |
358 |-----|-----|-----|
359 | `toggle.sh` | All startup/shutdown steps, errors, pre-flight results | Plain text with timestamps |
360 | `recover.sh` | Every recovery step (nuclear or post-wake) | Plain text |
361 | **WARP Watcher** | `WARP DOWN`, `WARP RECOVERED`, `WARP SHUTDOWN` | `[UTC timestamp]` prefix |
362 | **Host Leak Probe** | `LEAK`, `ROTATE`, `HOST-PROBE failed/timeout` | `[HH:MM:SS.mmm]` prefix |
363 | **DNS Watcher** | DNS re-hijack events | Appended inline |
364 | `docker compose` | Container logs on failure (last 100 lines of warp) | Dumped on ERR path |
365 | `routing-fix.sh` | (via `nullexit-logger` inside container) | Structured |
366
367 **Grep cheatsheet:**
368 ```bash
369 grep LEAK output.log           # Host-side warp-off events (real leaks)
370 grep ROTATE output.log         # Cloudflare edge IP rotations (harmless)
371 grep 'HOST-PROBE' output.log   # curl failures from probe
372 grep 'WARP DOWN' output.log    # Container-side tunnel drops
373 grep 'WARP SHUTDOWN' output.log # Auto-recovery triggered
374 grep 'WARP RECOVERED' output.log # Self-healed before threshold
375 ```

```

6 toggle.sh

```

1  #!/bin/bash
2  # toggle.sh nullexit gateway toggle script (macOS + Linux)
3  # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4  #
5  # One file, both OSes: the dispatch below runs the self-contained Linux
6  # implementation (shuf / ss / nmcli / ip / systemd) and exits; on macOS it
7  # falls through to the macOS implementation (jot / netstat / networksetup,
8  # launchd / pf).
9
10 set -e
11
12 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
13
14 #
15 # LINUX TOGGLE self-contained; runs the full toggle then exits before the
16 # macOS body below. Native Linux tooling (shuf / ss / nmcli / ip / systemd).
17 # (Body inlined verbatim at column 0 it contains multiline `bash -c ""`
18 # strings that must NOT be re-indented.)
19 #
20 if [[ "$OSTYPE" != "darwin"* ]]; then
21     # Define log file (used by the lifecycle breadcrumb + run_logged helper)
22     LOG_FILE="$PWD/output.log"
23
24     # --- LIFECYCLE PHASE TRACKING + EXIT BREADCRUMB ---
25     # TOGGLE_PHASE is updated as the script moves through STOP/START so that if the
26     # process is killed (terminal closed, pkill, OOM) or exits on error, the EXIT
27     # trap records the final exit code AND the phase it died in making an
28     # interrupted START traceable in output.log instead of ending abruptly.
29     TOGGLE_PHASE="init"
30     _log_exit_breadcrumb() {
31         local rc=$?
32         echo "[$(date '+%Y-%m-%d %H:%M:%S')] toggle.sh EXIT: code=$rc phase=$TOGGLE_PHASE (pid $$)" >> "$LOG_FILE"
33         2>/dev/null || true
34     }
35     trap '_log_exit_breadcrumb' EXIT

```

```

35
36 # --- DEBUG TRACE (opt-in via .env: DEBUG_TRACE=true) ---
37 # When enabled, mirror a full command-by-command xtrace to output.log. Off by
38 # default. Read directly from .env because common.sh (read_env_var) isn't
39 # sourced yet. Branch on bash version: BASH_XTRACEFD needs >= 4.1 (Linux
40 # usually has bash 5); fall back to teeing stderr on older bash. We never use
41 # the `exec {var}>>` dynamic-fd form (4.1+ only).
42 DEBUG_TRACE=$(grep -E '^DEBUG_TRACE=' .env 2>/dev/null | cut -d'=' -f2- | tr -d "\"" | tr '[:upper:]' '[:lower:]' | tr -d '[:space:]')
43 if [ "$DEBUG_TRACE" = "true" ]; then
44     echo "[$(date '+%Y-%m-%d %H:%M:%S')] DEBUG_TRACE enabled xtrace mirrored to output.log (bash ${BASH_VERSINFO
45     [0]}.${BASH_VERSINFO[1]})" >> "$LOG_FILE"
46     export PS4='+ [${SECONDS}s] ${BASH_SOURCE##*/}:${LINENO}:${FUNCNAME[0]:-main}: ' # subshell-free clock: a $
47     () in PS4 trips the bash 3.2 set -x/ERR heisenbug (devref 15.11.8)
48     if [ "${BASH_VERSINFO[0]}" -gt 4 ] || { [ "${BASH_VERSINFO[0]}" -eq 4 ] && [ "${BASH_VERSINFO[1]}" -ge 1 ];
49     }; then
50         exec 9>>"$LOG_FILE"
51         export BASH_XTRACEFD=9
52         set -x
53     else
54         # bash < 4.1: plain stderrfile redirect. NOT a process substitution under
55         # set -e, xtrace flowing through a backgrounded `tee` can abort the script
56         # (failed write/SIGPIPE trips set -e). See devref 15.11.8.
57         exec 2>>"$LOG_FILE"
58         set -x
59     fi
60 fi
61
62 # --- PROCEDURAL PORT GENERATION ---
63 # To prevent static port fingerprinting by malware, we randomize exposed ports
64 # on every boot and persist them to a hidden environment file.
65 PORTS_FILE="/tmp/nullexit-ports.env"
66 if [[ "$1" == "" || "$1" == "--restart" || "$1" == "restart" ]]; then
67
68     # Collision-avoidance function
69     GENERATED_PORTS=""
70     get_free_port() {
71         local port
72         while true; do
73             # Linux uses shuf or $RANDOM instead of jot
74             port=$(shuf -i 10000-65000 -n 1 2>/dev/null || echo $((10000 + RANDOM % 55000)))
75             # 1. Prevent self-collision within this loop
76             if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
77                 continue
78             fi
79             # 2. Prevent collision with ANY process on the machine (root daemons, terminal apps, etc)
80             # We use ss to read the kernel socket table directly.
81             if ! ss -ltn | awk '{print $4}' | grep -q ":$port$"; then
82                 echo "$port"
83                 return
84             fi
85         done
86     }
87
88     export TOR SOCKS_PORT=$(get_free_port)
89     GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
90
91     export TOR_CONTROL_PORT=$(get_free_port)
92     GENERATED_PORTS="$GENERATED_PORTS $TOR_CONTROL_PORT"
93
94     export SOCKS_PROXY_PORT=$(get_free_port)
95     GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
96
97     export DNS_PROXY_PORT=$(get_free_port)
98
99     echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
100     echo "export TOR_CONTROL_PORT=$TOR_CONTROL_PORT" >> "$PORTS_FILE"
101     echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
102     echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
103 elif [ -f "$PORTS_FILE" ]; then
104     # On STOP, source the existing ports so docker-compose down matches
105     source "$PORTS_FILE"
106 else
107     # Fallbacks if file is lost
108     export TOR SOCKS_PORT=9050
109     export TOR_CONTROL_PORT=9051
110     export SOCKS_PROXY_PORT=1080
111     export DNS_PROXY_PORT=5354
112 fi
113
114 # Start execution timer

```

```

112 TOGGLE_START_TIME=$SECONDS
113
114 if [[ "$1" == "restart" ]] || [[ "$1" == "--restart" ]]; then
115     echo "Executing Gateway Restart Sequence..."
116     # We must source common.sh here temporarily to check gateway state before we proceed
117     source "$SCRIPT_DIR/scripts/common.sh"
118     # Use persisted intent via gateway_state (echoes a string, never a return code;
119     # set -e-safe under set -x see devref 15.11.8). On Linux the marker is not
120     # maintained, so this degrades to the is_gateway_active fallback (fast native Docker).
121     if [ "$(gateway_state)" = "running" ]; then
122         echo "Gateway is currently running. Stopping it first..."
123         bash "$0"
124         echo "Gateway stopped. Now starting it..."
125     else
126         echo "Gateway is already stopped. Starting it..."
127     fi
128     bash "$0"
129     exit 0
130 fi
131
132 # Global flag to track if the script completed successfully
133 SUCCESS_RUN=false
134
135 # Global variable to track the currently running background command PID
136 CURRENT_BG_PID=""
137
138 # Source common bash functions
139 source "$SCRIPT_DIR/scripts/common.sh"
140
141
142 PID_FILE="/tmp/nullexit-cafeinate.pid"
143
144 # Start macOS caffeinate to prevent sleep while gateway is running
145 start_sleep_prevention() {
146     if ! command -v systemd-inhibit >/dev/null 2>&1; then
147         echo " [Warning] systemd-inhibit tool not found. Sleep prevention unavailable."
148         return 1
149     fi
150
151     # Stop any existing sleep prevention process first to avoid duplicates
152     stop_sleep_prevention
153
154     echo " Enabling system sleep prevention (systemd-inhibit)..."
155     # Run systemd-inhibit wrapped in a bash trap to prevent idle system sleep.
156     # Critically, this traps shutdown signals (SIGTERM) and automatically
157     # flushes hijacked DNS back to normal right before the PC powers off.
158     # We do not use recover.sh here because it is a heavy recovery script
159     # which takes too long and gets terminated before it can reset the DNS. Instead,
160     # we surgically and instantly restore normal DNS settings here.
161     nohup bash -c "
162         trap '
163             echo \"[Shutdown Trap] Reverting network settings...\" >> \"\$PWD/output.log\" 2>&1
164             kill \$! 2>/dev/null || true
165             sudo -n resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"\$PWD/output.log\" 2>&1 || true
166             sudo -n resolvectl revert \"\$ACTIVE_SERVICE\" >> \"\$PWD/output.log\" 2>&1 || true
167             resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"\$PWD/output.log\" 2>&1 || true
168             resolvectl revert \"\$ACTIVE_SERVICE\" >> \"\$PWD/output.log\" 2>&1 || true
169             if command -v tailscale >/dev/null 2>&1; then
170                 tailscale up --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 &
171                 TS_DOWN_ARGS=\"\"
172                 if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
173                 tailscale down \$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
174             fi
175             sleep 1
176             echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1
177             exit 0
178         ' SIGTERM SIGINT SIGHUP
179         systemd-inhibit --what=sleep --who=nullexit --why=\"gateway running\" --mode=block sleep infinity &
180         wait \$!
181         \" >> output.log 2>&1 &
182         local caffe_pid=$!
183         echo \"\$caffe_pid\" > \"$PID_FILE"
184         echo " Sleep prevention active (PID $caffe_pid). Your PC won't sleep while the gateway is running."
185     }
186
187 # Stop sleep prevention when gateway is stopped
188 stop_sleep_prevention() {
189     if [ -f "$PID_FILE" ]; then
190         local caffe_pid
191         caffe_pid=$(cat "$PID_FILE")

```

```

192     if [ -n "$caffe_pid" ] && kill -0 "$caffe_pid" 2>/dev/null && ps -p "$caffe_pid" -o command= 2>/dev/null |
193     grep -q -E "systemd-inhibit|recover.sh"; then
194         echo " Stopping system sleep prevention (systemd-inhibit wrapper PID $caffe_pid)..."
195         kill "$caffe_pid" 2>/dev/null || true
196     fi
197     rm -f "$PID_FILE"
198 fi
199 }
200 DNS_WATCHER_PID_FILE="/tmp/nullexit-dns-watcher.pid"
201
202 stop_dns_watcher() {
203     if [ -f "$DNS_WATCHER_PID_FILE" ]; then
204         local watcher_pid
205         watcher_pid=$(cat "$DNS_WATCHER_PID_FILE")
206         if [ -n "$watcher_pid" ] && kill -0 "$watcher_pid" 2>/dev/null; then
207             echo " Stopping background DNS Watcher (PID $watcher_pid)..."
208             kill -9 "$watcher_pid" 2>/dev/null || true
209         fi
210         rm -f "$DNS_WATCHER_PID_FILE"
211     fi
212 }
213
214 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
215 cleanup_handler() {
216     local exit_code=$?
217     local trigger_type="$1"
218     local line_no="$2"
219
220     if [ "$SUCCESS_RUN" = "false" ]; then
221         echo -e "\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1..."
222
223         if [ -n "$ACTIVE_SERVICE" ]; then
224             reset_dns
225         fi
226
227         if [ -n "$CURRENT_BG_PID" ]; then
228             echo "Terminating active command (PID $CURRENT_BG_PID)..."
229             kill -15 "$CURRENT_BG_PID" 2>> output.log || true
230         fi
231
232         # Disconnect host Tailscale and close the GUI application on failure
233         disconnect_tailscale_host
234
235         # Stop local DNS proxy if running
236         stop_local_dns_proxy
237
238         # Stop sleep prevention
239         stop_sleep_prevention
240         stop_dns_watcher
241
242         # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
243         if [ -n "$ACTIVE_SERVICE" ]; then
244             cleanup_network_state
245         fi
246
247         if [ "$trigger_type" = "ERR" ]; then
248             echo -e "\n=====
249             echo "ERROR: Script failed at line $line_no with exit code $exit_code."
250             echo "=====
251             echo "Troubleshooting steps:"
252             echo "1. Run 'colima status' to verify the VM is active."
253             echo "2. Run 'docker compose ps' to check container health."
254             echo "3. Run 'docker compose logs' to view application logs."
255             echo "4. Run 'tailscale status' to check host Tailscale status."
256             echo "=====
257         fi
258     fi
259 }
260
261 trap 'cleanup_handler ERR $LINENO' ERR
262 trap 'cleanup_handler INT' INT
263 trap 'cleanup_handler TERM' TERM
264 trap 'cleanup_handler HUP' HUP
265
266 # NOPASSWD sudo
267 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
268 # needed (resolvectl, ip, systemctl, systemd-inhibit, pkill, and
269 # python3 scripts/dns-proxy.py for the fallback DNS proxy).
270 # All privileged calls below use `sudo -n` which will run without prompting.

```

```

271 # No credential caching loop is needed.
272
273 # Add Homebrew and standard paths
274 export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
275
276 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
277 # Uses a Python DNS proxy (handles TCP wire format correctly).
278 # Python3 is built into macOS no external dependencies.
279 DNS_PROXY_BIN=""
280 if command -v python3 >/dev/null 2>&1; then
281     DNS_PROXY_BIN="python3 -I"
282 elif command -v python >/dev/null 2>&1; then
283     DNS_PROXY_BIN="python -I"
284 fi
285
286 # Path to the DNS proxy script (sibling of this script)
287 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/dns-proxy.py"
288
289 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
290 # (`brew install tailscale` + `systemctl start tailscaled`) NOT the
291 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
292 # to launch, no menu-bar click to perform, and no scutil Network Service to
293 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
294 # if you ran the install with sudo) and the control socket is always available.
295 TS_BIN=""
296 if command -v tailscale >> output.log 2>&1; then
297     TS_BIN="tailscale"
298 fi
299
300 # Baseline: disable Tailscale DNS management at the daemon level.
301 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
302 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
303 # re-enable `--accept-dns=true` and nullify this `set`.
304 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
305 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
306 #
307 # This initial `set` is still useful as a baseline for non-reset operations
308 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
309 # the brief window between this line and the first `--reset` call.
310 #
311 # Runs once at script start while tailscaled is still connected (if it was
312 # left running from a previous toggle), so the pref takes effect before any
313 # disconnection or reconnection logic.
314 if [ -n "$TS_BIN" ]; then
315     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
316 fi
317
318 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
319 # *entire* teardown no need to mess with system network managers.
320 # tailscaled keeps running (as a systemd service), which is intentional:
321 # the next `tailscale up` then reconnects in seconds rather than waiting for a
322 # slow daemon launch. The `tailscale down` call also retains the user's
323 # auth state, so we don't force a browser re-login every cycle.
324 #
325 # Defined early (before the if/else branches and referenced by cleanup_handler's
326 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
327 restart_tailscaled_daemon() {
328     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
329     echo " [Tailscale Recovery] Attempting to restart tailscaled service via systemctl..."
330     if run_with_timeout 15 sudo -n systemctl restart tailscaled >> output.log 2>&1; then
331         echo " [Tailscale Recovery] Successfully restarted tailscaled service."
332     else
333         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
334     fi
335     sleep 3
336 }
337 disconnect_tailscale_host() {
338     if [ -n "$TS_BIN" ]; then
339         echo "Disconnecting host Tailscale from mesh (tailscaled stays running as system service)..."
340         local ts_args=""
341         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
342         if ! run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1; then
343             restart_tailscaled_daemon
344             run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1 || true
345         fi
346     fi
347 }
348
349 ACTIVE_SERVICE=$(get_active_service)
350
351 # Helper to nuke leftover network state after teardown (proxy settings, routing

```

```

352 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
353 # problem where stale state kills internet after the gateway stops.
354 cleanup_network_state() {
355     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
356
357     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
358     # (Skipped for Linux since we don't set global SOCKS proxy)
359     echo " Proxies disabled."
360
361     # 2. Flush DNS cache
362     if command -v resolvectl >> output.log 2>&1; then
363         sudo -n resolvectl flush-caches 2>> output.log || true
364     elif command -v systemd-resolve >> output.log 2>&1; then
365         sudo -n systemd-resolve --flush-caches 2>> output.log || true
366     fi
367     echo " DNS cache flushed."
368
369     # 3. Flush stale routing table entries
370     if command -v ip >> output.log 2>&1; then
371         sudo -n ip route flush cache >> output.log 2>&1 || true
372     fi
373     echo " Routing table flushed."
374
375     # 4. Power-cycle Wi-Fi to clear any lingering interface state
376     echo " Restarting network interface..."
377     sudo -n ip link set dev "$ACTIVE_SERVICE" down 2>> output.log || true
378     sleep 1
379     sudo -n ip link set dev "$ACTIVE_SERVICE" up 2>> output.log || true
380     echo " Interface power-cycled ($ACTIVE_SERVICE)."
381     sleep 3
382
383     echo "Network state cleanup complete."
384 }
385
386 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
387 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
388 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
389 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
390 # Capture current DNS before resetting so we can check if it was hijacked
391 INITIAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head
392 -1 || true)
393
394 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
395 reset_dns
396
397 # Local Network Surveillance Check
398 echo -e "\n"
399 echo "Local Network Surveillance Check"
400 echo ""
401 # Count populated ARP cache entries to detect network scanning or high congestion
402 if command -v ip >/dev/null 2>&1; then
403     ARP_COUNT=$(ip neigh | grep -iv 'incomplete' | wc -l | tr -d ' ')
404 else
405     # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
406     ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
407 fi
408 if [ "$ARP_COUNT" -gt 15 ]; then
409     echo -e " \033[1;33m[!] WARNING: High ARP activity detected ($ARP_COUNT devices in cache).\033[0m"
410     echo -e " \033[1;33m[!] This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)
411     .\033[0m"
412     echo -e " [] Your traffic remains fully encrypted and invisible.\n"
413 else
414     echo -e " [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
415 fi
416
417 echo "Checking Gateway Status..."
418 # Decide STOP vs START from persisted intent via gateway_state, which echoes a
419 # string (never a return code) to stay set -e-safe under set -x. See devref 15.11.8.
420 # On Linux this falls back to is_gateway_active (fast native Docker).
421 if [ "$(gateway_state)" = "running" ]; then
422     TOGGLE_PHASE="stop"
423     echo -e "\n=====
424     echo "Gateway is RUNNING. Stopping it now..."
425     echo -e "=====
426
427 # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
428 disconnect_tailscale_host
429 echo ""
430
431 echo "Stopping Docker containers..."

```

```

431 # `|| true`: a disable must always complete even if the Docker daemon is down.
432 run_logged docker compose down -t 5 || true
433
434 echo -e "\nDocker daemon handles container teardown automatically."
435
436 # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
437 # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
438 # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
439 echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
440 reset_dns
441
442 # 3. Stop local DNS proxy if running
443 stop_local_dns_proxy
444
445 # Stop sleep prevention
446 stop_sleep_prevention
447 stop_dns_watcher
448
449 # 4. Nuke leftover network state so internet actually works after teardown
450 cleanup_network_state
451
452 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
453 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
454 else
455 TOGGLE_PHASE="start"
456 echo -e "\n=====
457 echo "Gateway is STOPPED. Starting it now..."
458 echo -e "=====
459
460 # 1. Prevent host exit-node deadlock during VM / Container startup
461 disconnect_tailscale_host
462
463 echo -e "\nUsing native Linux Docker..."
464
465 # 4. Clean up corrupted AdGuardHome configurations
466 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
467   rm -f "adguard/conf/AdGuardHome.yaml"
468   echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
469 fi
470
471
472
473 # 5. Start compose services
474 write_host_ips
475 # Procedurally generated ports have already been exported at the top of the script.
476 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
477
478 # --- HONEY-PORT TRIPWIRE ---
479 # Generate a random trap port. If any malware does a sequential port scan on localhost,
480 # it will inevitably hit this port. When it does, we instantly fire a desktop alert.
481 # Reap the honey-port from any previous toggle before arming a new one, so the
482 # `nc -l` tripwire never accumulates one idle listener per toggle-on (15.12.20).
483 pkill -f "nc -l .*127.0.0.1" 2>/dev/null || true
484 pkill -f "nc -l localhost" 2>/dev/null || true
485 HONEY_PORT=$(get_free_port)
486 (
487   if nc -l -p "$HONEY_PORT" 127.0.0.1 >/dev/null 2>&1 || nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
488     echo "[$(date +%Y-%m-%d %H:%M:%S)] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
489       just pinged the local Honey-Port ($HONEY_PORT)!" >> "$PWD/output.log"
490     if command -v notify-send >/dev/null 2>&1; then
491       notify-send -u critical "nullexit OPSEC Alert" "An unknown application is aggressively scanning your
492         local ports. Malware port-scan suspected."
493     fi
494   fi
495 ) &
496 disown %1 2>/dev/null || true
497 echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
498
499 echo -e "\nStarting Docker containers..."
500 run_logged docker compose up -d
501
502 # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
503 docker compose rm -s -f rule-compiler >> output.log 2>&1
504
505 # 6. Wait for the gateway container's Tailscale connection to be ready
506 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
507 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
508
509 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
510 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
511 connected=false

```

```

510 consecutive_failures=0
511 for i in {1..60}; do
512     # Run tailscale status and capture its output so the user can see what's happening.
513     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
514     ts_output=$(run_with_timeout 3 docker compose exec -T tailscale tailscale status 2>&1 || true)
515
516     if echo "$ts_output" | grep -q "offers exit node"; then
517         connected=true
518         break
519     fi
520
521     # Track consecutive failures to detect a stuck state.
522     # If we see 40+ consecutive 'NoState' responses, give up early
523     # the daemon is alive but can't connect to the coordination server.
524     if echo "$ts_output" | grep -q "NoState"; then
525         consecutive_failures=$((consecutive_failures + 1))
526         if [ "$consecutive_failures" -ge 40 ]; then
527             echo ""
528             echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
529             echo "     Tailscale daemon is running but can't reach the coordination server."
530             break
531         fi
532     else
533         consecutive_failures=0
534     fi
535
536     # Print a condensed status every 10 seconds so the user can see progress
537     if [ $((i % 10)) -eq 0 ]; then
538         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
539         echo " [attempt $i/60] $ts_short"
540     elif [ $((i % 5)) -eq 0 ]; then
541         echo -n "[i]"
542     else
543         echo -n "."
544     fi
545     sleep 1
546 done
547 echo ""
548
549 if [ "$connected" = "true" ]; then
550     echo "Gateway container is online on the Tailscale mesh."
551 elif [ "$BYPASS_PING" = "true" ]; then
552     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
553 else
554     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
555     echo " The container was seen in the following states during the wait:" >&2
556     echo "$ts_output" | sed 's/^/  /' >&2
557     echo "" >&2
558     echo " This could mean:" >&2
559     echo "   1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
560     echo "   2. Network issue inside the container check: docker compose logs tailscale" >&2
561     echo "   3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
562     echo "" >&2
563     echo " Diagnose manually:" >&2
564     echo "   docker compose exec -T tailscale tailscale status" >&2
565     echo "   docker compose exec -T tailscale tailscale netcheck" >&2
566 cleanup_handler ERR $LINENO
567 exit 1
568 fi
569
570 # 6b. Resolve the gateway's Tailscale IP.
571 # Dynamically query the container for the IP (handles IP changes after re-auth)
572 # and cache it in .gateway_ip for fast retrieval by recover.sh during Wi-Fi roaming.
573 #
574 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
575 # returns (\r) into captured output. If $TS_IP ends with \r, then
576 # `networksetup -setdnsservers` silently rejects the malformed IP and
577 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
578 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
579 # the first field of the first line.
580 TS_IP=""
581 if [ "$connected" = "true" ]; then
582     TS_IP=$(run_with_timeout 5 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
583         awk 'NR==1{print $1; exit}' || true)
584     if [ -n "$TS_IP" ]; then
585         echo "Resolved gateway Tailscale IP dynamically: $TS_IP"
586         echo "$TS_IP" > .gateway_ip
587     else
588         echo " [!] Failed to resolve gateway IP dynamically. Trying fallback cache..." >&2
589         TS_IP=$(read_adguard_ip || true)
590     fi

```

```

590 else
591     TS_IP=$(read_adguard_ip || true)
592 fi
593
594 # 7. Connect host Mac to Tailscale mesh.
595 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
596 # system LaunchDaemon via 'systemctl start tailscaled'), the previous five-
597 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
598 # no menu bar item to click, and no Accessibility permission required.
599 #
600 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
601 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
602 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
603 # are impossible without the host on the mesh).
604 #
605 # We connect in two phases:
606 # Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
607 # Phase B Set exit node only after pre-flight checks confirm the gateway works
608 HOST_ON_MESH=false
609 SKIP_EXIT_NODE=false
610 if [ -n "$TS_BIN" ]; then
611     echo -n "Verifying tailscaled is reachable"
612     daemon_ready=false
613     for i in {1..10}; do
614         # `tailscale status` returns exit code 1 when the daemon is alive but
615         # disconnected ("Tailscale is stopped."). We check the daemon socket
616         # directly as a fallback if the socket exists, tailscaled is running
617         # even if the host isn't connected to the mesh yet.
618         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
619             daemon_ready=true
620             break
621         fi
622         echo -n "."
623         sleep 1
624     done
625     echo ""
626
627     if [ "$daemon_ready" != "true" ]; then
628         echo -e "\033[0;33m[!] tailscaled daemon is not running. Attempting to start it...\033[0m"
629
630         # Try system-wide daemon (with timeout sudo may prompt for password)
631         if [ "$daemon_ready" != "true" ]; then
632             if run_with_timeout 10 sudo systemctl start tailscaled 2>> output.log; then
633                 echo "Started tailscaled as system LaunchDaemon."
634                 for i in {1..15}; do
635                     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
636                         daemon_ready=true
637                         break
638                     fi
639                     sleep 1
640                 done
641             fi
642         fi
643     fi
644
645     if [ "$daemon_ready" = "true" ]; then
646         echo "tailscaled is responsive (running as a system service)."
647
648         # Phase A: Join mesh without exit node
649         echo "Connecting host to Tailscale mesh (no exit node yet)..."
650         if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node=; then
651             HOST_ON_MESH=true
652             echo "Host is on Tailscale mesh."
653         else
654             echo -e "\033[0;33m[!] tailscale up failed even though the daemon is running.\033[0m"
655             echo "This usually means the host hasn't authenticated with your tailnet yet."
656             echo "Run this command in a terminal to authenticate:"
657             echo ""
658             echo "sudo tailscale up"
659             echo ""
660             echo "A browser will open. Log in to your Tailscale account, then re-run toggle.sh."
661         fi
662     else
663         echo -e "\033[0;31m[!] tailscaled could NOT be started automatically.\033[0m"
664         echo "Run this in a terminal to start and authenticate Tailscale:"
665         echo ""
666         echo "sudo systemctl start tailscaled"
667         echo "sudo tailscale up"
668         echo ""
669         echo "Then re-run toggle.sh."
670     fi

```

```

671 else
672     echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
673 fi
674
675 # Phase B: Pre-flight checks + enable exit node only if safe
676 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
677     echo -e "\n"
678     echo "Pre-flight connectivity check"
679     echo ""
680
681     # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
682     # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
683     # ("direct connection not established") even though the pong was received.
684     # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
685     echo -n " [1/3] Gateway reachable via Tailscale... "
686     ping_ok=false
687     # The host tailscaled needs a few seconds to propagate the new network map and
688     # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.
689     for attempt in {1..45}; do
690         if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
691             ping_ok=true
692             break
693         fi
694         sleep 1
695     done
696     if [ "$ping_ok" = "true" ]; then
697         echo "PASS"
698     else
699         echo "FAIL"
700         SKIP_EXIT_NODE=true
701     fi
702
703     # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
704     echo -n " [2/3] AdGuard DNS via localhost... "
705     if command -v dig >/dev/null 2>&1; then
706         if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 >/dev/null 2>&1; then
707             echo "PASS"
708         else
709             echo "FAIL"
710             SKIP_EXIT_NODE=true
711         fi
712     elif command -v nslookup >/dev/null 2>&1; then
713         if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 >/dev/null 2>&1; then
714             echo "PASS"
715         else
716             echo "FAIL"
717             SKIP_EXIT_NODE=true
718         fi
719     else
720         echo "SKIP (dig/nslookup not available)"
721     fi
722
723     # Check 3: Does the WARP container have external internet?
724     echo -n " [3/3] WARP container internet... "
725     if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace &>/dev/null;
726     then
727         echo "PASS"
728     else
729         echo "FAIL"
730         SKIP_EXIT_NODE=true
731     fi
732
733     # Act based on check results
734     if [ "$SKIP_EXIT_NODE" != "true" ]; then
735         echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
736         if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node="$TS_IP" --exit-node-allow-lan-access=
737         true; then
738             echo "Exit node enabled."
739         else
740             echo "[Warning] Failed to set exit node."
741             SKIP_EXIT_NODE=true
742         fi
743     fi
744
745     if [ "$SKIP_EXIT_NODE" = "true" ]; then
746         echo -e "\n\033[0;33m[!] Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED.\033[0m"
747         echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
748         echo " Troubleshoot with:"
749         echo "     docker compose logs warp | tail -20"
750         echo "     docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
751         echo ""

```

```

750     echo " Falling back to SOCKS5 proxy for traffic routing..."
751     fi
752     elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
753         echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
754         SKIP_EXIT_NODE=true
755     fi
756
757     # 8. Apply DNS Hijacking.
758     # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
759     # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
760     # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
761     if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
762         echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
763         echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
764         if force_dns_to_gateway "$TS_IP"; then
765             echo "DNS hijack verified: single server = $TS_IP."
766             start_dns_watcher "$TS_IP"
767         else
768             echo "[Warning] DNS hijack could not be verified."
769             echo " If DNS is broken, run manually:"
770             echo "     networksetup -setdnsservers \"\$ACTIVE_SERVICE\" $TS_IP"
771             echo "     networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" ts.net"
772         fi
773     elif [ -n "$TS_IP" ]; then
774         echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
775         echo " Trying local DNS proxy for ad-blocking..."
776         if start_local_dns_proxy; then
777             echo " Ad-blocking active via local DNS proxy through AdGuard."
778         else
779             echo " Local DNS proxy unavailable. DNS stays at 1.1.1.1."
780             echo " AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
781         fi
782
783         # Enable SOCKS5 proxy for traffic routing through WARP
784         # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
785         echo -e "\n Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
786         echo " (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
787         enable_socks_proxy "$ACTIVE_SERVICE"
788
789         # Verify the SOCKS5 proxy works through WARP
790         echo -n " Verifying SOCKS5 proxy routes through WARP... "
791         if curl --socks5-hostname 127.0.0.1:${SOCKS_PROXY_PORT} --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
792             trace 2>/dev/null | grep -q "warp=on"; then
793             echo "ok (warp=on)."
794             echo " All TCP traffic now encrypted through Cloudflare WARP tunnel."
795         else
796             echo "FAILED (proxy not routing through WARP)."
797             echo " Disabling SOCKS5 proxy internet stays on direct connection."
798             disable_socks_proxy
799             echo " SOCKS proxy available at localhost:${SOCKS_PROXY_PORT} for manual configuration."
800         fi
801     else
802         echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
803     fi
804
805     # 9. Verify host connectivity
806     if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
807         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
808             echo "Host is online on the Tailscale mesh."
809         else
810             echo "[Warning] Host is not responding on Tailscale."
811         fi
812     fi
813
814     ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
815     echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
816
817     # Prevent system from going to sleep while gateway is running
818     start_sleep_prevention
819 fi
820
821 SUCCESS_RUN=true
822
823 # Final DNS state summary
824 if [ -n "$ACTIVE_SERVICE" ]; then
825     FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
826 ' ' || true)
827     echo -e "\n"
828     echo -e "DNS STATE: $FINAL_DNS"
829     echo -e ""
830 fi

```

```

830
831 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
832     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
833         echo -e "\n[Warning] One or more blacklist URLs failed to download (404/Offline)."
834         echo "    The gateway gracefully fell back to yesterday's cached blacklist."
835         echo "    (Run 'docker logs rule-compiler' later to investigate which link died.)"
836     fi
837 fi
838
839 echo -e "\n"
840 read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
841 if [[ "${USER_CHOICE}" == "r" || "${USER_CHOICE}" == "R" ]]; then
842     echo "Reversing gateway state..."
843     exec bash "$0"
844 fi
845
846 echo -e "\nYou can close this terminal window now."
847
848 exit 0
849 fi
850
851 #
852 # macOS TOGGLE jot / netstat / networksetup, launchd / pf.
853 #
854
855 # Enforce Cryptographic Script Integrity
856 if [ -f "scripts/crypto.sh" ]; then
857     if ! bash scripts/crypto.sh --verify; then
858         exit 1
859     fi
860 fi
861
862 # Define log file
863 LOG_FILE="$PWD/output.log"
864
865 # Rotate log file if it exceeds 1GB (1073741824 bytes).
866 # Sized large on purpose: with DEBUG_TRACE the log holds a full command-by-command
867 # trace effectively the entire compilation/warning history of the framework
868 # which is valuable to keep for post-mortems. Raised from 50MB so always-on
869 # tracing doesn't churn that history away prematurely.
870 if [ -f "$LOG_FILE" ]; then
871     log_size=$(stat -f%z "$LOG_FILE" 2>/dev/null || stat -c%s "$LOG_FILE" 2>/dev/null || echo 0)
872     if [ "$log_size" -gt 1073741824 ]; then
873         mv "$LOG_FILE" "${LOG_FILE}.old" 2>/dev/null || rm -f "$LOG_FILE"
874         echo "[$(date '+%Y-%m-%d %H:%M:%S')] Log file exceeded 1GB; rotated to output.log.old" > "$LOG_FILE"
875     fi
876 fi
877
878 # --- LIFECYCLE PHASE TRACKING + EXIT BREADCRUMB ---
879 # TOGGLE_PHASE is updated as the script moves through STARTUP/STOP/START so that
880 # if the process is killed (window closed, pkill, OOM) or exits with an error,
881 # the EXIT trap records the final exit code AND the phase it died in. Previously
882 # an interrupted START left output.log ending abruptly after the teardown with no
883 # indication of what failed this makes every exit traceable.
884 TOGGLE_PHASE="init"
885 _log_exit_breadcrumb() {
886     local rc=$?
887     echo "[$(date '+%Y-%m-%d %H:%M:%S')] toggle.sh EXIT: code=$rc phase=$TOGGLE_PHASE (pid $$)" >> "$LOG_FILE"
888     2>/dev/null || true
889 }
890 trap '_log_exit_breadcrumb' EXIT
891
892 # --- DEBUG TRACE (opt-in via .env: DEBUG_TRACE=true) ---
893 # When enabled, mirror a full command-by-command xtrace to output.log for a
894 # complete post-mortem trail (e.g. a race during --restart while Wi-Fi is
895 # re-associating). Off by default. Read directly from .env here because
896 # common.sh (read_env_var) isn't sourced yet.
897 #
898 # BASH_XTRACEFD (send xtrace to its own fd, keeping the terminal clean) needs
899 # bash >= 4.1. macOS ships /bin/bash 3.2, so we branch:
900 #   - bash >= 4.1 (e.g. Linux): dedicate fd 9 to the log and point xtrace there;
901 #     the terminal stays clean and normal stderr is untouched.
902 #   - bash 3.2 (stock macOS): BASH_XTRACEFD is unsupported, so xtrace always goes
903 #     to stderr. We redirect stderr *straight* to the log file* with a plain
904 #     `exec 2>"$LOG_FILE"`. This deliberately does NOT use a process
905 #     substitution (`2>>(tee)`): under `set -e`, stderr flowing through the
906 #     backgrounded `tee` proved to abort the script mid-START on 3.2 (a failed
907 #     write / SIGPIPE in a pipeline returned non-zero and tripped `set -e`, with
908 #     $LINENO mis-blaming the enclosing `fi` see devref 15.11.8-A/E). The
909 #     tradeoff: on 3.2 the terminal no longer shows live progress while tracing
910 #     (everything is in output.log instead). Acceptable for an opt-in debug mode.

```

```

910 # We NEVER use the `exec {var}>>` dynamic-fd form (4.1+ only), which hard-fails on 3.2.
911 DEBUG_TRACE=$(grep -E '^DEBUG_TRACE=' .env 2>/dev/null | cut -d'=' -f2- | tr -d "\"" | tr '[:upper:]' '[:lower:]' | tr -d '[:space:]')
912 if [ "$DEBUG_TRACE" = "true" ]; then
913     echo "[$(date '+%Y-%m-%d %H:%M:%S')] DEBUG_TRACE enabled xtrace mirrored to output.log (bash ${BASH_VERSINFO[0]}.${BASH_VERSINFO[1]})" >> "$LOG_FILE"
914     # Timestamped, location-aware xtrace prompt (works on 3.2 and 4.x).
915     export PS4='+ [${SECONDS}s] ${BASH_SOURCE##*/}:${LINENO}:${FUNCNAME[0]:-main}: ' # subshell-free clock: a $
916     () in PS4 trips the bash 3.2 set -x/ERR heisenbug (devref 15.11.8)
917     if [ "${BASH_VERSINFO[0]}" -gt 4 ] || { [ "${BASH_VERSINFO[0]}" -eq 4 ] && [ "${BASH_VERSINFO[1]}" -ge 1 ]; }; then
918         exec 9>>"$LOG_FILE"
919         export BASH_XTRACEFD=9
920         set -x
921     else
922         # bash 3.2 (stock macOS): plain stderrfile redirect (no process substitution).
923         exec 2>>"$LOG_FILE"
924         set -x
925     fi
926 fi
927 # --- MAIN EXECUTION LOGGING ---
928 echo -e "\n===== " >> "$LOG_FILE"
929 echo "[$(date '+%Y-%m-%d %H:%M:%S')] toggle.sh invoked" >> "$LOG_FILE"
930 echo "===== " >> "$LOG_FILE"
931
932 # (Previously this script chmod 600'd .env at start to "unlock" it for docker compose,
933 # and chmod 000'd it on exit. That pattern broke docker compose on macOS/Colima when
934 # the umask produced 0600 the script removed its own ability to read .env before it could even proceed.)
935
936 # --- PROCEDURAL PORT GENERATION ---
937 # To prevent static port fingerprinting by malware, we randomize exposed ports
938 # on every boot and persist them to a hidden environment file.
939 PORTS_FILE="/tmp/nullexit-ports.env"
940 if [[ "$1" == "" || "$1" == "--restart" ]]; then
941
942     # Collision-avoidance function
943     GENERATED_PORTS=""
944     get_free_port() {
945         local port
946         while true; do
947             port=$(jot -r 1 10000 65000)
948             # 1. Prevent self-collision within this loop
949             if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
950                 continue
951             fi
952             # 2. Prevent collision with ANY process on the Mac (root daemons, terminal apps, etc)
953             # We use netstat instead of lsof because netstat reads the kernel socket table
954             # directly and doesn't require sudo to see ports bound by other users/root.
955             if ! netstat -an -p tcp | awk '{print $4}' | grep -q "\.${port}$"; then
956                 echo "$port"
957                 return
958             fi
959         done
960     }
961
962     export TOR SOCKS_PORT=$(get_free_port)
963     GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
964
965     export TOR_CONTROL_PORT=$(get_free_port)
966     GENERATED_PORTS="$GENERATED_PORTS $TOR_CONTROL_PORT"
967
968     export SOCKS_PROXY_PORT=$(get_free_port)
969     GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
970
971     export DNS_PROXY_PORT=$(get_free_port)
972
973     echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
974     echo "export TOR_CONTROL_PORT=$TOR_CONTROL_PORT" >> "$PORTS_FILE"
975     echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
976     echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
977     ADGUARD_PASS=$(grep -E "^ADGUARD_PASSWORD=" .env 2>/dev/null | cut -d'=' -f2- | tr -d "\"" )
978     echo "export TOR_PASSWORD=${ADGUARD_PASS:-nullexit}" >> "$PORTS_FILE"
979     elif [ -f "$PORTS_FILE" ]; then
980         # On STOP, source the existing ports so docker-compose down matches
981         source "$PORTS_FILE"
982     else
983         # Fallbacks if file is lost
984         export TOR SOCKS_PORT=9050
985         export TOR_CONTROL_PORT=9051
986         export SOCKS_PROXY_PORT=1080

```

```

987     export DNS_PROXY_PORT=5354
988 fi
989
990 # Start execution timer
991 TOGGLE_START_TIME=$SECONDS
992
993 if [[ "$1" == "--restart" ]]; then
994     echo "Executing Gateway Restart Sequence..."
995     # We must source common.sh here temporarily to check gateway state before we proceed
996     source "$SCRIPT_DIR/scripts/common.sh"
997     # Use persisted intent (marker) via gateway_state, which echoes a string (never
998     # a return code) to stay set -e-safe under set -x. See devref 15.11.8.
999     if [ "$(gateway_state)" = "running" ]; then
1000         echo "Gateway is currently running. Stopping it first..."
1001         bash "$0"
1002         echo "Gateway stopped. Now starting it..."
1003     else
1004         echo "Gateway is already stopped. Starting it..."
1005     fi
1006     bash "$0"
1007     exit 0
1008 fi
1009
1010 # Global flag to track if the script completed successfully
1011 SUCCESS_RUN=false
1012
1013 # Global variable to track the currently running background command PID
1014 CURRENT_BG_PID=""
1015
1016 # Source common bash functions
1017 source "$SCRIPT_DIR/scripts/common.sh"
1018
1019 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
1020 LOCK_FILE="/tmp/nullexit-toggle.lock"
1021 if [ -f "$LOCK_FILE" ]; then
1022     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
1023     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
1024         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
1025             die "Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running."
1026         fi
1027     fi
1028 fi
1029 echo "$$" > "$LOCK_FILE"
1030
1031 # Helper function to run a GUI command as the logged-in console user
1032 # (Prevents permission failures if the script is run with sudo)
1033 run_gui_cmd() {
1034     local console_user
1035     console_user=$(stat -f '%Su' /dev/console 2>> output.log || echo "$SUDO_USER")
1036     if [ -z "$console_user" ] || [ "$console_user" = "root" ]; then
1037         console_user=$(logname 2>> output.log || echo "$USER")
1038     fi
1039
1040     if [ -n "$console_user" ] && [ "$console_user" != "root" ] && [ "$EUID" -eq 0 ]; then
1041         sudo -u "$console_user" "$@"
1042     else
1043         "$@"
1044     fi
1045 }
1046
1047 # Active-state marker: written at end of START path so external watchers
1048 # (see launchd/com.nullexit.wake-recovery.plist + scripts/watcher.sh +
1049 # devref 10.29) know the gateway is currently up. They only fire
1050 # recover.sh --post-wake when this file is present. Cleared at the top
1051 # of the STOP path and inside cleanup_handler on any error/signal path
1052 # so a half-broken gateway never re-triggers post-wake refreshes.
1053 write_gateway_active_marker() {
1054     date -u +%FT%TZ > /tmp/nullexit-gateway-active.marker
1055     # Set the watcher debounce timestamp to now to prevent a redundant post-startup recovery run.
1056     date +%s > /tmp/nullexit-watcher.last-recovery 2>/dev/null || true
1057 }
1058
1059 clear_gateway_active_marker() {
1060     rm -f /tmp/nullexit-gateway-active.marker
1061     rm -f "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/TUNNEL_FAILED_CLOSED.marker"
1062 }
1063
1064 # Capture whether the gateway-active marker exists BEFORE the defensive clear
1065 # below wipes it. The STOP-vs-START decision (gateway_should_be_running) reads
1066 # this captured value persisted *intent* instead of a live Docker probe, so
1067 # a disable is instant and works even when Colima/Docker is down. Must be read

```

```

1068 # before clear_gateway_active_marker or it would always look "stopped".
1069 #
1070 # Written as default-assign + `if` WITHOUT an `else`: under `set -e` (with the
1071 # DEBUG_TRACE xtrace redirect active) an `if [ -f ]; then ; else ; fi` whose
1072 # condition is false was aborting the script here at init on macOS /bin/bash 3.2
1073 # the false `[ -f ]` status leaked through the compound. An `if` with no `else`
1074 # yields status 0 when the condition is false, so nothing leaks.
1075 GATEWAY_MARKER_AT_STARTUP="no"
1076 if [ -f "$MARKER_FILE" ]; then
1077     GATEWAY_MARKER_AT_STARTUP="yes"
1078 fi
1079
1080 # Defensive: clear any stale marker from a prior crashed/aborted run. If a
1081 # previous toggle.sh wrote the marker but was OOM-killed in the middle of the
1082 # START block (before the END-of-START write) or if the user rebooted mid
1083 # session this prevents the watcher from firing post-wake against a
1084 # non-running gateway on every wake event. Placed AFTER the function
1085 # definitions so bash resolves the call at parse time.
1086 clear_gateway_active_marker
1087
1088 PID_FILE="$PID_CAFFEINATE"
1089
1090 # Start macOS caffeinate to prevent sleep while gateway is running
1091 start_sleep_prevention() {
1092     if ! command -v caffeinate >/dev/null 2>&1; then
1093         echo " [Warning] caffeinate tool not found. Sleep prevention unavailable."
1094         return 1
1095     fi
1096
1097     # Stop any existing sleep prevention process first to avoid duplicates
1098     stop_pidfile_daemon "/tmp/nullexit-caffeinate.pid" "system sleep prevention"
1099
1100     echo " Enabling system sleep prevention (caffeinate)..."
1101     # Run caffeinate wrapped in a bash trap to prevent idle system sleep.
1102     # Critically, this traps macOS shutdown signals (SIGTERM) and automatically
1103     # flushes hijacked DNS back to normal right before the Mac powers off.
1104     # We do not use recover.sh here because it is a heavy recovery script (managing
1105     # containers, restarting tailscaled, etc.) which takes too long and gets
1106     # terminated by macOS before it can reset the DNS. Instead, we surgically and
1107     # instantly restore normal DNS and proxy settings here.
1108     nohup bash -c "
1109         trap '
1110             echo \"[Shutdown Trap] Reverting network settings...\" >> \"\$PWD/output.log\" 2>&1
1111             kill \$! 2>/dev/null || true
1112             sudo -n networksetup -setdnsservers \"\$ACTIVE_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
1113             sudo -n networksetup -setdnsservers \"\$ENO_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
1114             sudo -n networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
1115             sudo -n networksetup -setsearchdomains \"\$ENO_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
1116             sudo -n networksetup -setsocksfirewallproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 ||
1117             true
1118             sudo -n networksetup -setsocksfirewallproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
1119             sudo -n networksetup -setwebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
1120             sudo -n networksetup -setwebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
1121             sudo -n networksetup -setsecurewebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
1122             sudo -n networksetup -setsecurewebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
1123             if command -v tailscale >/dev/null 2>&1; then
1124                 tailscale up >> \"\$PWD/output.log\" 2>&1 || true
1125                 tailscale set --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 || true
1126                 TS_DOWN_ARGS=\"\"
1127                 if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
1128                 tailscale down \$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
1129             fi
1130             sleep 1
1131             echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1
1132             exit 0
1133         ' SIGTERM SIGINT SIGHUP
1134         caffeinate -i &
1135         wait \$!
1136     " >> output.log 2>&1 &
1137     local caffe_pid=$!
1138     echo "$caffe_pid" > "$PID_FILE"
1139     echo " Sleep prevention active (PID $caffe_pid). Your Mac won't sleep while the gateway is running."
1140 }
1141
1142 DNS_WATCHER_PID_FILE="$PID_DNS_WATCHER"
1143 WARP_WATCHER_PID_FILE="/tmp/nullexit-warp-watcher.pid"
1144
1145
1146
1147

```

```

1148 # Background WARP tunnel liveness monitor. Polls cdn-cgi/trace every 30s
1149 # while healthy, but accelerates to polling every 5s if a failure is detected.
1150 # Always logs state transitions to output.log. When warp=off persists for
1151 # WARP_FAIL_THRESHOLD consecutive polls (default 6 = 30s of downtime), the watcher
1152 # triggers a nuclear recovery: runs recover.sh to tear down the entire
1153 # gateway disconnecting Tailscale, stopping containers, resetting DNS,
1154 # and power-cycling Wi-Fi. Note: This minimizes exposure window, but
1155 # only the pf kill switch guarantees absolutely zero leakage on drop.
1156 #
1157 # The threshold (default 6) is statistically chosen: even at an
1158 # unrealistically high 10% false-positive rate per poll, P(6 consecutive
1159 # false positives) = 0.1^6 0.0001%. Real-world false-positive rates
1160 # are far lower, making 30s of continuous downtime overwhelmingly likely
1161 # to be a genuine outage.
1162 #
1163 # Configurable via .env: WARP_FAIL_THRESHOLD=3 (15s, more aggressive)
1164 start_warp_watcher() {
1165     stop_warp_watcher
1166     # Parse threshold from .env (same pattern as GATEWAY_MSS, GATEWAY_BYPASS_PING, etc.)
1167     local threshold
1168     threshold=$(read_env_var WARP_FAIL_THRESHOLD)
1169     if [ -z "$threshold" ]; then
1170         threshold="{WARP_FAIL_THRESHOLD:-6}"
1171     fi
1172     local trigger_file="/tmp/nullexit-warp-shutdown-triggered"
1173     rm -f "$trigger_file"
1174     echo " Starting background WARP Watcher (polling every 30s [accelerating to 5s on failure], shutdown after $
1175         {threshold} consecutive failures)..."
1176     nohup bash -c "
1177         trap 'exit 0' SIGTERM SIGINT SIGHUP
1178         last_state='on'
1179         consec_off=0
1180         threshold='$threshold'
1181         trigger_file='$trigger_file'
1182         out_log='$PWD/output.log'
1183         recover_bin='$PWD/recover.sh'
1184         inhibit_file='/tmp/nullexit-warp-inhibit.marker'
1185         while true; do
1186             # Inhibit check
1187             # recover.sh --post-wake writes this marker while force-recreating the
1188             # warp container. During that window, the container is intentionally
1189             # down don't count it as a failure or we'll fire nuclear recover.sh
1190             # and kill a gateway that was in the process of healing itself.
1191             if [ -f "$inhibit_file" ]; then
1192                 consec_off=0
1193                 sleep 5
1194                 continue
1195             fi
1196             state='off'
1197             if docker compose exec -T warp wget -qO- --timeout=5 \
1198                 https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null \
1199                 | grep -q 'warp=on'; then
1200                 state='on'
1201             fi
1202             # State-transition logging
1203             if [ "$state" != "$last_state" ]; then
1204                 if [ "$state" = 'off' ]; then
1205                     echo "[ $(date -u +%FT%TZ) ] WARP DOWN cdn-cgi/trace reports warp=off" >> "$out_log"
1206                 fi
1207                 last_state="$state"
1208             fi
1209             # Consecutive-failure tracking + auto-shutdown
1210             if [ "$state" = 'off' ]; then
1211                 consec_off=$((consec_off + 1))
1212                 if [ "$consec_off" -ge "$threshold" ] && [ ! -f "$trigger_file" ]; then
1213                     touch "$trigger_file"
1214                     echo "[ $(date -u +%FT%TZ) ] WARP SHUTDOWN $consec_off consecutive failures (threshold=$threshold)
1215 . Running recover.sh to kill the gateway and restore normal internet. Your IP is NO LONGER Cloudflare." >>
1216                     "$out_log"
1217                     # Nuclear recovery: tear down the entire gateway.
1218                     # recover.sh resets DNS 1.1.1.1, disconnects Tailscale,
1219                     # stops Docker containers, flushes routes, power-cycles Wi-Fi.
1220                     bash "$recover_bin" --auto >> "$out_log" 2>&1 || true
1221                     echo "[ $(date -u +%FT%TZ) ] WARP SHUTDOWN recover.sh completed, watcher exiting. Your IP is now
1222 your real ISP IP." >> "$out_log"
1223                     exit 0
1224                 fi
1225             else

```

```

1225         if [ \("${consec_off}" -gt 0 ); then
1226             echo \("${date -u +%FT%TZ}") WARP RECOVERED warp=on after \("${consec_off}" consecutive off readings (
threshold was \("${threshold}")" >> \("${out_log}"
1227         fi
1228         consec_off=0
1229     fi
1230
1231     if [ \("${state}" = "off" ); then
1232         sleep 5
1233     else
1234         sleep 30
1235     fi
1236 done
1237 " >> output.log 2>&1 &
1238 echo $! > "$WARP_WATCHER_PID_FILE"
1239 }
1240
1241 stop_warp_watcher() {
1242     if [ -f "$WARP_WATCHER_PID_FILE" ]; then
1243         local wp
1244         wp=$(cat "$WARP_WATCHER_PID_FILE")
1245         if [ -n "$wp" ] && kill -0 "$wp" 2>/dev/null; then
1246             echo " Stopping background WARP Watcher (PID $wp)..."
1247             kill "$wp" 2>/dev/null || true
1248         fi
1249         rm -f "$WARP_WATCHER_PID_FILE"
1250     fi
1251 }
1252
1253
1254
1255
1256
1257 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
1258 cleanup_handler() {
1259     local exit_code=$?
1260     local trigger_type="$1"
1261     local line_no="$2"
1262
1263     if [ "$SUCCESS_RUN" = "false" ]; then
1264         echo -e "\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1..."
1265
1266         if [ -n "$ACTIVE_SERVICE" ]; then
1267             reset_dns
1268         fi
1269
1270         if [ -n "$CURRENT_BG_PID" ]; then
1271             echo "Terminating active command (PID $CURRENT_BG_PID)..."
1272             kill -15 "$CURRENT_BG_PID" 2>> output.log || true
1273         fi
1274
1275         # Disconnect host Tailscale and close the GUI application on failure
1276         disconnect_tailscale_host
1277
1278         # Stop local DNS proxy if running
1279         stop_local_dns_proxy
1280
1281         # Stop sleep prevention
1282         stop_pidfile_daemon "/tmp/nullexit-cafeinate.pid" "system sleep prevention"
1283         stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
1284         stop_warp_watcher
1285         clear_gateway_active_marker
1286
1287         # Capture warp logs on failure for debugging before teardown
1288         if [ "$trigger_type" = "ERR" ]; then
1289             echo -e "\n--- WARP FAILURE LOGS ---" >> output.log
1290             docker compose logs warp --tail=100 >> output.log 2>&1 || true
1291             echo "-----" >> output.log
1292         fi
1293
1294         # Best-effort container teardown on failure
1295         docker compose down --remove-orphans -t 30 2>> output.log || true
1296
1297         # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
1298         if [ -n "$ACTIVE_SERVICE" ]; then
1299             cleanup_network_state
1300         fi
1301
1302         if [ "$trigger_type" = "ERR" ]; then
1303             echo -e "\n=====

```

```

1304     echo "ERROR: Script failed at line $line_no with exit code $exit_code."
1305     echo "=====
1306     echo "Troubleshooting steps:"
1307     echo "1. Run 'colima status' to verify the VM is active."
1308     echo "2. Run 'docker compose ps' to check container health."
1309     echo "3. Run 'docker compose logs' to view application logs."
1310     echo "4. Run 'tailscale status' to check host Tailscale status."
1311     echo "=====
1312 fi
1313 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
1314 fi
1315 }
1316
1317 trap 'cleanup_handler ERR $LINENO' ERR
1318 trap 'cleanup_handler INT' INT
1319 trap 'cleanup_handler TERM' TERM
1320 trap 'cleanup_handler HUP' HUP
1321
1322 # NOPASSWD sudo
1323 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
1324 # needed (networksetup, pfctl, route, ifconfig, dscacheutil, killall, pkill,
1325 # and python3 -I scripts/dns-proxy.py for the fallback DNS proxy).
1326 # All privileged calls below use `sudo -n` which will run without prompting.
1327 # No credential caching loop is needed.
1328
1329 # Add Homebrew and standard paths
1330 setup_standard_path
1331
1332 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
1333 # Uses a Python DNS proxy (handles TCP wire format correctly).
1334 # Python3 is built into macOS no external dependencies.
1335 DNS_PROXY_BIN=""
1336 if command -v python3 >> output.log 2>&1; then
1337     DNS_PROXY_BIN="python3 -I"
1338 elif command -v python >> output.log 2>&1; then
1339     DNS_PROXY_BIN="python -I"
1340 fi
1341
1342 # Path to the DNS proxy script (sibling of this script)
1343 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/dns-proxy.py"
1344
1345 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
1346 # (`brew install tailscale` + `brew services start tailscale`) NOT the
1347 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
1348 # to launch, no menu-bar click to perform, and no scutil Network Service to
1349 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
1350 # if you ran the install with sudo) and the control socket is always available.
1351 TS_BIN=""
1352 if command -v tailscale >> output.log 2>&1; then
1353     TS_BIN="tailscale"
1354 fi
1355
1356 # Baseline: disable Tailscale DNS management at the daemon level.
1357 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
1358 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
1359 # re-enable `--accept-dns=true` and nullify this `set`.
1360 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
1361 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
1362 #
1363 # This initial `set` is still useful as a baseline for non-reset operations
1364 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
1365 # the brief window between this line and the first `--reset` call.
1366 #
1367 # Runs once at script start while tailscaled is still connected (if it was
1368 # left running from a previous toggle), so the pref takes effect before any
1369 # disconnection or reconnection logic.
1370 if [ -n "$TS_BIN" ]; then
1371     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
1372 fi
1373
1374 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
1375 # *entire* teardown no scutil tunnel stop, no Tailscale.app GUI quit, no pkill.
1376 # tailscaled keeps running (as a LaunchAgent / LaunchDaemon), which is intentional:
1377 # the next `tailscale up` then reconnects in seconds rather than waiting for a
1378 # slow .app launch + Network Extension activation. The `tailscale down` call also
1379 # retains the user's auth state, so we don't force a browser re-login every cycle.
1380 #
1381 # Defined early (before the if/else branches and referenced by cleanup_handler's
1382 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
1383
1384

```

```

1385 # Wait for Network Readiness
1386 # On --restart, the gateway tears down while the host network interface may
1387 # still be reconnecting (Wi-Fi, Ethernet, USB tethering, etc.). If we proceed
1388 # too early, get_active_service returns nothing, routing targets the wrong
1389 # interface, and DNS/WARP setup silently fails.
1390 #
1391 # We use `route get default` to detect a valid default gateway rather than
1392 # checking a specific interface (e.g. en0) this generalizes across all
1393 # connection types: Wi-Fi, Ethernet, USB-C adapters, iPhone tethering, etc.
1394 # The moment the OS has any routable default gateway, we proceed.
1395 _net_wait_secs=0
1396 _net_timeout=60
1397 echo "Waiting for network connectivity before starting..."
1398 while true; do
1399     if route get default 2>/dev/null | grep -q 'gateway: '; then
1400         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
1401         echo "    Network ready (default gateway: $_gw)."
1402         break
1403     fi
1404     if [ "$_net_wait_secs" -ge "$_net_timeout" ]; then
1405         echo "    [Warning] No default gateway after ${_net_timeout}s proceeding anyway."
1406         break
1407     fi
1408     sleep 2
1409     _net_wait_secs=$(($_net_wait_secs + 2))
1410 done
1411
1412 ACTIVE_SERVICE=$(get_active_service)
1413
1414 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
1415 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
1416 # ignored unless en0's service is also updated.
1417 ENO_SERVICE=$(get_en0_service)
1418
1419 # Helper to add host-side bypass routes for the WARP WireGuard endpoints.
1420 # This prevents an infinite recursive routing loop (tunnel loop) where
1421 # the container's WireGuard packets exit the VM, reach the host, match the
1422 # default route (the exit node), and get routed back into the tunnel.
1423 # We route via the gateway IP (not interface) to ensure packets are routed
1424 # properly even when the host's default route changes to the Tailscale interface.
1425
1426 # Helper to manually override the host's default route to point through the
1427 # Tailscale utun* interface. Standalone tailscaled on macOS (Homebrew version)
1428 # lacks the platform-specific integration to override the default gateway
1429 # automatically when --exit-node is used.
1430 setup_exit_node_routing() {
1431     local ts_iface
1432     ts_iface=$(ifconfig | awk '/^[a-z0-9]+:/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ':')
1433
1434     if [ -n "$ts_iface" ]; then
1435         echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
1436         local host_mtu
1437         host_mtu=$(read_env_var HOST_MTU)
1438         host_mtu=${host_mtu:-1200}
1439
1440         # Lower the host Tailscale MTU (defaults to 1200) to prevent fragmentation when passing
1441         # through the container's WARP tunnel (which also has an MTU of 1280).
1442         sudo -n ifconfig "$ts_iface" mtu "$host_mtu" >> output.log 2>&1 || true
1443
1444         # DO NOT delete the physical default route! It crashes Colima.
1445         # Use the /1 trick to mathematically override the default route.
1446         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
1447         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
1448         sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
1449         sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
1450
1451         if netstat -nr | grep -E -q "^(0\.\0\.\0/1|0/1)[[:space:]]+.*$ts_iface"; then
1452             echo "    [] Default route successfully overridden via $ts_iface."
1453         else
1454             echo "    [!] Failed to verify exit node routing on $ts_iface."
1455         fi
1456     else
1457         echo "[Warning] Could not detect Tailscale utun interface for host routing."
1458     fi
1459 }
1460
1461 # Helper to nuke leftover network state after teardown (proxy settings, routing
1462 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
1463 # problem where stale state kills internet after the gateway stops.
1464 cleanup_network_state() {
1465     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."

```

```

1466
1467 # 1. Disable any leftover proxy settings (needs root on many macOS versions)
1468 for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
1469     disable_all_proxies "$svc"
1470 done
1471 echo "    Proxies disabled."
1472
1473 # 2. Flush DNS cache
1474 if command -v dscacheutil >> output.log 2>&1; then
1475     sudo -n dscacheutil -flushcache 2>> output.log || true
1476 fi
1477 sudo -n killall -HUP mDNSResponder 2>> output.log || true
1478 echo "    DNS cache flushed."
1479
1480 # 3. Flush stale routing table entries
1481 if command -v route >> output.log 2>&1; then
1482     sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
1483     sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
1484 fi
1485 remove_warp_bypass_routes
1486 disable_killswitch
1487 echo "    Routing table and firewall flushed."
1488
1489 # 4. Power-cycle Wi-Fi to clear any lingering interface state
1490 WIFI_PORT=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print $2
1491 }')
1492 if [ -n "$WIFI_PORT" ]; then
1493     sudo -n networksetup -setairportpower "$WIFI_PORT" off 2>> output.log || true
1494     sleep 2
1495     sudo -n networksetup -setairportpower "$WIFI_PORT" on 2>> output.log || true
1496     echo "    Wi-Fi power-cycled (interface $WIFI_PORT)."
1497     sleep 3
1498 else
1499     echo "    Wi-Fi interface not detected; bouncing en0..."
1500     sudo -n ifconfig en0 down 2>> output.log || true
1501     sudo -n ifconfig en0 up 2>> output.log || true
1502 fi
1503
1504 # Force restart sharing services to prevent AirDrop / AirPlay discovery freezes
1505 # after network interface/DNS changes.
1506 echo "    Resetting macOS sharing services (AirDrop/AirPlay)..."
1507 reset_sharing_services
1508
1509 echo "Network state cleanup complete."
1510 }
1511
1512 SOCKS_PROXY_PORT=${SOCKS_PROXY_PORT}
1513
1514 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
1515 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
1516 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
1517 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
1518
1519 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
1520 reset_dns
1521
1522
1523
1524
1525 echo "Checking Gateway Status..."
1526 HIJACK_HOST=$(read_env_var GATEWAY_HIJACK_HOST | tr '[:upper:]' '[:lower:]')
1527
1528 # Decide STOP vs START from persisted intent (marker), not a live Docker probe.
1529 # This makes *disable* instant and robust even when Colima/Docker is down.
1530 #
1531 # CRITICAL (set -e + set -x on bash 3.2): gateway_state ECHOES "running"/"stopped"
1532 # and always returns 0, so we branch on the STRING never on a function's exit
1533 # code. A helper that `return 1`'s trips a spurious `set -e` abort under `set -x`
1534 # (DEBUG_TRACE=true) even when guarded, hence the echo contract. See devref 15.11.8.
1535 if [ "$(gateway_state)" = "running" ]; then
1536     TOGGLE_PHASE="stop"
1537     echo -e "\n=====
1538     echo "Gateway is RUNNING. Stopping it now..."
1539     echo -e "=====
1540
1541 # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
1542 if [[ "$HIJACK_HOST" != "false" ]]; then
1543     disconnect_tailscale_host
1544     echo ""
1545 fi

```

```

1546 echo "Stopping Docker containers..."
1547 # `|| true`: a disable must always complete its teardown even if the Docker
1548 # daemon / Colima VM is already down (compose down would exit non-zero and,
1549 # under set -e, abort the STOP). The START path intentionally does NOT guard
1550 # its `docker compose up` a failed start SHOULD trigger cleanup.
1551 run_logged docker compose down --remove-orphans -t 30 || true
1552
1553
1554 # Only stop Colima if the user explicitly opted in (false by default) to prevent breaking their other Docker
1555 # dev projects
1556 STOP_COLIMA=$(read_env_var STOP_COLIMA_ON_EXIT | tr '[:upper:]' '[:lower:]')
1557 if [[ "$STOP_COLIMA" == "true" ]]; then
1558     echo -e "\nStopping Colima VM to free up host RAM and battery..."
1559     run_with_timeout 30 colima stop >> output.log 2>&1 || echo "Warning: Failed to stop Colima gracefully."
1560 else
1561     echo -e "\nLeaving Colima running. (Set STOP_COLIMA_ON_EXIT=true in .env to change this)"
1562 fi
1563
1564 if [[ "$HIJACK_HOST" != "false" ]]; then
1565     # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
1566     # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
1567     # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
1568     echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
1569     reset_dns
1570
1571     # 3. Stop local DNS proxy if running
1572     stop_local_dns_proxy
1573
1574     # Stop background daemons
1575     stop_pidfile_daemon "/tmp/nullexit-cafeinate.pid" "system sleep prevention"
1576     stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
1577     stop_warp_watcher
1578     clear_gateway_active_marker
1579
1580     # 4. Nuke leftover network state so internet actually works after teardown
1581     cleanup_network_state
1582 fi
1583
1584 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
1585 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
1586 else
1587     TOGGLE_PHASE="start"
1588     START_GATEWAY=true
1589     echo "[$(date +%Y-%m-%d %H:%M:%S)] Action: STARTUP GATEWAY" >> "$LOG_FILE"
1590     echo -e "\n=====
1591     echo "Gateway is STOPPED. Starting it now..."
1592     echo -e "=====
1593
1594 # 1. Prevent host exit-node deadlock during VM / Container startup
1595 if [[ "$HIJACK_HOST" != "false" ]]; then
1596     disconnect_tailscale_host
1597 fi
1598
1599 # 2. Wait for physical DHCP lease to settle (crucial for restarts/wake-up/roaming)
1600 wait_for_dhcp_settle
1601
1602 # 3. Boot Colima VM if it is not already running
1603 echo -e "\nChecking Colima VM status..."
1604 if ! run_with_timeout 15 colima status >> output.log 2>&1; then
1605     # Colima networking is gated on the SAME LAN-P2P signal routing-fix uses
1606     # (.lan_p2p_detected overrides TAILSCALE_ALLOW_LAN_P2P in .env). A LAN-reachable
1607     # VM (--network-address + bridged/shared) needs the socket_vmnet helper and is
1608     # only worthwhile on a trusted home network, so we request it only when P2P is
1609     # allowed; otherwise AP-isolated / enterprise, the common case we boot plain
1610     # user-mode networking. Either way host<->container P2P is UNAFFECTED: it rides
1611     # routing-fix's Docker-gateway / .host_ips RETURN rules, not the VM network mode
1612     # (see devref 15.3.5). colima_start_until_ready returns as soon as Docker answers
1613     # (~15s) instead of blocking on colima's ~2min post-provision hang (see 15.8.7);
1614     # that hang is colima-internal and mode-independent (it also occurs in user mode).
1615     allow_p2p=$(read_env_var "TAILSCALE_ALLOW_LAN_P2P")
1616     [ -z "$allow_p2p" ] && allow_p2p="false"
1617     if [ -f "$SCRIPT_DIR/.lan_p2p_detected" ]; then
1618         allow_p2p=$(tr -d '[:space:]' < "$SCRIPT_DIR/.lan_p2p_detected" 2>/dev/null || echo "false")
1619     fi
1620
1621     if [ "$allow_p2p" = "true" ]; then
1622         colima_network_mode="bridged"
1623         if [[ "$OSTYPE" == "darwin*" ]]; then
1624             security_type=$(system_profiler SPAirPortDataType 2>/dev/null | awk '/Current Network Information:/{
1625                 found=1} found && /Security:/{print $NF; exit}')
1626             if echo "$security_type" | grep -qi "Enterprise"; then

```

```

1625     echo -e "\n [!] WPA2-Enterprise Wi-Fi detected! Bridged mode will cause MAC-spoofing
deauthentication."
1626     echo "      Falling back to '--network-mode shared' for Colima."
1627     colima_network_mode="shared"
1628     fi
1629   fi
1630   echo "Colima is not running. Starting Colima (600MB RAM, vz VM, LAN P2P on: --network-address
$colima_network_mode)..."
1631   colima_start_until_ready --memory 0.6 --vm-type vz --network-address --network-mode "$colima_network_mode
" \
1632   || echo " [!] Colima Docker not confirmed ready; continuing (downstream docker-ps auto-heal will retry
)."
1633   else
1634     colima_network_mode="user"
1635     echo "Colima is not running. Starting Colima (600MB RAM, vz VM, LAN P2P off: fast user-mode networking)
..."
1636     colima_start_until_ready --memory 0.6 --vm-type vz \
1637     || echo " [!] Colima Docker not confirmed ready; continuing (downstream docker-ps auto-heal will retry
)."
1638     fi
1639     echo "$colima_network_mode" > /tmp/nullexit_colima_mode.txt
1640   else
1641     echo "Colima is already running."
1642   fi
1643
1644   # 3b. Auto-detect and fix Docker Subnet Collisions (e.g. if the user travels to a 172.17.x.x Wi-Fi)
1645   if [ -f "scripts/fix-docker-bridge-collision.sh" ]; then
1646     if bash scripts/fix-docker-bridge-collision.sh --dry | grep -q "collision detected"; then
1647       echo -e "\n[!] WARNING: Docker Subnet Collision Detected with your current Wi-Fi!"
1648       echo -e "      Auto-fixing Colima config to prevent internet jitter..."
1649       bash scripts/fix-docker-bridge-collision.sh --skip-toggle || true
1650     fi
1651   fi
1652
1653   # 3c. Configure swap file inside the VM to prevent OOM on low-memory limits
1654   # We also set vm.swappiness=10 so the Linux kernel strictly prefers physical RAM
1655   # and avoids unnecessarily wearing out the SSD with proactive background swapping.
1656   if ! run_with_timeout 15 colima ssh -- grep -q 'swapfile' /proc/swaps >> output.log 2>&1; then
1657     echo "Configuring 400MB swap file inside the VM to prevent OOM..."
1658     run_with_timeout 30 colima ssh -- sudo sh -c "if [ ! -f /swapfile ]; then dd if=/dev/zero of=/swapfile bs=1
M count=400 status=none && mkswap /swapfile; fi && swapon /swapfile && sysctl vm.swappiness=10" >> output.
log 2>&1 || echo "Warning: Failed to enable swap file inside the VM."
1659   fi
1660
1661   # 4. Clean up corrupted AdGuardHome configurations
1662   if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
1663     rm -f "adguard/conf/AdGuardHome.yaml"
1664     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
1665   fi
1666
1667   # 4b. Auto-heal stale Docker socket from hard reboots
1668   if ! run_with_timeout 15 docker ps >/dev/null 2>&1; then
1669     echo "Docker daemon is unresponsive (likely a stale socket from a crash). Auto-healing Colima..."
1670     run_with_timeout 120 colima restart >> output.log 2>&1
1671   fi
1672
1673   # 4c. Inject TCP MSS clamp from .env into post-rules.txt before starting
1674   if [ -f .env ] && grep -q "^GATEWAY_MSS=" .env; then
1675     GATEWAY_MSS=$(read_env_var GATEWAY_MSS)
1676     if [[ "$GATEWAY_MSS" =~ ^[0-9]+$ ]]; then
1677       # macOS sed syntax
1678       sed -i '' "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null || \
1679       # Linux sed fallback
1680       sed -i "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null
1681     fi
1682   fi
1683
1684   # 5. Start compose services
1685   write_host_ips
1686
1687   # Procedurally generated ports have already been exported at the top of the script.
1688   echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
1689
1690   # --- HONEY-PORT TRIPWIRE ---
1691   # Generate a random trap port. If any malware does a sequential port scan on localhost,
1692   # it will inevitably hit this port. When it does, we instantly fire a macOS alert.
1693   # Reap the honey-port from any previous toggle before arming a new one, so the
1694   # `nc -l` tripwire never accumulates one idle listener per toggle-on (15.12.20).
1695   pkill -f "nc -l localhost" 2>/dev/null || true
1696   HONEY_PORT=$(get_free_port)
1697   (

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```

1698     if nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
1699         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
1700             just pinged the local Honey-Port ($HONEY_PORT)!" >> "$LOG_FILE"
1701         osascript -e 'display notification "An unknown application is aggressively scanning your local ports.
1702             Malware port-scan suspected." with title "nullexit OPSEC Alert" sound name "Basso"' 2>/dev/null || true
1703     fi
1704     ) &
1705     disown %1 2>/dev/null || true
1706     echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
1707
1708     echo -e "\nStarting Docker containers..."
1709
1710     # --- Ad-block rule compilation: hash-gated one-off, off the critical path (15.8.8) ---
1711     # The rule-compiler (a `compile`-profiled container, the SOLE writer of adguard/work)
1712     # re-dedupes ~340k rules inside the 600MB VM (~28s) far too slow to run every toggle.
1713     # It is gated on a hash of the lists YOU control (black_list.txt + white_list.txt):
1714     # recompile only when they change, when compiled_rules.txt / ip_blocklist.ipset are
1715     # missing, or when they exceed RULES_MAX_AGE_DAYS (default 7, so the remote blocklists
1716     # still refresh occasionally). Otherwise AdGuard/routing-fix boot on the existing
1717     # artifacts and we skip the ~28s they no longer `depends_on` the rule-compiler
1718     # container (see docker-compose.yml). A compile failure is non-fatal: we keep the
1719     # previous compiled rules rather than bricking the tunnel (ad-block != the kill-switch).
1720     RULES_HASH_FILE="$SCRIPT_DIR/.rules_hash"
1721     rules_hash=$(compute_rules_hash)
1722     stored_rules_hash=""
1723     [ -f "$RULES_HASH_FILE" ] && stored_rules_hash=$(cat "$RULES_HASH_FILE" 2>/dev/null)
1724     rules_max_age_days=$(read_env_var RULES_MAX_AGE_DAYS)
1725     [ -z "$rules_max_age_days" ] && rules_max_age_days=7
1726     need_compile=false
1727     if [ ! -f adguard/work/userfilters/compiled_rules.txt ] || [ ! -f adguard/work/userfilters/ip_blocklist.ipset
1728         ]; then
1729         need_compile=true
1730     elif [ -z "$rules_hash" ] || [ "$rules_hash" != "$stored_rules_hash" ]; then
1731         need_compile=true
1732     elif [ -n "$(find adguard/work/userfilters/compiled_rules.txt -mtime +"$rules_max_age_days" 2>/dev/null)" ];
1733         then
1734         need_compile=true
1735     fi
1736     if [ "$need_compile" = "true" ]; then
1737         echo " Ad-block lists changed (or artifacts missing/stale) compiling rules (one-off, ~28s)..."
1738         if run_logged docker compose run --build --rm rule-compiler >> output.log 2>&1; then
1739             if [ -n "$rules_hash" ]; then echo "$rules_hash" > "$RULES_HASH_FILE"; fi
1740         else
1741             echo " [!] Rule compile failed continuing with existing compiled ad-block rules."
1742         fi
1743     else
1744         echo " Ad-block lists unchanged reusing compiled rules (skipping the ~28s recompile)."
1745     fi
1746
1747     # --- Gate `--build` for the long-running images (routing-fix, tor) ---
1748     # BuildKit re-checking these on the 600MB VM costs ~30s even when every layer is
1749     # CACHED (15.8.8). Only pass --build when the hashed inputs change or an image is
1750     # missing; otherwise plain `up -d` reuses cached images. `up` starts everything
1751     # except the `compile`-profiled rule-compiler.
1752     BUILD_HASH_FILE="$SCRIPT_DIR/.build_hash"
1753     build_inputs_hash=$(compute_build_inputs_hash)
1754     stored_build_hash=""
1755     [ -f "$BUILD_HASH_FILE" ] && stored_build_hash=$(cat "$BUILD_HASH_FILE" 2>/dev/null)
1756     local_images_present=true
1757     for _img in nullexit-routing-fix:v1.0.0 nullexit-tor:v1.0.0; do
1758         docker image inspect "$_img" >> output.log 2>&1 || local_images_present=false
1759     done
1760     if [ -n "$build_inputs_hash" ] && [ "$build_inputs_hash" = "$stored_build_hash" ] && [ "$local_images_present"
1761         = "true" ]; then
1762         echo " Build inputs unchanged skipping image build (reusing cached images)."
1763         run_logged docker compose up -d
1764     else
1765         echo " Build inputs changed or a local image is missing building images..."
1766         run_logged docker compose up -d --build
1767         # Record the hash only after a successful build (set -e aborts above on failure).
1768         if [ -n "$build_inputs_hash" ]; then echo "$build_inputs_hash" > "$BUILD_HASH_FILE"; fi
1769     fi
1770
1771     RULE_COUNT=$(grep "! Total Block Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
1772     NATIVE_COUNT=$(grep "! Native AdGuard Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
1773
1774     if [ "$RULE_COUNT" != "0" ] && [ "$RULE_COUNT" != "" ]; then
1775         if [ "$NATIVE_COUNT" != "0" ] && [ "$NATIVE_COUNT" != "" ]; then
1776             TOTAL_COUNT=$((RULE_COUNT + NATIVE_COUNT))

```

```

1772     # Format with commas for readability (e.g. 441,578)
1773     if command -v printf &> /dev/null; then
1774         FORMATTED_RULE=$(printf "%'d" "$RULE_COUNT")
1775         FORMATTED_TOTAL=$(printf "%'d" "$TOTAL_COUNT")
1776         echo "   DNS: Compiled $FORMATTED_RULE unique custom rules (Total active in AdGuard: ~$FORMATTED_TOTAL
rules)."
1777     else
1778         echo "   DNS: Compiled $RULE_COUNT unique custom rules (Total active in AdGuard: ~$TOTAL_COUNT rules)."
1779     fi
1780 else
1781     echo "   DNS: Compiled and loaded $RULE_COUNT optimized rules."
1782 fi
1783 fi
1784
1785 # Report IP blocklist compilation results
1786 IP_COUNT=$(grep "^# Entries:" adguard/work/userfilters/ip_blocklist.ipset 2>/dev/null | awk -F': ' '{print $2
}' || echo "0")
1787 if [ "$IP_COUNT" != "0" ] && [ "$IP_COUNT" != "" ]; then
1788     if command -v printf &> /dev/null; then
1789         FORMATTED_IP=$(printf "%'d" "$IP_COUNT")
1790         echo "   IP:   Loaded $FORMATTED_IP threat intelligence IPs/CIDRs into kernel firewall."
1791     else
1792         echo "   IP:   Loaded $IP_COUNT threat intelligence IPs/CIDRs into kernel firewall."
1793     fi
1794 fi
1795
1796 # 6. Wait for the gateway container's Tailscale connection to be ready
1797 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
1798 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
1799
1800 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
1801 echo "   (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
1802 connected=false
1803 consecutive_failures=0
1804 for i in {1..60}; do
1805     # Run tailscale status and capture its output so the user can see what's happening.
1806     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
1807     ts_output=$(run_with_timeout 3 docker compose exec -T tailscale tailscale status 2>&1 || true)
1808
1809     if echo "$ts_output" | grep -q "offers exit node"; then
1810         connected=true
1811         break
1812     fi
1813
1814     # Track consecutive failures to detect a stuck state.
1815     # If we see 40+ consecutive 'NoState' responses, give up early
1816     # the daemon is alive but can't connect to the coordination server.
1817     if echo "$ts_output" | grep -q "NoState"; then
1818         consecutive_failures=$((consecutive_failures + 1))
1819         if [ "$consecutive_failures" -ge 40 ]; then
1820             echo ""
1821             echo "   [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
1822             echo "   Tailscale daemon is running but can't reach the coordination server."
1823             break
1824         fi
1825     else
1826         consecutive_failures=0
1827     fi
1828
1829     # Print a condensed status every 10 seconds so the user can see progress
1830     if [ $((i % 10)) -eq 0 ]; then
1831         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
1832         echo "   [attempt $i/60] $ts_short"
1833     elif [ $((i % 5)) -eq 0 ]; then
1834         echo -n "$[i]"
1835     else
1836         echo -n "."
1837     fi
1838     sleep 1
1839 done
1840 echo ""
1841
1842 if [ "$connected" = "true" ]; then
1843     echo "Gateway container is online on the Tailscale mesh."
1844 elif [ "$BYPASS_PING" = "true" ]; then
1845     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
1846 else
1847     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
1848     echo "   The container was seen in the following states during the wait:" >&2
1849     echo "$ts_output" | sed 's/^/    /' >&2
1850     echo "" >&2

```

```

1851 echo " This could mean:" >&2
1852 echo " 1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
1853 echo " 2. Network issue inside the container check: docker compose logs tailscale" >&2
1854 echo " 3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
1855 echo "" >&2
1856 echo " Diagnose manually:" >&2
1857 echo " docker compose exec -T tailscale tailscale status" >&2
1858 echo " docker compose exec -T tailscale tailscale netcheck" >&2
1859 cleanup_handler ERR $LINENO
1860 exit 1
1861 fi
1862
1863 # 6b. Resolve the gateway's Tailscale IP.
1864 # Dynamically query the container for the IP (handles IP changes after re-auth)
1865 # and cache it in .gateway_ip for fast retrieval by recover.sh during Wi-Fi roaming.
1866 #
1867 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
1868 # returns (\r) into captured output. If $TS_IP ends with \r, then
1869 # `networksetup -setdnsservers` silently rejects the malformed IP and
1870 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
1871 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
1872 # the first field of the first line.
1873 TS_IP=""
1874 if [ "$connected" = "true" ]; then
1875     TS_IP=$(run_with_timeout 5 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
1876     awk 'NR==1{print $1; exit}' || true)
1877     if [ -n "$TS_IP" ]; then
1878         echo "Resolved gateway Tailscale IP dynamically: $TS_IP"
1879         echo "$TS_IP" > .gateway_ip
1880     else
1881         echo " [!] Failed to resolve gateway IP dynamically. Trying fallback cache..." >&2
1882         TS_IP=$(read_adguard_ip || true)
1883     fi
1884 else
1885     TS_IP=$(read_adguard_ip || true)
1886 fi
1887
1888 # 7. Connect host Mac to Tailscale mesh.
1889 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
1890 # system LaunchDaemon via 'brew services start tailscale'), the previous five-
1891 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
1892 # no menu bar item to click, and no Accessibility permission required.
1893 #
1894 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
1895 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
1896 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
1897 # are impossible without the host on the mesh).
1898 #
1899 # We connect in two phases:
1900 # Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
1901 # Phase B Set exit node only after pre-flight checks confirm the gateway works
1902 HOST_ON_MESH=false
1903 SKIP_EXIT_NODE=false
1904 if [ "$HIJACK_HOST" = "false" ]; then
1905     echo -e "\n[Info] HIJACK_HOST is false (Headless Mode). Host networking will remain untouched."
1906     SKIP_EXIT_NODE=true
1907 elif [ -n "$TS_BIN" ]; then
1908     echo -n "Verifying tailscaled is reachable"
1909     daemon_ready=false
1910     status_exit=0
1911     for i in {1..10}; do
1912         # Test if the daemon responds to status check.
1913         # If status returns non-zero, check if it was a normal "stopped/disconnected" state
1914         # (exit code 1 is normal for disconnected). If the command exited quickly (not killed by timeout),
1915         # and the error is just "Tailscale is stopped", then the daemon is alive and ready.
1916         # But if it timed out (exit code 143) or is completely unresponsive, it's wedged.
1917         status_exit=0
1918         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1919             daemon_ready=true
1920             break
1921         else
1922             status_exit=$?
1923             if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
1924                 daemon_ready=true
1925                 break
1926             fi
1927         fi
1928     done
1929     echo -n "."
1930     sleep 1
1931 done
1932 echo ""

```

```

1931
1932 # If the daemon was unresponsive or socket check failed, attempt auto-restart
1933 if [ "$daemon_ready" != "true" ]; then
1934     if [ "$status_exit" -eq 143 ]; then
1935         warn "tailscaled daemon is wedged/unresponsive. Restarting it..."
1936     else
1937         warn "tailscaled daemon is not running. Attempting to start it..."
1938     fi
1939     restart_tailscaled_daemon
1940
1941     # Re-verify
1942     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
1943         daemon_ready=true
1944     fi
1945 fi
1946
1947 if [ "$daemon_ready" = "true" ]; then
1948     echo "tailscaled is responsive (running as a system service)."
1949
1950     # Phase A: Join mesh without exit node
1951     echo "Connecting host to Tailscale mesh (no exit node yet)..."
1952     if $TS_BIN up; then
1953         $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node= || true
1954         HOST_ON_MESH=true
1955         echo "Host is on Tailscale mesh."
1956     else
1957         warn "tailscale up failed even though the daemon is running."
1958         echo " This usually means the host hasn't authenticated with your tailnet yet."
1959         echo " Run this command in a terminal to authenticate:"
1960         echo ""
1961         echo "     sudo tailscale up"
1962         echo ""
1963         echo " A browser will open. Log in to your Tailscale account, then re-run toggle.sh."
1964     fi
1965 else
1966     fail "tailscaled could NOT be started automatically."
1967     echo " Run this in a terminal to start and authenticate Tailscale:"
1968     echo ""
1969     echo "     sudo brew services start tailscale"
1970     echo "     sudo tailscale up"
1971     echo ""
1972     echo " Then re-run toggle.sh."
1973 fi
1974 else
1975     echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
1976 fi
1977
1978 # Phase B: Pre-flight checks + enable exit node only if safe
1979 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
1980     echo -e "\n"
1981     echo "Pre-flight connectivity check"
1982     echo ""
1983
1984     # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
1985     # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
1986     # ("direct connection not established") even though the pong was received.
1987     # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
1988     echo -n " [1/3] Gateway reachable via Tailscale... "
1989     ping_ok=false
1990     # The host tailscaled needs a few seconds to propagate the new network map and
1991     # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.
1992     for attempt in {1..45}; do
1993         if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
1994             ping_ok=true
1995             break
1996         fi
1997         sleep 1
1998     done
1999     if [ "$ping_ok" = "true" ]; then
2000         echo "PASS"
2001     else
2002         echo "FAIL"
2003         SKIP_EXIT_NODE=true
2004     fi
2005
2006     # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
2007     # Colima's SSH tunnel only forwards TCP (not UDP), so we use +tcp.
2008     echo -n " [2/3] AdGuard DNS via localhost... "
2009     if command -v dig &>/dev/null; then
2010         if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 &>/dev/null; then
2011             echo "PASS"

```

```

2012     else
2013         echo "FAIL"
2014         SKIP_EXIT_NODE=true
2015     fi
2016 elif command -v nslookup &>/dev/null; then
2017     if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 &>/dev/null; then
2018         echo "PASS"
2019     else
2020         echo "FAIL"
2021         SKIP_EXIT_NODE=true
2022     fi
2023 else
2024     echo "SKIP (dig/nslookup not available)"
2025 fi
2026
2027 # Check 3: Does the WARP container have external internet?
2028 echo -n " [3/3] WARP container internet... "
2029 tmp_err=$(mktemp)
2030 if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace >/dev/null 2>
2031 "$tmp_err"; then
2032     echo "PASS"
2033 else
2034     echo "FAIL"
2035     if [ -s "$tmp_err" ]; then
2036         err_msg=$(cat "$tmp_err" | tr -d '\r')
2037         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Check 3] WARP container check failed: $err_msg" >> output.log
2038     fi
2039     SKIP_EXIT_NODE=true
2040 fi
2041 rm -f "$tmp_err"
2042
2043 # Act based on check results
2044 if [ "$SKIP_EXIT_NODE" != "true" ]; then
2045     echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
2046     $TS_BIN up || true
2047     if $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node="$TS_IP" --exit-node-allow-
2048     lan-access=true; then
2049         echo "Exit node enabled."
2050         add_warp_bypass_routes
2051         setup_exit_node_routing
2052         enable_killswitch
2053     else
2054         echo "[Warning] Failed to set exit node."
2055         SKIP_EXIT_NODE=true
2056     fi
2057 fi
2058
2059 if [ "$SKIP_EXIT_NODE" = "true" ]; then
2060     warn "Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED."
2061     echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
2062     echo " Troubleshoot with:"
2063     echo "     docker compose logs warp | tail -20"
2064     echo "     docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
2065     echo ""
2066     echo " Falling back to SOCKS5 proxy for traffic routing..."
2067 fi
2068 elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
2069     echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
2070     SKIP_EXIT_NODE=true
2071 fi
2072
2073 # 8. Apply DNS Hijacking.
2074 # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
2075 # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
2076 # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
2077 if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
2078     echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
2079     echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
2080     if force_dns_to_gateway "$TS_IP"; then
2081         echo "DNS hijack verified: single server = $TS_IP."
2082     else
2083         echo "[Warning] DNS hijack could not be verified."
2084         echo " If DNS is broken, run manually:"
2085         echo "     networksetup -setdnsservers \"$ACTIVE_SERVICE\" \"$TS_IP"
2086         echo "     networksetup -setsearchdomains \"$ACTIVE_SERVICE\" ts.net"
2087     fi
2088 elif [ -n "$TS_IP" ] && [ "$HIJACK_HOST" != "false" ]; then
2089     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
2090     echo " Trying local DNS proxy for ad-blocking..."
2091     if start_local_dns_proxy; then
2092         echo " Ad-blocking active via local DNS proxy through AdGuard."

```

```

2091     else
2092         echo "    Local DNS proxy unavailable. DNS stays at 1.1.1.1."
2093         echo "    AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
2094     fi
2095
2096     # Enable SOCKS5 proxy for traffic routing through WARP
2097     # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
2098     echo -e "\n    Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
2099     echo "    (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
2100     enable_socks_proxy "$ACTIVE_SERVICE"
2101
2102     # Verify the SOCKS5 proxy works through WARP
2103     echo -n "    Verifying SOCKS5 proxy routes through WARP... "
2104     if curl --socks5-hostname 127.0.0.1:${SOCKS_PROXY_PORT} --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
2105         trace 2>/dev/null | grep -q "warp=on"; then
2106         echo "ok (warp=on)."

```

```

2171     echo -e ""
2172 fi
2173
2174 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
2175     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
2176         echo -e "\n[Warning] One or more blocklist URLs failed to download (404/Offline)."
```

```

2177         echo "    The gateway gracefully fell back to yesterday's cached blocklist."
2178         echo "    (Run 'docker logs rule-compiler' later to investigate which link died.)"
2179     fi
2180 fi
2181
2182 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
2183 if [ -t 0 ]; then
2184     read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
2185     if [[ "${USER_CHOICE}" == "r" || "${USER_CHOICE}" == "R" ]]; then
2186         echo "Reversing gateway state..."
2187         exec bash "$0"
2188     fi
2189 fi
2190
2191 echo -e "\nYou can close this terminal window now."

```

7 recover.sh

```

1  #!/bin/bash
2  # recover.sh nullexit KILL-SWITCH recovery script (macOS + Linux)
3  # Fixes internet when toggle.sh leaves you stranded.
4  #
5  # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6  # 1. Disconnecting host Tailscale from the mesh (exit-node off)
7  # 2. Resetting DNS to default on ALL network services
8  # 3. Disabling rogue proxy settings (SOCKS, web, secure web)
9  # 4. Flushing the DNS cache
10 # 5. Flushing stale routing table entries
11 # 6. Stopping gateway Docker containers
12 # 7. Power-cycling Wi-Fi
13 # 8. Verifying internet is working
14 #
15 # One file, both OSes: the dispatch below runs the self-contained Linux
16 # recovery and exits; on macOS it falls through to the richer dual-mode
17 # implementation (default teardown / --post-wake refresh).
18 #
19 # Usage: double-click Recover-Gateway.applescript
20 #         OR  ./recover.sh
21 #         OR  ./recover.sh --post-wake    (macOS only lighter refresh, keeps gateway live)
22
23 set -e
24
25 # Resolve script directory (used for path-relative refs)
26 # recover.sh lives at the repo root; SCRIPT_DIR lets us launch toggle.sh
27 # (and reference any other repo-root file) from the same directory no
28 # matter where the user invoked recover.sh from.
29 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
30 cd "$SCRIPT_DIR" || exit 1
31
32 #
33 # LINUX RECOVERY self-contained; runs the full teardown then exits before the
34 # macOS body below. Native Linux tooling (resolvectl / nmcli / ip) throughout.
35 #
36 if [[ "$OSTYPE" != "darwin"* ]]; then
37     # Source common formatting and helper functions
38     source "$SCRIPT_DIR/scripts/common.sh"
39
40     # Always add Homebrew path
41     export PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:$PATH"
42
43     echo ""
44     echo -e "${RED}${BOLD}${NC}"
45     echo -e "${RED}${BOLD}          Nullexit Gateway  RECOVERY MODE          ${NC}"
46     echo -e "${RED}${BOLD}${NC}"
47     echo ""
48     echo "This script will tear down the gateway and restore normal internet."
49     echo ""
50
51     # Cache sudo credentials upfront
52     # Prevents the script from hanging halfway through when it needs to run

```

```

53 # privileged commands (route flush, Wi-Fi power cycle, etc.).
54 echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
55 sudo -v
56 (
57     while true; do
58         sudo -n true 2>/dev/null
59         sleep 60
60     done
61 ) &
62 SUDO_KEEPER_PID=$!
63 trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
64
65 # 1. Disconnect host Tailscale
66 step "Disconnecting host Tailscale from mesh"
67 if command -v tailscale >> output.log 2>&1; then
68     # Check if actually connected first (with timeout tailscaled could be wedged)
69     if run_with_timeout 5 tailscale status >> output.log 2>&1; then
70         # Also explicitly reset any exit-node so it doesn't linger.
71         if run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= >> output.log
72         2>&1; then
73             ok "Exit-node preference cleared"
74         else
75             warn "tailscale up --reset didn't respond (tailscaled may be wedged)"
76         fi
77         ts_args=""
78         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
79         if run_with_timeout 10 tailscale down $ts_args >> output.log 2>&1; then
80             ok "Tailscale disconnected"
81         else
82             warn "tailscale down didn't respond"
83         fi
84     else
85         warn "tailscaled not reachable skipping (timeout after 5s)"
86     fi
87 else
88     warn "tailscale CLI not found skipping"
89 fi
90
91 # 2. Reset DNS to default on ALL network services
92 step "Resetting DNS to default on active interface"
93
94 ACTIVE_SERVICE=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
95 if [ -n "$ACTIVE_SERVICE" ]; then
96     sudo resolvectl revert "$ACTIVE_SERVICE" 2>> output.log || true
97     ok "DNS reset on $ACTIVE_SERVICE"
98 else
99     warn "Could not determine active network interface"
100 fi
101
102 # 3. Disable rogue proxy settings
103 step "Disabling proxy settings (SOCKS, web, secure web)"
104 echo " (Linux global SOCKS proxy configuration skipped.)"
105 ok "Proxy settings cleared"
106
107 # 4. Flush DNS cache
108 step "Flushing DNS cache"
109 if command -v resolvectl >> output.log 2>&1; then
110     sudo resolvectl flush-caches 2>> output.log || true
111     ok "DNS cache flushed"
112 else
113     warn "resolvectl not found"
114 fi
115
116 # 5. Clear stale routing table
117 step "Flushing stale routing table entries"
118 sudo ip route flush cache >> output.log 2>&1 || true
119 ok "Routing table flushed"
120
121 # 6. Stop gateway Docker containers
122 step "Stopping gateway Docker containers"
123 if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
124     if [ -f "docker-compose.yml" ]; then
125         docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
126     else
127         warn "docker-compose.yml not found in current directory"
128     fi
129 else
130     warn "Docker not available skipping"
131 fi
132

```

```

133 # 6b. Stopping sleep prevention (systemd-inhibit)
134 step "Stopping sleep prevention"
135 PID_FILE="/tmp/nullexit-cafeinate.pid"
136 if [ -f "$PID_FILE" ]; then
137     INHIBIT_PID=$(cat "$PID_FILE")
138     if [ -n "$INHIBIT_PID" ] && kill -0 "$INHIBIT_PID" 2>/dev/null; then
139         kill "$INHIBIT_PID" 2>/dev/null || true
140         ok "Sleep prevention stopped (PID $INHIBIT_PID)"
141     else
142         warn "Sleep prevention process not running, cleaning up stale PID file"
143         fi
144         rm -f "$PID_FILE"
145     else
146         ok "Sleep prevention was not active"
147     fi
148
149 # 7. Power-cycle Wi-Fi
150 step "Power-cycling Wi-Fi"
151 if command -v nmcli &>/dev/null; then
152     sudo nmcli radio wifi off 2>> output.log || true
153     sleep 2
154     sudo nmcli radio wifi on 2>> output.log || true
155     ok "Wi-Fi bounced via nmcli"
156     sleep 3
157 else
158     warn "nmcli not found. Falling back to ip link..."
159     WIFI_IFACE=$(ip link | grep -E '^[0-9]+: (wl[a-zA-Z0-9]+|wlan[0-9]+):' | awk -F': ' '{print $2}')
160     if [ -n "$WIFI_IFACE" ]; then
161         sudo ip link set "$WIFI_IFACE" down 2>> output.log || true
162         sleep 2
163         sudo ip link set "$WIFI_IFACE" up 2>> output.log || true
164         ok "Wi-Fi bounced via ip link (interface $WIFI_IFACE)"
165         sleep 3
166     else
167         warn "No Wi-Fi interface detected."
168     fi
169 fi
170
171 # 8. Verify internet connectivity
172 verify_internet_connectivity
173
174 # Summary
175 echo ""
176 echo -e "${BOLD}${NC}"
177 echo -e "${BOLD}RECOVERY SUMMARY${NC}"
178 echo -e "${BOLD}${NC}"
179
180 # Show final DNS state
181 FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr
182 '\n' ' ' || echo "N/A")
183 echo " DNS ($ACTIVE_SERVICE): $FINAL_DNS"
184
185 echo ""
186
187 if [ "$INTERNET_OK" = "true" ]; then
188     echo -e " ${GREEN}${BOLD} Internet is working.${NC}"
189 else
190     echo -e " ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
191     echo " This might mean:"
192     echo " - Your Wi-Fi is disconnected from the router"
193     echo " - Tailscale is still interfering (try restarting tailscaled:"
194     echo "     systemctl restart tailscaled)"
195     echo " - You need to re-connect to Wi-Fi in the menu bar"
196     echo " - Try toggling Wi-Fi off/on in the menu bar"
197     echo ""
198     echo " Or try manually:"
199     echo "     resolvectl revert $ACTIVE_SERVICE"
200     extra_args=""
201     if is_kill_switch_enabled; then extra_args="--accept-risk=lose-ssh"; fi
202     echo "     tailscale down $extra_args"
203     echo "     sudo ip route flush cache"
204     echo "     systemctl restart tailscaled"
205 fi
206
207 echo ""
208 echo -e "${GREEN}${BOLD} Recovery complete.${NC}"
209 echo ""
210
211 exit 0
212 fi

```

```

213 #
214 # macOS RECOVERY (dual-mode: default teardown / --post-wake refresh)
215 #
216
217 # Enforce Cryptographic Script Integrity
218 if [ -f "scripts/crypto.sh" ]; then
219     if ! bash scripts/crypto.sh --verify; then
220         exit 1
221     fi
222 fi
223
224 # Argument parsing
225 # --post-wake: lightweight refresh after sleep/wake or Wi-Fi roam.
226 #     Keeps the gateway live while HUPing DNS caches, refreshing the
227 #     Tailscale exit-node leak, re-hijacking host DNS, and force-
228 #     recreating the warp container if its gluetun healthcheck failed.
229 #     Does NOT tear down Docker, Tailscale, sleep prevention, or Wi-Fi.
230 #     Called from scripts/watcher.sh on every wake / network-change event
231 #     while /tmp/nullexit-gateway-active.marker is present (i.e. the
232 #     gateway is currently up). See devref.md 10.29 for the why.
233 POST_WAKE=false
234 AUTO_MODE=false
235 for arg in "$@"; do
236     if [ "$arg" = "--post-wake" ]; then
237         POST_WAKE=true
238     elif [ "$arg" = "--auto" ] || [ "$arg" = "--non-interactive" ]; then
239         AUTO_MODE=true
240     fi
241 done
242
243 rm -f "$SCRIPT_DIR/TUNNEL_FAILED_CLOSED.marker"
244
245 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
246 LOCK_FILE="/tmp/nullexit-toggle.lock"
247 if [ -f "$LOCK_FILE" ]; then
248     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
249     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
250         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
251             if [ "$POST_WAKE" = "true" ]; then
252                 echo "[$(date -u +%FT%TZ)] POST-WAKE skipped: another lifecycle script (PID $LOCK_PID) is running." >>
253                     output.log
254                 exit 0
255             else
256                 echo "Error: Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running." >&2
257                 exit 1
258             fi
259         fi
260     fi
261     echo "$$" > "$LOCK_FILE"
262
263 # Clear the active-state marker in nuclear recovery mode since the gateway is being
264 # completely torn down. This prevents scripts/watcher.sh from triggering post-wake
265 # recoveries in response to network changes caused by the teardown itself.
266 if [ "$POST_WAKE" = "false" ]; then
267     rm -f "/tmp/nullexit-gateway-active.marker"
268 fi
269
270 # WARP Watcher inhibit marker: written immediately in --post-wake mode so the
271 # background WARP Watcher (in toggle.sh) skips failure counting while we are
272 # intentionally tearing down / recreating the warp container. Without this, the
273 # watcher accumulates 6 consecutive warp=off readings during force-recreate (~35s)
274 # and fires nuclear recover.sh, killing a gateway that would have been healthy.
275 WARP_INHIBIT_FILE="/tmp/nullexit-warp-inhibit.marker"
276 if [ "$POST_WAKE" = "true" ]; then
277     touch "$WARP_INHIBIT_FILE"
278     echo "[$(date -u +%FT%TZ)] POST-WAKE started WARP Watcher inhibited to prevent spurious shutdown during warp
279         recreate." >> output.log
280 fi
281
282 # Ensure the inhibit marker and lock file are always removed when the script exits (success or error)
283 cleanup_recover() {
284     rm -f "$WARP_INHIBIT_FILE"
285     if [ -f "$LOCK_FILE" ] && [ "$(cat "$LOCK_FILE" 2>/dev/null)" = "$$" ]; then
286         rm -f "$LOCK_FILE"
287     fi
288 }
289 trap cleanup_recover EXIT
290
291 # Source common formatting and helper functions
292 source "$SCRIPT_DIR/scripts/common.sh"

```

```

292
293 # Always add Homebrew path
294 setup_standard_path
295
296 echo ""
297 if [ "$POST_WAKE" = "true" ]; then
298     echo -e "${YELLOW}${BOLD}${NC}"
299     echo -e "${YELLOW}${BOLD}  Nullexit  POST-WAKE / POST-ROAM LIGHT ${NC}"
300     echo -e "${YELLOW}${BOLD}  (keeps the gateway live, refreshes)  ${NC}"
301     echo -e "${YELLOW}${BOLD}${NC}"
302     echo ""
303     echo "Re-hijacking DNS, refreshing Tailscale exit-node, force-recreating"
304     echo "warp if unhealthy, and resetting sharing services. The gateway stays"
305     echo "up. docker compose / Wi-Fi / caffeinate are NOT touched."
306     echo ""
307 else
308     echo -e "${RED}${BOLD}${NC}"
309     echo -e "${RED}${BOLD}  Nullexit Gateway  RECOVERY MODE  ${NC}"
310     echo -e "${RED}${BOLD}${NC}"
311     echo ""
312     echo "This script will tear down the gateway and restore normal internet."
313     echo ""
314 fi
315
316 # Cache sudo credentials upfront (default mode ONLY)
317 # Skip in --post-wake: the LaunchAgent-launched watcher has no TTY, so `sudo -v`
318 # would block forever waiting for a password. All post-wake commands use `sudo -n`
319 # which is non-blocking and relies on /etc/sudoers.d/nullexit NOPASSWD entries.
320 SUDO_KEEPER_PID=""
321 if [ "$POST_WAKE" = "false" ] && [ "$AUTO_MODE" = "false" ] && [ -t 0 ]; then
322     echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
323     sudo -v
324     (
325         while true; do
326             sudo -n true 2>/dev/null
327             sleep 60
328         done
329     ) &
330     SUDO_KEEPER_PID=$!
331     trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
332 fi
333
334 # Resolve physical default interface and gateway upfront
335 PHYSICAL_IFACE=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
    getline; print $2; exit}')
336 [ -z "$PHYSICAL_IFACE" ] && PHYSICAL_IFACE="en0"
337
338 if [ "$POST_WAKE" = "true" ]; then
339     wait_for_dhcp_settle
340 else
341     # Quick resolution in non-post-wake mode (non-blocking)
342     PHYSICAL_GW=$(ipconfig getpacket "$PHYSICAL_IFACE" 2>> output.log | awk -F'[{}]' '/router/{print $2}')
343 fi
344
345 # 1. Disconnect host Tailscale (default) / Refresh exit-node (post-wake)
346
347 if [ "$POST_WAKE" = "true" ]; then
348     step "Refreshing host Tailscale exit-node routing (post-wake)"
349     if command -v tailscale >> output.log 2>&1; then
350         # Re-issue the SAME exit-node up command toggle.sh START uses. This refreshes
351         # the DERP relay mapping and re-asserts the exit-node preference without
352         # dropping the host's mesh connection (which `tailscale down` would).
353         TS_IP=""
354         for src in .gateway_ip; do
355             [ -f "$src" ] || continue
356             TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
357             [ -n "$TS_IP" ] && break
358         done
359
360         if [ -n "$TS_IP" ]; then
361             step "Re-establishing exit node $TS_IP via 'tailscale set --exit-node'"
362             # `tailscale set` only mutates the named preference (exit-node here).
363             # Crucially it does NOT call --reset, which would re-apply ALL flags and
364             # trigger a sub-second route-table re-evaluation cycle each post-wake.
365             # On already-connected nodes, the lighter `set` keeps the existing tunnel
366             # alive while only changing the exit-node preference.
367             if run_with_timeout 15 tailscale set --exit-node="$TS_IP" \
368                 --exit-node-allow-lan-access=true \
369                 >> output.log 2>&1; then
370                 ok "Exit-node $TS_IP re-asserted via 'tailscale set'"
371             else

```

```

372     warn "tailscale set --exit-node=$TS_IP didn't respond; falling back to mesh-only"
373     run_with_timeout 10 tailscale set --exit-node= \
374         >> output.log 2>&1 || true
375 fi
376 else
377     warn ".gateway_ip missing cannot re-assert exit node"
378     run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= \
379         >> output.log 2>&1 || true
380 fi
381 else
382     warn "tailscale CLI not found"
383 fi
384 else
385     step "Disconnecting host Tailscale from mesh"
386     disconnect_tailscale_host "tailscale"
387 fi
388
389 # 2. Reset DNS to default on ALL network services (default) / Re-hijack gateway DNS (post-wake)
390 if [ "$POST_WAKE" = "true" ]; then
391     step "Re-hijacking host DNS to gateway IP (post-wake / post-roam)"
392     TS_IP=""
393     for src in .gateway_ip; do
394         [ -f "$src" ] || continue
395         TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
396         [ -n "$TS_IP" ] && break
397     done
398
399     if [ -n "$TS_IP" ]; then
400         # Detect the active network service (using helper in common.sh)
401         ACTIVE_SVC=$(get_active_service)
402         ENO_SVC=$(get_en0_service)
403
404         networksetup -setsearchdomains "$ACTIVE_SVC" "ts.net" 2>> output.log || true
405         networksetup -setdnsservers "$ACTIVE_SVC" "$TS_IP" 2>> output.log || true
406         networksetup -setsearchdomains "$ENO_SVC" "ts.net" 2>> output.log || true
407         networksetup -setdnsservers "$ENO_SVC" "$TS_IP" 2>> output.log || true
408
409         HITS=$(networksetup -getdnsservers "$ACTIVE_SVC" 2>> output.log | grep -Fx "$TS_IP" || true)
410         if [ -n "$HITS" ]; then
411             ok "DNS re-hijacked to $TS_IP on services: $ACTIVE_SVC, $ENO_SVC"
412         else
413             warn "networksetup accepted but DNS read-back didn't include $TS_IP"
414         fi
415     else
416         warn ".gateway_ip missing cannot re-hijack DNS (user must run toggle.sh START)"
417     fi
418 else
419     step "Resetting DNS to default on all network services (empty = DHCP)"
420
421     # Get ALL network services (handles spaces in names, skips disabled ones)
422     SERVICES=$(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" || true)
423
424     if [ -z "$SERVICES" ]; then
425         # Fallback to common names
426         SERVICES="Wi-Fi Ethernet Thunderbolt Ethernet USB 10/100 LAN"
427     fi
428
429     RESET_COUNT=0
430     while IFS= read -r service; do
431         # Strip leading asterisk (disabled services)
432         service_clean=$(echo "$service" | sed 's/^\*//')
433         [ -z "$service_clean" ] && continue
434
435         # Check if this service has a hardware port (skip VPN/service-only entries)
436         if networksetup -listallhardwareports 2>> output.log | grep -A1 "Port: $service_clean$" | grep -q "Device:"
437             || [ "$service_clean" = "Wi-Fi" ]; then
438             (
439                 networksetup -setsearchdomains "$service_clean" "Empty" 2>> output.log
440                 # Set to "empty" so macOS uses DHCP-assigned DNS (not hardcoded 1.1.1.1)
441                 sudo networksetup -setdnsservers "$service_clean" "empty" 2>> output.log
442             ) && RESET_COUNT=$((RESET_COUNT + 1)) || true
443         fi
444     done <<< "$SERVICES"
445
446     ok "DNS reset on $RESET_COUNT service(s)"
447 fi
448
449 # 3. Disable rogue proxy settings (default only gateway uses proxy in non-exit-node path)
450 if [ "$POST_WAKE" = "false" ]; then
451     step "Disabling proxy settings (SOCKS, web, secure web)"

```

```

451 for svc in $(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" ||
452     echo "Wi-Fi"); do
453     svc_clean=$(echo "$svc" | sed 's/\^*//')
454     [ -z "$svc_clean" ] && continue
455     disable_all_proxies "$svc_clean"
456 done
457 ok "Proxy settings cleared"
458 else
459     step "Proxy settings: leaving untouched (post-wake keeps gateway SOCKS wiring alive)"
460 fi
461 # 4. Flush macOS DNS cache (always safe and useful in both modes)
462 step "Flushing DNS cache"
463 if command -v dscacheutil >> output.log 2>&1; then
464     flush_dns_cache
465     ok "DNS cache flushed (mDNSResponder HUP)"
466 else
467     warn "dscacheutil not found"
468 fi
469 # 5. Clear stale routing table (always safe in both modes)
470 step "Flushing stale routing table entries"
471 # Resolve physical default gateway IP before flushing
472 # We cannot trust `route get default` here because TailScale may have already hijacked
473 # the default route to utunX. We must read the actual DHCP router assignment from the hardware.
474 # PHYSICAL_IFACE and PHYSICAL_GW resolved upfront
475 # Instead of full route -n flush (which destroys macOS loopback/multicast and wedges the OS until reboot),
476 # we cleanly delete the TailScale overrides and Cloudflare WARP bypass routes in ALL modes.
477 # This ensures tailscaled isn't blocked from reaching its control plane when waking from sleep.
478 sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
479 sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
480 remove_warp_bypass_routes
481 ok "Routing overrides cleared"
482 if [ "$POST_WAKE" = "false" ]; then
483     disable_killswitch
484     ok "Routing table flush completed via targeted deletes and firewall disabled"
485 else
486     ok "Routing overrides cleared (post-wake mode to preserve Colima bridge100)"
487 fi
488 # In post-wake mode we don't bounce Wi-Fi, so we MUST manually restore the physical default route
489 if [ "$POST_WAKE" = "true" ] && [ -n "$PHYSICAL_GW" ]; then
490     # Ensure physical gateway is set just in case it was dropped
491     sudo -n route add default "$PHYSICAL_GW" >> output.log 2>&1 || true
492 fi
493 # CONDITIONAL BYPASS ROUTE RESTORATION:
494 # If this was a post-wake recovery and the exit node is active, the route flush
495 # just wiped the static bypass routes for the WARP endpoints. We must re-add
496 # them immediately to prevent a routing loop when TailScale starts sending traffic.
497 if [ "$POST_WAKE" = "true" ]; then
498     if [ -z "${ACTIVE_IF:-}" ]; then
499         ACTIVE_IF=$(route get default 2>> output.log | awk '/interface:/{print $2; exit}')
500         if [ -z "$ACTIVE_IF" ] || [[ "$ACTIVE_IF" =~ ^utun ]] || [[ "$ACTIVE_IF" =~ ^tun ]]; then
501             for i in en0 en1 en2 en3; do
502                 if ifconfig "$i" 2>> output.log | grep -q 'status: active'; then
503                     ACTIVE_IF="$i"; break
504                 fi
505             done
506         fi
507         [ -z "$ACTIVE_IF" ] && ACTIVE_IF="en0"
508     fi
509     echo "Re-adding host bypass routes for Cloudflare WARP endpoints..."
510     remove_warp_bypass_routes
511     if [ -n "${PHYSICAL_GW:-}" ]; then
512         add_warp_bypass_routes "$PHYSICAL_GW"
513     else
514         add_warp_bypass_routes "$ACTIVE_IF"
515     fi
516 # Also re-apply the default route pointing to the TailScale interface
517 ts_iface=$(ifconfig 2>> output.log | grep -B4 "inet 100." | grep -E '[a-z0-9]+' | cut -d: -f1 | head -n 1)
518 if [ -n "$ts_iface" ]; then
519     echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
520     # DO NOT delete the physical default route! It crashes Colima.
521     # Use the /1 trick to mathematically override the default route.
522     sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
523     sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true

```

```

531 sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
532 sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
533 enable_killswitch
534 fi
535 fi
536
537 # 6. Stop gateway Docker containers (default) / Force-recreate warp if unhealthy (post-wake)
538 if [ "$POST_WAKE" = "true" ]; then
539 # 6a. Check for Roaming Network Mismatch (Bridged vs Enterprise Wi-Fi)
540 if [ -f "/tmp/nullexit_colima_mode.txt" ] && [ -f "$SCRIPT_DIR/../../lan_p2p_detected" ]; then
541 current_colima_mode=$(cat /tmp/nullexit_colima_mode.txt)
542 p2p_allowed=$(cat "$SCRIPT_DIR/../../lan_p2p_detected")
543 required_colima_mode="bridged"
544 [ "$p2p_allowed" == "false" ] && required_colima_mode="shared"
545
546 if [ "$current_colima_mode" != "$required_colima_mode" ]; then
547 warn "Network roaming mismatch detected!"
548 warn "Colima is running in '$current_colima_mode' mode but new network requires '$required_colima_mode'."
549 warn "Executing full gateway restart to protect against MAC-spoofing deauthentication..."
550 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Roam] Mismatch ($current_colima_mode vs $required_colima_mode).
Triggering toggle.sh --restart." >> output.log
551 nohup bash "$SCRIPT_DIR/../../toggle.sh" --restart >/dev/null 2>&1 &
552 exit 0
553 fi
554 fi
555
556 step "Checking Colima VM state"
557 colima_status=$(colima status 2>&1)
558 if echo "$colima_status" | grep -qi "running"; then
559 ok "Colima VM is running"
560 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima status check passed." >> output.log
561 else
562 warn "Colima VM is NOT running or unresponsive"
563 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima is unhealthy or dead! Output: $colima_status"
>> output.log
564 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Attempting to restart Colima..." >> output.log
565 colima restart >> output.log 2>&1 || warn "Failed to restart Colima"
566 fi
567
568 step "Checking warp container health (gluetun UDP tunnel)"
569 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Inspecting warp container state..." >> output.log
570 # The warp container is the most failure-prone link in the chain at wake/roam:
571 # its UDP NAT binding to Cloudflare's edge (162.159.192.1:2408) silently dies
572 # during a Wi-Fi roam. We verify the tunnel is actually passing traffic via curl.
573 WARP_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
574 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container status: $WARP_STATE" >> output.log
575
576 if [ "$WARP_STATE" = "missing" ]; then
577 warn "warp container is missing leaving gateway containers alone (full relaunch requires \"$SCRIPT_DIR/
toggle.sh\")"
578 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container missing from Docker engine. Requires
full toggle.sh launch." >> output.log
579 else
580 tmp_err=$(mktemp)
581 if docker compose exec -T warp wget -q0- --timeout=3 https://www.cloudflare.com/cdn-cgi/trace 2>"$tmp_err"
| grep -q "warp=on"; then
582 ok "warp container is healthy and actively tunneling traffic (no recreate needed)"
583 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container passed live traffic healthcheck (
Cloudflare trace successful)." >> output.log
584 else
585 warn "warp container tunnel is dead (Docker status: $WARP_STATE) force-recreating all gateway containers
to restore network namespace"
586 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container is DEAD or STUCK. Forcing recreation
of gateway stack..." >> output.log
587 if [ -s "$tmp_err" ]; then
588 err_msg=$(cat "$tmp_err" | tr -d '\r')
589 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Healthcheck failure details: $err_msg" >> output.
log
590 fi
591 docker compose up -d --force-recreate >> output.log 2>&1 || warn "force-recreate failed"
592
593 NEW_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
594 ok "warp container health after recreate: $NEW_STATE"
595 echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container health after recreation: $NEW_STATE"
>> output.log
596 fi
597 rm -f "$tmp_err"
598 fi
599 else
600 step "Stopping gateway Docker containers"
601 if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then

```

```

602     if [ -f "docker-compose.yml" ]; then
603         docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
604     else
605         warn "docker-compose.yml not found in current directory"
606     fi
607 else
608     warn "Docker not available skipping"
609 fi
610 fi
611
612 # 6b. Stopping sleep prevention (caffeinate) default only
613 if [ "$POST_WAKE" = "false" ]; then
614     step "Stopping sleep prevention"
615     stop_pidfile_daemon "$PID_CAFFEINATE" "system sleep prevention"
616 else
617     step "Sleep prevention: leaving untouched (caffeinate already re-acquired by parent shell)"
618 fi
619
620 # 6c. Stopping DNS Watcher default only
621 if [ "$POST_WAKE" = "false" ]; then
622     step "Stopping DNS Watcher"
623     stop_pidfile_daemon "$PID_DNS_WATCHER" "background DNS Watcher"
624 else
625     step "DNS Watcher: leaving untouched (it keeps re-hijacking DNS on every roam)"
626 fi
627
628 # 7. Power-cycle Wi-Fi default only
629 if [ "$POST_WAKE" = "false" ]; then
630     step "Power-cycling Wi-Fi"
631     bounce_wifi_interfaces
632 else
633     step "Wi-Fi: leaving untouched (post-wake keeps the current Wi-Fi association)"
634 fi
635
636 # 8. Verify (default checks direct internet; post-wake verifies the gateway)
637 if [ "$POST_WAKE" = "true" ]; then
638     step "Verifying gateway is healthy after post-wake refresh"
639
640     # Wait a moment for tailscale + warp healthcheck to settle
641     sleep 3
642
643     GATEWAY_OK=false
644     if command -v dig >> output.log 2>&1; then
645         # Source procedural ports if available
646         if [ -f "/tmp/nullexit-ports.env" ]; then
647             source "/tmp/nullexit-ports.env"
648         fi
649         DNS_PORT=${DNS_PROXY_PORT:-5354}
650
651         # AdGuard is exposed at localhost:${DNS_PORT}. If this resolves, the warp tunnel
652         # is functioning properly because AdGuard routes upstream to warp.
653         if dig +tcp @127.0.0.1 -p "$DNS_PORT" google.com +short +timeout=5 >> output.log 2>&1; then
654             ok "Gateway DNS (AdGuard via localhost:${DNS_PORT}) works"
655             GATEWAY_OK=true
656         else
657             warn "Gateway DNS not responding yet"
658         fi
659     fi
660
661     # Direct host gateway check via Tailscale ping (relayed pong counts as success)
662     if command -v tailscale >> output.log 2>&1; then
663         TS_IP=""
664         [ -f .gateway_ip ] && TS_IP=$(read_adguard_ip || true)
665         if [ -n "$TS_IP" ] && tailscale ping --until-direct=false -c 1 --timeout 4s "$TS_IP" 2>> output.log | grep
        -q pong; then
666             ok "Gateway $TS_IP reachable via Tailscale"
667             GATEWAY_OK=true
668         else
669             warn "Tailscale ping to gateway did not pong exit-node may still be re-establishing"
670         fi
671     fi
672
673     if [ "$GATEWAY_OK" = "true" ]; then
674         echo -e " ${GREEN}${BOLD} Gateway is healthy after post-wake refresh.${NC}"
675     else
676         echo -e " ${YELLOW}${BOLD} Gateway recovery is in-progress.${NC}"
677         echo " The route may need another 30s before queries succeed."
678         echo " If it stays broken for >60s, run: bash \"${SCRIPT_DIR}/toggle.sh\""
679     fi
680 else
681     verify_internet_connectivity

```

```

682 fi
683
684 # Summary
685 echo ""
686 echo -e "${BOLD}${NC}"
687 if [ "$POST_WAKE" = "true" ]; then
688     echo -e "${BOLD}POST-WAKE SUMMARY${NC}"
689 else
690     echo -e "${BOLD}RECOVERY SUMMARY${NC}"
691 fi
692 echo -e "${BOLD}${NC}"
693
694 # Show final DNS state
695 FINAL_DNS=$(networksetup -getdnsservers "Wi-Fi" 2>/dev/null || echo "N/A")
696 echo "  DNS (Wi-Fi):  $FINAL_DNS"
697
698 FINAL_DNS2=$(networksetup -getdnsservers "Ethernet" 2>/dev/null || true)
699 if [[ -z "$FINAL_DNS2" || "$FINAL_DNS2" == *"not a recognized"* || "$FINAL_DNS2" == *"Error"* ]]; then
700     FINAL_DNS2="N/A (Not configured)"
701 fi
702 echo "  DNS (Ethernet): $FINAL_DNS2"
703
704 echo ""
705
706 if [ "$POST_WAKE" = "false" ]; then
707     if [ "$INTERNET_OK" = "true" ]; then
708         echo -e "  ${GREEN}${BOLD} Internet is working.${NC}"
709     else
710         echo -e "  ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
711         echo "    This might mean:"
712         echo "    - Your Wi-Fi is disconnected from the router"
713         echo "    - Tailscale is still interfering (try restarting tailscaled:"
714         echo "      brew services restart tailscale)"
715         echo "    - You need to re-connect to Wi-Fi in the menu bar"
716         echo "    - Try toggling Wi-Fi off/on in the menu bar"
717         echo ""
718         echo "    Or try manually:"
719         echo "      networksetup -setdnsservers Wi-Fi empty"
720         echo "      tailscale down"
721         echo "      sudo route delete -net 0.0.0.0/1"
722         echo "      sudo route delete -net 128.0.0.0/1"
723         echo "      brew services restart tailscale"
724     fi
725 fi
726
727 # Reset sharing services to prevent AirDrop freezes after interface changes
728 # This is the SAME step in BOTH modes (post-wake inherits the previous fix from 10.27).
729 echo -e "  Resetting macOS sharing services (AirDrop/AirPlay)..."
730 reset_sharing_services
731
732 echo ""
733 if [ "$POST_WAKE" = "true" ]; then
734     echo -e "${YELLOW}${BOLD}Post-wake refresh complete.${NC}"
735     # Remove the WARP Watcher inhibit marker now that the gateway is fully healthy.
736     # The watcher will resume normal liveness monitoring on its next 5s poll.
737     rm -f "$WARP_INHIBIT_FILE"
738     echo "[$(date -u +%FT%TZ)] POST-WAKE complete WARP Watcher inhibit released." >> output.log
739 else
740     echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
741 fi
742 echo ""

```

8 setup.sh

```

1 #!/bin/bash
2 # setup.sh nullexit one-shot setup script (macOS)
3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "$(dirname "$0")/scripts/common.sh"
8
9 # Run common installation logic
10 source "$(dirname "$0")/scripts/setup-common.sh"
11
12 # Compile macOS Shortcuts

```

```

13 echo -e "\n${BOLD}Compiling macOS Desktop Shortcuts...${NC}"
14 if command -v osacompile &> /dev/null; then
15     osacompile -o "Toggle Gateway.app" "Toggle-Gateway.applescript" >/dev/null 2>&1
16     osacompile -o "Recover Gateway.app" "Recover-Gateway.applescript" >/dev/null 2>&1
17     echo "    Created 'Toggle Gateway.app' and 'Recover Gateway.app'"
18 fi
19
20 # Done
21 echo ""
22 echo -e "${GREEN}${BOLD}${NC}"
23 echo -e "${GREEN}${BOLD}                Setup complete!                ${NC}"
24 echo -e "${GREEN}${BOLD}${NC}"
25 echo ""
26 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
27 echo "    https://login.tailscale.com/admin/machines"
28 echo "    Find your gateway '...' 'Edit route settings'  enable Exit Node"
29 echo ""
30
31 echo -e "${YELLOW}${BOLD}Sudoers / Background Execution Requirements:${NC}"
32 echo "    To allow silent, non-interactive execution of the firewall and network routes (especially during sleep/
wake recovery), you must configure passwordless sudo."
33 echo "    Run this exact command in your terminal (grants are scoped to least privilege):"
34 echo "    echo \"\$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /sbin/route, /sbin/ifconfig,
/usr/bin/dscacheutil -flushcache, /usr/bin/killall -HUP mDNSResponder, /usr/bin/killall sharingd rapportd,
/usr/bin/pkill -9 tailscale, /usr/bin/pkill -f dns-proxy.py, /bin/kill, /usr/bin/true, /opt/homebrew/bin/
brew services * tailscale, /opt/homebrew/bin/brew uninstall tailscale, /usr/local/bin/brew services *
tailscale, /usr/local/bin/brew uninstall tailscale, /usr/bin/python3 -I \$PWD/scripts/dns-proxy.py, /opt/
homebrew/bin/python3 -I \$PWD/scripts/dns-proxy.py, /usr/local/bin/python3 -I \$PWD/scripts/dns-proxy.py\"
| sudo tee /etc/sudoers.d/nullexit >/dev/null && sudo visudo -cf /etc/sudoers.d/nullexit"
35 echo ""
36
37 if [[ -n "${TS_IP:-}" ]]; then
38     echo -e "${BOLD}AdGuard Home dashboard:${NC}  http://${TS_IP}:3000  (username: admin)"
39     echo ""
40 fi
41
42 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
43 echo "    macOS:  double-click the 'Toggle Gateway' app icon!"
44 echo "    Windows: double-click Toggle-Gateway.bat"
45 echo ""
46
47 if [[ "$OSTYPE" == "darwin"* ]]; then
48     # LuLu localhost filtering check
49     if [ -d "/Applications/LuLu.app" ] || systemextensionsctl list 2>/dev/null | grep -q "com.objective-see.
lulu"; then
50         echo -e "${YELLOW}${BOLD}LuLu Firewall Detected:${NC}"
51         echo "    LuLu's localhost (loopback) filtering is OFF by default. To properly protect the"
52         echo "    local nullexit proxy ports from malicious local processes, you must manually enable"
53         echo "    'Allow Loopback' (or block it selectively) in LuLu's Preferences -> Rules."
54         echo ""
55     fi
56
57     echo -e "${YELLOW}${BOLD}Note on macOS Tailscale Sandboxing:${NC}"
58     echo "    The Tailscale GUI app (tailscale-app) installed on macOS is sandboxed."
59     echo "    While you cannot run a Tailscale SSH server on this Mac host, you can"
60     echo "    still SSH outbound to other devices on the mesh using their MagicDNS"
61     echo "    addresses directly (e.g. ssh user@device.tailnet-name.ts.net)."
62     echo ""
63 fi

```

9 scripts/setup-linux.sh

```

1 #!/bin/bash
2 # scripts/setup-linux.sh nullexit one-shot setup script (Linux)
3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "$(dirname "$0")/common.sh"
8
9 # Run common installation logic
10 source "$(dirname "$0")/setup-common.sh"
11
12 # Compile Linux Desktop Shortcuts
13 echo -e "\n${BOLD}Creating Linux Desktop Shortcuts...${NC}"
14 DIR="$(pwd)"

```

```

15 cat <<EOF > "Toggle Gateway.desktop"
16 [Desktop Entry]
17 Version=1.0
18 Name=Toggle Gateway
19 Comment=Start or Stop Nullexit Gateway
20 Exec=bash -c "cd '$DIR' && ./toggle.sh; read -p 'Press Enter to close...' dummy"
21 Icon=utilities-terminal
22 Terminal=true
23 Type=Application
24 Categories=Network;Security;
25 EOF
26
27 cat <<EOF > "Recover Gateway.desktop"
28 [Desktop Entry]
29 Version=1.0
30 Name=Recover Gateway
31 Comment=Emergency DNS and Network Recovery
32 Exec=bash -c "cd '$DIR' && ./recover.sh; read -p 'Press Enter to close...' dummy"
33 Icon=utilities-terminal
34 Terminal=true
35 Type=Application
36 Categories=Network;Security;
37 EOF
38 echo "    Created 'Toggle Gateway.desktop' and 'Recover Gateway.desktop'"
39
40 # Done
41 echo ""
42 echo -e "${GREEN}${BOLD}${NC}"
43 echo -e "${GREEN}${BOLD}                Setup complete!                ${NC}"
44 echo -e "${GREEN}${BOLD}${NC}"
45 echo ""
46 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
47 echo "    https://login.tailscale.com/admin/machines"
48 echo "    Find your gateway  '...'  'Edit route settings'  enable Exit Node"
49 echo ""
50
51 if [[ -n "${TS_IP:-}" ]]; then
52     echo -e "${BOLD}AdGuard Home dashboard:${NC}    http://${TS_IP}:3000    (username: admin)"
53     echo ""
54 fi
55
56 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
57 echo "    Linux: double-click the 'Toggle Gateway' desktop icon!"
58 echo "    (or run ./toggle.sh in terminal)"
59 echo ""

```

10 docker-compose.yml

```

1 services:
2   warp:
3     image: qmcgaw/gluetun:v3.41.1
4     container_name: warp
5     hostname: nullexit-gateway
6     cap_add:
7       - NET_ADMIN
8     devices:
9       - /dev/net/tun:/dev/net/tun
10    ports:
11      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/udp"    # AdGuard DNS: host binds 5354 (the mapped DNS-proxy
12        port); 5335 is AdGuard's in-container port
13      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/tcp"    # AdGuard DNS (TCP)
14      - "41642:41642/udp"    # Tailscale WireGuard (direct connection to host)
15      - "127.0.0.1:${SOCKS_PROXY_PORT:-1080}:1080/tcp"    # SOCKS5 proxy (routed through WARP tunnel)
16      - "127.0.0.1:${TOR SOCKS PORT:-9050}:9050/tcp"    # Tor SOCKS5 proxy (procedurally generated)
17      - "127.0.0.1:${TOR_CONTROL_PORT:-9051}:9051/tcp"    # Tor Control port (procedurally generated)
18    environment:
19      - TZ=America/New_York
20      - VPN_SERVICE_PROVIDER=custom
21      - VPN_TYPE=wireguard
22      - WIREGUARD_PRIVATE_KEY=${WIREGUARD_PRIVATE_KEY}
23      - WIREGUARD_PUBLIC_KEY=${WIREGUARD_PUBLIC_KEY}
24      - WIREGUARD_ADDRESSES=${WIREGUARD_ADDRESSES}
25      - WIREGUARD_ENDPOINT_IP=${WARP_ENDPOINT_1:-162.159.192.1}
26      - WIREGUARD_ENDPOINT_PORT=2408
27      # Set to 25s (WireGuard default) to beat standard 30s hardware NAT/firewall UDP timeouts
28      - WIREGUARD_PERSISTENT_KEEPLIVE_INTERVAL=25s

```

```

28 # MTU Fix (Critical): Prevent packet fragmentation from double-encryption
29 - WIREGUARD_MTU=1280
30 # Give WARP more time to establish the connection before internal restart (default is 6s which is too
short)
31 - HEALTH_VPN_DURATION_INITIAL=30s
32 # Use plaintext DNS resolvers instead of DNS-over-TLS (DoT) to prevent startup failures on networks where
port 853 is blocked
33 - DNS_UPSTREAM_RESOLVER_TYPE=plain
34 - DNS_UPSTREAM_PLAIN_ADDRESSES=1.1.1.1:53,8.8.8.8:53
35 # Allow local network traffic to bypass WARP so Tailscale can establish fast, direct peer-to-peer
connections (no DERP relays) on the LAN
36 - FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8
37 # Built-in DNS blocking for ads, malicious domains, and tracking
38 # NOTE: This is the massive "Hagezi-style" built-in blocklist.
39 # You can turn these to 'on' for maximum security if you have spare RAM
40 # or are running this on a cloud exit node with adequate autonomous resources (requires approx 1-2GB of
RAM).
41 # Disabled intentionally: AdGuard Home provides the DNS filtering layer instead.
42 # If AdGuard fails, there is no fallback filtering, but the healthcheck
43 # dependency chain ensures Tailscale won't come up if AdGuard is down.
44 - BLOCK_MALICIOUS=off
45 - BLOCK_SURVEILLANCE=off
46 - BLOCK_ADS=off
47 dns:
48 - 1.1.1.1
49 sysctls:
50 - net.ipv4.ip_forward=1
51 volumes:
52 - ./post-rules.txt:/iptables/post-rules.txt:ro
53 restart: unless-stopped
54 healthcheck:
55   test: ["CMD-SHELL", "/gluetun-entrypoint healthcheck"]
56   interval: 2s
57   timeout: 5s
58   retries: 15
59   start_period: 60s
60 logging:
61   driver: "json-file"
62   options:
63     max-size: "10m"
64     max-file: "3"
65
66 tailscale:
67   image: tailscale/tailscale:v1.98.4
68   container_name: tailscale
69   network_mode: service:warp
70   depends_on:
71     warp:
72       condition: service_healthy
73     adguardhome:
74       condition: service_healthy
75   cap_add:
76     - NET_ADMIN
77     - NET_RAW
78     - SYS_MODULE
79   sysctls:
80     - net.ipv4.ip_forward=1
81     - net.ipv6.conf.all.forwarding=1
82   environment:
83     - TZ=America/New_York
84     - TS_EXTRA_ARGS=--advertise-exit-node
85     - PORT=41642
86     - TS_USERSPACE=false
87     - TS_AUTHKEY=${TS_AUTHKEY}
88     - TS_STATE_DIR=/var/lib/tailscale
89     # (Note: For headless/cloud deployments, you can bypass this footgun by
90     # using Tailscale Ephemeral Keys which auto-clean themselves).
91     - TS_AUTH_ONCE=true
92     # MTU Optimization: Force Tailscale packets to be smaller than WARP's 1280 MTU
93     # to prevent WireGuard-in-WireGuard fragmentation, completely eliminating bufferbloat
94     - TS_DEBUG_MTU=1200
95     # Enable Tailscale's built-in SOCKS5 proxy to route out of the WARP tunnel
96     - TS_SOCKS5_SERVER=:1080
97   devices:
98     - /dev/net/tun:/dev/net/tun
99   volumes:
100     - tailscale_state:/var/lib/tailscale
101   restart: unless-stopped
102   logging:
103     driver: "json-file"
104     options:

```

```

105     max-size: "10m"
106     max-file: "3"
107
108 routing-fix:
109     image: nullexit-routing-fix:v1.0.0
110     build:
111         context: ./scripts
112         dockerfile: Dockerfile.routing-fix
113     container_name: routing-fix
114     network_mode: service:warp
115     # NOTE: Deliberately NOT sharing tailscale's PID namespace the shared
116     # namespace caused routing-fix to be killed whenever tailscale restarted
117     # (e.g. during gluetun health-check flaps), which dropped all routing
118     # table changes and left traffic leaking through eth0 instead of tun0.
119     cap_add:
120         - NET_ADMIN
121         - SYSLOG
122     depends_on:
123         warp:
124             condition: service_healthy
125             # rule-compiler dependency removed: it is now a hash-gated one-off run by
126             # toggle.sh before `up` (15.8.8). This service boots on the existing
127             # compiled_rules.txt / ip_blocklist.ipset already present in ./adguard/work.
128     environment:
129         - TZ=America/New_York
130         - WARP_ENDPOINT=${WARP_ENDPOINT_1:-162.159.192.1}
131         - BLOCKED_COUNTRIES=${BLOCKED_COUNTRIES}
132         - TAILSCALE_ALLOW_LAN_P2P=${TAILSCALE_ALLOW_LAN_P2P:-false}
133     volumes:
134         - ./scripts/routing-fix.sh:/routing-fix.sh:ro
135         # SECURITY: ./adguard/work/ is bind-mounted into this container.
136         # Never write to it from the host or another container. The
137         # single allowed writer is `rule-compiler` container manual
138         # edits corrupt the AdGuard cache + BoltDB session store.
139         - ./adguard/work/userfilters:/userfilters:ro
140         - ./adguard/work:/adguard_work:ro
141         - ./app
142     restart: unless-stopped
143     stop_grace_period: 1s
144     logging:
145         driver: "json-file"
146         options:
147             max-size: "1m"
148             max-file: "1"
149
150 adguardhome:
151     # WARNING: AdGuard is pre-configured with hardcoded credentials (admin/nullexit).
152     # This is fine for a local Mac running behind a NAT, but if you are deploying
153     # this on a public cloud VPS with port 80/3000 exposed, change this immediately!
154     image: adguard/adguardhome:v0.107.77
155     container_name: adguardhome
156     network_mode: service:warp
157     depends_on:
158         warp:
159             condition: service_healthy
160             # rule-compiler dependency removed: it is now a hash-gated one-off run by
161             # toggle.sh before `up` (15.8.8). This service boots on the existing
162             # compiled_rules.txt / ip_blocklist.ipset already present in ./adguard/work.
163     environment:
164         - TZ=America/New_York
165     healthcheck:
166         test: ["CMD-SHELL", "nc -z 127.0.0.1 5335 || exit 1"]
167         interval: 2s
168         timeout: 5s
169         retries: 10
170     volumes:
171         # SECURITY: ./adguard/work/ is bind-mounted into this container.
172         # Never write to it from the host or another container. The
173         # single allowed writer is `rule-compiler` container manual
174         # edits corrupt the AdGuard cache + BoltDB session store.
175         - ./adguard/work:/opt/adguardhome/work
176         - ./adguard/conf:/opt/adguardhome/conf
177     restart: unless-stopped
178     stop_grace_period: 2s
179     logging:
180         driver: "json-file"
181         options:
182             max-size: "10m"
183             max-file: "3"
184
185 rule-compiler:

```

```

186 image: nullexit-rule-compiler:v1.0.0
187 build:
188     context: ./scripts
189     dockerfile: Dockerfile.rule-compiler
190 container_name: rule-compiler
191 # `compile` profile: excluded from `docker compose up`. toggle.sh runs it
192 # on-demand via `docker compose run --build --rm rule-compiler`, hash-gated on
193 # black_list.txt/white_list.txt, so ~340k rules are re-deduped only when a list
194 # actually changes instead of every toggle (~28s saved on the common path, 15.8.8).
195 profiles: ["compile"]
196 volumes:
197     - ./app
198
199 tor:
200     build: ./docker/tor
201     image: nullexit-tor:v1.0.0
202     container_name: tor
203     network_mode: service:warp
204     depends_on:
205         warp:
206             condition: service_healthy
207     environment:
208         - TZ=America/New_York
209         - TOR_USE_BRIDGES=${TOR_USE_BRIDGES:-false}
210         - TOR_BRIDGE_LINES_FILE=${TOR_BRIDGE_LINES_FILE:-./tor-bridges.txt}
211         - TOR_PASSWORD=${ADGUARD_PASSWORD:-nullexit}
212         - TOR_EXCLUDE_EXIT_NODES=${TOR_EXCLUDE_EXIT_NODES:-{us}}
213     working_dir: /nullexit
214     volumes:
215         - ./nullexit:ro
216         - ./output.log:/output.log:rw
217     restart: unless-stopped
218     logging:
219         driver: "json-file"
220         options:
221             max-size: "10m"
222             max-file: "3"
223
224 volumes:
225     tailscale_state:
226
227 networks:
228     default:
229         ipam:
230             config:
231                 - subnet: 10.200.1.0/24

```

11 scripts/common.sh

```

1 #!/bin/bash
2 # common.sh shared bash utilities for nullexit scripts
3
4 # Formatting
5 RED='\033[0;31m'
6 GREEN='\033[0;32m'
7 YELLOW='\033[1;33m'
8 BLUE='\033[0;34m'
9 BOLD='\033[1m'
10 NC='\033[0m'
11
12 # Constants & Environment
13 export MARKER_FILE="/tmp/nullexit-gateway-active.marker"
14 export PID_CAFFEINATE="/tmp/nullexit-caffeinate.pid"
15 export PID_DNS_WATCHER="/tmp/nullexit-dns-watcher.pid"
16 export DEFAULT_WARP_ENDPOINT_1="162.159.192.1"
17 export DEFAULT_WARP_ENDPOINT_2="162.159.193.1"
18
19 setup_standard_path() {
20     export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
21 }
22
23 step() { echo -e "\n[ $(date '+%Y-%m-%d %H:%M:%S') ] ${BLUE}${BOLD} ${*}${NC}"; }
24 ok() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${GREEN} ${*}${NC}"; }
25 warn() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${YELLOW} ${*}${NC}"; }
26 fail() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${RED} ${*}${NC}"; }
27 die() { echo -e "\n[ $(date '+%Y-%m-%d %H:%M:%S') ] ${RED} ${*}${NC}\n"; exit 1; }

```

```

28
29 # Helper Functions
30
31 # Append a timestamped line to output.log (best-effort; never fails the caller).
32 # Used to record lifecycle breadcrumbs (EXEC/EXIT markers, phase changes) so a
33 # START/STOP that is killed or window-closed still leaves a retraceable trail.
34 log_line() {
35     echo "[$(date '+%Y-%m-%d %H:%M:%S')] $*" >> "${LOG_FILE:-output.log}" 2>/dev/null || true
36 }
37
38 # Run a lifecycle command with FULL observability: log the command, stream its
39 # combined stdout+stderr to the terminal AND append it to output.log via tee,
40 # then log the exit code explicitly. Returns the command's real exit code (not
41 # tee's). This closes the blind spot where `docker compose up` / `colima start`
42 # printed only to the terminal so when they fail (or the run is interrupted),
43 # output.log shows exactly what the last command was and whether it succeeded.
44 #
45 # Usage: run_logged docker compose up -d --build
46 run_logged() {
47     local logf="${LOG_FILE:-output.log}"
48     log_line "EXEC: $*"
49     # PIPESTATUS[0] is the command's exit code; tee (PIPESTATUS[1]) is ignored.
50     "$@" 2>&1 | tee -a "$logf"
51     local rc=${PIPESTATUS[0]}
52     log_line "EXIT $rc: $*"
53     return "$rc"
54 }
55
56 # Pure-bash timeout (no dependency on GNU coreutils' timeout)
57 # Runs a command with a safety cutoff so a wedged daemon can't hang the script.
58 run_with_timeout() {
59     local timeout_sec="$1"
60     shift
61     if [ $# -eq 0 ]; then return 1; fi
62
63     "$@" &
64     local cmd_pid=$!
65     CURRENT_BG_PID="$cmd_pid"
66
67     (
68         sleep "$timeout_sec"
69         if kill -0 "$cmd_pid" 2>> output.log; then
70             echo -e "\n[Timeout] Command '$*' exceeded $timeout_sec seconds. Terminating..." >&2
71             kill -15 "$cmd_pid" 2>> output.log || true
72             sleep 2
73             if kill -0 "$cmd_pid" 2>> output.log; then
74                 kill -9 "$cmd_pid" 2>> output.log || true
75             fi
76         fi
77     ) &
78     local watcher_pid=$!
79
80     # Disable set -e temporarily to capture exit status of wait
81     set +e
82     wait "$cmd_pid" 2>> output.log
83     local exit_status=$?
84     set -e
85
86     # Kill the watcher since the command finished
87     kill "$watcher_pid" 2>> output.log && wait "$watcher_pid" 2>> output.log || true
88
89     CURRENT_BG_PID=""
90     return "$exit_status"
91 }
92
93 # Start Colima and return the moment Docker is actually reachable, instead of
94 # blocking on colima 0.10.3's post-provision hang. Observed on macOS/vz: the VM
95 # reaches READY and creates the `colima` docker context in ~10-15s, then
96 # `colima start` frequently fails to return for ~110s (until a watchdog kills it)
97 # even though Docker is fully usable the entire time. We poll the colima docker
98 # context and proceed as soon as it answers, then reap the (usually hung) starter.
99 # The VM keeps running and `colima status` stays accurate. $@ = colima start args.
100 # Returns 0 once Docker responds, 1 if it never does within the cap.
101 colima_start_until_ready() {
102     local max_wait=90 waited=0 ready=false start_pid
103     echo "[$(date '+%Y-%m-%d %H:%M:%S')] EXEC(bg): colima start $*" >> output.log
104     colima start "$@" >> output.log 2>&1 &
105     start_pid=$!
106     while [ "$waited" -lt "$max_wait" ]; do
107         if run_with_timeout 5 docker --context colima info >/dev/null 2>&1; then
108             ready=true

```

```

109     break
110 fi
111 # If colima start exited on its own (genuine finish or hard failure), stop waiting.
112 if ! kill -0 "$start_pid" 2>/dev/null; then
113     run_with_timeout 5 docker --context colima info >/dev/null 2>&1 && ready=true
114     break
115 fi
116 sleep 3
117 waited=$((waited + 3))
118 done
119 # Reap the starter (harmless if it already exited); this does NOT stop the VM.
120 kill "$start_pid" 2>/dev/null || true
121 wait "$start_pid" 2>/dev/null || true
122 if [ "$ready" = "true" ]; then
123     docker context use colima >/dev/null 2>&1 || true
124     echo "[$(date '+%Y-%m-%d %H:%M:%S')] Colima Docker ready after ${waited}s (starter reaped)." >> output.log
125     return 0
126 fi
127 echo "[$(date '+%Y-%m-%d %H:%M:%S')] Colima Docker NOT ready after ${max_wait}s." >> output.log
128 return 1
129 }
130
131 # Hash the files that feed the long-running locally-built images that `docker
132 # compose up` starts (routing-fix, tor). toggle.sh uses this to skip
133 # `docker compose --build` when nothing changed: BuildKit re-evaluating those
134 # images on the 600MB Colima VM costs ~30s per toggle even when every layer is
135 # CACHED (15.8.8). Includes the Dockerfiles, everything they COPY (routing-fix.sh,
136 # tor/entrypoint.sh, the logger/ Go source), and docker-compose.yml. The
137 # rule-compiler image is NOT here it is a `compile`-profiled one-off, built+run
138 # by the gated compile step (15.8.8). Prints a sha256 hex digest, or empty.
139 compute_build_inputs_hash() {
140     local root="${SCRIPT_DIR:-}" f
141     {
142         for f in \
143             "$root/scripts/Dockerfile.routing-fix" \
144             "$root/scripts/routing-fix.sh" \
145             "$root/docker/tor/Dockerfile" \
146             "$root/docker/tor/entrypoint.sh" \
147             "$root/docker-compose.yml"; do
148             printf ':%s: ' "$f"; cat "$f" 2>/dev/null
149         done
150         find "$root/scripts/logger" -type f 2>/dev/null \
151             | LC_ALL=C sort | while IFS= read -r f; do printf ':%s: ' "$f"; cat "$f" 2>/dev/null; done
152     } | openssl dgst -sha256 2>/dev/null | awk '{print $NF}'
153 }
154
155 # Hash the ad-block lists the user controls: black_list.txt (custom blocks) and
156 # white_list.txt (the allow-list that governs what DNS resolves through). toggle.sh
157 # gates the ~28s rule recompile on this editing either list changes the digest
158 # and triggers a one-off recompile of compiled_rules.txt + ip_blocklist.ipset;
159 # an unchanged toggle skips it entirely (15.8.8). Prints a sha256 hex digest.
160 compute_rules_hash() {
161     local root="${SCRIPT_DIR:-}"
162     { printf ':black: '; cat "$root/black_list.txt" 2>/dev/null
163       printf ':white: '; cat "$root/white_list.txt" 2>/dev/null
164     } | openssl dgst -sha256 2>/dev/null | awk '{print $NF}'
165 }
166
167 # Check if the gateway is active (either containers are running, or host DNS is hijacked)
168 is_gateway_active() {
169     # 1. Check if containers are running (suppress stderr to avoid errors if docker is down)
170     if run_with_timeout 15 docker compose ps --status running 2>/dev/null | grep -q 'warp'; then
171         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (warp container is running)" >> "$LOG_FILE"
172         return 0
173     fi
174
175     # 2. Check if host DNS was hijacked (not 1.1.1.1 and not default/empty)
176     local current_dns=""
177     local active_svc=""
178     if [[ "$OSTYPE" == "darwin"* ]]; then
179         active_svc=$(get_active_service)
180         current_dns=$(networksetup -getdnsservers "$active_svc" 2>> output.log || true)
181     else
182         current_dns=$(resolvectl dns 2>/dev/null | awk '/Global|Link/ {for(i=4;i<NF;i++) print $i}' | head -n1)
183     fi
184
185     if [[ -n "$current_dns" && "$current_dns" != "1.1.1.1" && ! "$current_dns" =~ "There aren't any DNS Servers"
186         ]]; then
187         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (DNS was hijacked: $current_dns)" >> "
188             $LOG_FILE"
189         return 0
190     fi

```

```

188 fi
189
190 # 3. Check if SOCKS proxy is enabled (macOS only)
191 if [[ "$OSTYPE" == "darwin"* ]]; then
192     local socks_proxy
193     socks_proxy=$(networksetup -getsocksfirewallproxy "$active_svc" 2>> output.log || true)
194     if echo "$socks_proxy" | grep -q "Enabled: Yes"; then
195         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (SOCKS proxy enabled)" >> "$LOG_FILE"
196         return 0
197     fi
198 fi
199
200 echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: FALSE (Containers down, DNS clean, SOCKS disabled)"
201 >> "$LOG_FILE"
202 return 1
203 }
204
205 # Decide whether the gateway SHOULD be running, from persisted *intent* rather
206 # than a live health probe. This is the correct signal for toggle.sh's
207 # STOP-vs-START decision: a disable should tear down whatever the last START
208 # left behind it should NOT depend on whether Docker/Colima happens to be
209 # reachable right now.
210 #
211 # Why this exists: using is_gateway_active() for the decision means every toggle
212 # (including *disable*) runs `docker compose ps` first. When Colima is down the
213 # socket is absent, that call blocks for the full run_with_timeout window, and
214 # under `set -e` + the ERR trap a non-zero result at the `if then else fi`
215 # boundary could abort the script BEFORE the START body runs so the very path
216 # that would boot Colima never executed (observed as `EXIT: code=1 phase=start`).
217 #
218 # Signal source: MARKER_FILE is written at the end of a successful START
219 # (write_gateway_active_marker) and cleared at the top of STOP and in
220 # cleanup_handler. Reading it is instant and cannot hang or abort.
221 #
222 # ECHOES "running" or "stopped" to stdout (and ALWAYS returns 0). Callers read
223 # the string: [ "$(gateway_state)" = "running" ] &&
224 #
225 # CRITICAL why this echoes instead of using a 0/1 return code: on macOS
226 # /bin/bash 3.2, with `set -e` + an `ERR` trap + `set -x` (DEBUG_TRACE=true) all
227 # active, a function that does `return 1` triggers a SPURIOUS `set -e` abort in
228 # the caller even when the call is guarded (`if f; then`, or even `f || rc=$?`).
229 # The `return N` from the function, traced by `set -x`, trips the ERR trap. This
230 # is a genuine bash-3.2 interaction (see devref 15.11.8). Echoing a result and
231 # always returning 0 sidesteps `return`/`set -e`/`set -x` entirely the function
232 # can never contribute a non-zero status, so no caller can be aborted by it.
233 gateway_state() {
234     # Primary: persisted intent. Marker present => a START completed and no STOP
235     # has cleared it yet.
236     #
237     # NOTE: toggle.sh runs a "defensive stale-marker clear" near the top of every
238     # invocation (before this decision) to stop the wake-watcher firing against a
239     # crashed gateway. That would wipe the marker before we read it, so toggle.sh
240     # captures the marker's existence into GATEWAY_MARKER_AT_STARTUP *before* that
241     # clear and we honor it here. If the variable is unset (other callers, or the
242     # --restart pre-check which runs before the defensive clear), fall back to the
243     # live marker file.
244     if [ -n "${GATEWAY_MARKER_AT_STARTUP:-}" ]; then
245         if [ "$GATEWAY_MARKER_AT_STARTUP" = "yes" ]; then echo "running"; return 0; fi
246     elif [ -f "$MARKER_FILE" ]; then
247         echo "running"; return 0
248     fi
249
250     # Marker absent. Normally this means "stopped". But cover the edge case where
251     # the marker was lost (e.g. /tmp cleared, or gateway started by an older build)
252     # yet containers are genuinely up reconcile via is_gateway_active, but ONLY
253     # when Docker is actually reachable, so a dead-socket probe can never hang. On
254     # macOS Docker runs in the Colima VM; if that socket is absent, the gateway
255     # definitively is not running.
256     local docker_up="no"
257     if [[ "$OSTYPE" == "darwin"* ]]; then
258         if [ -S /var/run/docker.sock ] || [ -S "$HOME/.colima/default/docker.sock" ]; then docker_up="yes"; fi
259     else
260         if [ -S /var/run/docker.sock ]; then docker_up="yes"; fi
261     fi
262
263     if [ "$docker_up" = "yes" ] && is_gateway_active; then
264         echo "running"; return 0
265     fi
266     echo "stopped"
267     return 0
268 }

```

```

268
269 # Reads an environment variable from a file (default: .env), strips quotes and whitespace
270 read_env_var() {
271     local var_name="$1"
272     local env_file="${2:-.env}"
273     grep -E "$var_name" "$env_file" 2>/dev/null | cut -d '=' -f2- | tr -d "\"\`" | sed -e 's/^[[:space:]]*//'
274     -e 's/[[:space:]]*$//' | echo ""
275 }
276
277 # Load KILL_SWITCH variable from .env (defaults to false if not found/empty)
278 export KILL_SWITCH
279 KILL_SWITCH=$(read_env_var "KILL_SWITCH" 2>/dev/null | tr '[:upper:]' '[:lower:]')
280 if [ -z "$KILL_SWITCH" ]; then
281     KILL_SWITCH="false"
282 fi
283
284 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
285 get_active_service() {
286     if [[ "$OSTYPE" == "darwin*" ]]; then
287         local iface
288         iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
289
290         # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
291         if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
292             for i in en0 en1 en2 en3; do
293                 if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "
294                     inet "; then
295                     iface="$i"
296                     break
297                 fi
298             done
299         fi
300
301         # Default fallback if still empty or not enX
302         if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
303             iface="en0"
304         fi
305
306         # Map interface (e.g., en0) to service name (e.g., Wi-Fi)
307         local service
308         service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B 1 "Device: $iface" | head -n 1 |
309             sed -E 's/^\([0-9\*]+\)/ /')
310
311         if [ -n "$service" ]; then
312             echo "$service"
313         else
314             # Get the interface used for default internet routing
315             local iface
316             iface=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
317             if [ -z "$iface" ]; then
318                 echo ""
319                 return 1
320             fi
321             echo "$iface"
322         fi
323     }
324
325     get_en0_service() {
326         local service
327         service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B1 "Device: en0" | head -1 | sed -E 's
328             /\([0-9\*]+\)/ /' || true)
329         if [ -z "$service" ]; then
330             echo "Wi-Fi"
331         else
332             echo "$service"
333         fi
334     }
335
336     get_warp_endpoint_1() { read_env_var "WARP_ENDPOINT_1" ".env" | grep -Eo '[0-9\.]+' || echo "
337         $DEFAULT_WARP_ENDPOINT_1"; }
338     get_warp_endpoint_2() { read_env_var "WARP_ENDPOINT_2" ".env" | grep -Eo '[0-9\.]+' || echo "
339         $DEFAULT_WARP_ENDPOINT_2"; }
340
341     restart_tailscaled_daemon() {
342         echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
343         echo " [Tailscale Recovery] Attempting to restart tailscaled service..."
344
345         if run_with_timeout 15 brew services restart tailscale >> output.log 2>&1; then

```

```

343     echo " [Tailscale Recovery] Successfully restarted tailscaled (user service)."
344 elif run_with_timeout 15 sudo -n brew services restart tailscale >> output.log 2>&1; then
345     echo " [Tailscale Recovery] Successfully restarted tailscaled (system service)."
346 else
347     echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
348 fi
349 sleep 3
350 }
351
352 disconnect_tailscale_host() {
353     local ts_bin="${1:-tailscale}"
354     if command -v "$ts_bin" >> output.log 2>&1; then
355         echo " Disconnecting host Tailscale from mesh..."
356
357         # Check if actually connected first (with timeout tailscaled could be wedged)
358         local status_ok=false
359         if run_with_timeout 5 "$ts_bin" status >> output.log 2>&1; then
360             status_ok=true
361         else
362             local status_exit=$?
363             # If it was a quick exit code 1 (disconnected), it is still reachable
364             if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
365                 status_ok=true
366             fi
367         fi
368
369         if [ "$status_ok" = "false" ]; then
370             restart_tailscaled_daemon
371         fi
372
373         # Explicitly reset any exit-node so it doesn't linger.
374         "$ts_bin" up >> output.log 2>&1 || true
375         if run_with_timeout 10 "$ts_bin" set --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1; then
376             echo " [ ] Exit-node preference cleared."
377         else
378             echo " [!] tailscale up didn't respond (tailscaled may be wedged)"
379         fi
380
381         local ts_args=""
382         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
383         if run_with_timeout 10 "$ts_bin" down $ts_args >> output.log 2>&1; then
384             echo " [ ] Tailscale disconnected."
385         else
386             echo " [!] tailscale down didn't respond"
387         fi
388     else
389         echo " [!] tailscale CLI not found skipping"
390     fi
391 }
392
393 stop_pidfile_daemon() {
394     local pid_file="$1"
395     local label="$2"
396     if [ -f "$pid_file" ]; then
397         local pid
398         pid=$(cat "$pid_file")
399         if kill -0 "$pid" 2>/dev/null; then
400             echo " Stopping $label (PID $pid)..."
401             sudo -n kill "$pid" 2>> output.log || kill "$pid" 2>> output.log || true
402             sleep 0.5
403             if kill -0 "$pid" 2>/dev/null; then
404                 sudo -n kill -9 "$pid" 2>> output.log || kill -9 "$pid" 2>> output.log || true
405             fi
406         fi
407         rm -f "$pid_file"
408     fi
409 }
410
411 disable_all_proxies() {
412     local svc="$1"
413     if [ -n "$svc" ]; then
414         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
415         sudo -n networksetup -setwebproxystate "$svc" off 2>> output.log || true
416         sudo -n networksetup -setsecurewebproxystate "$svc" off 2>> output.log || true
417     fi
418 }
419
420 flush_dns_cache() {
421     if [[ "$OSTYPE" == "darwin"* ]]; then
422         sudo -n dscacheutil -flushcache 2>> output.log || true
423         sudo -n killall -HUP mDNSResponder 2>> output.log || true

```

```

424 else
425     resolvectl flush-caches 2>> output.log || true
426 fi
427 }
428
429 wait_for_dhcp_settle() {
430     # Resolve physical interface (Wi-Fi or Ethernet)
431     local physical_iface
432     physical_iface=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
433         getline; print $2; exit}')
434     [ -z "$physical_iface" ] && physical_iface="en0"
435
436     echo -n "Waiting for physical network interface ($physical_iface) DHCP lease to settle..."
437     local settled=false
438     for attempt in {1..60}; do
439         PHYSICAL_GW=$(ipconfig getpacket "$physical_iface" 2>> output.log | awk -F'[{}]' '/router/{print $2}')
440         local local_ip
441         local_ip=$(ifconfig "$physical_iface" 2>/dev/null | awk '/inet/{print $2}')
442
443         if [ -n "$PHYSICAL_GW" ] && [[ "$PHYSICAL_GW" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
444             [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
445             [[ "$local_ip" != "169.254.*" ]]; then
446             echo "settled (Interface IP: $local_ip, Router IP: $PHYSICAL_GW)."
447             settled=true
448             break
449         fi
450
451         if [ "$attempt" -gt 15 ] && [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
452             [[ "$local_ip" != "169.254.*" ]]; then
453             local fallback_gw
454             fallback_gw=$(route get default 2>> output.log | awk '/gateway:/{print $2}')
455             if [ -n "$fallback_gw" ] && [[ "$fallback_gw" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]]; then
456                 PHYSICAL_GW="$fallback_gw"
457                 fi
458                 echo "static/settled detected (Interface IP: $local_ip, Gateway IP: ${PHYSICAL_GW:-unknown})."
459                 settled=true
460                 break
461             fi
462             echo -n "."
463             sleep 0.5
464         done
465         if [ "$settled" = "false" ]; then
466             echo "timed out or no active link. Proceeding anyway."
467         fi
468     }
469
470 reset_sharing_services() {
471     if [[ "$OSTYPE" == "darwin"* ]]; then
472         sudo -n killall sharingd rapportd 2>> output.log || true
473     fi
474 }
475
476 read_adguard_ip() {
477     if [ -f ".gateway_ip" ]; then
478         tr -d '\r' < ".gateway_ip" | awk 'NR==1{print $1;exit}'
479     fi
480 }
481
482 is_kill_switch_enabled() {
483     [ "$KILL_SWITCH" = "true" ]
484 }
485
486 add_warp_bypass_routes() {
487     local target="${1:-}"
488     local ep1
489     ep1=$(get_warp_endpoint_1)
490     local ep2
491     ep2=$(get_warp_endpoint_2)
492
493     if [[ "$OSTYPE" == "darwin"* ]]; then
494         local routing_arg=""
495         local msg_via=""
496         if [ -n "$target" ]; then
497             if [[ "$target" =~ ^en[0-9]+$ || "$target" == "bridge"* ]]; then
498                 routing_arg="-interface $target"
499                 msg_via="interface $target"
500             else
501                 routing_arg="$target"
502                 msg_via="gateway $target"
503             fi
504         fi
505     else
506

```

```

503     local gateway_ip
504     gateway_ip=$(route get default 2>> output.log | awk '/gateway:/ {print $2}')
505     if [ -z "$gateway_ip" ]; then
506         local iface
507         iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
508         if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
509             iface="en0"
510         fi
511         routing_arg="-interface $iface"
512         msg_via="interface $iface"
513     else
514         routing_arg="$gateway_ip"
515         msg_via="gateway $gateway_ip"
516     fi
517 fi
518
519 echo -e "\nAdding host bypass routes for Cloudflare WARP endpoints via $msg_via..."
520 sudo -n route delete -host "$sep1" 2>/dev/null || true
521 sudo -n route delete -host "$sep2" 2>/dev/null || true
522 sudo -n route add -host "$sep1" $routing_arg >> output.log 2>&1 || true
523 sudo -n route add -host "$sep2" $routing_arg >> output.log 2>&1 || true
524
525 echo -e "Adding host bypass routes for Tailscale control plane via $msg_via..."
526 sudo -n route delete -net 192.200.0.0/24 2>/dev/null || true
527 sudo -n route add -net 192.200.0.0/24 $routing_arg >> output.log 2>&1 || true
528
529 echo -e "Adding host bypass routes for Tailscale DERP relays via $msg_via..."
530 local derp_ips
531 derp_ips=$(curl -s --connect-timeout 5 https://login.tailscale.com/derpmap/default | grep -oE '"IPv4"[[:
532 space:]]*:[[:space:]]*" [0-9.]+"' | cut -d'"' -f4 | grep -E '^ [0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$' || true)
533 if [ -n "$derp_ips" ]; then
534     echo "$derp_ips" > /tmp/nullexit-derp-ips.txt
535     for ip in $derp_ips; do
536         sudo -n route delete -host "$ip" 2>/dev/null || true
537         sudo -n route add -host "$ip" $routing_arg >> output.log 2>&1 || true
538     done
539 fi
540 else
541     local gateway_ip="${target}"
542     if [ -z "$gateway_ip" ]; then
543         gateway_ip=$(ip route show default 2>/dev/null | awk '/default/ {print $3}')
544     fi
545     if [ -n "$gateway_ip" ]; then
546         sudo ip route add "$sep1" via "$gateway_ip" >> output.log 2>&1 || true
547         sudo ip route add "$sep2" via "$gateway_ip" >> output.log 2>&1 || true
548     fi
549 fi
550 }
551
552 remove_warp_bypass_routes() {
553     local ep1
554     ep1=$(get_warp_endpoint_1)
555     local ep2
556     ep2=$(get_warp_endpoint_2)
557
558     if [[ "$OSTYPE" == "darwin"* ]]; then
559         echo -e "\nRemoving host bypass routes for Cloudflare WARP endpoints..."
560         sudo -n route delete -host "$sep1" >> output.log 2>&1 || true
561         sudo -n route delete -host "$sep2" >> output.log 2>&1 || true
562         echo -e "Removing host bypass routes for Tailscale control plane..."
563         sudo -n route delete -net 192.200.0.0/24 >> output.log 2>&1 || true
564         if [ -f /tmp/nullexit-derp-ips.txt ]; then
565             echo -e "Removing host bypass routes for Tailscale DERP relays..."
566             while read -r ip; do
567                 sudo -n route delete -host "$ip" >> output.log 2>&1 || true
568             done < /tmp/nullexit-derp-ips.txt
569             rm -f /tmp/nullexit-derp-ips.txt
570         fi
571     else
572         sudo ip route del "$sep1" >> output.log 2>&1 || true
573         sudo ip route del "$sep2" >> output.log 2>&1 || true
574     fi
575 }
576
577 bounce_wifi_interfaces() {
578     if [[ "$OSTYPE" == "darwin"* ]]; then
579         local wifi_port
580         wifi_port=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print
581 $2}')
582         if [ -n "$wifi_port" ]; then
583             sudo -n networksetup -setairportpower "$wifi_port" off 2>> output.log || true

```

```

582     sleep 2
583     sudo -n networksetup -setairportpower "$wifi_port" on 2>> output.log || true
584     echo "Wi-Fi bounced (interface $wifi_port)"
585     sleep 3
586 else
587     echo "Could not detect Wi-Fi interface. Bouncing fallback interfaces..."
588     set +e
589     for iface in en0 en1 en2 en3 en4 en5; do
590         if ifconfig "$iface" >> output.log 2>&1; then
591             sudo -n ifconfig "$iface" down 2>> output.log || true
592             sudo -n ifconfig "$iface" up 2>> output.log || true
593         fi
594     done
595     set -e
596     echo "Fallback interfaces bounced"
597 fi
598 fi
599 }
600
601 get_active_interface() {
602     local iface
603     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
604
605     # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
606     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
607         for i in en0 en1 en2 en3; do
608             if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "
inet "; then
609                 iface="$i"
610                 break
611             fi
612         done
613     fi
614
615     # Default fallback if still empty or not enX
616     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
617         iface="en0"
618     fi
619
620     echo "$iface"
621 }
622
623 enable_killswitch() {
624     if [[ "$OSTYPE" == "darwin"* ]]; then
625         if ! is_kill_switch_enabled; then
626             return 0
627         fi
628
629         local ext_if
630         ext_if=$(get_active_interface)
631         local ep1
632         ep1=$(get_warp_endpoint_1)
633         local ep2
634         ep2=$(get_warp_endpoint_2)
635         local pf_conf_path
636         pf_conf_path="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/pf.conf"
637
638         echo -e "\nEnabling macOS Packet Filter (PF) Kill-Switch on $ext_if..."
639
640         # Enable PF and capture the reference token
641         local token_out
642         token_out=$(sudo -n pfctl -E 2>> output.log || true)
643         local token
644         token=$(echo "$token_out" | awk '/Token/{print $NF}')
645         if [ -n "$token" ]; then
646             echo "$token" > /tmp/nullexit-pf-token.txt
647             echo " [] PF enabled with token $token."
648         else
649             echo " [!] WARNING: Could not capture PF reference token (may already be enabled or sudo failed)."
650         fi
651
652         # Load the rules into the anchor
653         if sudo -n pfctl -D "ext_if=$ext_if" -a com.apple/nullexit -f "$pf_conf_path" >> output.log 2>&1; then
654             echo " [] Ruleset loaded into anchor com.apple/nullexit."
655         else
656             echo " [!] ERROR: Failed to load ruleset into anchor."
657         fi
658
659         # Populate tables in the anchor
660         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T flush >> output.log 2>&1 || true
661         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$ep1" >> output.log 2>&1 || true

```

```

662 sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$ep2" >> output.log 2>&1 || true
663
664 if [ -f /tmp/nullexit-derp-ips.txt ]; then
665     local derp_ips
666     derp_ips=$(tr '\n' ' ' < /tmp/nullexit-derp-ips.txt)
667     if [ -n "$derp_ips" ]; then
668         sudo -n pfctl -a com.apple/nullexit -t derp_relays -T flush >> output.log 2>&1 || true
669         sudo -n pfctl -a com.apple/nullexit -t derp_relays -T add $derp_ips >> output.log 2>&1 || true
670     fi
671 fi
672 echo " [] Kill-switch tables populated."
673 fi
674 }
675
676 disable_killswitch() {
677     if [[ "$OSTYPE" == "darwin"* ]]; then
678         if ! is_kill_switch_enabled; then
679             return 0
680         fi
681         echo -e "\nDisabling macOS PF Kill-Switch..."
682
683         # Flush rules from anchor com.apple/nullexit
684         sudo -n pfctl -a com.apple/nullexit -F all >> output.log 2>&1 || true
685
686         # Release the enable reference if token is present
687         if [ -f /tmp/nullexit-pf-token.txt ]; then
688             local token
689             token=$(cat /tmp/nullexit-pf-token.txt)
690             if [ -n "$token" ]; then
691                 sudo -n pfctl -X "$token" >> output.log 2>&1 || true
692                 echo " [] Released PF enable reference token $token."
693             fi
694             rm -f /tmp/nullexit-pf-token.txt
695         else
696             echo " [] Rules flushed from anchor com.apple/nullexit."
697         fi
698     fi
699 }
700
701 write_host_ips() {
702     # Gather all active IPv4 addresses on the host and write to .host_ips
703     # routing-fix.sh will read this file to explicitly whitelist them for direct Tailscale P2P
704     local repo_dir
705     repo_dir="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
706
707     if [[ "$OSTYPE" == "darwin"* ]]; then
708         ifconfig | awk '/inet /{print $2}' | grep -v '127.0.0.1' > "$repo_dir/.host_ips" 2>/dev/null || true
709     else
710         ip -4 addr show 2>/dev/null | awk '/inet /{print $2}' | cut -d/ -f1 | grep -v '127.0.0.1' > "$repo_dir/.
711         host_ips" 2>/dev/null || true
712     fi
713 }
714
715 start_dns_watcher() {
716     if [[ "$OSTYPE" == "darwin"* ]]; then
717         local target_ip=$1
718         stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
719         echo " Starting background DNS Watcher for seamless Wi-Fi roaming..."
720         nohup bash -c "
721             source \"$SCRIPT_DIR/scripts/common.sh\"
722             trap 'exit 0' SIGTERM SIGINT SIGHUP
723             while true; do
724                 ACTIVE_IF=$(get_active_service)
725                 if [ -n \"$ACTIVE_IF\" ]; then
726                     CURRENT_DNS=$(networksetup -getdnsservers \"$ACTIVE_IF\" 2>/dev/null)
727                     if [ \"$CURRENT_DNS\" != \"$target_ip\" ]; then
728                         networksetup -setdnsservers \"$ACTIVE_IF\" \"$target_ip\" >/dev/null 2>&1
729                     fi
730                 fi
731                 sleep 30
732             done
733         " >> output.log 2>&1 &
734         echo $! > "$DNS_WATCHER_PID_FILE"
735     else
736         local target_ip=$1
737         stop_dns_watcher
738         echo " Starting background DNS Watcher for seamless Wi-Fi roaming..."
739         nohup bash -c "
740             trap 'exit 0' SIGTERM SIGINT SIGHUP
741             while true; do

```

```

742         if [ -n "\$ACTIVE_SERVICE" ]; then
743             CURRENT_DNS=\$(resolvectl dns "\$ACTIVE_SERVICE" 2>/dev/null | grep -oE
              '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head -1)
744             if [ "\$CURRENT_DNS" != "\$target_ip" ]; then
745                 resolvectl dns "\$ACTIVE_SERVICE" "\$target_ip" >/dev/null 2>&1 || true
746             fi
747         fi
748         sleep 30
749     done
750     " >> output.log 2>&1 &
751     echo \$! > "\$DNS_WATCHER_PID_FILE"
752 fi
753 }
754
755 # SOCKS5 Proxy (WARP Tunnel)
756 # When Tailscale's exit node can't route through WARP (due to userspace
757 # forwarding), we use a SOCKS5 proxy running inside the warp container's
758 # network namespace. Connections created by this proxy go through the
759 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
760 # userspace exit node which bypasses it.
761 #
762 # The SOCKS5 proxy is exposed on localhost:\${SOCKS_PROXY_PORT} via Docker port mapping.
763 # `networksetup -setsocksfirewallproxy` on macOS routes system TCP traffic
764 # through the proxy, which then goes through WARP.
765 #
766 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
767 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
768
769 enable_socks_proxy() {
770     if [[ "$OSTYPE" == "darwin"* ]]; then
771         local svc="$1"
772         if [ -z "$svc" ]; then
773             svc="\$ACTIVE_SERVICE"
774         fi
775
776         echo -n " Enabling system-wide SOCKS5 proxy on $svc (localhost:\$SOCKS_PROXY_PORT)... "
777         sudo -n networksetup -setsocksfirewallproxy "$svc" 127.0.0.1 \$SOCKS_PROXY_PORT 2>> output.log || true
778         sudo -n networksetup -setsocksfirewallproxystate "$svc" on 2>> output.log || true
779
780         # Also set on en0 service for macOS resolver consistency
781         if [ "$svc" != "\$ENO_SERVICE" ]; then
782             sudo -n networksetup -setsocksfirewallproxy "\$ENO_SERVICE" 127.0.0.1 \$SOCKS_PROXY_PORT 2>> output.log ||
              true
783             sudo -n networksetup -setsocksfirewallproxystate "\$ENO_SERVICE" on 2>> output.log || true
784         fi
785
786         # IMPORTANT: Exclude local LAN and mDNS traffic from the proxy so P2P works natively
787         # without source-IP masking from the Colima VM bridge.
788         sudo -n networksetup -setproxybypassdomains "$svc" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
              172.16.0.0/12 "*.local" 2>> output.log || true
789         if [ "$svc" != "\$ENO_SERVICE" ]; then
790             sudo -n networksetup -setproxybypassdomains "\$ENO_SERVICE" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
              172.16.0.0/12 "*.local" 2>> output.log || true
791         fi
792
793         echo "done."
794         echo " All TCP traffic now routed through gateway -> WARP tunnel -> internet."
795         echo " (CLI tools: export ALL_PROXY=socks5://127.0.0.1:\$SOCKS_PROXY_PORT)"
796     else
797         local svc="$1"
798         echo " (Linux global SOCKS proxy configuration is desktop-environment specific, skipping.)"
799         echo " SOCKS5 proxy is active and listening on 127.0.0.1:\$SOCKS_PROXY_PORT"
800     fi
801 }
802
803 disable_socks_proxy() {
804     if [[ "$OSTYPE" == "darwin"* ]]; then
805         for svc in "\$ACTIVE_SERVICE" "\$ENO_SERVICE"; do
806             sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
807         done
808         echo " SOCKS5 proxy disabled."
809     else
810         local svc="$1"
811         echo " (Linux global SOCKS proxy configuration skipped.)"
812     fi
813 }
814
815 # Helper to restore host DNS to a clean state (used on script start, on failure,
816 # and after a successful STOP). Without this, a successful toggle-off leaves the
817 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
818 # DNS timeout (-5s) before falling through to whatever next server is configured.

```

```

819 # With it, we always leave the host in `1.1.1.1 / no search domain`.
820 #
821 # We set DNS on BOTH the active service AND the en0 service because macOS's scutil DNS
822 # resolver is commonly scoped to en0 (Wi-Fi). A per-service change on e.g.
823 # "USB 10/100 LAN" is ignored by the system resolver unless the en0 service is also updated.
824 reset_dns() {
825     if [[ "$OSTYPE" == "darwin"* ]]; then
826         if [ -n "$ACTIVE_SERVICE" ]; then
827             for dns_svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
828                 networksetup -setsearchdomains "$dns_svc" "Empty" 2>> output.log || true
829                 networksetup -setdnsservers "$dns_svc" 1.1.1.1 2>> output.log || true
830             done
831         fi
832     else
833         if [ -n "$ACTIVE_SERVICE" ]; then
834             resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
835             resolvectl revert "$ACTIVE_SERVICE" 2>/dev/null || true
836         fi
837     fi
838 }
839
840 # Verify basic direct-internet connectivity after a recovery/teardown: DNS via
841 # 1.1.1.1, then HTTP (curl) or ping to 1.1.1.1. Sets the global INTERNET_OK to
842 # "true" on any successful check (callers read $INTERNET_OK in their summary).
843 # Always returns 0 so a bare call can't trip `set -e`. Used by recover.sh's macOS
844 # and Linux recovery paths (extracted from a formerly-duplicated block).
845 verify_internet_connectivity() {
846     step "Verifying internet connectivity"
847
848     # Wait a moment for the network to settle
849     sleep 2
850
851     INTERNET_OK=false
852
853     # Try DNS resolution first
854     if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
855         ok "DNS resolution works (google.com via 1.1.1.1)"
856         INTERNET_OK=true
857     elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
858         ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
859         INTERNET_OK=true
860     else
861         warn "DNS resolution check failed will still try ping/curl"
862     fi
863
864     # Try actual HTTP connectivity
865     if command -v curl >> output.log 2>&1; then
866         if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
867             ok "Internet reachable via HTTP"
868             INTERNET_OK=true
869         elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
870             ok "Internet reachable via HTTP (plain)"
871             INTERNET_OK=true
872         else
873             warn "HTTP check failed"
874         fi
875     elif command -v ping >> output.log 2>&1; then
876         if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
877             ok "Internet reachable via ping"
878             INTERNET_OK=true
879         else
880             warn "Ping check failed"
881         fi
882     fi
883
884     return 0
885 }
886
887 # Local DNS Proxy (Python)
888 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
889 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
890 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.
891 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
892 # correctly unlike socat which just forwards raw bytes.
893 #
894 # Python3 is built into macOS no external dependencies.
895 DNS_PROXY_PID=""
896
897 # Kill any leftover DNS proxy process
898 stop_local_dns_proxy() {
899     if [ -n "$DNS_PROXY_PID" ]; then

```

```

900 kill "$DNS_PROXY_PID" 2>/dev/null || true
901 wait "$DNS_PROXY_PID" 2>/dev/null || true
902 DNS_PROXY_PID=""
903 fi
904 # Clean up any stale Python DNS proxy processes
905 sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
906 }
907
908 # Start local DNS proxy (Python)
909 start_local_dns_proxy() {
910     if [ -z "$DNS_PROXY_BIN" ]; then
911         echo " Python not found. Python3 is built into macOS this shouldn't happen."
912         return 1
913     fi
914
915     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
916         echo " DNS proxy script not found at $DNS_PROXY_SCRIPT"
917         return 1
918     fi
919
920     # Kill any leftover process first
921     stop_local_dns_proxy
922
923     echo -n " Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
924     sudo -n bash -c "TARGET_PORT=$DNS_PROXY_PORT \"$DNS_PROXY_BIN\" \"$DNS_PROXY_SCRIPT\" \" &
925     DNS_PROXY_PID=$!
926     disown "$DNS_PROXY_PID" 2>/dev/null || true
927
928     # Give it a moment to bind
929     sleep 0.5
930
931     if kill -0 "$DNS_PROXY_PID" 2>/dev/null; then
932         echo "started (PID $DNS_PROXY_PID).\"
933
934         # Hijack host DNS to localhost
935         echo -n " Hijacking host DNS to 127.0.0.1 for ad-blocking... \"
936         if [[ "$OSTYPE" == "darwin"* ]]; then
937             networksetup -setsearchdomains "$ACTIVE_SERVICE" "Empty" 2>/dev/null || true
938             networksetup -setsearchdomains "$ENO_SERVICE" "Empty" 2>/dev/null || true
939             if ! networksetup -setdnsservers "$ACTIVE_SERVICE" 127.0.0.1; then
940                 echo "FAILED (networksetup error check permissions).\"
941                 stop_local_dns_proxy
942                 return 1
943             fi
944             networksetup -setdnsservers "$ENO_SERVICE" 127.0.0.1 2>/dev/null || true
945             sudo -n dscacheutil -flushcache 2>/dev/null || true
946             sudo -n killall -HUP mDNSResponder 2>/dev/null || true
947         else
948             resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
949             resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
950             if ! sudo -n resolvectl dns "$ACTIVE_SERVICE" 127.0.0.1; then
951                 echo "FAILED (resolvectl error check permissions).\"
952                 stop_local_dns_proxy
953                 return 1
954             fi
955             resolvectl dns "$ACTIVE_SERVICE" 127.0.0.1 2>/dev/null || true
956             sudo -n resolvectl flush-caches 2>/dev/null || true
957         fi
958
959         # Verify DNS actually works through the proxy
960         echo -n " Verifying DNS resolution... \"
961         if dig @127.0.0.1 google.com +short +timeout=5 &>/dev/null; then
962             echo "ok.\"
963             return 0
964         else
965             echo "FAILED (DNS queries not reaching AdGuard).\"
966             echo " Restoring DNS to 1.1.1.1...\"
967             reset_dns
968             stop_local_dns_proxy
969             return 1
970         fi
971     else
972         echo "FAILED (could not bind port 53 is it in use?).\"
973         DNS_PROXY_PID=""
974         return 1
975     fi
976 }
977
978 # Helper to FORCIBLY hijack host DNS to a single server (the gateway's AdGuard IP)
979 # and VERIFY via `networksetup -getdnsservers` that the only name server is exactly
980 # $1. Returns 0 iff the live state matches; non-zero if it can't be confirmed.

```

```

981 #
982 # This deliberately does NOT append a 1.1.1.1 fallback: macOS's resolver queries
983 # the list in order and falls back to the next entry on timeout, so on a
984 # misbehaving AdGuard (filter compile crash, OOM kill, etc.) the resolver still
985 # leaks to 1.1.1.1 and silently bypasses ad-blocking. With no fallback, a broken
986 # gateway manifests as a *visible* DNS outage that prompts the user to
987 # investigate which is the correct failure mode.
988 force_dns_to_gateway() {
989     if [[ "$OSTYPE" == "darwin"* ]]; then
990         local target_ip="$1"
991         if [ -z "$target_ip" ] || [ -z "$ACTIVE_SERVICE" ]; then
992             echo " [force_dns] ERROR: target_ip or ACTIVE_SERVICE is empty" >&2
993             return 1
994         fi
995
996         local attempt
997         for attempt in 1 2 3; do
998             echo -n " [force_dns] Attempt $attempt/3: setting DNS to $target_ip on \"\$ACTIVE_SERVICE\" + \"\$ENO_SERVICE\"... "
999
1000             # Apply to BOTH the active service AND the en0 service the scutil DNS resolver
1001             # is scoped to en0 (Wi-Fi) and ignores per-service changes that don't include it.
1002             # Fail fast on ACTIVE_SERVICE; best-effort on ENO_SERVICE (it may not exist).
1003             networksetup -setsearchdomains "$ACTIVE_SERVICE" "ts.net" || true
1004             if ! networksetup -setdnsservers "$ACTIVE_SERVICE" "$target_ip"; then
1005                 echo "FAILED (networksetup error check permissions)"
1006                 sleep 1
1007                 continue
1008             fi
1009             networksetup -setsearchdomains "$ENO_SERVICE" "ts.net" || true
1010             networksetup -setdnsservers "$ENO_SERVICE" "$target_ip" || true
1011
1012             local entries
1013             entries=$(networksetup -getdnsservers "$ACTIVE_SERVICE" 2>>> output.log \
1014                 | awk '/^(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\. (25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\.(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\.(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)$/ {print}')
1015             if echo "$entries" | grep -qFx "$target_ip"; then
1016                 echo "VERIFIED"
1017                 return 0
1018             fi
1019
1020             local current
1021             current=$(echo "$entries" | tr '\n' ' ' | sed 's/ $//')
1022             echo "NOT YET (current DNS: [${current:-none}])"
1023
1024             if [ "$attempt" = "3" ]; then sleep 2; else sleep 1; fi
1025             done
1026
1027             echo " [force_dns] FAILED after 3 attempts"
1028             return 1
1029         else
1030             local target_ip="$1"
1031
1032             for attempt in {1..3}; do
1033                 echo -n " [force_dns] Attempt $attempt/3: setting DNS to $target_ip on $ACTIVE_SERVICE... "
1034                 sudo -n resolvectl domain "$ACTIVE_SERVICE" "-ts.net" 2>/dev/null || true
1035                 if ! sudo -n resolvectl dns "$ACTIVE_SERVICE" "$target_ip"; then
1036                     echo "FAILED (resolvectl error)"
1037                 else
1038                     # Verify
1039                     entries=$(resolvectl dns "$ACTIVE_SERVICE" 2>>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+')
1040                     if [ "$entries" == "$target_ip" ]; then
1041                         echo "VERIFIED"
1042                         return 0
1043                     else
1044                         echo "MISMATCH (Resolver refused lock)"
1045                     fi
1046                 fi
1047                 sleep 2
1048             done
1049             return 1
1050         fi
1051     }

```

12 scripts/watcher.sh

```

1 #!/bin/bash
2 # scripts/watcher.sh nullexit post-wake / post-roam recovery watcher
3 #
4 # Long-running daemon. Launched by launchd as a LaunchAgent at
5 # ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
6 # (label com.nullexit.wake-recovery).
7 #
8 # Two independent listeners run as backgrounded pipelines:
9 #
10 # (1) WAKE listener `log stream` live-follows the macOS unified log for
11 # power-management events (lid closewake, manual wake, DarkWake from
12 # clamshell). Each event triggers run_recover (with a global debounce so
13 # a single wake that emits 2-3 log lines only causes ONE post-wake run).
14 #
15 # (2) NETWORK listener `scutil n.watch` on `State:/Network/Global/IPv4`
16 # fires on every Wi-Fi roam, ethernet swap, hotspot join, captive-portal
17 # re-bind, VPN up/down, etc. Same debounce.
18 #
19 # Both listeners call run_recover, which:
20 # - checks /tmp/nullexit-gateway-active.marker (only act when toggle.sh left
21 # the gateway up)
22 # - debounces (10s default) so multiple events don't pile up
23 # - shells out to `bash <repo>/recover.sh --post-wake` directly (no GUI auth prompt needed due to NOPASSWD
24 # sudoers)
25 #
26 # See devref.md 10.29 for the full Apple power management primer and the
27 # rationale behind using `log stream` + `scutil n.watch` from a shell daemon.
28 #
29 # NOTE: errexit (`set -e`) is intentionally NOT enabled. This is a long-running
30 # daemon, not a discrete-step script, and errexit actively breaks the reconnect
31 # loops below: any failed pipe (log stream on rotation, scutil on state churn)
32 # would kill the outer subshell before `echo "reconnecting..."; sleep 5;` runs,
33 # AND the parent's final `wait` would propagate a non-zero child exit and kill
34 # the entire watcher. Every error path is already wrapped in `|| true` /
35 # `2>/dev/null` fallbacks.
36 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
37 source "$SCRIPT_DIR/common.sh"
38 setup_standard_path
39 LOG="$SCRIPT_DIR/../output.log"
40 # Resolve our own directory; recover.sh lives one level up at the repo root.
41 # This makes the daemon install-agnostic no need to template the path
42 # in launchd, no need to keep a deploy-specific hardcoded location in sync.
43 RECOVER="$SCRIPT_DIR/../recover.sh"
44 MARKER="$MARKER_FILE"
45 DEBOUNCE_FILE="/tmp/nullexit-watcher.last-recovery"
46 DEBOUNCE_SECONDS="${NULLEXIT_DEBOUNCE_SECONDS:-10}"
47 # Post-recovery SETTLE cooldown. A post-wake run re-asserts the exit node
48 # (recover.sh: `tailscale set --exit-node`), which re-writes the host default
49 # route and that route IS State:/Network/Global/IPv4, the exact key Listener 2
50 # watches. The self-triggered change lands ~15-45s AFTER the run finishes, past
51 # the 10s debounce, so without a longer window the watcher re-fires post-wake in
52 # a ~40s self-feedback loop that only stops once routing happens to reach a fixed
53 # point (devref 15.12.21). This cooldown is armed ONLY after a recovery runs, so
54 # a genuine first wake/roam is still handled immediately; only the change we
55 # caused ourselves gets absorbed. A real new roam during the window is delayed at
56 # most this many seconds, which is acceptable.
57 COOLDOWN_FILE="/tmp/nullexit-watcher.settle-until"
58 POSTWAKE_SETTLE_SECONDS="${NULLEXIT_POSTWAKE_SETTLE_SECONDS:-45}"
59
60 exec >> "$LOG" 2>&1
61
62 # Single-instance lock
63 # Without this, repeated lid-closewake cycles under macOS Sonoma+ accumulate
64 # orphan watcher.sh processes: launchd suspends on sleep and SIGCONT-resumes
65 # on wake, and KeepAlive.Crashed=true caused relaunch on any non-zero exit;
66 # the listener grandchildren (log stream | while ... and (echo n.add ...;
67 # sleep 86400) | scutil | while ...) stay alive because pipe producers don't
68 # reliably respond to plain SIGTERM if the consumer is in a half-closed state.
69 #
70 # We solve it with a PID-file lock (POSIX-portable; macOS does NOT ship `flock`).
71 # The trap below removes the lock file on every exit so a stale lock never
72 # outlives a crashed watcher. On launch: read the existing PID, check it is
73 # alive AND is actually a `scripts/watcher.sh` process; if so, exit 0; else
74 # overwrite with our PID and proceed. The check-then-write window has a TOCTOU
75 # race but is microseconds wide and only matters for hand-launched duplicate
76 # test invocations; production is single-launch via launchctl load -w.
77 LOCK_FILE="/tmp/nullexit-watcher.lock"
78 LOCK_PID=""
79 if [ -e "$LOCK_FILE" ]; then
80 LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")

```

```

81 if [ -n "$LOCK_PID" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
82     # Verify the holder is actually a scripts/watcher.sh process (not some
83     # unrelated process that got the recycled PID). The exact-launchd pattern
84     # `bash <abs-path>/scripts/watcher.sh` is hard to accidentally match with
85     # a `grep scripts/watcher.sh .` invocation because we anchor on `^bash`
86     # followed by whitespace and end-of-line on the script path.
87     if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -qE "(^|[:space:]))bash[:space:]]+.*scripts/watcher\\.sh$"; then
88         echo "[$(date -u +%FT%TZ)] watcher.sh: another instance holds $LOCK_FILE (pid=$LOCK_PID), exiting"
89         exit 0
90     fi
91 fi
92 fi
93 echo $$ > "$LOCK_FILE"
94
95 echo "[$(date -u +%FT%TZ)] watcher.sh start (pid=$$ ppid=$PPID user=$(id -un) lock=acquired)"
96
97 # Treat launchd-resume as a wake signal. Apple's launchd SUSPENDS LaunchAgents
98 # during system sleep and restores them on wake; during that suspended gap,
99 # macOS's `log stream` cannot catch the wake event that prompted the resume.
100 # The next thing that happens after wake IS our own startup, so we always
101 # fire one post-wake run unconditionally here. The marker check + debounce
102 # in run_recover() make this idempotent: if the gateway isn't up or we just
103 # debounced, it no-ops. Without this, the FIRST wake-after-install goes
104 # unseen and the watcher appears broken until the SECOND wake.
105 # Helper: run recover.sh --post-wake if the gateway is active
106 run_recover() {
107     local why="$1"
108     if [ ! -f "$MARKER" ]; then
109         echo "[$(date -u +%FT%TZ)] $why no $MARKER, skip"
110         return 0
111     fi
112     local now last settle_until
113     now=$(date +%s)
114     # Post-recovery settle window: ignore the Global/IPv4 change our own last
115     # post-wake run caused (exit-node re-assert) so we don't self-feedback loop.
116     settle_until=$(cat "$COOLDOWN_FILE" 2>/dev/null || echo 0)
117     if [ "$now" -lt "$settle_until" ]; then
118         echo "[$(date -u +%FT%TZ)] $why settling ($((settle_until - now))s left in post-recovery cooldown), skip"
119         return 0
120     fi
121     last=$(cat "$DEBOUNCE_FILE" 2>/dev/null || echo 0)
122     if [ "$((now - last))" -lt "$DEBOUNCE_SECONDS" ]; then
123         echo "[$(date -u +%FT%TZ)] $why debounced ($((now - last))s since last, threshold ${DEBOUNCE_SECONDS}s)"
124         return 0
125     fi
126     echo "$now" > "$DEBOUNCE_FILE"
127     echo "[$(date -u +%FT%TZ)] $why bash $RECOVER --post-wake"
128     bash "$RECOVER" --post-wake
129     local exit_code=$?
130     now=$(date +%s)
131     echo "$now" > "$DEBOUNCE_FILE"
132     # Arm the settle cooldown so the route change this run just caused can't re-fire us.
133     echo "$((now + POSTWAKE_SETTLE_SECONDS))" > "$COOLDOWN_FILE"
134     echo "[$(date -u +%FT%TZ)] $why exit=$exit_code (settle until +${POSTWAKE_SETTLE_SECONDS}s, now=$now)"
135     return $exit_code
136 }
137
138 if [ -f "$MARKER" ]; then
139     run_recover "initial post-wake (launchd resume = wake signal)"
140 fi
141
142 # LAN P2P Auto-Detection
143 # Detects whether the current network allows safe Tailscale LAN P2P by:
144 # 1. Checking WPA2-Enterprise (802.1x) via system_profiler SPAirPortDataType always forces P2P=false
145 #    because enterprise networks almost universally have AP Client Isolation.
146 # 2. AP Isolation probe sends a broadcast ping and checks if ANY LAN host responds.
147 #    If no LAN hosts respond on a non-empty network, AP Isolation is active.
148 # Writes 'true' or 'false' to .lan_p2p_detected in the repo root.
149 # routing-fix.sh reads this file dynamically every 30s loop iteration.
150 P2P_OVERRIDE_FILE="$SCRIPT_DIR/../../lan_p2p_detected"
151
152
153 detect_lan_p2p_mode() {
154     local reason="unknown" allow="false"
155
156     # Step 1: Detect Wi-Fi security type via system_profiler (airport binary removed in macOS 14.4+)
157     # SPAirPortDataType lists all known networks; the first "Security:" line is the current association.
158     local security
159     security=$(system_profiler SPAirPortDataType 2>/dev/null \
160         | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF; exit}')

```

```

161
162 if [ -z "$security" ]; then
163     # Not on Wi-Fi or system_profiler unavailable fail safe
164     reason="no-wifi" allow="false"
165 elif echo "$security" | grep -qi "Enterprise"; then
166     # WPA2/WPA3 Enterprise = 802.1x always AP isolated
167     reason="802.1x-enterprise" allow="false"
168 elif echo "$security" | grep -qi "Personal\\None\\Open"; then
169     # WPA2/WPA3 Personal or open probe for AP isolation
170     local gw
171     gw=$(route -n get 1.1.1.1 2>/dev/null | awk '/gateway:/{print $2}')
172     if [ -z "$gw" ]; then
173         reason="no-default-route" allow="false"
174     else
175         local responses
176         responses=$(ping -c 3 -t 1 -q "$gw" 2>/dev/null | awk '/packets transmitted/{print $4}')
177         if [ "${responses:-0}" -gt 0 ]; then
178             reason="wpa-personal-lan-reachable" allow="true"
179         else
180             reason="wpa-personal-ap-isolated" allow="false"
181         fi
182     fi
183 else
184     # Unknown security type fail safe
185     reason="unknown-security-type" allow="false"
186 fi
187
188 echo "[$(date -u +%FT%TZ)] P2P detect: security='${security:-none}' allow=${allow} (reason: ${reason})"
189 echo "$allow" > "$P2P_OVERRIDE_FILE"
190 write_host_ips
191 }
192
193 # Run once on startup
194 detect_lan_p2p_mode
195
196 # Listener 1: WAKE events via unified log
197 # Predicate picks up:
198 # * "didWake"          regular kernel wake events (lid open, RTC alarm)
199 # * "Wake from Sleep"  powerd-driven normal wake
200 # * "DarkWake"         clamshell-display wake that didn't fully wake the
201 #                       Mac (relevant when lid opens during display sleep)
202 # * "Waking from"      generic catch-all for variations in newer macOS
203 # `[c]` makes the match case-insensitive (the unified log uses mixed-case
204 # phrases). --info reduces noise by dropping debug/trace log levels.
205 # Subsystem-based predicate catches every powermanagement emission (willSleep,
206 # didWake, didDim, didUndim, DarkWake, etc.) regardless of message phrasing
207 # across kernel versions. Phrasing-based predicates ('didWake', 'Wake from Sleep')
208 # are fragile across macOS releases. We accept the extra log noise because the
209 # run_recover() debounce + marker check filter out anything that isn't a real
210 # gateway-relevant event.
211 (
212     # Reconnect loop: macOS rotates the unified log buffer periodically; when
213     # that happens, this `log stream` subscriber's pipe EOFs, the inner
214     # `while read` consumer terminates, and without this wrapper the listener
215     # would be silently dead for the rest of the watcher's lifetime.
216     while ;; do
217         log stream --predicate \
218             'subsystem == "com.apple.powermanagement"' \
219             --style compact --info 2>/dev/null \
220             | while IFS= read -r line; do
221                 [ -z "$line" ] && continue
222                 run_recover "WAKE: $(echo "$line" | head -c 100)"
223             done
224         echo "[$(date -u +%FT%TZ)] log stream subshell exited; reconnecting in 5s"
225         sleep 5
226     done
227 ) &
228
229 # Listener 2: NETWORK state changes via scutil
230 # `scutil n.watch` on a state key prints SCEventUpdate lines whenever the key
231 # changes. We watch Global/IPv4 because that's the umbrella that catches link,
232 # address, route, and DNS changes for ALL interfaces in one namespace.
233 # The pipe is kept alive by an `sleep 86400` loop so scutil never sees EOF
234 # from us (which would silently terminate the watch).
235 (
236     # Reconnect loop: scutil can exit noisily on certain state changes (rare,
237     # but observed). Without the wrapper, the network listener's pipe EOFs and
238     # every subsequent roam goes unseen until launchd restarts us.
239     while ;; do
240         (
241             # Watch both IPv4 and IPv6 umbrella state IPv6-only or dual-stack

```

```

242 # networks never fire on the IPv4 key alone, and the user's first roam
243 # in a hotel / airport / on cellular hotspot is often IPv6-first under
244 # NAT64.
245 echo "n.add State:/Network/Global/IPv4"
246 echo "n.add State:/Network/Global/IPv6"
247 echo "n.watch"
248 while true; do sleep 86400; done
249 ) | script -q /dev/null scutil 2>/dev/null \
250 | while IFS= read -r line; do
251     case "$line" in
252         *changedKey*|*State:/Network/Global/IPv*|*n.state*|*SCEventUpdate*)
253             detect_lan_p2p_mode
254             run_recover "NET: $(echo "$line" | head -c 100)"
255             ;;
256         *) ;;
257     esac
258 done
259 echo "[$(date -u +%FT%TZ)] scutil pipe exited; reconnecting in 5s"
260 sleep 5
261 done
262 ) &
263 # Listener 3: TUNNEL_FAILED_CLOSED.marker Watcher
264 # Polls for the fail-closed marker dropped by the routing-fix container.
265 (
266     last_notified=0
267     while :; do
268         if [ -f "$SCRIPT_DIR/../TUNNEL_FAILED_CLOSED.marker" ]; then
269             now=$(date +%s)
270             # Notify once every 5 minutes (300s) if it stays failed
271             if [ "$(now - last_notified)" -ge 300 ]; then
272                 osascript -e 'display notification "VPN Tunnel Health Check Failed. Internet traffic is blackholed to prevent IP leak." with title "NullExit Alert"' || true
273                 last_notified=$now
274             fi
275         else
276             last_notified=0
277         fi
278         sleep 3
279     done
280 ) &
281
282 # Lifetime
283 # `wait` blocks until both background pipelines exit (which they normally
284 # never do). launchctl `bootout`/SIGTERM triggers the trap, which kills the
285 # whole process group so the sleep-forever in the scutil pipe also dies.
286 # We catch TERM/INT/HUP for explicit signals and EXIT for any normal-exit path
287 # (so listeners always get cleaned up even if `exit 0` runs somewhere).
288 # `rm -f $LOCK_FILE` first so a stale lock can never block the next launch.
289 # `pkill -TERM -P $$` then targets direct listener-children, then `kill 0`
290 # takes care of any backgrounded group-mates not reachable via the parent.
291 trap 'echo "[$(date -u +%FT%TZ)] watcher.sh terminating (signal/exit=$?)"; rm -f "$LOCK_FILE" 2>/dev/null || true; pkill -TERM -P $$ 2>/dev/null || true; kill 0 2>/dev/null || true; exit 0' TERM INT HUP EXIT
292 wait

```

13 scripts/crypto.sh

```

1 #!/usr/bin/env bash
2 # crypto.sh - Cryptographic Integrity Check
3
4 set -euo pipefail
5 cd "$(dirname "$0")/.."
6
7 if [ "$#" -ne 1 ] || { [ "$1" != "--sign" ] && [ "$1" != "--verify" ]; }; then
8     echo "Usage: $0 --sign | --verify"
9     exit 1
10 fi
11
12 if [ ! -f .env ]; then
13     echo "Error: .env not found."
14     exit 1
15 fi
16
17 SEED=$(grep "^NULLEXIT_SEED=" .env | cut -d '=' -f2)
18 if [ -z "$SEED" ]; then
19     echo "Error: NULLEXIT_SEED not found in .env."
20     exit 1

```

```

21 fi
22
23 if [ "$1" == "--sign" ]; then
24     echo "Generating cryptographic signatures using seed..."
25     rm -f .signatures
26     for file in toggle.sh recover.sh scripts/common.sh scripts/routing-fix.sh scripts/diagnose-host-leak.sh
27         scripts/sweep.sh scripts/watcher.sh scripts/pf.conf post-rules.txt docker-compose.yml black_list.txt
28         white_list.txt; do
29         if [ -f "$file" ]; then
30             hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
31             echo "$file:$hash" >> .signatures
32             echo " [Signed] $file"
33         else
34             echo " [Skipped] $file not found"
35         fi
36     done
37     echo "Signatures saved to .signatures"
38     exit 0
39 fi
40
41 if [ "$1" == "--verify" ]; then
42     if [ ! -f .signatures ]; then
43         echo "Error: .signatures file missing. Run $0 --sign first."
44         exit 1
45     fi
46     FAIL=0
47     while IFS=: read -r file expected_hash; do
48         if [ -z "$file" ]; then continue; fi
49         if [ ! -f "$file" ]; then
50             echo "[FAIL] $file is missing!"
51             FAIL=1
52             continue
53         fi
54         actual_hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
55         if [ "$actual_hash" != "$expected_hash" ]; then
56             echo "[FAIL] $file has been modified! Hash mismatch."
57             FAIL=1
58         fi
59     done < .signatures
60
61     if [ "$FAIL" -eq 1 ]; then
62         echo ""
63         echo "CRITICAL: Script integrity verification failed!"
64         echo "One or more core files have been modified without authorization."
65         echo "To authorize these changes, run: ./scripts/crypto.sh --sign"
66         echo ""
67         exit 1
68     fi
69 fi

```

14 .aiignore

```

1 # Environment variables and secrets
2 .env
3 tor-bridges.txt

```

15 .claude/settings.local.json

```

1 {
2   "permissions": {
3     "allow": [
4       "Bash(grep *)"
5     ]
6   }
7 }

```

16 .gitignore

```

1 # Environment variables and secrets
2 .env
3
4 # wgcf generated files containing keys and account info
5 wgcf-account.toml
6 wgcf-profile.conf
7
8 # Tailscale persistent state containing node identity
9 tailscale-state/
10
11 # Binaries
12 wgcf
13
14 # macOS App bundles
15 *.app/
16 *.app
17
18 # AdGuard Home persistent state
19 adguard/
20 .gateway_ip
21 logs.txt
22
23 #
24 stability_incident_report.md
25 output.log
26 .DS_Store
27 *.desktop
28 blocked.log
29 wtf.md
30 host-leak-diagnostic-*.txt
31 nullexit_review.md
32 nullexit_unified.tex
33 nullexit_unified.txt
34
35 nullexit_unified.synctex.gz
36 tor-bridges.txt
37
38 # Runtime state files
39 .host_ips
40 .lan_p2p_detected
41 .build_hash
42 .rules_hash
43 .signatures
44 TUNNEL_FAILED_CLOSED.marker
45
46 # Python cache
47 __pycache__/_
48 *.py[cod]
49
50 # LaTeX build artifacts
51 *.aux
52 *.log
53 *.out
54 *.toc
55 *.xdv
56 *.fdb_latexmk
57 *.fls
58 *.synctex*
59 refactor.md
60 sweep.md

```

17 Recover-Gateway.applescript

```

1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; bash recover.sh"
10 end tell

```

18 Toggle-Gateway.applescript

```
1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; ./toggle.sh"
10 end tell
```

19 benchmarks/benchmark.sh

```
1 #!/usr/bin/env bash
2
3 # benchmark.sh
4 # Automates the full benchmarking suite (both Python download test and Lighthouse HTML test)
5 # Runs the suite with Gateway ON, toggles OFF, runs again, then toggles back ON.
6
7 set -e
8
9 # Change to project root so we can access toggle.sh reliably
10 cd "$(dirname "$0")/.."
11
12 # Ensure jq is installed
13 if ! command -v jq &> /dev/null; then
14     echo "Error: 'jq' is not installed. Please run 'brew install jq'"
15     exit 1
16 fi
17
18 run_lighthouse() {
19     local url=$1
20     local file_prefix=$2
21     echo " [Lighthouse] Testing $url ..."
22
23     # Run lighthouse headlessly, silencing standard output
24     npx -y lighthouse "$url" --chrome-flags="--headless" --only-categories=performance --output=json --output-path="${file_prefix}.json" >/dev/null 2>&1
25
26     # Parse the results
27     local score=$(jq -r '.categories.performance.score' "${file_prefix}.json")
28     local tti=$(jq -r '.audits["interactive"].displayValue' "${file_prefix}.json")
29     local si=$(jq -r '.audits["speed-index"].displayValue' "${file_prefix}.json")
30
31     # Convert score to percentage
32     local score_pct=$(awk "BEGIN {print $score*100}")
33
34     echo "    -> Performance Score: ${score_pct}%"
35     echo "    -> Time to Interactive: $tti"
36     echo "    -> Speed Index: $si"
37
38     # Cleanup
39     rm -f "${file_prefix}.json"
40 }
41
42 echo "=====
43 echo " PHASE 1: Testing with Gateway ON"
44 echo "=====
45 # 1. Run raw Python download test
46 python3 benchmarks/test_load.py
47
48 # 2. Run Lighthouse HTML render test
49 echo ""
50 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
51 run_lighthouse "https://www.cnet.com/" "on_cnet"
52 run_lighthouse "https://www.independent.co.uk/" "on_ind"
53
54
55 echo ""
56 echo "=====
57 echo " PHASE 2: Toggling Gateway OFF"
58 echo "=====
59 ./toggle.sh
```

```

60 echo "Waiting 10 seconds for macOS Wi-Fi to stabilize after DNS reset..."
61 sleep 10
62
63
64 echo ""
65 echo "=====
66 echo "    PHASE 3: Testing with Gateway OFF"
67 echo "=====
68 # 1. Run raw Python download test
69 python3 benchmarks/test_load.py
70
71 # 2. Run Lighthouse HTML render test
72 echo ""
73 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
74 run_lighthouse "https://www.cnet.com/" "off_cnet"
75 run_lighthouse "https://www.independent.co.uk/" "off_ind"
76
77
78 echo ""
79 echo "=====
80 echo "    PHASE 4: Restoring Gateway ON"
81 echo "=====
82 ./toggle.sh
83
84 echo ""
85 echo "=====
86 echo " BENCHMARKS COMPLETE!"
87 echo "You can scroll up to compare Phase 1 (AdGuard ON) vs Phase 3 (AdGuard OFF)."
88 echo "=====

```

20 benchmarks/test_load.py

```

1 import urllib.request
2 import urllib.error
3 import time
4 from html.parser import HTMLParser
5 import concurrent.futures
6
7 class ResourceParser(HTMLParser):
8     def __init__(self):
9         super().__init__()
10        self.urls = set()
11
12    def handle_starttag(self, tag, attrs):
13        attrs = dict(attrs)
14        url = None
15        if tag in ['img', 'script']:
16            url = attrs.get('src')
17        elif tag == 'link' and attrs.get('rel') == 'stylesheet':
18            url = attrs.get('href')
19
20        if url and url.startswith('http'):
21            self.urls.add(url)
22
23 def fetch_url(url):
24     try:
25         req = urllib.request.Request(url, headers={'User-Agent': 'Mozilla/5.0'})
26         urllib.request.urlopen(req, timeout=3)
27         return True
28     except Exception:
29         return False
30
31 def measure_url(target_url):
32     print(f"\nTesting {target_url} Load...")
33     start_total = time.time()
34
35     # Fetch main HTML
36     try:
37         req = urllib.request.Request(target_url, headers={'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36'})
38         response = urllib.request.urlopen(req, timeout=5)
39         html = response.read().decode('utf-8', errors='ignore')
40     except Exception as e:
41         print(f"Failed to fetch {target_url}: {e}")
42     return
43

```

```

44     parser = ResourceParser()
45     parser.feed(html)
46     urls = list(parser.urls)
47     print(f"Found {len(urls)} subresources to fetch (scripts, images, css)...")
48
49     # Concurrently fetch subresources
50     success_count = 0
51     fail_count = 0
52     max_threads = min(64, max(1, len(urls)))
53     with concurrent.futures.ThreadPoolExecutor(max_workers=max_threads) as executor:
54         results = list(executor.map(fetch_url, urls))
55
56     success_count = sum(1 for r in results if r)
57     fail_count = len(results) - success_count
58
59     total_time = time.time() - start_total
60     print(f"Time taken: {total_time:.2f} seconds")
61     print(f"Successfully fetched: {success_count}, Failed/Blocked: {fail_count}")
62
63     urls_to_test = [
64         "https://www.dailymail.co.uk/home/index.html",
65         "https://www.cnet.com/",
66         "https://www.independent.co.uk/"
67     ]
68
69     for u in urls_to_test:
70         measure_url(u)

```

21 black_list.txt

```

1  ad.doubleclick.net
2  adclick.g.doubleclick.net
3  adeventtracker.spotify.com
4  adfstat.yandex.ru
5  ads-api.tiktok.com
6  ads-api.twitter.com
7  ads-fa.spotify.com
8  ads-sg.tiktok.com
9  ads.tiktok.com
10 adserver.unityads.unity3d.com
11 adsfs.oppomobile.com
12 an.facebook.com$important
13 connect.facebook.net$important
14 pixel.facebook.com$important
15 analytics.query.yahoo.com
16 analytics.s3.amazonaws.com
17 analytics.spotify.com
18 analyticsengine.s3.amazonaws.com
19 appmetrica.yandex.ru
20 audio2.spotify.com
21 bdapi-ads.realmemobile.com
22 bdapi-in-ads.realmemobile.com
23 bounceexchange.com
24 bs.serving-sys.com
25 business-api.tiktok.com
26 ck.ads.oppomobile.com
27 crashdump.spotify.com
28 data.ads.oppomobile.com
29 data.mistat.india.xiaomi.com
30 data.mistat.rus.xiaomi.com
31 data.mistat.xiaomi.com
32 desktop.spotify.com
33 doubleclick.net
34 ds.metric.spotify.com
35 gemini.yahoo.com
36 googleads.g.doubleclick.net
37 googleadservices.com
38 googlesyndication.com
39 grs.hicloud.com
40 gtm.spotify.com
41 iot-eu-logser.realme.com
42 iot-logser.realme.com
43 log.byteoversea.com
44 log.fc.yahoo.com
45 log.spotify.com
46 m.doubleclick.net

```

```

47 mediavisor.doubleclick.net
48 metrics.spotify.com
49 metrika.yandex.ru
50 omaze.com
51 pagead2.googlesyndication.com
52 securepubads.g.doubleclick.net
53 static.doubleclick.net
54 stats.g.doubleclick.net
55 tracking.rus.miui.com
56 udcml.yahoo.com
57 v.jwpcdn.com
58 weblb-wg.gslb.spotify.com
59 webview.unityads.unity3d.com
60 diagassets.apple.com
61 iphonesubmissions.apple.com
62 radarsubmissions.apple.com
63 metrics.apple.com
64 xp.apple.com
65 heads-fa.spotify.com
66 heads4.spotify.com
67 pixel.spotify.com
68 segs.spotify.com
69 audio-fa.spotify.com
70 events.data.microsoft.com
71 datarouter-prod.ak.epicgames.com
72 nel.cloudflare.com
73 eu-mobile.events.data.microsoft.com
74 stocks-analytics-events.apple.com

```

22 cloud-init.yaml

```

1 #cloud-config
2 # Generic Cloud-Init Bootstrap Script for Remote Exit Nodes (Ubuntu/Debian)
3 # Paste this into the "User Data" or "Initialization Script" section when creating a VPS.
4
5 package_update: true
6 package_upgrade: true
7
8 packages:
9   - apt-transport-https
10  - ca-certificates
11  - curl
12  - gnupg
13  - lsb-release
14  - git
15  - python3
16  - python3-pip
17
18 runcmd:
19   # 1. Install Docker natively
20   - curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-
     archive-keyring.gpg
21   - echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
     https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/
     docker.list > /dev/null
22   - sudo apt-get update
23   - sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
24
25   # 2. Add default user (ubuntu) to docker group
26   - usermod -aG docker ubuntu || true
27   - usermod -aG docker debian || true
28
29   # 3. Create the installation directory
30   - mkdir -p /opt/nullexit
31   - chown -R ubuntu:ubuntu /opt/nullexit || chown -R debian:debian /opt/nullexit
32
33   # 4. Enable IPv4 and IPv6 forwarding for VPN routing
34   - echo "net.ipv4.ip_forward=1" >> /etc/sysctl.d/99-tailscale.conf
35   - echo "net.ipv6.conf.all.forwarding=1" >> /etc/sysctl.d/99-tailscale.conf
36   - sysctl -p /etc/sysctl.d/99-tailscale.conf
37
38 final_message: "The remote exit node has been fully bootstrapped. SSH in, clone the repo to /opt/nullexit, add
     your .env, and run ./setup-linux.sh!"

```

23 docker/tor/Dockerfile

```
1 FROM debian:bookworm-slim
2
3 RUN apt-get update && \
4     apt-get install -y tor obfs4proxy gosu bash && \
5     rm -rf /var/lib/apt/lists/*
6
7 COPY entrypoint.sh /entrypoint.sh
8 RUN chmod +x /entrypoint.sh
9
10 ENTRYPOINT ["/entrypoint.sh"]
```

24 docker/tor/entrypoint.sh

```
1 #!/bin/bash
2 set -e
3
4 TORRC="/etc/tor/torrc"
5 mkdir -p /etc/tor /var/lib/tor
6
7 # Base configuration: SOCKS 9050, Control 9051, TransPort 9040, DNSPort 5353 all bound to
8 # 0.0.0.0 (required for Docker/Colima host port-mapping). The ControlPort is secured via a
9 # dynamic HashedControlPassword (added below when TOR_PASSWORD is set); see devref.md 15.10.2.
10 cat <<EOF > $TORRC
11 SocksPort 0.0.0.0:9050
12 ControlPort 0.0.0.0:9051
13 CookieAuthentication 0
14 Log notice stdout
15 DataDirectory /var/lib/tor
16
17 # Transparent proxying & DNS resolution for .onion mapping
18 TransPort 0.0.0.0:9040
19 DNSPort 0.0.0.0:5353
20 AutomapHostsOnResolve 1
21 VirtualAddrNetworkIPv4 198.18.0.0/15
22 EOF
23
24 if [ -n "$TOR_PASSWORD" ]; then
25     HASH=$(tor --hash-password "$TOR_PASSWORD" | tail -n 1)
26     echo "HashedControlPassword $HASH" >> $TORRC
27 fi
28
29 if [ -n "$TOR_EXCLUDE_EXIT_NODES" ]; then
30     echo "ExcludeExitNodes $TOR_EXCLUDE_EXIT_NODES" >> $TORRC
31     echo "StrictNodes 1" >> $TORRC
32 fi
33
34 if [ "${TOR_USE_BRIDGES:-false}" = "true" ]; then
35     BRIDGE_FILE="${TOR_BRIDGE_LINES_FILE:-/tor-bridges.txt}"
36
37     if ! grep -vE "^[[:space:]]*#" "$BRIDGE_FILE" | grep -q "^[[:space:]]"; then
38         MSG="[(date '+%Y-%m-%d %H:%M:%S')] [CRITICAL] [Tor Proxy] TOR_USE_BRIDGES is enabled, but no bridge
39         lines found at $BRIDGE_FILE. Proxy startup ABORTED. Get bridges from https://bridges.torproject.org."
40         echo "$MSG"
41         if [ -w "/output.log" ]; then
42             echo "$MSG" >> /output.log
43         fi
44         # Sleep indefinitely to prevent docker-compose 'unless-stopped' from triggering a crash loop
45         exec sleep infinity
46     fi
47
48     echo "UseBridges 1" >> $TORRC
49     echo "ClientTransportPlugin obfs4 exec /usr/bin/obfs4proxy" >> $TORRC
50
51     while IFS= read -r line; do
52         # Ignore comments and empty lines
53         [[ -z "$line" || "$line" == \#* ]] && continue
54         echo "Bridge $line" >> $TORRC
55     done < "$BRIDGE_FILE"
56 fi
57
58 # Debian tor package uses 'debian-tor' user
59 chown -R debian-tor:debian-tor /var/lib/tor /etc/tor
```

```
60 exec gosu debian-tor /usr/bin/tor -f $TORRC
```

25 launchd/com.nullexit.wake-recovery.plist

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <!--DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
3 <plist version="1.0">
4 <dict>
5   <key>Label</key>
6   <string>com.nullexit.wake-recovery</string>
7
8   <!-- WorkingDirectory + relative program arg keeps this plist install-agnostic.
9        The source-of-truth for the install path is __NULLEXIT_HOME__, which
10       setup.sh or the user must substitute with their absolute nullexit
11       install root at install time. Program arg is `scripts/watcher.sh`
12       RELATIVE to WorkingDirectory, so launchd's evaluated cwd IS the
13       install root and BASH_SOURCE inside scripts/watcher.sh still
14       resolves to the real <install>/scripts/watcher.sh, preserving the
15       SCRIPT_DIR pattern. After install-time `sed -i ' ' 's|__NULLEXIT_HOME__|<abs_path>|'`,
16       the plist is portable to any clone location. -->
17   <key>WorkingDirectory</key>
18   <string>__NULLEXIT_HOME__</string>
19
20   <key>ProgramArguments</key>
21   <array>
22     <string>/bin/bash</string>
23     <string>scripts/watcher.sh</string>
24   </array>
25
26   <!-- Start the daemon immediately on launchctl load (instead of waiting
27        for the next user login). Combined with keepAlive below this means
28        once installed: load once, runs forever across logouts/reboots. -->
29   <key>RunAtLoad</key>
30   <true/>
31
32   <!-- Restart policy:
33        SuccessfulExit=false    SIGTERM / clean exit does NOT relaunch (so
34                                `launchctl bootout` cleanly stops, not loops).
35        Crashed=false          HARD crashes ALSO do NOT relaunch. We've seen
36                                that repeated sleep/wake cycles on macOS
37                                Sonoma+ accumulate orphan watcher.sh PIDs
38                                because SIGCONT/resume doesn't reliably kill
39                                the prior instance. The watcher.sh script
40                                now enforces single-instance via a manual
41                                PID-file and process-identity check at the
42                                top, so a re-launch on crash is
43                                actively dangerous (it would skip the lock
44                                and exit; better to surface the crash in
45                                logs than to silently 0-process). -->
46   <key>KeepAlive</key>
47   <dict>
48     <key>SuccessfulExit</key>
49     <false/>
50     <key>Crashed</key>
51     <false/>
52   </dict>
53
54   <!-- Background hint to App Nap so it doesn't get suspended when
55        the user is inactive. We're the entire reason this LaunchAgent
56        needs to NOT nap. -->
57   <key>ProcessType</key>
58   <string>Background</string>
59
60   <!-- launchd-level stdout/stderr (the bash launcher shell). The script
61        itself writes to /tmp/nullexit-watcher.log via exec >>. These
62        separate channels help us tell "launchd crashed" from "the bash
63        wrapper piped to recover.sh errored". -->
64   <key>StandardOutPath</key>
65   <string>/tmp/nullexit-watcher.out.log</string>
66   <key>StandardErrorPath</key>
67   <string>/tmp/nullexit-watcher.err.log</string>
68 </dict>
69 </plist>
```

26 post-rules.txt

```
1 iptables -A FORWARD -i tailscale0 -o tun0 -j ACCEPT
2 iptables -A FORWARD -i tun0 -o tailscale0 -j ACCEPT
3 iptables -A INPUT -i tailscale0 -j ACCEPT
4 iptables -A INPUT -i eth0 -p udp --dport 41642 -j ACCEPT
5 iptables -t nat -A POSTROUTING -o tun0 -j MASQUERADE
6 # Clamp MSS to 1120. With Tailscale overhead on top of WARP (each WireGuard encapsulation adds ~60 bytes),
7 # the safe MSS for double-tunneled traffic is 1120 to prevent strict-MSS endpoints from stalling.
8 # NOTE: this 1120 integer is rewritten at boot by toggle.sh from GATEWAY_MSS in .env
9 # (Gluetun's parser can't interpolate variables) change GATEWAY_MSS in .env, not this line.
10 iptables -t mangle -A POSTROUTING -p tcp --tcp-flags SYN,RST SYN -j TCPMSS --set-mss 1120
11
12 iptables -t nat -A PREROUTING -i tailscale0 -p udp --dport 53 -j REDIRECT --to-port 5335
13 iptables -t nat -A PREROUTING -i tailscale0 -p tcp --dport 53 -j REDIRECT --to-port 5335
14
15 # Also redirect DNS from Docker bridge interface so the host can reach AdGuard via localhost:53
16 iptables -t nat -A PREROUTING -i eth0 -p udp --dport 53 -j REDIRECT --to-port 5335
17 iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 53 -j REDIRECT --to-port 5335
18
19 # Block IPv6 exit-node traffic to prevent leakage (Gluetun/WARP is IPv4-only)
20 ip6tables -A FORWARD -i tailscale0 -j DROP
21
22 # Block DNS-over-TLS (Port 853) to force Android Private DNS to fallback to Port 53
23 iptables -A FORWARD -i tailscale0 -p tcp --dport 853 -j REJECT --reject-with tcp-reset
24 iptables -A FORWARD -i tailscale0 -p udp --dport 853 -j REJECT
25
26 # Block QUIC (UDP 443) to force Facebook/Google to fallback to TCP.
27 # UDP bypasses TCP MSS clamping, causing fragmentation. Over weak Wi-Fi (far from gateway),
28 # dropping a single fragment destroys the entire packet, stalling image downloads.
29 iptables -A FORWARD -i tailscale0 -p udp --dport 443 -j REJECT
```

27 scripts/Dockerfile.routing-fix

```
1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY logger/ ./logger/
4 RUN cd logger && go build -o logger main.go
5
6 FROM alpine:3.20
7 RUN apk add --no-cache iptables iproute2 curl ipset
8 COPY --from=builder /app/logger/logger /usr/local/bin/nullexit-logger
9 COPY routing-fix.sh /routing-fix.sh
10 CMD ["sh", "/routing-fix.sh"]
```

28 scripts/Dockerfile.rule-compiler

```
1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY rule-compiler/ ./rule-compiler/
4 RUN cd rule-compiler && go build -o sync-rules main.go
5
6 FROM alpine:3.20
7 COPY --from=builder /app/rule-compiler/sync-rules /usr/local/bin/sync-rules
8 WORKDIR /app
9 CMD ["sync-rules"]
```

29 scripts/Toggle-Gateway.bat

```
1 <# :
2 @echo off
3 powershell -NoProfile -ExecutionPolicy Bypass -Command "Invoke-Expression ([System.IO.File]::ReadAllText('%-f0
4 ')) -replace '^(?s).*?#s*>s*\r?\n',''"
5 exit /b
6 #>
7 # Toggle-Gateway.bat (Hybrid Batch-PowerShell Toggler) nullexit Gateway Toggler for Windows
```

```

7 # Mirrors the sophisticated error handling, checks, and automation of toggle.sh
8
9 # 1. Administrator Check & Elevation
10 $isAdmin = ([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([
    Security.Principal.WindowsBuiltInRole]::Administrator)
11 if (-not $isAdmin) {
12     Write-Host "Requesting Administrator privileges..." -ForegroundColor Yellow
13     Start-Process PowerShell -ArgumentList "-NoProfile -ExecutionPolicy Bypass -Command `\"Invoke-Expression ([
        System.IO.File]::ReadAllText('$PSCommandPath') -replace '^(?s).*?#\\s*>\\s*\\r?\\n', '')`\" -Verb RunAs
14     Exit
15 }
16
17 # Set console output encoding to UTF8 for clean symbols
18 [Console]::OutputEncoding = [System.Text.Encoding]::UTF8
19 Clear-Host
20
21 Write-Host "" -ForegroundColor Green
22 Write-Host "          nullexit Windows Toggler          " -ForegroundColor Green
23 Write-Host "" -ForegroundColor Green
24 Write-Host ""
25
26 # 2. Helpers
27 function Reset-HostDNS {
28     Write-Host "Restoring default DNS (DHCP/DHCP-assigned DNS)..." -ForegroundColor Cyan
29     $ActiveAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
30     foreach ($adapter in $ActiveAdapters) {
31         Write-Host "    -> Resetting DNS on adapter: $($adapter.Name)" -ForegroundColor Gray
32         Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ResetServerAddresses -ErrorAction
            SilentlyContinue
33     }
34 }
35
36 function Hijack-HostDNS ($ip) {
37     Write-Host "Hijacking DNS to route through AdGuard ($ip)..." -ForegroundColor Yellow
38     $ActiveAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
39     foreach ($adapter in $ActiveAdapters) {
40         Write-Host "    -> Pointing DNS to $ip on adapter: $($adapter.Name)" -ForegroundColor Gray
41         Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ServerAddresses $ip -ErrorAction
            SilentlyContinue
42     }
43 }
44
45 function Get-HostTailscalePath {
46     $paths = @(
47         "$env:ProgramFiles\\Tailscale\\tailscale.exe",
48         "$env:ProgramFiles (x86)\\Tailscale\\tailscale.exe",
49         "tailscale.exe"
50     )
51     foreach ($path in $paths) {
52         if (Test-Path $path) { return $path }
53     }
54     return $null
55 }
56
57 # 3. Detect State
58 # Prevent deadlocks by resetting DNS to default temporarily during checks
59 Reset-HostDNS
60
61 Write-Host "Checking Docker daemon status..." -ForegroundColor Cyan
62 & docker info >$null 2>&1
63 if ($LASTEXITCODE -ne 0) {
64     Write-Host "Docker is not running. Starting Docker Desktop..." -ForegroundColor Yellow
65     $dockerPath = "C:\\Program Files\\Docker\\Docker\\Docker Desktop.exe"
66     if (Test-Path $dockerPath) {
67         Start-Process $dockerPath
68         Write-Host "Waiting for Docker daemon to become responsive..." -ForegroundColor Gray
69         $timeout = 60
70         while ($timeout -gt 0) {
71             Start-Sleep -Seconds 2
72             & docker info >$null 2>&1
73             if ($LASTEXITCODE -eq 0) {
74                 Write-Host "Docker is ready!" -ForegroundColor Green
75                 break
76             }
77             $timeout -= 2
78         }
79         if ($timeout -le 0) {
80             Write-Error "Docker failed to start in time. Aborting."
81             Exit 1
82         }
83     } else {

```

```

84     Write-Error "Docker Desktop executable not found at standard path. Please start Docker manually."
85     Exit 1
86 }
87 }
88
89 # Detect if gateway is already running
90 $runningWarp = docker compose ps --status running --format json | ConvertFrom-Json -ErrorAction
    SilentlyContinue | Where-Object { $_.Service -eq "warp" }
91 $isGatewayActive = ($null -ne $runningWarp)
92
93 # 4. Execution
94 if ($isGatewayActive) {
95     # STOP PATH
96     Write-Host "Gateway is currently RUNNING. Disabling now..." -ForegroundColor Yellow
97     Write-Host ""
98
99     # Disconnect host Tailscale from exit node if installed
100    $tsCli = Get-HostTailscalePath
101    if ($null -ne $tsCli) {
102        Write-Host "Disconnecting host Tailscale from exit node..." -ForegroundColor Cyan
103        & $tsCli up --exit-node= --accept-dns=true >$null 2>&1
104    }
105
106    Write-Host "Stopping Docker containers..." -ForegroundColor Cyan
107    docker compose down -t 5
108
109    Reset-HostDNS
110    Write-Host "Gateway has been successfully STOPPED." -ForegroundColor Green
111 } else {
112     # START PATH
113     Write-Host "Gateway is currently STOPPED. Enabling now..." -ForegroundColor Yellow
114     Write-Host ""
115
116     # Clean up corrupted AdGuardHome configurations
117     $confFile = "adguard\conf\AdGuardHome.yaml"
118     if (Test-Path $confFile) {
119         $fileSize = (Get-Item $confFile).Length
120         if ($fileSize -eq 0) {
121             Remove-Item $confFile -Force
122             Write-Host "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop." -
                ForegroundColor Yellow
123         }
124     }
125
126     Write-Host "Starting Docker containers..." -ForegroundColor Cyan
127     docker compose up -d
128
129     Write-Host "Waiting for container's Tailscale connection to offer Exit Node..." -ForegroundColor Cyan
130     Write-Host " (This can take up to 60 seconds..." -ForegroundColor Gray
131
132     $elapsed = 0
133     $maxWait = 60
134     $resolvedIP = $null
135
136     while ($elapsed -lt $maxWait) {
137         $statusJson = docker compose exec -T tailscale tailscale status --json 2>$null
138         if ($null -ne $statusJson) {
139             $status = $statusJson | ConvertFrom-Json -ErrorAction SilentlyContinue
140             if ($null -ne $status -and $null -ne $status.Self) {
141                 # Check if it has a valid Tailscale IP and is running
142                 if ($status.Self.Online -and $status.Self.TailscaleIPs.Count -gt 0) {
143                     $resolvedIP = $status.Self.TailscaleIPs[0]
144                     break
145                 }
146             }
147         }
148         Start-Sleep -Seconds 2
149         $elapsed += 2
150     }
151
152     if ($null -ne $resolvedIP) {
153         Write-Host "Tailscale IP resolved: $resolvedIP" -ForegroundColor Green
154         Write-Host ""
155
156         # Configure host Tailscale client to route through this exit node
157         $tsCli = Get-HostTailscalePath
158         if ($null -ne $tsCli) {
159             Write-Host "Configuring host Tailscale to use Exit Node ($resolvedIP)..." -ForegroundColor Cyan
160             & $tsCli up --exit-node=$resolvedIP --accept-dns=false >$null 2>&1
161         }
162     }

```

```

163     # Hijack DNS
164     Hijack-HostDNS $resolvedIP
165     Write-Host ""
166     Write-Host "Gateway has been successfully STARTED." -ForegroundColor Green
167 } else {
168     Write-Host "Warning: Could not resolve gateway Tailscale IP. DNS hijacking skipped." -ForegroundColor
169     Red
170     # Fallback to .gateway_ip if it exists
171     if (Test-Path ".gateway_ip") {
172         $fallbackIP = Get-Content ".gateway_ip" -TotalCount 1
173         if (-not [string]::IsNullOrEmpty($fallbackIP)) {
174             Write-Host "[Fallback] Using static IP from .gateway_ip: $fallbackIP" -ForegroundColor Yellow
175             Hijack-HostDNS $fallbackIP
176             Write-Host "Gateway STARTED (with static fallback IP)." -ForegroundColor Green
177         }
178     } else {
179         Write-Host "Gateway started in offline/unauthenticated state. Run setup.sh if this is a fresh setup
180         ." -ForegroundColor Yellow
181     }
182 }
183 }
184 Write-Host ""
185 Write-Host "Press any key to exit..."
186 [void]$Host.UI.RawUI.ReadKey("NoEcho,IncludeKeyDown")

```

30 scripts/check_circuits.py

```

1  #!/usr/bin/env python3
2  import socket
3  import os
4  import re
5
6  def query_tor(port, command, password=""):
7      try:
8          s = socket.socket()
9          s.settimeout(2)
10         s.connect(('127.0.0.1', port))
11         s.sendall(f"AUTHENTICATE \"{password}\"\\r\\n{command}\\r\\nQUIT\\r\\n".encode())
12         response = b""
13         while True:
14             chunk = s.recv(4096)
15             if not chunk:
16                 break
17             response += chunk
18         return response.decode()
19     except Exception as e:
20         return f"Error connecting to Tor Control Port: {e}"
21
22 def get_ports():
23     ports = {}
24     ports_file = "/tmp/nullexit-ports.env"
25     if os.path.exists(ports_file):
26         with open(ports_file, 'r') as f:
27             for line in f:
28                 if line.startswith("export "):
29                     # Split at first '=' to handle potential multiple '=' in value
30                     parts = line.replace("export ", "").strip().split("=", 1)
31                     if len(parts) == 2:
32                         key, val = parts
33                         try:
34                             ports[key] = int(val)
35                         except ValueError:
36                             ports[key] = val
37     return ports
38
39 def get_password(ports):
40     # 1. Try ports.env
41     password = ports.get("TOR_PASSWORD")
42     if password:
43         return password
44
45     # 2. Try .env
46     if os.path.exists(".env"):
47         try:
48             with open(".env", "r") as f:

```

```

49         for line in f:
50             if line.startswith("ADGUARD_PASSWORD="):
51                 return line.split("=", 1)[1].strip().strip('"\'')
52             if line.startswith("TOR_PASSWORD="):
53                 return line.split("=", 1)[1].strip().strip('"\'')
54         except Exception:
55             pass
56
57     # 3. Fallback
58     return "nullexit"
59
60 def main():
61     ports = get_ports()
62     control_port = ports.get("TOR_CONTROL_PORT", 9051)
63     password = get_password(ports)
64
65     print(f"Connecting to Tor Control Port on 127.0.0.1:{control_port}...")
66     status = query_tor(control_port, "GETINFO circuit-status", password)
67
68     if "Error" in status:
69         print(status)
70         return
71
72     print("\nActive Tor Circuits:")
73     circuits = re.findall(r"(\d+)\s+BUILT\s+(.+) \s+PURPOSE", status)
74     if not circuits:
75         print("No active built circuits found yet. Tor might still be bootstrapping.")
76         return
77
78     for circ_id, path_str in circuits:
79         print(f"\nCircuit #{circ_id}:")
80         nodes = path_str.split(",")
81         for idx, node in enumerate(nodes):
82             fingerprint = node.split("~")[0].replace("$", "")
83             nickname = node.split("~")[1] if "~" in node else "unknown"
84
85             # Fetch node IP
86             ns_info = query_tor(control_port, f"GETINFO ns/id/{fingerprint}", password)
87             ip = "unknown"
88             for line in ns_info.split("\n"):
89                 if line.startswith("r "):
90                     parts = line.split(" ")
91                     if len(parts) >= 7:
92                         ip = parts[6]
93                     break
94
95             # Fetch country
96             country = "unknown"
97             if ip != "unknown":
98                 country_info = query_tor(control_port, f"GETINFO ip-to-country/{ip}", password)
99                 for line in country_info.split("\n"):
100                     if line.startswith("250-ip-to-country/"):
101                         country = line.split("=")[1].strip().upper()
102                     break
103
104             role = "Guard" if idx == 0 else ("Exit" if idx == len(nodes) - 1 else "Middle")
105             print(f"    [{role}] {nickname} ({ip}) - Country: {country}")
106
107 if __name__ == "__main__":
108     main()

```

31 scripts/diagnose-host-leak.sh

```

1  #!/bin/bash
2  # scripts/diagnose-host-leak.sh  nullexit host-routing diagnostic
3  #
4  # Symptom this addresses: tests like https://www.whatismyip.com on the
5  # *HOST MAC* return the underlying physical ISP/ASN
6  # local ISP / campus network") even though the nullexit gateway is active and remote
7  # tailnet clients correctly egress via Cloudflare WARP. The container and
8  # the gateway itself are healthy  only the host's own routing is leaking.
9  #
10 # This script runs 7 checks in sequence, classifies the result into ONE
11 # of the three known leak scenarios, writes the full report to a
12 # timestamped file in the project root, and prints a verdict + a
13 # ready-to-run fix command on stdout. Pass --fix to apply the matching

```

```

14 # remediation in the same invocation and re-verify egress afterwards.
15 # See devref.md 10.30 for the deep-dive rationale.
16 #
17 # Usage:
18 #   bash scripts/diagnose-host-leak.sh           # diagnose + write report
19 #   bash scripts/diagnose-host-leak.sh --fix      # diagnose + fix + re-verify
20 #   bash scripts/diagnose-host-leak.sh --watch    # full baseline then loop checks every 60s
21 #   bash scripts/diagnose-host-leak.sh --watch 30 # same, with custom interval (seconds)
22 #   bash scripts/diagnose-host-leak.sh --help    # this message
23 #
24 # Multiple runs produce timestamped files (one per run) so historical
25 # reports can be diffed. The newest file is the most recent.
26
27 set -uo pipefail
28 # NOTE: errexit (`set -e`) is intentionally NOT enabled. Several checks
29 # below are EXPECTED to fail and that's diagnostic data (e.g. `tailscale
30 # status` returns non-zero when the daemon is wedged). Every command
31 # pipes to a helper that records the exit code without killing the script.
32
33 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
34 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
35 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
36 source "$SCRIPT_DIR/common.sh"
37 LOG_FILE="$PROJECT_ROOT/output.log"
38 REPORT_FILE="$PROJECT_ROOT/host-leak-diagnostic-$TIMESTAMP.txt"
39
40 # Argument parsing
41 DO_FIX=false
42 DO_WATCH=false
43 WATCH_INTERVAL=60
44 WATCH_LOG="$PROJECT_ROOT/host-leak-watch.log"
45 case "${1:-}" in
46     --fix) DO_FIX=true ;;
47     --watch)
48         DO_WATCH=true
49         # Optional second argument: custom interval in seconds
50         if [ -n "${2:-}" ] && [[ "${2}" =~ ^[0-9]+$ ]]; then
51             WATCH_INTERVAL="${2}"
52         fi
53         ;;
54     --help|-h)
55         sed -n '2,31p' "$0"
56         exit 0
57         ;;
58     *)
59         : # default: diagnose only
60         ;;
61     *)
62         echo "Unknown argument: $1" >&2
63         echo "Run with --help for usage." >&2
64         exit 2
65         ;;
66 esac
67
68 # Colours (auto-disabled when stdout is not a tty, e.g. piped)
69 if [ -t 1 ]; then
70     RED='\033[0;31m'; GREEN='\033[0;32m'; YELLOW='\033[1;33m'
71     BOLD='\033[1m'; NC='\033[0m'
72 else
73     RED=''; GREEN=''; YELLOW=''; BOLD=''; NC=''
74 fi
75
76 # Output helpers write to BOTH stdout and the report file
77 exec 3>> "$REPORT_FILE" || {
78     printf '\n[FATAL] cannot open %s for write\n' "$REPORT_FILE" >&2
79     exit 1
80 }
81
82 section() {
83     local title="$1"
84     printf '\n%b %s %b\n' "$BOLD" "$title" "$NC" >&3
85     printf '\n%b %s %b\n' "$BOLD" "$title" "$NC"
86 }
87
88 line() { printf ' %b\n' "$1" >&3; printf ' %b\n' "$1"; }
89 ok() { line "${GREEN}${NC} $1"; }
90 warn() { line "${YELLOW}${NC} $1"; }
91 fail() { line "${RED}${NC} $1"; }
92 note() { line "(no ${1})"; }
93
94 # Watch-mode header (only shown when entering watch loop)
95 if [ "${DO_WATCH}" = "true" ]; then
96     echo ""

```

```

95 echo ""
96 echo " n u l l e x i t   w a t c h   m o d e "
97 echo ""
98 echo " Interval:          ${WATCH_INTERVAL}s"
99 echo " Watch log:          $WATCH_LOG"
100 echo ""
101 else
102 echo ""
103 echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c "
104 echo ""
105 echo " Timestamp (UTC): $TIMESTAMP"
106 echo " Project root:     $PROJECT_ROOT"
107 echo " Report file:       $REPORT_FILE"
108 echo " Mode:              ${[ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only (pass --
    fix to remediate)"}"
109 echo ""
110 echo "" >&3
111 echo "" >&3
112 echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c " >&3
113 echo "" >&3
114 echo " Timestamp (UTC): $TIMESTAMP" >&3
115 echo " Mode:              ${[ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only")" >&3
116 fi
117
118 # Always add Homebrew to PATH (same as toggle.sh / recover.sh)
119 export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"
120
121 # Pure-bash timeout (macOS lacks GNU `timeout`)
122 run_with_timeout() {
123     local timeout_sec="$1"
124     shift
125     if [ "$#" -eq 0 ]; then return 1; fi
126     "$@" &
127     local cmd_pid=$!
128     (
129         sleep "$timeout_sec"
130         if kill -0 "$cmd_pid" 2>> "$LOG_FILE"; then
131             kill -9 "$cmd_pid" 2>> "$LOG_FILE" || true
132         fi
133     ) >> "$LOG_FILE" 2>&1 &
134     local watcher_pid=$!
135     set +e
136     wait "$cmd_pid"
137     local exit_status=$?
138     set -uo pipefail
139     kill "$watcher_pid" 2>> "$LOG_FILE" || true
140     return "$exit_status"
141 }
142
143 #
144 # Quick-check functions (used by --watch loop; return simple status strings)
145 # Each echoes its result so the caller can capture and compare.
146 #
147
148 quick_warp() {
149     # Returns: "on", "off", "unknown"
150     local out
151     out=$(run_with_timeout 8 curl -s --max-time 6 \
152         https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null || echo "")
153     local warp
154     warp=$(printf '%s' "$out" | awk -F=' ' /^warp={print $2; exit}')
155     printf '%s' "${warp:-unknown}"
156 }
157
158 quick_ipv6() {
159     # Returns: the IPv6 address if leaking, or empty string if clean
160     local out
161     out=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>/dev/null || echo "")
162     if [ -n "$out" ] && printf '%s' "$out" | grep -qE '^[0-9a-fA-F:]+$'; then
163         printf '%s' "$out"
164     fi
165 }
166
167 quick_default_route() {
168     # Returns: "utun" if default route is via Tailscale utun*,
169     #           "wifi" if via physical Wi-Fi (192.168.x.x),
170     #           "other" otherwise
171     local route
172     route=$(netstat -rn 2>/dev/null | awk '/^default/ {print $0; exit}')
173     if [ -z "$route" ]; then
174         printf '%s' "none"

```

```

175 elif printf '%s' "$route" | grep -qE 'utun[0-9]+'; then
176     printf '%s' "utun"
177 elif printf '%s' "$route" | grep -qE '^(default|0\.\0\.\0\.\0).*\b192\.\168\.'; then
178     printf '%s' "wifi"
179 else
180     printf '%s' "other"
181 fi
182 }
183
184 # Watch loop (runs after baseline diagnostic when --watch is passed)
185 run_watch_loop() {
186     local baseline_warp="$1"
187     local baseline_default="$2"
188     local baseline_ipv6_leak="$3"
189     local interval="$4"
190
191     echo ""
192     echo ""
193     echo " Baseline established. Monitoring every ${interval}s..."
194     echo " warp:      $baseline_warp"
195     echo " default:    $baseline_default"
196     echo ""
197
198     # Safety: warn if interval is very short (avoids hammering APIs)
199     if [ "$interval" -lt 30 ]; then
200         printf '%b' Short interval (%ss) rate-limiting risk on external APIs%b\n' "$YELLOW" "$interval" "$NC"
201         printf ' Consider 30s or longer for production use.\n'
202     fi
203     printf '[%s] WATCH START warp=%s default=%s interval=%ss\n' \
204         "$(date -u +%Y-%m-%dT%H:%M:%SZ)" "$baseline_warp" "$baseline_default" "$interval" \
205         >> "$WATCH_LOG"
206
207     local iteration=0
208     local last_warp="$baseline_warp"
209     local last_default="$baseline_default"
210     local last_ipv6_ok=$(( [ "$baseline_ipv6_leak" = "false" ] && echo true || echo false ))
211
212     trap 'printf "\n%bWatch stopped by user. Log at: %s%b\n" "$YELLOW" "$WATCH_LOG" "$NC"; exit 0' INT TERM
213
214     while true; do
215         iteration=$((iteration + 1))
216
217         local warp_now ipv6_now default_now
218         warp_now=$(quick_warp)
219         ipv6_now=$(quick_ipv6)
220         default_now=$(quick_default_route)
221
222         local ts
223         ts=$(date -u +%H:%M:%S)
224
225         # WARP check
226         if [ "$warp_now" != "$last_warp" ]; then
227             if [ "$warp_now" = "off" ] || [ "$warp_now" = "unknown" ]; then
228                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"
229                 printf '%b[%s] %bWARP FLIPPED: %s %s%b YOUR IP IS LEAKING!\n' \
230                     "$RED" "$ts" "$BOLD" "$last_warp" "$warp_now" "$NC"
231                 printf '[%s] ALERT warp flipped %s%s\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
232             else
233                 printf '\n%b[%s] warp recovered: %s %s%b\n' \
234                     "$GREEN" "$ts" "$last_warp" "$warp_now" "$NC"
235                 printf '[%s] OK warp recovered %s%s\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
236             fi
237             last_warp="$warp_now"
238         fi
239
240         # IPv6 check
241         if [ -n "$ipv6_now" ]; then
242             if [ "$last_ipv6_ok" = true ]; then
243                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"
244                 printf '%b[%s] %bIPv6 LEAK DETECTED %s%b\n' \
245                     "$RED" "$ts" "$BOLD" "$ipv6_now" "$NC"
246                 printf '[%s] ALERT IPv6 leak %s\n' "$ts" "$ipv6_now" >> "$WATCH_LOG"
247                 last_ipv6_ok=false
248             fi
249         else
250             if [ "$last_ipv6_ok" = false ]; then
251                 printf '%b[%s] IPv6 leak resolved%b\n' "$GREEN" "$ts" "$NC"
252                 printf '[%s] OK IPv6 resolved\n' "$ts" >> "$WATCH_LOG"
253             fi
254             last_ipv6_ok=true
255         fi

```

```

256 # Default route check
257 if [ "$default_now" != "$last_default" ]; then
258     if [ "$default_now" != "utun" ]; then
259         printf '\n\b ALERT %b\n' "$RED$BOLD" "$NC"
260         printf '%b[%s] %bDEFAULT ROUTE CHANGED: %s %s\b traffic may bypass WARP!\n' \
261             "$RED" "$ts" "$BOLD" "$last_default" "$default_now" "$NC"
262         printf '[%s] ALERT route changed %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
263     else
264         printf '%b[%s] default route recovered: %s %s\b\n' \
265             "$GREEN" "$ts" "$last_default" "$default_now" "$NC"
266         printf '[%s] OK route recovered %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
267     fi
268     last_default="$default_now"
269 fi
270
271 # Heartbeat (every 10th iteration or if all OK)
272 if [ $((iteration % 10)) -eq 0 ]; then
273     printf '[%s] #d warp=%s ipv6=%s route=%s\n' \
274         "$ts" "$iteration" "$warp_now" "${ipv6_now:-none}" "$default_now" >> "$WATCH_LOG"
275     printf '\r\b[%s] #d warp=%s route=%s (Ctrl+C to stop)%b\n' \
276         "$GREEN" "$ts" "$iteration" "$warp_now" "$default_now" "$NC"
277 fi
278
279 sleep "$interval"
280 done
281 }
282
283 #
284 # Begin main diagnostic (skipped in non-watch modes if we're just watching)
285 #
286
287 ACTIVE_SERVICE=$(get_active_service)
288 ENO_SERVICE=$(networksetup -listnetworkserviceorder 2>> "$LOG_FILE" \
289     | grep -B 1 "Device: en0" | head -1 \
290     | sed -E 's/^\([0-9\]*\)+\)/' || true)
291 [ -z "$ENO_SERVICE" ] && ENO_SERVICE="Wi-Fi"
292
293 # Resolve the gateway's Tailscale IP.
294 GATEWAY_TS_IP=""
295 if [ -f "$PROJECT_ROOT/.gateway_ip" ]; then
296     GATEWAY_TS_IP=$(cat "$PROJECT_ROOT/.gateway_ip" 2>> "$LOG_FILE" \
297         | tr -d '\r' | awk 'NR==1{print $1; exit}')
298 fi
299
300 if [ -z "$GATEWAY_TS_IP" ] && command -v docker >> "$LOG_FILE" 2>&1 \
301     && docker compose ps --status running 2>/dev/null | grep -q 'tailscale'; then
302     GATEWAY_TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale \
303         tailscale ip -4 2>> "$LOG_FILE" \
304         | tr -d '\r' | awk 'NR==1{print $1; exit}')
305 fi
306
307 # Scenario classification state
308 SCENARIO=""
309 WARP_STATUS=""
310 SOCKS5_ENABLED=false
311 TS_ON_MESH=false
312 DEFAULT_VIA_UTUN=false
313 IPV6_LEAK=false
314
315 #
316 section "1/8 Host Tailscale mesh status"
317 #
318 if ! command -v tailscale >> "$LOG_FILE" 2>&1; then
319     fail "tailscale CLI not on PATH install via 'brew install tailscale' then re-run"
320     echo " (a Tailscale daemon is required for the exit-node path to work)"
321 else
322     set +e
323     TS_OUT=$(run_with_timeout 5 tailscale status 2>&1)
324     TS_EXIT=$?
325     set -uo pipefail
326     TS_HEAD=$(printf '%s\n' "$TS_OUT" | head -5 | tr -d '\r')
327     if [ "$TS_EXIT" -eq 137 ]; then
328         fail "tailscaled daemon is wedged/unresponsive (5s timeout)"
329         echo " Fix path: 'sudo brew services restart tailscale'"
330     elif printf '%s' "$TS_OUT" | grep -qE 'Logged out|not logged in|expired'; then
331         fail "tailscale is logged out / auth expired on this host"
332         echo " Fix path: 'sudo tailscale up' (browser auth once)"
333     elif printf '%s' "$TS_OUT" | grep -q '100\.'; then
334         ok "Host is on tailnet (100.x.x.x visible in status)"
335         TS_ON_MESH=true
336         if printf '%s' "$TS_OUT" | grep -q "$GATEWAY_TS_IP"; then

```

```

337     ok "Gateway $GATEWAY_TS_IP visible in host's peer list"
338     elif [ -n "$GATEWAY_TS_IP" ]; then
339         warn "Gateway $GATEWAY_TS_IP NOT in host's peer list possible auth/key mismatch"
340     fi
341     line " first 5 lines of 'tailscale status' "
342     line "${(printf '%s' "$TS_HEAD" | awk '{print "    "$0}')} "
343 else
344     fail "tailscale output unrecognized:"
345     line "${(printf '%s' "$TS_OUT" | head -3 | awk '{print "    "$0}')} "
346 fi
347 fi
348
349 #
350 section "2/8 Active SOCKS5 proxy on Wi-Fi"
351 #
352 SOCKS_OUT=$(networksetup -getsocksfirewallproxy "$ACTIVE_SERVICE" 2>> "$LOG_FILE" \
353             || networksetup -getsocksfirewallproxy Wi-Fi 2>> "$LOG_FILE" \
354             || echo "Could not query SOCKS5 proxy state")
355 if printf '%s' "$SOCKS_OUT" | grep -q "Enabled: Yes"; then
356     fail "SOCKS5 proxy IS enabled on $ACTIVE_SERVICE toggle.sh fell back to SOCKS5 mode last run"
357     SOCKS5_ENABLED=true
358     line " Full state:"
359     line "${(printf '%s' "$SOCKS_OUT" | awk '{print "    "$0}')} "
360     line " Browse on macOS ignores system SOCKS5 by default traffic bypasses WARP."
361 else
362     ok "SOCKS5 proxy is OFF on $ACTIVE_SERVICE exit-node path should be active"
363 fi
364
365 #
366 section "3/8 Egress IP + WARP tunnel verification (definitive test)"
367 #
368 CDN_OUT=$(run_with_timeout 8 curl -s --max-time 6 \
369           https://www.cloudflare.com/cdn-cgi/trace 2>&1 || echo "(curl failed)")
370 CDN_WARP=$(printf '%s' "$CDN_OUT" | awk -F=' ' '/warp/{print $2; exit}')
371 CDN_IP=$(printf '%s' "$CDN_OUT" | awk -F=' ' '/ip/{print $2; exit}')
372 CDN_COLO=$(printf '%s' "$CDN_OUT" | awk -F=' ' '/colo/{print $2; exit}')
373 WARP_STATUS="$CDN_WARP"
374 if [ "$CDN_WARP" = "on" ]; then
375     ok "warp=on tunnel is hot, host traffic IS going through Cloudflare WARP"
376     line " Public IP: $CDN_IP"
377     line " Cloudflare: ${CDN_COLO:-unknown colo}"
378     line " whatismyip.com's 'local ISP' result is its ISP-database error, not yours."
379 elif [ "$CDN_WARP" = "off" ]; then
380     fail "warp=off host traffic is NOT going through WARP"
381     line " Public IP: $CDN_IP (NOT Cloudflare)"
382     warn "This is the actual leak."
383 else
384     warn "could not determine warp status (curl failed or returned unexpected shape)"
385     line " Raw output: $(printf '%s' "$CDN_OUT" | head -3 | tr '\n' '|') "
386 fi
387
388 #
389 section "4/8 IPv6 leak probe (bypasses IPv4 exit-node on campus networks)"
390 #
391 IPV6_OUT=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
392 if [ -n "$IPV6_OUT" ] && printf '%s' "$IPV6_OUT" | grep -qE '[0-9a-fA-F:]+$'; then
393     fail "host has live IPv6 egress $IPV6_OUT (A6 record bypasses WARP)"
394     IPV6_LEAK=true
395     line " Tailscale exit-node advertisement is IPv4-only, so IPv6 traffic skips it."
396     line " Quick-fix: 'sudo networksetup -setv6off $ACTIVE_SERVICE'"
397 else
398     ok "no IPv6 egress detected (good IPv6 won't bypass the tunnel)"
399 fi
400
401 #
402 section "5/8 Host egress route for real traffic (route -n get 1.1.1.1)"
403 #
404 # WHY NOT 'netstat -rn | grep default':
405 # On macOS, Tailscale does NOT replace the physical default route installed by
406 # DHCP. Instead it adds a second interface-scoped route bound to the utun
407 # interface that the kernel prefers for real traffic forwarding. The literal
408 # "default" entry (192.168.x.x via en0) is left untouched in the table as a
409 # BSD routing quirk it's not lying, it's just answering a different question.
410 # The correct question is: "where does a real destination actually resolve?"
411 # 'route -n get 1.1.1.1' asks the kernel's forwarding logic directly.
412 ROUTE_GET=$(route -n get 1.1.1.1 2>/dev/null)
413 ROUTE_IFACE=$(echo "$ROUTE_GET" | awk '/interface/{print $2}')
414 ROUTE_GATEWAY=$(echo "$ROUTE_GET" | awk '/gateway/{print $2}')
415 if [ -z "$ROUTE_IFACE" ]; then
416     warn "route -n get 1.1.1.1 returned no interface routing table may be empty"
417 else

```

```

418 line " interface: $ROUTE_IFACE gateway: ${ROUTE_GATEWAY:-n/a}"
419 if echo "$ROUTE_IFACE" | grep -qE '^utun[0-9]+'; then
420     ok "1.1.1.1 resolves via $ROUTE_IFACE Tailscale interface-scoped route is winning"
421     DEFAULT_VIA_UTUN=true
422 elif echo "$ROUTE_IFACE" | grep -qE '^en[0-9]+'; then
423     fail "1.1.1.1 resolves via $ROUTE_IFACE (physical Wi-Fi) exit-node interface route not active"
424 else
425     warn "1.1.1.1 resolves via unexpected interface: $ROUTE_IFACE"
426 fi
427 fi
428
429 #
430 section "6/8 Last toggle.sh START-relevant lines in output.log (if any)"
431 #
432 if [ ! -f "$LOG_FILE" ]; then
433     note "output.log"
434     line " toggle.sh was never run from this directory. Run './toggle.sh' first,"
435     line " then re-run this diagnostic."
436 else
437     HITS=$(grep -E 'Exit node enabled|SKIP_EXIT_NODE|Pre-flight|Falling back to SOCKS|Starting|Successfully' \
438             "$LOG_FILE" 2>/dev/null | tail -8 || true)
439     if [ -z "$HITS" ]; then
440         note "matching lines in output.log"
441     else
442         line "$(printf '%s\n' "$HITS" | awk '{print "    "$0}')"
443     fi
444 fi
445
446 #
447 section "7/8 Gateway IP we should be pointing at"
448 #
449 if [ -n "$GATEWAY_TS_IP" ]; then
450     ok "gateway Tailscale IP: $GATEWAY_TS_IP"
451 else
452     warn "could not resolve gateway Tailscale IP"
453     line " (looked in .gateway_ip and tried docker compose exec tailscale ip -4)"
454 fi
455
456 #
457 section "8/8 Tailscale System Extension (App Store conflict check)"
458 #
459 SE_PENDING_UNINSTALL=false
460 SE_ACTIVE=false
461 if command -v systemextensionsctl >/dev/null 2>&1; then
462     SE_OUT=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale')
463     if [ -n "$SE_OUT" ]; then
464         line " Tailscale System Extension(s) detected:"
465         line "$(printf '%s' "$SE_OUT" | awk '{print "    "$0}')"
466         if printf '%s' "$SE_OUT" | grep -q 'waiting to uninstall'; then
467             SE_PENDING_UNINSTALL=true
468             warn "System Extension is pending uninstall on next reboot."
469         else
470             SE_ACTIVE=true
471             warn "Active Tailscale System Extension detected. This will conflict with Homebrew Tailscale."
472         fi
473     else
474         ok "no Tailscale System Extension detected"
475     fi
476 else
477     note "systemextensionsctl not available (not on macOS or permission restricted)"
478 fi
479
480 #
481 section "CLASSIFICATION applies ONE of the four known scenarios"
482 #
483
484 if [ "$WARP_STATUS" = "on" ] && [ "$IPV6_LEAK" = "false" ]; then
485     SCENARIO="OK"
486 elif [ "$SE_PENDING_UNINSTALL" = "true" ]; then
487     SCENARIO="SE_PENDING"
488 elif [ "$SOCKS5_ENABLED" = "true" ]; then
489     SCENARIO="A"
490 elif [ "$IPV6_LEAK" = "true" ]; then
491     SCENARIO="B"
492 elif [ "$DEFAULT_VIA_UTUN" = "false" ] && [ "$TS_ON_MESH" = "true" ]; then
493     SCENARIO="C"
494 else
495     SCENARIO="UNKNOWN"
496 fi
497
498 case "$SCENARIO" in

```

```

499 SE_PENDING)
500 echo ""
501 echo -e " ${RED}${BOLD}VERDICT: Scenario SE_PENDING Tailscale System Extension Pending Uninstall${NC}"
502 echo " A prior App Store Tailscale System Extension is pending uninstall."
503 echo " Since System Integrity Protection (SIP) is enabled, macOS blocks manual"
504 echo " uninstallation of terminated system extensions until the next reboot."
505 echo " Brew-managed tailscaled cannot engage the Network Extension slot until this is resolved."
506 echo ""
507 echo " ${BOLD}Recommended fix:${NC}"
508 echo " sudo shutdown -r now"
509 echo " # After reboot, run: ./toggle.sh"
510 ;;
511 A)
512 echo ""
513 echo -e " ${RED}${BOLD}VERDICT: Scenario A SOCKS5 fallback lane${NC}"
514 echo " toggle.sh's pre-flight checks failed last run. The script activated"
515 echo " SOCKS5 + local DNS proxy as a fallback. DNS gets hijacked but"
516 echo " browsers on macOS ignore the system SOCKS5 traffic egresses"
517 echo " direct through the local network."
518 echo ""
519 echo " ${BOLD}Recommended fix:${NC}"
520 echo " echo 'GATEWAY_BYPASS_PING=true' >> $PROJECT_ROOT/.env"
521 echo " sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \\"
522 if [ -n "$GATEWAY_TS_IP" ]; then
523 echo " --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true"
524 else
525 echo " --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true # fill in $GATEWAY_TS_IP"
526 fi
527 echo " sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
528 echo " ./toggle.sh"
529 ;;
530 B)
531 echo ""
532 echo -e " ${RED}${BOLD}VERDICT: Scenario B IPv6 leak over dual-stack campus/local IPv6${NC}"
533 echo " Tailscale exit-node advertises only IPv4. macOS sends AAAA queries"
534 echo " straight out en0, which is dual-stack on most campus APs IPv6"
535 echo " traffic bypasses the WARP tunnel entirely."
536 echo ""
537 echo " ${BOLD}Recommended fix:${NC}"
538 echo " sudo networksetup -setv6off $ACTIVE_SERVICE"
539 echo " sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
540 echo " # Re-enable IPv6 later with: sudo networksetup -setv6automatic $ACTIVE_SERVICE"
541 ;;
542 C)
543 echo ""
544 echo -e " ${RED}${BOLD}VERDICT: Scenario C Tailscale route-freeze${NC}"
545 echo " Host's default route points at the Wi-Fi gateway instead of the"
546 echo " Tailscale utun* interface. The exit-node preference is set but the"
547 echo " routing-table assertion did not take effect (devref 10.26)."
548 echo ""
549 echo " ${BOLD}Recommended fix:${NC}"
550 echo " sudo brew services restart tailscale"
551 echo " sudo route delete -net 0.0.0.0/1"
552 echo " sudo route delete -net 128.0.0.0/1"
553 echo " sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
554 if [ -n "$GATEWAY_TS_IP" ]; then
555 echo " sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \\"
556 echo " --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true"
557 else
558 echo " sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true --exit-node=$TS_IP"
559 echo " --exit-node-allow-lan-access=true"
560 fi
561 echo " ./toggle.sh"
562 ;;
563 OK)
564 echo ""
565 echo -e " ${GREEN}${BOLD}VERDICT: No host leak detected tunnel is working${NC}"
566 echo " The local ISP string from whatismyip.com is its IP-database"
567 echo " misclassifying the Cloudflare egress IP. CDN-CGI's trace endpoint"
568 echo " (which is what we use here) reports warp=on because that endpoint"
569 echo " is hosted by Cloudflare itself and can see its own edge."
570 echo " If you want a third-party confirmation:"
571 echo " curl -s https://api.ipify.org; echo"
572 ;;
573 *)
574 echo ""
575 echo -e " ${YELLOW}${BOLD}VERDICT: Could not classify automatically${NC}"
576 echo " The diagnostic data did not match any known scenario cleanly."
577 echo " Likely causes:"
578 echo " - tailscaled is logged out / expired (see Section 1)"
579 echo " - Docker is not running (gateway containers down)"

```

```

579     echo "    - .gateway_ip missing AND docker compose exec failed"
580     echo "    Inspect the report at: $REPORT_FILE"
581 ;;
582 esac
583 echo ""
584
585 #
586 # Write the verdict block to the report file
587 #
588 SECOND_BLOCK=$(cat <<EOF
589
590
591 C L A S S I F I C A T I O N   R E S U L T
592
593 Scenario: $SCENARIO
594 WARP egress: $WARP_STATUS (cdn-cgi/trace)
595 SOCKS5 lane: $([ "$SOCKS5_ENABLED" = true ] && echo "ENABLED (browsers bypass)" || echo "off")
596 IPv6 leak: $([ "$IPV6_LEAK" = true ] && echo "YES (campus IPv6 egress)" || echo "no")
597 Default via: $([ "$DEFAULT_VIA_UTUN" = true ] && echo "utun* (Tailscale)" || echo "Wi-Fi gateway")
598 Host on mesh: $([ "$TS_ON_MESH" = true ] && echo "yes" || echo "no")
599 SE pending: $([ "$SE_PENDING_UNINSTALL" = true ] && echo "yes" || echo "no")
600 Gateway IP: ${GATEWAY_TS_IP:-unresolved}
601 EOF
602 )
603 printf '%s\n' "$SECOND_BLOCK" >&3
604
605 #
606 # --fix mode
607 #
608 if [ "$DO_FIX" = "true" ] && [ "$SCENARIO" != "OK" ] && [ "$SCENARIO" != "UNKNOWN" ]; then
609     echo ""
610     echo ""
611     echo " A P P L Y I N G   F I X (scenario $SCENARIO)"
612     echo ""
613     echo ""
614
615     case "$SCENARIO" in
616         SE_PENDING)
617             warn "System Extension uninstall is pending on next reboot."
618             line "Please run 'sudo shutdown -r now' to reboot the system."
619             ;;
620         A)
621             if [ -n "$GATEWAY_TS_IP" ]; then
622                 line " writing GATEWAY_BYPASS_PING=true to .env"
623                 ENV="$PROJECT_ROOT/.env"
624                 [ -f "$ENV" ] || touch "$ENV"
625                 if grep -q '^GATEWAY_BYPASS_PING=' "$ENV" 2>/dev/null; then
626                     sed -i 's/^GATEWAY_BYPASS_PING=.*GATEWAY_BYPASS_PING=true/' "$ENV"
627                 else
628                     printf '\nGATEWAY_BYPASS_PING=true\n' >> "$ENV"
629                 fi
630                 line " re-asserting tailscale exit-node (with --reset)"
631                 sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
632                     --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
633                     >> "$LOG_FILE" 2>&1 \
634                     && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
635                     || warn "tailscale up did not return success (check $LOG_FILE)"
636             else
637                 warn "no gateway Tailscale IP resolved skipping tailscale up"
638                 warn "fix manually: 'sudo tailscale up --reset ... --exit-node=<TS_IP>'"
639             fi
640             ;;
641         B)
642             line " disabling IPv6 on $ACTIVE_SERVICE while gateway is up"
643             sudo -n networksetup -setv6off "$ACTIVE_SERVICE" >> "$LOG_FILE" 2>&1 \
644                 && ok "IPv6 disabled on $ACTIVE_SERVICE (re-enable with 'setv6automatic')" \
645                 || warn "networksetup -setv6off failed (check $LOG_FILE)"
646             ;;
647         C)
648             line " restarting tailscaled (route-table fix)"
649             run_with_timeout 15 brew services restart tailscale >> "$LOG_FILE" 2>&1 \
650                 || run_with_timeout 15 sudo -n brew services restart tailscale >> "$LOG_FILE" 2>&1
651             sleep 3
652             line " flushing stale host routes + DNS cache"
653             sudo -n route delete -net 0.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
654             sudo -n route delete -net 128.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
655             sudo -n dscacheutil -flushcache >> "$LOG_FILE" 2>&1 || true
656             sudo -n killall -HUP mDNSResponder >> "$LOG_FILE" 2>&1 || true
657             if [ -n "$GATEWAY_TS_IP" ]; then
658                 line " re-asserting exit-node after restart"
659                 sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \

```

```

660         --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
661         >> "$LOG_FILE" 2>&1 \
662         && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
663         || warn "tailscale up did not return success (check $LOG_FILE)"
664     fi
665     ;;
666 esac
667
668 echo ""
669 echo "Re-verifying after fix..."
670 sleep 3
671 CDN_WARP_AFTER=$(run_with_timeout 8 curl -s --max-time 6 \
672     https://www.cloudflare.com/cdn-cgi/trace 2>&1 \
673     | awk -F=' ' /^warp=/{print $2; exit}')
674 IPV6_AFTER=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
675 printf '\n[AFTER FIX]\n' >&3
676 printf ' warp status:      %s\n' "$CDN_WARP_AFTER" >&3
677 printf ' IPv6 egress:       %s\n' "${IPV6_AFTER:-none}" >&3
678 line " warp status:      ${CDN_WARP_AFTER:-unknown}"
679 line " IPv6 egress:      ${IPV6_AFTER:-none}"
680 if [ "$CDN_WARP_AFTER" = "on" ] && [ -z "$IPV6_AFTER" ]; then
681     echo ""
682     ok "Fix verified host is now routing through WARP. Remote clients unaffected."
683 else
684     echo ""
685     warn "Fix did not fully take effect on first try. Likely causes:"
686     line " tailscaled needed a longer settling window (re-run diagnostic in 30s)"
687     line " The classified scenario was wrong; paste this report for triage."
688     line " Tailscale needs a full auth refresh: 'sudo tailscale up'"
689 fi
690 fi
691
692 echo ""
693 echo ""
694 echo " Full report written to:"
695 echo " $REPORT_FILE"
696 echo ""
697 echo ""
698 #
699 # --watch mode: enter the continuous monitoring loop after baseline diagnostic
700 #
701 if [ "$DO_WATCH" = "true" ]; then
702     # Build baseline from the full diagnostic just completed
703     BASELINE_WARP="$WARP_STATUS"
704     BASELINE_DEFAULT=$( [ "$DEFAULT_VIA_UTUN" = "true" ] && echo "utun" || echo "wifi/other" )
705
706     if [ "$SCENARIO" != "OK" ]; then
707         warn "Baseline diagnostic found issues (scenario: $SCENARIO)."
708         line " Watch loop will still run alerts fire on any STATE CHANGE from current baseline."
709         line " Fix the issue first? Re-run with: bash scripts/diagnose-host-leak.sh --fix"
710     fi
711     echo ""
712 fi
713
714 run_watch_loop "$BASELINE_WARP" "$BASELINE_DEFAULT" "$IPV6_LEAK" "$WATCH_INTERVAL"
715 fi
716
717 exit 0

```

32 scripts/dns-proxy.py

```

1  #!/usr/bin/env python3
2  """
3  dns-proxy.py Local DNS proxy for nullexit gateway.
4
5  Listens on UDP port 53, receives DNS queries, forwards them over TCP
6  to AdGuard at 127.0.0.1:5354 (the Docker port mapping), and returns
7  the response via UDP. Handles the DNS-over-TCP wire format (2-byte
8  length prefix) correctly unlike socat which just forwards raw bytes.
9  """
10
11 import socket
12 import struct
13 import sys
14 import os
15 from concurrent.futures import ThreadPoolExecutor

```

```

16
17 LISTEN_ADDR = os.environ.get("LISTEN_ADDR", "127.0.0.1")
18 LISTEN_PORT = int(os.environ.get("LISTEN_PORT", 53))
19 TARGET_HOST = os.environ.get("TARGET_HOST", "127.0.0.1")
20 TARGET_PORT = int(os.environ.get("TARGET_PORT", 5354))
21
22
23 def handle_query(data: bytes, client_addr: tuple, sock: socket.socket) -> None:
24     """Forward a single DNS query over TCP and send the response back via UDP."""
25     try:
26         tcp = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
27         tcp.settimeout(5)
28         tcp.connect((TARGET_HOST, TARGET_PORT))
29
30         # DNS over TCP: prepend 2-byte length (big-endian)
31         tcp.sendall(struct.pack("!H", len(data)) + data)
32
33         # Read the 2-byte response length
34         raw_len = tcp.recv(2)
35         if len(raw_len) < 2:
36             tcp.close()
37             return
38         resp_len = struct.unpack("!H", raw_len)[0]
39
40         # Read the full response
41         response = b""
42         while len(response) < resp_len:
43             chunk = tcp.recv(resp_len - len(response))
44             if not chunk:
45                 break
46             response += chunk
47
48         tcp.close()
49
50         # Send the response back over UDP (strip TCP length prefix)
51         if response:
52             sock.sendto(response, client_addr)
53     except Exception:
54         # Silently drop failed queries (malformed, timeout, etc.)
55         pass
56
57
58 def main() -> None:
59     # Create UDP socket
60     sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
61     sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
62
63     try:
64         sock.bind((LISTEN_ADDR, LISTEN_PORT))
65     except PermissionError:
66         print(f"dns-proxy: ERROR need root to bind port {LISTEN_PORT}", flush=True)
67         sys.exit(1)
68     except OSError as e:
69         print(f"dns-proxy: ERROR {e}", flush=True)
70         sys.exit(1)
71
72     print(f"dns-proxy: listening on UDP {LISTEN_ADDR}:{LISTEN_PORT} TCP {TARGET_HOST}:{TARGET_PORT}", flush=True)
73
74     # Use a thread pool to handle concurrent DNS queries efficiently without thread exhaustion
75     executor = ThreadPoolExecutor(max_workers=64)
76
77     while True:
78         try:
79             data, client_addr = sock.recvfrom(4096) # Support EDNS0 (up to 4096 bytes) to prevent truncation
80             executor.submit(handle_query, data, client_addr, sock)
81         except KeyboardInterrupt:
82             break
83         except Exception:
84             pass
85
86     executor.shutdown(wait=False)
87     sock.close()
88
89
90 if __name__ == "__main__":
91     main()

```

33 scripts/fix-docker-bridge-collision.sh

```
1 #!/bin/bash
2 # scripts/fix-docker-bridge-collision.sh nullexit fix for devref 9.13
3 #
4 # Symptom: host default route points at Docker's bridge gateway (172.17.0.1)
5 # instead of the real Wi-Fi gateway. ALL host internet egress fails. Remote
6 # tailnet clients route through the gateway correctly (their tunnel is
7 # independent of the host's broken route table), so they look healthy while
8 # the host itself cannot reach anything.
9 #
10 # Root cause: Docker's default bridge claims 172.17.0.0/16 by default. When the
11 # host's physical Wi-Fi also lands in the same /16 extremely common on
12 # university networks which hand out 172.16.0.0/12 the kernel
13 # installs Docker's bridge as the default route, and every packet bound for
14 # the internet is silently absorbed by docker0 instead of egressing.
15 #
16 # This script:
17 # 1. Detects the actual collision (Docker's 172.17.0.0/16 vs the host's
18 # Wi-Fi subnet, queried live via the ACTIVE interface not hardcoded en0)
19 # 2. Picks a non-colliding Docker bip using /8-family routing (works for
20 # 172.16.0.0/12 campus networks; naive prefix-match would pick 172.26/24
21 # which still lives INSIDE a /12 of 172.16.0.0/12)
22 # 3. Backs up ~/.colima/default/colima.yaml with an ISO-8601 timestamp
23 # 4. Edits the file in place, idempotent (replace existing docker.bip /
24 # append new docker.bip without disturbing other keys; skip YAML comments)
25 # 5. Restarts Colima so the VM rebuilds with the new bridge subnet
26 # 6. Rebinds Wi-Fi DHCP so the host re-learns its real Wi-Fi gateway
27 # 7. Verifies the new default route is no longer the Docker bridge
28 # 8. Re-runs toggle.sh to bring the gateway back up cleanly
29 # 9. Re-runs scripts/diagnose-host-leak.sh and prints the new verdict
30 #
31 # Usage:
32 # bash scripts/fix-docker-bridge-collision.sh # apply the fix
33 # bash scripts/fix-docker-bridge-collision.sh --dry # show what would change, then exit
34 # bash scripts/fix-docker-bridge-collision.sh --skip-toggle # fix but don't restart gateway
35 # bash scripts/fix-docker-bridge-collision.sh --help # usage
36
37 set -uo pipefail
38 # NOTE: errexit (`set -e`) is intentionally NOT enabled so intermediate
39 # `command || warn` patterns can continue past non-fatal failures.
40 # Every critical failure (colima restart, sed, cp backup, toggle.sh)
41 # uses an explicit `|| die` so a half-applied state never escapes.
42
43 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
44 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
45 LOG_FILE="$PROJECT_ROOT/output.log"
46 source "$SCRIPT_DIR/common.sh"
47 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
48 COLIMA_YAML="$HOME/.colima/default/colima.yaml"
49 PLATFORM="$(uname -s)"
50
51 # Argument parsing
52 DRY_RUN=false
53 SKIP_TOGGLE=false
54 case "${1:-}" in
55     --dry|-n) DRY_RUN=true ;;
56     --skip-toggle) SKIP_TOGGLE=true ;;
57     --help|-h)
58         sed -n '28,37p' "$0"
59         exit 0
60     ;;
61     *)
62         : # default: apply
63     ;;
64 *)
65     printf 'Unknown argument: %s\n' "$1" >&2
66     exit 2
67     ;;
68 esac
69
70
71 # Helper: print "[dry-run] $cmd" or actually run $cmd. Wraps the redirect
72 # INSIDE the function so outer `>> $FILE` statements never accidentally
73 # append dry-run noise to the real file. (Earlier draft didn't and the
74 # dry-run polluted $COLIMA_YAML.)
75 run() {
76     if [ "$DRY_RUN" = "true" ]; then
77
```

```

78     printf '        [dry-run] %s\n' "$*"
79     return 0
80 fi
81 printf '    $ %s\n' "$*"
82 "$@"
83 }
84
85 # Platform-aware sed in-place. macOS BSD sed needs an empty argument;
86 # GNU sed (Linux) does not. Earlier draft used `-i ''` everywhere and
87 # would have failed on Linux silently.
88 sed_inplace() {
89     local expr="$1"
90     local file="$2"
91     case "$PLATFORM" in
92         Darwin) run sed -i '' "$expr" "$file" ;;
93         *)      run sed -i "$expr" "$file" ;;
94     esac
95 }
96
97 # 0. Resolve the ACTIVE physical interface once (used everywhere)
98 # Replaces the earlier `en0` hardcoding. Same algorithm recover.sh uses:
99 # `route get default` if utun/tun, walk en0..en5 first with `status: active`
100 # + `inet` fall back to en0. This means the script also works on hosts
101 # whose primary NIC is en1/en2 (Thunderbolt Ethernet, USB-LAN, etc.).
102 resolve_active_iface() {
103     local iface
104     iface=$(route get default 2>> "$LOG_FILE" | awk '/interface:/{print $2; exit}')
105     if [ -z "$iface" ] || [ "${iface#utun}" != "$iface" ] || [ "${iface#tun}" != "$iface" ]; then
106         for i in en0 en1 en2 en3 en4 en5; do
107             if ifconfig "$i" 2>> "$LOG_FILE" | grep -q "status: active" \
108                 && ifconfig "$i" 2>> "$LOG_FILE" | grep -q "inet "; then
109                 iface="$i"; break
110             fi
111         done
112     fi
113     [ -z "$iface" ] && iface="en0"
114     printf '%s\n' "$iface"
115 }
116 ACTIVE_IFACE=$(resolve_active_iface)
117
118 # 1. Detect collision
119 step "1/9 Detecting Docker-bridge vs Wi-Fi subnet collision"
120
121 # Host subnet on the ACTIVE interface (not hardcoded en0).
122 HOST_SUBNET=$(ifconfig "$ACTIVE_IFACE" 2>> "$LOG_FILE" \
123     | awk '/inet /{print $2; exit}')
124 HOST_GATEWAY=$(route get default 2>> "$LOG_FILE" \
125     | awk '/gateway:/{print $2; exit}')
126 DEFAULT_DEV=$(route get default 2>> "$LOG_FILE" \
127     | awk '/interface:/{print $2; exit}')
128
129 printf '    active iface: %s\n' "$ACTIVE_IFACE"
130 printf '    iface inet: %s\n' "${HOST_SUBNET:-none}"
131 printf '    default via: %s\n' "${HOST_GATEWAY:-none}"
132 printf '    default dev: %s\n' "${DEFAULT_DEV:-none}"
133
134 # Collision fires when the host's default-route gateway is 172.17.0.1 (Docker's
135 # bridge) OR when the host subnet is in 172.17.0.0/16. The full-bridge-collision
136 # case (gateway is literally the docker bridge IP) is the smoking gun.
137 COLLISION=false
138 if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#172.17.}" != "$HOST_GATEWAY" ]; then
139     COLLISION=true
140     fail "default gateway $HOST_GATEWAY collides with Docker's default bridge (172.17.0.0/16)"
141 elif [ -n "$HOST_SUBNET" ] && [ "${HOST_SUBNET#172.17.}" != "$HOST_SUBNET" ]; then
142     COLLISION=true
143     fail "host subnet $HOST_SUBNET overlaps Docker's 172.17.0.0/16 bridge"
144 else
145     ok "no collision detected between host Wi-Fi and Docker's default bridge"
146 fi
147
148 if [ "$COLLISION" = "false" ]; then
149     echo ""
150     echo "Nothing to fix. If you still see egress issues, run:"
151     echo "    bash scripts/diagnose-host-leak.sh"
152     exit 0
153 fi
154
155 # 2. Pick non-colliding bip via /8-family routing
156 step "2/9 Picking a non-colliding Docker bip for ~/.colima/default/colima.yaml"
157
158 # Earlier draft tried naive prefix-match: "if candidate's 3-octet prefix is

```

```

159 # not a literal prefix of host gateway, pick it". That breaks on /12 campus
160 # networks (172.16.0.0/12 covers 172.16-172.31): a candidate like
161 # 172.26.0.1/24 LIVES INSIDE the host's /12 even though the prefix test
162 # passes. Fix: classify the host's address family FIRST, then choose ALL
163 # candidates from a DIFFERENT RFC1918 family. /8 boundaries are coarse
164 # enough that no /24 picked this way can ever collide.
165 case "$HOST_GATEWAY" in
166     172.16.*|172.17.*|172.18.*|172.19.*|172.20.*|172.21.*|172.22.*|172.23.*|\
167     172.24.*|172.25.*|172.26.*|172.27.*|172.28.*|172.29.*|172.30.*|172.31.*)
168     HOST_FAMILY="172"
169     # Host is in 172.x jump past 172.31 entirely. Pick from 10.x / 192.168.x.
170     CANDIDATES="10.200.0.1/24 10.201.0.1/24 192.168.99.1/24"
171     ;;
172 10.*)
173     HOST_FAMILY="10"
174     # Host is in 10.x pick from 172.x (well outside any commonly-deployed
175     # 10.0.0.0/8 slice) / 192.168.x.
176     CANDIDATES="172.26.0.1/24 172.28.0.1/24 192.168.99.1/24"
177     ;;
178 192.168.*)
179     HOST_FAMILY="192"
180     CANDIDATES="172.26.0.1/24 10.200.0.1/24 10.201.0.1/24"
181     ;;
182 *)
183     HOST_FAMILY="unknown"
184     warn "host gateway $HOST_GATEWAY is not in any RFC1918 range (/8 family)"
185     warn "falling back to abstract candidate set; verify collision-free manually"
186     CANDIDATES="172.26.0.1/24 10.200.0.1/24 192.168.99.1/24"
187     ;;
188 esac
189
190 printf ' host address family: %s\n' "$HOST_FAMILY"
191 printf ' candidate pool: %s\n' "$CANDIDATES"
192
193 # Sanity-check the chosen pool has at least one entry that doesn't literally
194 # share a /24 with the host gateway. Naive check (just /24 prefix) is fine
195 # here because we already picked across families.
196 CHOSEN=""
197 for cand in $CANDIDATES; do
198     PREFIX3=$(printf '%s' "$cand" | cut -d. -f1-3)
199     if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#"$PREFIX3"}" != "$HOST_GATEWAY" ]; then
200         warn "skip $cand (literal /24 collision with host gateway $HOST_GATEWAY)"
201         continue
202     fi
203     CHOSEN="$cand"
204     ok "chose bip=$CHOSEN (cross-family, no /24 literal collision)"
205     break
206 done
207
208 if [ -z "$CHOSEN" ]; then
209     CHOSEN="172.26.0.1/24"
210     warn "could not find a clean candidate; defaulting to $CHOSEN"
211     warn "verify with 'route get default' after restart"
212 fi
213
214 # 3. Backup colima.yaml
215 step "3/9 Backing up $COLIMA_YAML"
216 mkdir -p "$(dirname "$COLIMA_YAML")"
217 BACKUP="$HOME/.colima/default/colima.yaml.bak.$TIMESTAMP"
218 if [ -f "$COLIMA_YAML" ]; then
219     # `cp` is safe in dry-run because run() suppresses the inner command
220     # entirely no chance of accidentally writing a real backup.
221     run cp -p "$COLIMA_YAML" "$BACKUP" || die "backup failed"
222     if [ "$DRY_RUN" = "false" ]; then ok "backup: $BACKUP"; fi
223 else
224     warn "no existing $COLIMA_YAML (will create fresh)"
225 fi
226
227 # 4. Edit colima.yaml in place (idempotent)
228 step "4/9 Setting docker.bip=$CHOSEN in colima.yaml (in place)"
229
230 # Resolve the existing docker.bip (if any) without picking up commented-out
231 # YAML keys. awk-based extraction; the negated-comment check (`! /^[[:space:]]*#`)
232 # ensures `# bip: 1.2.3.4/24` is treated as a comment, not the active value.
233 EXISTING_BIP=""
234 if [ -f "$COLIMA_YAML" ]; then
235     EXISTING_BIP=$(awk '
236         /^[[:space:]]*#/{next}
237         /^[[:space:]]*bip:[[:space:]]*/{print $2; exit}
238         "$COLIMA_YAML" 2>> "$LOG_FILE" || true)
239 fi

```

```

240
241 if [ -n "$EXISTING_BIP" ] && [ "$EXISTING_BIP" = "$CHOSEN" ]; then
242     ok "colima.yaml already has docker.bip=$CHOSEN no edit needed"
243 elif [ -n "$EXISTING_BIP" ]; then
244     printf ' (replacing existing docker.bip=%s with %s)\n' "$EXISTING_BIP" "$CHOSEN"
245     # Use sed_inplace so Linux / macOS get the right syntax. Skip YAML comment
246     # lines so existing `# bip:` comments aren't rewired.
247     sed_inplace "s|~\|([[:space:]]*bip:[[:space:]]*\|)\|.*$|\|1$CHOSEN|" "$COLIMA_YAML" \
248     || die "sed replacement failed (colima.yaml malformed?)"
249     ok "colima.yaml updated"
250 else
251     # No existing `bip:` key. Branch:
252     # - file has `docker:` block append ` bip:` under it
253     # - file has no `docker:` block append new docker: section
254     # - file does not exist create fresh
255     # Each branch is dry-run-aware so no branch can accidentally leak noise
256     # into $COLIMA_YAML during a preview.
257     HAS_DOCKER_BLOCK=false
258     if [ -f "$COLIMA_YAML" ] && grep -q '^docker:' "$COLIMA_YAML"; then
259         HAS_DOCKER_BLOCK=true
260     fi
261
262     if [ "$HAS_DOCKER_BLOCK" = "true" ]; then
263         if [ "$DRY_RUN" = "true" ]; then
264             printf ' [dry-run] would append to %s:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
265         else
266             printf '\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
267             || die "append failed"
268             ok "appended bip under existing docker: block"
269         fi
270     elif [ -f "$COLIMA_YAML" ]; then
271         if [ "$DRY_RUN" = "true" ]; then
272             printf ' [dry-run] would append to %s:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
273         else
274             printf '\ndocker:\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
275             || die "append failed"
276             ok "appended new docker: block at end of file"
277         fi
278     else
279         if [ "$DRY_RUN" = "true" ]; then
280             printf ' [dry-run] would create %s with:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
281         else
282             printf 'docker:\n bip: %s\n' "$CHOSEN" > "$COLIMA_YAML" \
283             || die "create failed"
284             ok "created fresh $COLIMA_YAML with docker.bip=$CHOSEN"
285         fi
286     fi
287 fi
288
289 # Always show the final colima.yaml docker section for visibility
290 printf '\n%b colima.yaml `docker:` section now reads:%b\n' "$BOLD" "$NC"
291 awk '
292 /~docker:/{p=1}
293 p && /[~ ]/ && !/~docker:/{exit}
294 p
295 ' "$COLIMA_YAML" 2>/dev/null | sed 's/~ / /' || true
296
297 if [ "$DRY_RUN" = "true" ]; then
298     printf '\n %b(dry-run: exiting before colima restart / dhcp rebind / toggle.sh)%b\n' "$YELLOW" "$NC"
299     exit 0
300 fi
301
302 # 5. Restart Colima
303 step "5/9 Restarting Colima VM with new bip"
304 if ! command -v colima >> "$LOG_FILE" 2>&1; then
305     die "colima not on PATH install with 'brew install colima' first"
306 fi
307 # No timeout wrapper (would mask actual VM startup time). If colima hangs,
308 # the user kills with Ctrl+C and can manually `colima restart` from a fresh
309 # terminal the rest of this script is idempotent so re-running works.
310 colima restart >> "$LOG_FILE" 2>&1 || die "colima restart failed; check $LOG_FILE"
311 ok "colima restarted"
312
313 # 6. Rebind DHCP on the ACTIVE interface (not hardcoded en0)
314 step "6/9 Rebinding DHCP on $ACTIVE_IFACE so host re-learns its real gateway"
315
316 # Map ifname macOS network service name (Wi-Fi, USB 10/100 LAN, etc.).
317 # Same logic as the diagnostic + toggle.sh.
318 ACTIVE_SVC=""
319 if [ "$PLATFORM" = "Darwin" ]; then
320     ACTIVE_SVC=$(get_active_service)

```

```

321 fi
322
323 case "$PLATFORM" in
324     Darwin)
325         sudo -n networksetup -setdhcp "$ACTIVE_SVC" renew >> "$LOG_FILE" 2>&1 \
326             && ok "DHCP rebind issued on $ACTIVE_SVC" \
327             || warn "setdhcp renew failed; try 'sudo ipconfig set $ACTIVE_IFACE DHCP' manually"
328         ;;
329     Linux)
330         sudo -n dhclient -r "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 || true
331         sudo -n dhclient "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 \
332             && ok "dhclient rebind on $ACTIVE_IFACE" \
333             || warn "dhclient rebind failed"
334         ;;
335     *)
336         warn "unsupported platform $PLATFORM; skip DHCP rebind"
337         ;;
338 esac
339
340 # 7. Verify new default route
341 step "7/9 Verifying the new default route is NOT the Docker bridge"
342 sleep 2 # give DHCP + kernel a beat
343 NEW_DEFAULT=$(netstat -rn 2>> "$LOG_FILE" | awk '/^default/ {print $0; exit}')
344 printf ' %s\n' "${NEW_DEFAULT:-no default route detected}"
345 NEW_GATEWAY=$(printf '%s' "$NEW_DEFAULT" | awk '{print $2; exit}')
346
347 if [ -n "$NEW_GATEWAY" ] && [ "${NEW_GATEWAY#172.17.}" != "$NEW_GATEWAY" ]; then
348     fail "default route STILL points at 172.17.x.x DHCP rebind may need manual touch"
349     warn "try: sudo ipconfig set $ACTIVE_IFACE DHCP (or reconnect Wi-Fi in the menu bar)"
350 elif [ -n "$NEW_GATEWAY" ]; then
351     ok "default route is now via $NEW_GATEWAY (no longer Docker's bridge)"
352 else
353     warn "could not determine new gateway; verify with 'route get default'"
354 fi
355
356 # 8. Re-run toggle.sh to bring the gateway back up
357 if [ "$SKIP_TOGGLE" = "true" ]; then
358     step "8/9 Skipping toggle.sh (--skip-toggle)"
359     warn "gateway is currently down run ./toggle.sh manually when ready"
360 else
361     step "8/9 Re-running toggle.sh to bring the gateway back up"
362     if bash "$PROJECT_ROOT/toggle.sh" 2>&1 | tee -a "$LOG_FILE"; then
363         ok "toggle.sh completed"
364     else
365         warn "toggle.sh returned non-zero see $LOG_FILE for details"
366     fi
367 fi
368
369 # 9. Re-run the diagnostic + show verdict
370 step "9/9 Re-running host-leak diagnostic to confirm fix"
371 if bash "$SCRIPT_DIR/diagnose-host-leak.sh" 2>&1 | tee -a "$LOG_FILE"; then
372     LATEST_REPORT=$(ls -t "$PROJECT_ROOT"/host-leak-diagnostic-*.txt 2>/dev/null | head -1)
373     if [ -n "$LATEST_REPORT" ]; then
374         VERDICT=$(awk '/^ Scenario:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
375         WARP=$(awk -F: '*/' /^ WARP egress:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
376         printf '\n Most recent verdict: Scenario=%s WARP=%s\n' \
377             "${VERDICT:-?}" "${WARP:-?}"
378         if [ "$VERDICT" = "OK" ]; then
379             ok "fix confirmed by diagnostic host traffic is going through WARP"
380         fi
381     fi
382 else
383     warn "diagnostic exited non-zero; verify manually"
384 fi
385
386 printf '\n%b%b\n' "$BOLD" "$NC"
387 printf '%bDone.%b Full transcript at %s\n' "$BOLD" "$NC" "$LOG_FILE"
388 printf '%b%b\n' "$BOLD" "$NC"
389 exit 0

```

34 scripts/fix-ssh-delay.sh

```

1 #!/bin/bash
2 # scripts/fix-ssh-delay.sh Fix ~20s SSH connection delay caused by UseDNS
3 #
4 # macOS OpenSSH defaults UseDNS to "yes". When Tailscale SSH connects from

```

```

5 # a peer (phone/laptop), sshd performs a reverse-DNS (PTR) lookup on the
6 # client's 100.x.x.x Tailscale IP. Since DNS is hijacked to the nullexit
7 # gateway AdGuard Cloudflare WARP, and 100.64.0.0/10 is CGNAT private
8 # space, the PTR query times out (~15-20s) before SSH proceeds.
9 #
10 # This script uncomments "UseDNS no" in /etc/ssh/sshd_config and reloads
11 # sshd. Existing SSH sessions are NOT dropped.
12 #
13 # Usage:
14 #   sudo bash scripts/fix-ssh-delay.sh
15
16 set -euo pipefail
17
18 SSHD_CONFIG="/etc/ssh/sshd_config"
19
20 # Check we're root
21 if [ "$(id -u)" -ne 0 ]; then
22     echo "ERROR: This script must be run with sudo (it edits $SSHD_CONFIG)"
23     echo "Usage: sudo bash scripts/fix-ssh-delay.sh"
24     exit 1
25 fi
26
27 # Apply config (idempotent)
28 if grep -qE '^UseDNS no' "$SSHD_CONFIG" 2>/dev/null; then
29     echo "UseDNS is already set to 'no' in $SSHD_CONFIG"
30 elif grep -qE '^#UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
31     echo "Uncommenting 'UseDNS no' in $SSHD_CONFIG"
32     sed -i 's/^#UseDNS no/UseDNS no/' "$SSHD_CONFIG"
33 elif grep -qEi '^UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
34     echo "Changing existing UseDNS line to 'UseDNS no'"
35     sed -i 's/^UseDNS.*/UseDNS no/' "$SSHD_CONFIG"
36 else
37     echo "Appending 'UseDNS no' to $SSHD_CONFIG"
38     echo "" >> "$SSHD_CONFIG"
39     echo "UseDNS no" >> "$SSHD_CONFIG"
40 fi
41
42 # Verify
43 if ! grep -qE '^UseDNS no' "$SSHD_CONFIG"; then
44     echo "Failed to apply UseDNS no check $SSHD_CONFIG manually"
45     exit 1
46 fi
47
48 # Reload sshd (does NOT drop existing connections)
49 echo "Reloading sshd..."
50
51 SSHD_PID=$(pgrep -f /usr/sbin/sshd 2>/dev/null | head -1 || true)
52 if [ -n "$SSHD_PID" ]; then
53     kill -HUP "$SSHD_PID"
54     echo "Sent SIGHUP to sshd (PID $SSHD_PID) config reloaded"
55 else
56     echo "No running sshd found. macOS starts sshd on-demand per connection."
57     echo "The new 'UseDNS no' config will take effect on the next SSH connection."
58 fi
59
60 echo ""
61 echo "Done. SSH connections should now connect instantly."
62 echo "Existing sessions were not interrupted."

```

35 scripts/generate_tex.py

```

1 #!/usr/bin/env python3
2 import os
3
4 PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), '..'))
5
6 IGNORE_DIRS = {'.git', '.github', 'adguard', '__pycache__', 'Recover Gateway.app', 'Toggle Gateway.app'}
7 IGNORE_FILES = {'.env', 'nullexit_unified.tex', 'nullexit_unified.pdf', 'output.log', 'blocked.log', '.DS_Store',
8                 '.lan_p2p_detected', '.signatures', '.gateway_ip', 'TUNNEL_FAILED_CLOSED.marker', '.host_ips', 'refactor.md',
9                 'sweep.md', '.build_hash', '.rules_hash'}
10 IGNORE_EXTS = {'.pdf', '.tex', '.png', '.jpg', '.jpeg', '.zip', '.tar', '.gz', '.log', '.aux', '.fls', '.fdb_latexmk',
11                '.out', '.toc', '.xdv', '.synctex(busy)'}
12
13 # Exclude from code block inclusion, but keep in tree
14 SKIP_CONTENT_FILES = {'LICENSE'}

```

```

13 PRIORITY_FILES = [
14     'README.md',
15     'agent.md',
16     'devref.md',
17     'diagrams.md',
18     'toggle.sh',
19     'recover.sh',
20     'setup.sh',
21     'scripts/setup-linux.sh',
22     'docker-compose.yml',
23     'scripts/common.sh',
24     'scripts/watcher.sh',
25     'scripts/crypto.sh'
26 ]
27
28 def get_language(filename):
29     ext = os.path.splitext(filename)[1].lower()
30     if ext == '.sh' or ext == '.applescript':
31         return 'bash'
32     elif ext == '.py':
33         return 'Python'
34     elif ext == '.go':
35         return 'Go'
36     elif ext in {'.yaml', '.yml'}:
37         return 'bash'
38     elif ext == '.plist':
39         return 'XML'
40     elif ext == '.md':
41         return ''
42     return ''
43
44 def escape_tex(text):
45     return text.replace('_', '\\_')
46
47 def generate_tree(dir_path, prefix=""):
48     tree_str = ""
49     entries = sorted(os.listdir(dir_path))
50     entries = [e for e in entries if e not in IGNORE_DIRS and e not in IGNORE_FILES and os.path.splitext(e)[1].
51                lower() not in IGNORE_EXTS]
52
53     for i, entry in enumerate(entries):
54         path = os.path.join(dir_path, entry)
55         is_last = (i == len(entries) - 1)
56         connector = "-- " if is_last else "|-- "
57         tree_str += f"{prefix}{connector}{entry}\n"
58
59         if os.path.isdir(path):
60             extension = " " if is_last else "|  "
61             tree_str += generate_tree(path, prefix + extension)
62
63     return tree_str
64
65 def main():
66     tex_path = os.path.join(PROJECT_ROOT, 'nullexit_unified.tex')
67
68     preamble = r"""
69 \documentclass{article}
70 \usepackage[utf8]{inputenc}
71 \usepackage[margin=1in]{geometry}
72 \usepackage{listings}
73 \usepackage{xcolor}
74 \usepackage{hyperref}
75 \usepackage{tocloft}
76
77 \definecolor{codegreen}{rgb}{0,0.6,0}
78 \definecolor{codegray}{rgb}{0.5,0.5,0.5}
79 \definecolor{codepurple}{rgb}{0.58,0,0.82}
80 \definecolor{backcolour}{rgb}{0.97,0.97,0.96}
81
82 \lstdefinestyle{mystyle}{
83     backgroundcolor=\color{backcolour},
84     commentstyle=\color{codegreen},
85     keywordstyle=\color{blue}\bfseries,
86     numberstyle=\tiny\color{codegray},
87     stringstyle=\color{codepurple},
88     basicstyle=\ttfamily\scriptsize,
89     breakatwhitespace=false,
90     breaklines=true,
91     captionpos=b,
92     keepspaces=true,
93     numbers=left,
94     numbersep=5pt,

```

```

93     showspace=false,
94     showstringspaces=false,
95     showtabs=false,
96     tabsize=2,
97     frame=single,
98     rulecolor=\color{codegray},
99     extendedchars=true,
100     literate={-}1 {}{=}1 {}{{OK}}2 {}{-}1 {}{}1 {}{+}1 {}{+}1 {}{+}1 {}{+}1 {}{+}1 {}{+}1 {}{+}1 {}{+}1 {}{+}1
101 }
102 \lstset{style=mystyle}
103
104 \title{Nullexit: Unified Project Document}
105 \author{Omar Alaaeldin}
106 \date{\today}
107
108 \begin{document}
109 \maketitle
110
111 \tableofcontents
112 \newpage
113
114 \section{Project Directory Tree}
115 \begin{verbatim}
116 nullexit/
117 """
118
119     tree = generate_tree(PROJECT_ROOT)
120     midamble = r"""\end{verbatim}
121 \newpage
122 """
123
124     all_relpaths = []
125     for root, dirs, files in os.walk(PROJECT_ROOT):
126         dirs[:] = sorted([d for d in dirs if d not in IGNORE_DIRS])
127         for file in sorted(files):
128             if file in IGNORE_FILES or file in SKIP_CONTENT_FILES or os.path.splitext(file)[1].lower() in
129             IGNORE_EXTS:
130                 continue
131
132             filepath = os.path.join(root, file)
133             relpath = os.path.relpath(filepath, PROJECT_ROOT)
134             all_relpaths.append(relpath)
135
136     def sort_key(path):
137         if path in PRIORITY_FILES:
138             return (0, PRIORITY_FILES.index(path))
139         return (1, path)
140
141     all_relpaths.sort(key=sort_key)
142
143     with open(tex_path, 'w', encoding='utf-8') as f:
144         f.write(preamble)
145         f.write(tree)
146         f.write(midamble)
147
148         for relpath in all_relpaths:
149             lang = get_language(relpath)
150             lang_attr = f"[language={lang}]" if lang else ""
151
152             f.write(f"\\section{{{escape_tex(relpath)}}}\n")
153             f.write(f"\\begin{{{lstlisting}}}{lang_attr}\n")
154             try:
155                 with open(os.path.join(PROJECT_ROOT, relpath), 'r', encoding='utf-8', errors='ignore') as src:
156                     content = src.read()
157                     import re
158                     # Strip all non-ASCII characters (emojis, box drawings, em dashes, etc)
159                     content = re.sub(r'[\x00-\x7F]+', '', content)
160                     f.write(content)
161                     if not content.endswith('\n'):
162                         f.write('\n')
163             except Exception as e:
164                 f.write(f"Error reading file: {e}\n")
165             f.write("\\end{{{lstlisting}}}\n\n")
166
167         f.write("\\end{document}\n")
168
169     print(f"Generated {tex_path}")
170
171 if __name__ == '__main__':
172     main()

```

36 scripts/logger/go.mod

```
1 module nullexit/logger
2
3 go 1.22
```

37 scripts/logger/main.go

```
1 package main
2
3 import (
4     "bufio"
5     "encoding/json"
6     "fmt"
7     "io"
8     "log"
9     "os"
10    "regexp"
11    "strings"
12    "time"
13 )
14
15 const (
16     queryLogPath = "/adguard_work/data/querylog.json"
17     procKmsgPath = "/proc/kmsg"
18     outputLogPath = "/app/blocked.log"
19 )
20
21 func logMessage(msg string) {
22     timestamp := time.Now().Format("2006-01-02 15:04:05")
23     formattedMsg := fmt.Sprintf("%s %s\n", timestamp, msg)
24     fmt.Print(formattedMsg)
25
26     f, err := os.OpenFile(outputLogPath, os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
27     if err != nil {
28         log.Printf("Error writing to log file: %v\n", err)
29         return
30     }
31     defer f.Close()
32     f.WriteString(formattedMsg)
33 }
34
35 func tailFile(filepath string, out chan<- string) {
36     for {
37         if _, err := os.Stat(filepath); os.IsNotExist(err) {
38             time.Sleep(1 * time.Second)
39             continue
40         }
41         break
42     }
43
44     f, err := os.Open(filepath)
45     if err != nil {
46         log.Printf("Error opening file %s: %v\n", filepath, err)
47         return
48     }
49     defer func() {
50         if f != nil {
51             f.Close()
52         }
53     }()
54
55     // Seek to end
56     f.Seek(0, io.SeekEnd)
57     reader := bufio.NewReader(f)
58
59     for {
60         line, err := reader.ReadString('\n')
61         if err != nil {
62             if err == io.EOF {
```

```

63     stat, err := os.Stat(filepath)
64     if err == nil {
65         currentOffset, _ := f.Seek(0, io.SeekCurrent)
66         if stat.Size() < currentOffset {
67             // File truncated or rotated
68             f.Close()
69             for {
70                 if _, err := os.Stat(filepath); os.IsNotExist(err) {
71                     time.Sleep(1 * time.Second)
72                     continue
73                 }
74                 break
75             }
76             f, _ = os.Open(filepath)
77             reader = bufio.NewReader(f)
78             continue
79         }
80     }
81     time.Sleep(500 * time.Millisecond)
82     continue
83 }
84 log.Printf("Error reading file %s: %v\n", filepath, err)
85 time.Sleep(1 * time.Second)
86 continue
87 }
88 out <- line
89 }
90 }
91
92 type adguardLogEntry struct {
93     IP      string `json:"IP"`
94     QH      string `json:"QH"`
95     QT      string `json:"QT"`
96     Result struct {
97         IsFiltered bool `json:"IsFiltered"`
98         Reason      int  `json:"Reason"`
99         Rules       []struct {
100             Text string `json:"Text"`
101         } `json:"Rules"`
102     } `json:"Result"`
103 }
104
105 func monitorDNS() {
106     logMessage("[System] Starting DNS block logger...")
107     lines := make(chan string)
108     go tailFile(queryLogPath, lines)
109
110     reasonMap := map[int]string{
111         2: "CustomRule",
112         3: "BlockList",
113         4: "SafeBrowsing",
114         5: "ParentalControl",
115         6: "SafeSearch",
116         7: "BlockedService",
117     }
118
119     for line := range lines {
120         var data adguardLogEntry
121         if err := json.Unmarshal([]byte(line), &data); err != nil {
122             continue
123         }
124
125         res := data.Result
126         if res.IsFiltered || (res.Reason != 0 && res.Reason != 1) {
127             qh := data.QH
128             if qh == "" {
129                 qh = "unknown"
130             }
131             qt := data.QT
132             if qt == "" {
133                 qt = "unknown"
134             }
135             clientIP := data.IP
136             if clientIP == "" {
137                 clientIP = "unknown"
138             }
139
140             ruleText := "unknown rule"
141             if len(res.Rules) > 0 && res.Rules[0].Text != "" {
142                 ruleText = res.Rules[0].Text
143             }

```

```

144     reasonStr, ok := reasonMap[res.Reason]
145     if !ok {
146         reasonStr = fmt.Sprintf("Reason-%d", res.Reason)
147     }
148
149     logMessage(fmt.Sprintf("[DNS] Blocked %s (Type: %s) for client %s | Reason: %s | Rule: %s", qh, qt,
150         clientIP, reasonStr, ruleText))
151 }
152 }
153 }
154
155 func monitorIPs() {
156     logMessage("[System] Starting IP block logger...")
157
158     prefixRe := regexp.MustCompile(`(IP_BLOCK_[A-Z_]+):`)
159     srcRe := regexp.MustCompile(`SRC=([0-9a-fA-F\.:]+)`)
160     dstRe := regexp.MustCompile(`DST=([0-9a-fA-F\.:]+)`)
161     protoRe := regexp.MustCompile(`PROTO=([A-Z0-9]+)`)
162     sptRe := regexp.MustCompile(`SPT=(\d+)`)
163     dptRe := regexp.MustCompile(`DPT=(\d+)`)
164     inRe := regexp.MustCompile(`IN=([a-zA-Z0-9\.\_-]+)`)
165     outRe := regexp.MustCompile(`OUT=([a-zA-Z0-9\.\_-]+)`)
166
167     f, err := os.Open(procKmsgPath)
168     if err != nil {
169         if os.IsPermission(err) {
170             logMessage("[System] ERROR: Insufficient permissions to read /proc/kmsg. Enable CAP_SYSLOG.")
171         } else {
172             logMessage(fmt.Sprintf("[System] ERROR in IP logger: %v", err))
173         }
174         return
175     }
176     defer f.Close()
177
178     reader := bufio.NewReader(f)
179     for {
180         line, err := reader.ReadString('\n')
181         if err != nil {
182             if err == io.EOF {
183                 time.Sleep(100 * time.Millisecond)
184                 continue
185             }
186             logMessage(fmt.Sprintf("[System] ERROR reading kmsg: %v", err))
187             time.Sleep(1 * time.Second)
188             continue
189         }
190
191         if strings.Contains(line, "IP_BLOCK") {
192             match := prefixRe.FindStringSubmatch(line)
193             if len(match) > 1 {
194                 prefix := match[1]
195
196                 src := "unknown"
197                 if m := srcRe.FindStringSubmatch(line); len(m) > 1 {
198                     src = m[1]
199                 }
200
201                 dst := "unknown"
202                 if m := dstRe.FindStringSubmatch(line); len(m) > 1 {
203                     dst = m[1]
204                 }
205
206                 proto := "unknown"
207                 if m := protoRe.FindStringSubmatch(line); len(m) > 1 {
208                     proto = m[1]
209                 }
210
211                 spt := ""
212                 if m := sptRe.FindStringSubmatch(line); len(m) > 1 {
213                     spt = m[1]
214                 }
215
216                 dpt := ""
217                 if m := dptRe.FindStringSubmatch(line); len(m) > 1 {
218                     dpt = m[1]
219                 }
220
221                 inIf := ""
222                 if m := inRe.FindStringSubmatch(line); len(m) > 1 {
223                     inIf = m[1]

```

```

224     }
225
226     outIf := ""
227     if m := outRe.FindStringSubmatch(line); len(m) > 1 {
228         outIf = m[1]
229     }
230
231     direction := "inbound"
232     if strings.HasSuffix(prefix, "DST") {
233         direction = "outbound"
234     }
235
236     listType := "MALICIOUS"
237     if !strings.Contains(prefix, "MALICIOUS") {
238         parts := strings.Split(prefix, "_")
239         if len(parts) >= 3 {
240             listType = "GEO_" + parts[2]
241         }
242     }
243
244     portStr := ""
245     if dpt != "" {
246         portStr = ":" + dpt
247     }
248
249     srcPortStr := ""
250     if spt != "" {
251         srcPortStr = ":" + spt
252     }
253
254     ifStr := ""
255     if inIf != "" && outIf != "" {
256         ifStr = fmt.Sprintf("IF: %s->%s", inIf, outIf)
257     } else if inIf != "" || outIf != "" {
258         ifStr = fmt.Sprintf("IF: %s%s", inIf, outIf)
259     }
260
261     logMessage(fmt.Sprintf("[IP] Blocked %s to %s%s (Proto: %s) from %s%s (%s) | List: %s", direction, dst,
portStr, proto, src, srcPortStr, ifStr, listType))
262 }
263 }
264 }
265 }
266
267 func main() {
268     logMessage("[System] Nullexit Block Logger starting...")
269
270     go monitorDNS()
271     monitorIPs()
272 }

```

38 scripts/pf.conf

```

1 # scripts/pf.conf - Packet Filter configuration for nullexit killswitch
2 # Dynamically parameterized by pfctl macro override
3
4 # ext_if must be passed via command line, e.g. -D ext_if=en0
5 # If not passed, default to en0
6 ext_if = "en0"
7 ts_if = "utun+"
8
9 # Tables populated dynamically via pfctl
10 table <vpn_endpoints> persist
11 table <derp_relays> persist
12
13 # Core Rules
14 set skip on lo0                # Ensure local loopback is never blocked
15 scrub out all max-mss 1160    # TCP MSS Clamping: Prevents fragmentation on host TCP flows (like Tailscale
control plane)
16 pass quick on $ts_if all      # Allow all Tailscale traffic (essential to prevent SSH lockout)
17
18 # Default deny outbound WAN traffic
19 block out on $ext_if all
20
21 # Block inbound Screen Sharing (VNC) and SSH from local Wi-Fi, forcing them to use Tailscale
22 block in on $ext_if proto tcp from any to any port { 22, 5900 }

```

```

23
24 # Exceptions for tunnel handshakes and coordination
25 pass out quick on $ext_if proto udp to <vpn_endpoints> port 2408
26 pass out quick on $ext_if proto udp to <derp_relays>
27 pass out quick on $ext_if proto tcp to 192.200.0.0/24 port 443
28
29 # Allow local LAN traffic (AirDrop, local Tailscale P2P, etc)
30 pass out quick on $ext_if proto tcp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
31 pass out quick on $ext_if proto udp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
32 pass out quick on $ext_if proto icmp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
33
34 # Allow DHCP outbound (client port 68 to server port 67) so we can obtain/renew IP address
35 pass out quick on $ext_if proto udp from any port 68 to any port 67

```

39 scripts/reinstall-host-tailscale.sh

```

1 #!/usr/bin/env bash
2 #
3 # scripts/reinstall-host-tailscale.sh
4 #
5 # Complete uninstall + reinstall of the host-side Homebrew Tailscale.
6 #
7 # Use this when the brew LaunchAgent is wedged, Login Items entries are
8 # duplicated, "Tailscale is stopped" persists despite
9 # `brew services restart tailscale`, or you simply want a clean baseline
10 # before re-engaging the gateway as exit node.
11 #
12 # Idempotent. Safe to re-run. Use --dry to preview every step with zero
13 # changes. Use --yes to skip the per-step confirmation prompts.
14 #
15 # This script ONLY touches the HOST-side Tailscale (the one installed by
16 # Homebrew `brew install tailscale` and managed by launchd as
17 # homebrew.mxcl.tailscale). The nullexit gateway container's tailscaled
18 # (which lives inside the Colima VM at 100.100.21.8 and offers the exit
19 # node to this host) is unaffected.
20
21 set -u
22
23 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
24
25 DRY=0
26 YES=0
27
28 for arg in "$@"; do
29     case "$arg" in
30         --dry) DRY=1 ;;
31         --yes) YES=1 ;;
32         --help|-h) # NOTE: heredoc is UNQUOTED so $SCRIPT_DIR expands in the body. If you
33             # add new $-style placeholders here, they'll expand the same way.
34             cat <<USAGE
35 Usage: bash "$SCRIPT_DIR/reinstall-host-tailscale.sh" [--dry] [--yes]
36
37 --dry Print every step, make zero changes. Use to audit before
38 committing.
39 --yes Auto-confirm the prompts at every destructive step. Useful in
40 CI / fully-scripted recovery; do not use until you've read
41 the script.
42
43 After this script completes cleanly, you still must (MANUALLY):
44
45 1. Open System Settings Privacy & Security Local Network and
46 enable Tailscale. macOS will prompt automatically the first time
47 the fresh install of `tailscale` is invoked.
48
49 2. Re-engage the tailnet + exit node:
50     sudo tailscale up --ssh=true --accept-dns=false \
51       --exit-node="$(cat .gateway_ip | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
52       --exit-node-allow-lan-access=true
53
54 3. Re-run ./toggle.sh from the project root.
55
56 4. Final verification:
57     curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)= '
58     expected: warp=on, ip=<Cloudflare>, colo=YYZ/YUL
59 USAGE
60     exit 0

```

```

61     ;;
62     *)
63     echo "Unknown flag: $arg" >&2
64     exit 2
65     ;;
66     esac
67 done
68
69 # ----- helpers -----
70
71 confirm() {
72     if [ "$YES" = 1 ] || [ "$DRY" = 1 ]; then
73         return 0
74     fi
75     printf "      %s [y/N] " "$1"
76     read -r ans
77     case "$ans" in
78         y|Y|yes|YES) return 0 ;;
79         *)
80             echo "      aborted by user"
81             exit 1
82             ;;
83     esac
84 }
85
86 header() {
87     printf '\n %s\n' "$1"
88 }
89
90 # with_timeout <seconds> <cmd...>
91 # Run a command in the background and kill it if it exceeds <seconds>.
92 # Returns the command's exit code if it finished in time, 124 (matching
93 # GNU `timeout` convention) if it timed out. macOS doesn't ship GNU
94 # coreutils `timeout`, so this is the bash-3.2-friendly replacement.
95 # stdout + stderr are discarded (we only care about exit code); redirect
96 # or pipe externally if you need the output.
97 with_timeout() {
98     local timeout_s="${1:-30}"
99     shift
100     "$@" >/dev/null 2>&1 &
101     local cmdpid=$!
102     local elapsed=0
103     while kill -0 "$cmdpid" 2>/dev/null; do
104         if [ "$elapsed" -ge "$timeout_s" ]; then
105             kill -9 "$cmdpid" 2>/dev/null
106             wait "$cmdpid" 2>/dev/null
107             return 124
108         fi
109         sleep 1
110         elapsed=$((elapsed+1))
111     done
112     wait "$cmdpid" 2>/dev/null
113     return $?
114 }
115
116 # ----- 0. Pre-flight -----
117 header "Pre-flight current host tailscale state"
118 echo "      brew: $(command -v brew >/dev/null 2>&1 && brew --version | head -1 || echo 'not on
119 PATH')"
119 echo "      tailscaled running: $(pgrep -lf tailscaled 2>/dev/null | wc -l | tr -d ' ') process(es)"
120 echo "      brew service tailscale: $(brew services list 2>/dev/null | awk '1=="tailscale"{print $2, $3}' | tr
121 '\n' ' ' | sed 's/ $//')
122 echo "      LaunchAgent plists (under ~/Library/LaunchAgents):"
123 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's/~/ /' || echo "      (none)"
124
125 # Detect Tailscale's macOS Network Extension (residual from prior App Store
126 # Tailscale.app installs). Lives under /Library/Apple/SystemExtensions/ and
127 # is managed by `systemextensionsctl` NOT launchd. If loaded, brew
128 # tailscaled will NOT engage the Network Extension framework slot because
129 # the SE still has it claimed. Symptom: 'sudo tailscale status' returns
130 # "Tailscale is stopped" even when the daemon is alive and listening.
131 se_found=0
132 if command -v systemextensionsctl >/dev/null 2>&1; then
133     # Grab any line matching tailscale and network-extension
134     se_line=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale\..*network-extension' | head -
135 n 1)
136 if [ -n "$se_line" ]; then
137     se_found=1
138     # Extract the teamID (10-char uppercase/numeric word)
139     se_teamID=$(echo "$se_line" | tr -s ' ' '\t' '\n' | grep -E '^[A-Z0-9]{10}$' | head -n 1)
140     # Extract the bundleID (starts with io.tailscale or com.tailscale)

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```

139 se_bundleID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^(io|com)\.tailscale\. ' | head -n 1 | sed 's
140 /(.*/)/')
141
142 # If teamID is not found, fallback to '-'
143 if [ -z "$se_teamID" ]; then
144     se_teamID="-"
145 fi
146 fi
147
148 if [ "$se_found" = 1 ] && [ -n "$se_bundleID" ]; then
149     echo "    Tailscale Network Extension System Extension is loaded (residual from App Store Tailscale.app
150         install)"
151     echo "    Team ID: $se_teamID, Bundle ID: $se_bundleID"
152     echo "    brew tailscale cannot engage the NE slot while this SE has it claimed."
153     if [ "$DRY" = 1 ]; then
154         echo "    [dry] (would prompt) sudo systemextensionsctl uninstall $se_teamID $se_bundleID"
155     else
156         confirm "Run 'sudo systemextensionsctl uninstall $se_teamID $se_bundleID'? (frees the NE slot for brew
157             tailscale; non-fatal if it errors)"
158         sudo systemextensionsctl uninstall "$se_teamID" "$se_bundleID" 2>&1 \
159             || echo "    ! uninstall returned non-zero (may require reboot, non-fatal)"
160     fi
161 else
162     echo "    no Tailscale System Extension loaded"
163 fi
164
165 # Pattern used by the Step-1 bootout loops, in both dry + real branches.
166 # Widened from a bare /tailscale/ so the same loops also pick up any
167 # nullexit-labeled LaunchAgents (e.g., com.nullexit.wake-recovery, the
168 # caffeinate + post-wake recovery hook). The wake-recovery LaunchAgent
169 # is re-installed by ./toggle.sh / setup.sh on re-engage, so wiping it
170 # here is non-fatal and is part of restoring a clean baseline.
171 TAILSCALE_LABEL_RE='tailscale'
172 NULLEXIT_LABEL_RE='nullexit'
173
174 # ----- 1. brew services stop -----
175 header "Step 1 Stop the brew-managed service + unload LaunchAgents"
176 # Detect whether the brew-managed service is registered as $USER (gui/$UID
177 # domain) or as root (system domain). NOPASSWD sudoers can leave a stale
178 # root-owned registration after a prior 'sudo brew services ...' round;
179 # 'brew services stop' without sudo refuses to touch a root-owned plist
180 # (Error: Service 'tailscale' is started as `root`).
181 brew_status_line=$(brew services list 2>/dev/null | awk '$1=="tailscale"')
182 brew_status_user=$(echo "$brew_status_line" | awk '{print $3}')
183 if [ -n "$brew_status_line" ]; then
184     if [ "$DRY" = 1 ]; then
185         echo "    [dry] brew services stop tailscale (user=$brew_status_user)"
186     else
187         if [ "$brew_status_user" = "root" ]; then
188             confirm "Run 'sudo brew services stop tailscale' (service is registered as root)?"
189             sudo brew services stop tailscale 2>&1 || echo "    (sudo brew services stop returned non-zero non-fatal
190                 )"
191         else
192             echo "    brew services stop tailscale (user=$brew_status_user, no sudo needed)"
193             brew services stop tailscale 2>&1 || echo "    (brew reported it wasn't actually running)"
194         fi
195     fi
196 fi
197 echo "    unload any tailscale-related LaunchAgents (prevents launchd from respawning after kill)"
198 if [ "$DRY" = 1 ]; then
199     echo "    [dry] for each 'tailscale|nullexit'-matched label in gui/$UID domain:"
200     while IFS= read -r label; do
201         [ -z "$label" ] && continue
202         echo "    [dry] launchctl bootout gui/$UID/$label"
203     done <<(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 &&
204         ($3 ~ ts || $3 ~ nx) {print $3}')
205     echo "    [dry] for each 'tailscale|nullexit'-matched label in system domain (root-owned brew plist):"
206     while IFS= read -r label; do
207         [ -z "$label" ] && continue
208         echo "    [dry] (would prompt) sudo launchctl bootout system/$label"
209     done <<(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" '
210         NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
211     if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
212         echo "    [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macos"
213     fi
214     if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
215         echo "    [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macsys"
216     fi
217 else
218     # Bootout every gui-domain LaunchAgent whose label matches 'tailscale'
219     # OR 'nullexit'. The nullexit arm catches com.nullexit.wake-recovery,

```

```

214 # which holds caffeinate/watcher.sh alive across a daemon reset not
215 # desired while the script is establishing a fresh baseline. Awake-
216 # recovery is re-installed by ./toggle.sh / setup.sh on re-engage, so
217 # wiping here is non-fatal. The pattern stays narrower than a bare
218 # /tail/ or /null/ so unrelated Apple services are not unloaded.
219 while IFS= read -r label; do
220     [ -z "$label" ] && continue
221     echo "        launchctl bootout gui/$UID/$label"
222     launchctl bootout "gui/$UID/$label" 2>/dev/null || true
223 done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 &&
($3 ~ ts || $3 ~ nx) {print $3}')
224 # Bootout every system-domain LaunchDaemon/Agent whose label matches
225 # 'tailscale' OR 'nullexit'. This catches the root-owned brew plist left
226 # behind by prior 'sudo brew services ...' invocations that was the
227 # case that left tailscaled PID 47447 57734 respawning between sudo
228 # pkill runs. Defensive nullexit catch in case the user previously
229 # `sudo ln -s`d the wake-recovery plist into /Library/LaunchDaemons.
230 while IFS= read -r label; do
231     [ -z "$label" ] && continue
232     confirm "Run 'sudo launchctl bootout system/$label'?"
233     sudo launchctl bootout "system/$label" 2>/dev/null || true
234 done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" '
NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
235 # Legacy Tailscale.app system-domain entries. Each one is checked and
236 # confirmed separately so we don't sudo-promp or sudo-bootout for nothing.
237 if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
238     confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macos'?"
239     sudo launchctl bootout system/com.tailscale.ipn.macos 2>/dev/null || true
240 fi
241 if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
242     confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macsys'?"
243     sudo launchctl bootout system/com.tailscale.ipn.macsys 2>/dev/null || true
244 fi
245 fi
246 echo "        waiting up to 10s for tailscaled to exit"
247 for i in 1 2 3 4 5 6 7 8 9 10; do
248     if [ "$DRY" = 1 ]; then
249         echo "        [dry] sleep 1; loop check would terminate here"
250         break
251     fi
252     sleep 1
253     if ! pgrep -lf tailscaled >/dev/null 2>&1; then
254         echo "        tailscaled exited in ${i}s"
255         break
256     fi
257     [ "$i" = 10 ] && echo "        tailscaled still alive after 10s Step 2 will deal with it"
258 done
259 else
260     echo "        brew reports no tailscale service loaded"
261 fi
262
263 # ----- 2. Kill any orphan tailscaled -----
264 header "Step 2 Kill any orphan tailscaled process(es) (may escalate to sudo)"
265 if ! pgrep -lf tailscaled >/dev/null 2>&1; then
266     echo "        no orphans"
267 else
268     echo "        Stray process(es):"
269     pgrep -lf tailscaled | sed 's/^/    /'
270     echo "        no-sudo: pkill tailscaled"
271     if [ "$DRY" = 1 ]; then
272         echo "        [dry] pkill tailscaled on failure: sudo pkill -9 tailscaled"
273     else
274         if pkill tailscaled 2>/dev/null; then
275             echo "        no-sudo pkill succeeded"
276         else
277             echo "        ! no-sudo pkill insufficient; cannot escalate yet"
278         fi
279         sleep 1
280     fi
281 fi
282 # Recheck; if still alive, offer sudo escalation
283 if { [ "$DRY" = 0 ] && pgrep -lf tailscaled >/dev/null 2>&1; } || [ "$DRY" = 1 ]; then
284     if [ "$DRY" = 1 ]; then
285         echo "        [dry] sudo pkill -9 tailscaled (escalation would fire here)"
286     else
287         confirm "Run 'sudo pkill -9 tailscaled' to escalate?"
288         sudo pkill -9 tailscaled 2>&1
289         sleep 1
290     fi
291 fi
292

```

```

293 if [ "$DRY" = 0 ]; then
294     if pgrep -lf tailscaled >/dev/null 2>&1; then
295         echo "      Failed to kill stragglers bailing out"
296         pgrep -lf tailscaled | sed 's/^/      /'
297         exit 3
298     fi
299     echo "      all tailscaled processes gone"
300 fi
301 fi
302
303 # ----- 3. brew uninstall -----
304 header "Step 3 brew uninstall tailscale"
305 if ! brew list tailscale >/dev/null 2>&1; then
306     echo "      not currently installed via brew skip"
307 else
308     # Detect whether the keg directory is owner=$USER or owner=root. If the
309     # keg is root-owned (typically a residual of some prior 'sudo brew ...'
310     # cycle that did perms fixups), 'brew uninstall' without sudo refuses
311     # to remove the files and prints: 'Error: Could not remove tailscale
312     # keg! Do so manually: sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>'.
313     # Detect that and escalate to a sudoed uninstall, with a manual rm -rf
314     # fallback if `sudo brew uninstall` itself errors out (e.g., partial
315     # state where the keg dir is gone but brew still thinks it's installed).
316     # Detect keg state. Treat `(dir absent)` AND `(dir present + empty)` as the
317     # same `<unknown>` bucket so both escalate to sudo. The empty-parent case
318     # is what you get after a manual `sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>`
319     # from inside: brew's parent dir is left present-but-empty and brew still
320     # thinks the package is installed.
321     keg_dir="/opt/homebrew/Cellar/tailscale"
322     if [ -d "$keg_dir" ] && [ -n "$(ls -A "$keg_dir" 2>/dev/null)" ]; then
323         keg_owner="$(stat -f '%Su' "$keg_dir" 2>/dev/null || echo '<unknown>')"
324     else
325         keg_owner="<unknown>"
326     fi
327     if [ "$DRY" = 1 ]; then
328         echo "      [dry] brew uninstall tailscale (keg_owner=$keg_owner)"
329     else
330         # 'root' AND '<unknown>' both require sudo escalation. The '<unknown>'
331         # case is the partial-state path: keg dir was already removed (manually
332         # or by a prior failed uninstall) but brew's internal state still says
333         # installed. Without this branch the script would fall through to a
334         # non-sudo brew uninstall that fails again with the same 'Could not
335         # remove tailscale keg!' error.
336         if [ "$keg_owner" = "root" ] || [ "$keg_owner" = "<unknown>" ]; then
337             confirm "Run 'sudo brew uninstall tailscale' (keg=$keg_owner)?"
338             sudo brew uninstall tailscale 2>&1 || {
339                 echo "      ! sudo brew uninstall failed; offering manual rm -rf fallback"
340                 # Guard the glob so an empty/absent dir doesn't literal-expand the
341                 # '*' and confuse the user with a sudo error about a non-existent path.
342                 if [ -d /opt/homebrew/Cellar/tailscale ] && [ -n "$(ls -A /opt/homebrew/Cellar/tailscale 2>/dev/null)" ]; then
343                     confirm "Run 'sudo rm -rf /opt/homebrew/Cellar/tailscale/*'?"
344                     sudo rm -rf /opt/homebrew/Cellar/tailscale/* 2>&1
345                 else
346                     echo "      keg dir already empty or absent nothing more to rm"
347                 fi
348             }
349         else
350             confirm "Run 'brew uninstall tailscale' (removes LaunchAgent plist + linked symlinks)?"
351             brew uninstall tailscale 2>&1
352         fi
353     fi
354 fi
355
356 # ----- 4. Wipe state directories -----
357 header "Step 4 Wipe state state"
358 # User-owned
359 declare -a USER_PATHS
360 USER_PATHS=(
361     "$HOME/.local/share/tailscale"
362     "$HOME/.local/run/tailscaled.socket"
363     "$HOME/.local/run/tailscaled.state"
364     "$HOME/.local/state/tailscale"
365     "$HOME/Library/Application Support/Tailscale"
366     "$HOME/Library/Caches/com.tailscale.ipn.macos"
367     "$HOME/Library/Caches/Tailscale"
368     "$HOME/Library/Logs/Tailscale"
369 )
370 echo "      User-owned paths (no sudo):"
371 for p in "${USER_PATHS[@]}; do
372     if [ -e "$p" ]; then

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```

373     if [ "$DRY" = 1 ]; then
374         echo "        [dry] rm -rf '$p'"
375     else
376         rm -rf "$p"
377         echo "        rm -rf '$p'"
378     fi
379 fi
380 done
381
382 # System
383 declare -a SUDO_PATHS
384 SUDO_PATHS=(
385     "/var/lib/tailscale"
386     "/var/cache/tailscale"
387 )
388 echo "    System paths under /var (will prompt for sudo if any are present):"
389 eligible_sudo=0
390 for p in "${SUDO_PATHS[@]}; do
391     if [ -e "$p" ]; then
392         eligible_sudo=1
393         if [ "$DRY" = 1 ]; then
394             echo "        [dry] sudo rm -rf '$p'"
395         else
396             if confirm "Run 'sudo rm -rf $p'?" ; then
397                 sudo rm -rf "$p" && echo "        rm -rf '$p'"
398             fi
399         fi
400     fi
401 done
402 [ "$eligible_sudo" = 0 ] && echo "        (none of /var/lib/tailscale, /var/cache/tailscale are present)"
403
404 # ----- 5. Wipe stale LaunchAgent plists -----
405 header "Step 5 Wipe stale LaunchAgent plist files"
406 declare -a PLIST_FILES
407 PLIST_FILES=(
408     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
409     "$HOME/Library/LaunchAgents/com.tailscale.ipn.macos.plist"
410     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist.bak"
411     "$HOME/Library/LaunchAgents/com.nullexit.wake-recovery.plist"
412 )
413 for f in "${PLIST_FILES[@]}; do
414     if [ -e "$f" ]; then
415         if [ "$DRY" = 1 ]; then
416             echo "        [dry] rm -f '$f'"
417         else
418             rm -f "$f"
419             echo "        rm -f '$f'"
420         fi
421     fi
422 done
423 echo "    Remaining tailscale|nullexit plists (should be empty):"
424 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's/^/' | echo "        (none)"
425
426 # ----- 5.5. Drain Background-Items (Login Items) -----
427 header "Step 5.5 Drain Background-Items (Login Items)"
428 # macOS 13+ keeps a parallel "Background Items" database alongside
429 # classic LaunchAgents. Even when we wipe ~/Library/LaunchAgents/homebrew.mxcl.tailscale.plist
430 # in Step 5, the .btm file at
431 # ~/Library/Application Support/com.apple.backgroundtaskmanagementagent.backgrounditems.btm
432 # can still hold a Background-Items entry pointing at the (now-dead) plist
433 # path. Next login: macOS sees the entry, tries to bootstrap, finds the
434 # plist missing, and either silently regenerates it via brew's post-install
435 # hook (causing tailscaled respawn at next login) or refuses outright.
436 # `sfltool reset-login-items` is Apple's canonical API for wiping the
437 # entire .btm file. It DOES wipe all Login Items the user re-adds what
438 # they want via System Settings General Login Items. Acceptable here
439 # because the goal of this script is "fresh baseline for Tailscale".
440 BTM="$HOME/Library/Application Support/com.apple.backgroundtaskmanagementagent/backgrounditems.btm"
441 if [ -f "$BTM" ]; then
442     if [ "$DRY" = 1 ]; then
443         echo "        [dry] found $BTM; would prompt sfltool reset-login-items"
444     else
445         confirm "Run 'sfltool reset-login-items' (wipes ALL Login Items; rebuild via System Settings General Login Items)?"
446         if sfltool reset-login-items 2>&1; then
447             echo "        Login Items drained"
448         else
449             echo "        ! sfltool reset-login-items failed (non-fatal; do manually: System Settings General Login Items)"
450         fi
451     fi

```

```

452 else
453     echo "        no backgrounditems.btm present"
454 fi
455
456 # ----- 6. brew install -----
457 header "Step 6 brew install tailscale (fresh)"
458 if brew list tailscale >/dev/null 2>&1; then
459     echo "        already installed skip"
460 else
461     confirm "Run 'brew install tailscale'?"
462     if [ "$DRY" = 1 ]; then
463         echo "        [dry] brew install tailscale"
464     else
465         brew install tailscale 2>&1
466     fi
467 fi
468
469 # ----- 6.5. Strip quarantine xattr from keg -----
470 header "Step 6.5 Strip quarantine xattr from tailscale install"
471 # brew's freshly poured bottles don't carry com.apple.quarantine (the
472 # bottles are content-addressed and Homebrew-verified), but the upstream
473 # installer (Tailscale.app from App Store) and any manual download do.
474 # Strip the xattr from the entire keg as a defensive safety net so the
475 # freshly-installed tailscaled binary and any helper binaries are not
476 # Gatekeeper-blocked on first launch. Idempotent no-op if no quarantine
477 # xattr is present.
478 quarantine_count="$(xattr -lr /opt/homebrew/Cellar/tailscale 2>/dev/null | grep -c '^[^']* com.apple.quarantine
479 ' || echo 0)"
479 if [ "${quarantine_count:-0}" -gt 0 ]; then
480     if [ "$DRY" = 1 ]; then
481         echo "        [dry] found $quarantine_count file(s) with com.apple.quarantine; would prompt xattr -dr"
482     else
483         confirm "Run 'xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale'? (strips quarantine xattr from
484 $quarantine_count file(s); non-fatal if it errors)"
485         if xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale 2>&1; then
486             echo "        quarantine xattr stripped"
487         else
488             echo "        ! xattr -dr failed (non-fatal; open /opt/homebrew/Cellar/tailscale in Finder, right-click
489 tailscaled Open Anyway)"
490         fi
491     fi
492 else
493     echo "        no com.apple.quarantine xattr on tailscale install"
494 fi
495
496 # ----- 7. brew services start (NO sudo!) -----
497 header "Step 7 Start the service as $USER (NO sudo, with multi-tier self-healing)"
498 if ! command -v tailscale >/dev/null 2>&1; then
499     echo "        tailscale not on PATH after install brew install errored; aborting"
500     exit 4
501 fi
502
503 # Aliveness fan-out: ANY one of these is taken as proof of life. None of
504 # them alone is reliable pgrep misses tailscaled if it does setproctitle
505 # or fork-detach; launchctl's reported PID can be a shim that exited
506 # within 1s; tailscale-cli hangs while macOS Local-Network permission is
507 # unresolved. Combined, they cover each other's blind spots.
508
509 tailscaled_api_up() {
510     # Strategy A: tailscale CLI can talk to the local API. Canonical proof.
511     # Bound at 8s tailscale status itself returns quickly on success OR
512     # failure, but if the post-install state is wedged it can hang.
513     with_timeout 8 /opt/homebrew/bin/tailscale status >/dev/null 2>&1
514 }
515
516 tailscaled_proc_up() {
517     # Strategy B: any process has 'tailscaled' substring in argv.
518     pgrep -lf tailscaled >/dev/null 2>&1
519 }
520
521 tailscaled_lc_up() {
522     # Strategy C: launchctl-managed service has a non-dash, non-zero PID.
523     local lc_pid
524     lc_pid="$(launchctl list 2>/dev/null | awk '$3 == "homebrew.mxcl.tailscale" {print $1; exit}')"
525     if [ -n "$lc_pid" ] && [ "$lc_pid" != "-" ] && [ "$lc_pid" -gt 0 ] 2>/dev/null; then
526         return 0
527     fi
528     return 1
529 }
530
531 tailscaled_is_alive() {

```

```

530     tailscaled_api_up  && return 0
531     tailscaled_proc_up && return 0
532     tailscaled_lc_up   && return 0
533     return 1
534 }
535
536 confirm "Run 'brew services start tailscale'?"
537 if [ "$DRY" = 1 ]; then
538     echo "    [dry] brew services start tailscale"
539     echo "    [dry] (then poll up to 30s for tailscaled_is_alive)"
540     echo "    [dry] (recovery tier 1: launchctl kickstart)"
541     echo "    [dry] (recovery tier 2: launchctl bootout + bootstrap of explicit plist)"
542     echo "    [dry] (recovery tier 3: brew services restart (stop+start cycle))"
543     echo "    [dry] (recovery tier 4: direct /opt/homebrew/bin/tailscaled spawn (bypasses launchd))"
544 else
545     brew services start tailscale 2>&1
546     # Primary poll tailscaled_is_alive covers all 3 detection strategies.
547     echo "    polling for tailscaled to come up (up to ~30s)"
548     tailscaled_up=0
549     for i in 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15; do
550         if tailscaled_is_alive; then
551             tailscaled_up=1
552             echo "        tailscaled alive after ${i}*2s"
553             break
554         fi
555         sleep 2
556     done
557
558     # Recovery tier 1: kickstart the launchd-managed service. This is the
559     # cheapest and most-aligned-with-launchd fix when brew has loaded the
560     # LaunchAgent but tailscaled didn't fork. Kickstart tells launchd
561     # "re-spawn if not running" without forcing a teardown.
562     if [ "$tailscaled_up" = 0 ]; then
563         echo "        ! Recovery tier 1: launchctl kickstart"
564         if launchctl kickstart -kp "gui/$UID/homebrew.mxcl.tailscale" 2>&1; then
565             for i in 1 2 3 4 5 6 7 8 9 10; do
566                 if tailscaled_is_alive; then
567                     tailscaled_up=1
568                     echo "        tailscaled alive after kickstart (${i}*2s)"
569                     break
570                 fi
571                 sleep 2
572             done
573         else
574             echo "        ! kickstart returned non-zero (LaunchAgent may not be loaded)"
575         fi
576     fi
577
578     # Recovery tier 2: explicit plist bootout + bootstrap. Forces a fresh
579     # launchd registration cycle for the brew-managed service, which fixes
580     # the wider class of "brew says started but daemon never spawned"
581     # wedge states that kickstart alone can't dig out of.
582     if [ "$tailscaled_up" = 0 ]; then
583         echo "        ! Recovery tier 2: launchctl bootout + bootstrap of explicit plist"
584         launchctl bootout "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null || true
585         sleep 1
586         # Brew installs the canonical plist template at $HOMEBREW_PREFIX/Library/.
587         # On Apple Silicon that's /opt/homebrew/Library/LaunchAgents/. On Intel
588         # it's /usr/local/Library/LaunchAgents/. Detect which and pick the
589         # right one so this script works across both architectures.
590         if [ -f /opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
591             plist_path="/opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
592         elif [ -f /usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
593             plist_path="/usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
594         else
595             plist_path=""
596         fi
597         if [ -n "$plist_path" ]; then
598             if launchctl bootstrap "gui/$UID" "$plist_path" 2>&1; then
599                 for i in 1 2 3 4 5 6 7 8 9 10; do
600                     if tailscaled_is_alive; then
601                         tailscaled_up=1
602                         echo "        tailscaled alive after bootstrap (${i}*2s)"
603                         break
604                     fi
605                     sleep 2
606                 done
607             else
608                 echo "        ! explicit bootstrap returned non-zero"
609             fi
610         else

```

```

611     echo "          ! could not locate brew plist template; skipping bootstrap tier"
612 fi
613 fi
614
615 # Recovery tier 3: brew services restart (forces a stop + start cycle,
616 # which unsticks LaunchAgents with stale 'started' state from prior
617 # half-cycles). This is the canonical brew-doc fix for 'services don't
618 # actually start even though brew says started'.
619 if [ "$tailscaled_up" = 0 ]; then
620     echo "          ! Recovery tier 3: brew services restart (forces stop+start cycle)"
621     if brew services restart tailscale 2>&1; then
622         for i in 1 2 3 4 5 6 7 8 9 10; do
623             if tailscaled_is_alive; then
624                 tailscaled_up=1
625                 echo "          tailscaled alive after brew restart (${i}*2s)"
626                 break
627             fi
628             sleep 2
629         done
630     else
631         echo "          ! brew services restart returned non-zero"
632     fi
633 fi
634
635 # Recovery tier 4: last-resort direct spawn of the daemon binary,
636 # bypassing launchd entirely. If tailscaled crashes here too, we'll
637 # capture its actual stderr in /tmp for the user to read (NOPASSWD
638 # sudoers often have local-network permission denied the captured
639 # stderr makes it obvious what to fix in System Settings).
640 if [ "$tailscaled_up" = 0 ]; then
641     echo "          ! Recovery tier 4: direct spawn (bypasses launchd; tail stderr to /tmp)"
642     /opt/homebrew/bin/tailscaled >/tmp/nullexit-tailscaled-spawn.log 2>&1 &
643     spawned_pid=$!
644     disown "$spawned_pid" 2>/dev/null || true
645     sleep 3
646     if tailscaled_is_alive; then
647         tailscaled_up=1
648         echo "          tailscaled alive via direct spawn (pid=$spawned_pid)"
649         echo "          (NOTE: not under launchd supervision a reboot, re-run of"
650         echo "          this script, or ./toggle.sh is required to recreate."
651         echo "          Last 5 lines of stderr captured to /tmp/nullexit-tailscaled-spawn.log:)"
652         tail -n 5 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/^/' '/' || true
653     fi
654 fi
655
656 if [ "$tailscaled_up" = 0 ]; then
657     echo "          HARD FAIL: tailscaled refuses to stay alive across every recovery tier"
658     echo "          launchctl-managed state (if available):"
659     launchctl print "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null \
660     | grep -E 'state =|last exit code =|runs =|path =' \
661     | sed 's/^/' '/' \
662     || echo "          (no launchctl state available for homebrew.mxcl.tailscale)"
663     echo "          Last 10 lines of /tmp/nullexit-tailscaled-spawn.log (if any):"
664     tail -n 10 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/^/' '/' || echo "          (no log)"
665     echo "          Most likely remaining cause: macOS Local Network permission denied."
666     echo "          System Settings Privacy & Security Local Network toggle Tailscale ON."
667     exit 7
668 fi
669 fi
670 sleep 3
671
672 # ----- 8. Verify -----
673 header "Step 8 Verify fresh install"
674 echo "          brew service tailscale: $(brew services list 2>/dev/null | awk '$1=="tailscale"{print $2, $3}' | tr
675 '\n' ' ' | sed 's/ $//')
676 echo "          tailscaled process:      $(pgrep -lf tailscaled 2>/dev/null | head -1 || echo 'NONE (check brew logs)
677 ')
678 ts_ver=""
679 if [ -x /opt/homebrew/bin/tailscale ]; then
680     ts_ver="$(/opt/homebrew/bin/tailscale version 2>/dev/null | head -1 | tr -d '\n')"
681 elif command -v tailscale >/dev/null 2>&1; then
682     ts_ver="$(tailscale version 2>/dev/null | head -1 | tr -d '\n')"
683 fi
684 echo "          tailscale version:      ${ts_ver:-binary missing}"
685 ts_status=""
686 if [ -x /opt/homebrew/bin/tailscale ]; then
687     ts_status="$(/opt/homebrew/bin/tailscale status 2>/dev/null | head -3)"
688 fi
689 if [ -n "$ts_status" ]; then
690     echo "          tailscale status:"
691     printf '%s\n' "$ts_status" | sed 's/^/' '/'

```

```

690 else
691     echo "    tailscale status:      (binary missing or errored)"
692 fi
693
694 if { [ "$DRY" = 0 ] && /opt/homebrew/bin/tailscale status 2>/dev/null | grep -q '^Tailscale is stopped'; };
695 then
696     echo "    NOTICE: 'tailscale status' still reports stopped after a fresh install."
697     echo "    Two equally likely causes:"
698     echo "    (a) macOS hasn't yet shown the first-run 'Local Network' prompt for"
699     echo "    Tailscale open System Settings Privacy & Security Local"
700     echo "    Network. Toggle Tailscale ON."
701     echo "    (b) macOS denied Local Network permission at some prior point."
702     echo "    Either way: System Settings Privacy & Security Local Network."
703     echo "    If the entry isn't present, invoking any tailscale command will"
704     echo "    re-trigger the prompt automatically."
705 fi
706
707 # ----- 9. Done -----
708 header "Done"
709 if [ "$DRY" = 1 ]; then
710     echo "    (dry-mode finished; no follow-up commands were issued.)"
711     exit 0
712 fi
713
714 cat <<'DONE'
715
716 Manual follow-ups (all of these; the script stops short of running sudo
717 tailscale up so you can verify the brew install is actually live first):
718
719 1. System Settings Privacy & Security Local Network Tailscale ON
720 macOS will normally pop the prompt the first time you run `tailscale`.
721 If it didn't, trigger it via the menu bar check.
722
723 2. Re-engage the tailnet + exit node:
724     sudo tailscale up --ssh=true --accept-dns=false \
725         --exit-node="$(cat .gateway_ip | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
726         --exit-node-allow-lan-access=true
727
728 3. Re-run: ./toggle.sh
729 (re-installs the com.nullexit.wake-recovery LaunchAgent wiped
730 in Step 5 to clear the slate and re-engages the gateway)
731
732 4. Final verification:
733     curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
734     expect: warp=on, ip=<Cloudflare IP>, colo=YYZ/YUL
735
736 Container-side (gateway / 100.100.21.8) Tailscale is unaffected by this
737 script. Other tailnet devices (S24, iPhone, Windows) are unaffected.
738
739 DONE

```

40 scripts/routing-fix.sh

```

1 #!/bin/sh
2 # routing-fix: Maintains routing for SOCKS5 proxy traffic through WARP (tun0)
3 #               and FORWARD rules for Tailscale exit-node return traffic.
4 #
5 # Runs inside the warp container's network namespace. This script ensures:
6 # 1. Table 200 has default via tun0 (for SOCKS5 proxy traffic from 172.18.0.2)
7 #    with the WARP endpoint exception via eth0 (prevents tunnel loop)
8 # 2. The CGNAT ip rule (100.64.0.0/10 table 52 tailscale routes) is maintained for Tailscale
9 # 3. FORWARD RELATED,ESTABLISHED rule allows return traffic for exit-node
10 #    forwarded connections (S24 tailscale0 eth0 host internet)
11 #
12 # CRITICAL: Does NOT touch the main table's default route. Gluetun handles that
13 # internally via its own routing rules and WireGuard fwmark (0xca6c table 51820).
14 # Overriding the main table default would create a loop: encrypted WireGuard
15 # packets exiting tun0 would match default dev tun0 and re-enter the tunnel.
16
17 set -e
18
19 trap 'echo "routing-fix: Caught signal, exiting..."; exit 0' TERM INT QUIT
20
21 sleep 15 &
22 wait $! || true

```

```

23
24 echo "routing-fix: Setting up routing for SOCKS5 proxy through WARP (tun0)..."
25
26 # Dynamically detect Docker subnet from eth0
27 DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
28 DOCKER_GW=$(ip route show default | head -1 | awk '{print $3}')
29 WARP_ENDPOINT="${WARP_ENDPOINT}"
30 TAILSCALE_ALLOW_LAN_P2P="${TAILSCALE_ALLOW_LAN_P2P:-false}"
31 IP_BLOCKLIST_FILE="/userfilters/ip_blocklist.ipset"
32 LAST_IP_MTIME=""
33 echo "routing-fix: Docker network: ${DOCKER_NET}, gateway: ${DOCKER_GW}"
34
35 # Table 200: Used by ip rule 100 (from 172.18.0.2) for SOCKS5 proxy traffic
36 # WARP endpoint goes via eth0 to avoid tunnel loop, everything else via tun0
37 ip route add "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || \
38   ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
39 ip route replace default dev tun0 table 200 2>/dev/null || true
40
41 # Main table: Just ensure the Docker subnet route exists via eth0
42 # (gluetun owns the default route here do NOT touch it)
43 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
44
45 # FORWARD RELATED, ESTABLISHED: Allow return traffic for exit-node connections.
46 # The container has BOTH iptables (nftables backend) and iptables-legacy
47 # loaded. Gluetun's post-rules.txt uses the nftables backend, while Tailscale
48 # uses the legacy backend. Packets go through BOTH stacks, so we add the rule
49 # to both to ensure it survives regardless of which backend has policy DROP.
50 #
51 # Packet flow: S24 outgoing (tailscale0->eth0) ACCEPTed by first rule.
52 # Return traffic (eth0->eth0 via table 52 to host) needs RELATED, ESTABLISHED
53 # to match the conntrack entry created by the outgoing flow.
54 add_fwd_related_established() {
55   # nftables backend (used by gluetun's post-rules.txt)
56   if ! iptables -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
57     iptables -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
58   fi
59   # legacy backend (used by Tailscale's ts-forward)
60   if command -v iptables-legacy >/dev/null 2>&1; then
61     if ! iptables-legacy -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
62       iptables-legacy -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
63     fi
64   fi
65 }
66
67 add_fwd_related_established
68
69 # Policy Routing for Tailscale P2P:
70 # Tailscale sends its encrypted mesh UDP packets from port 41642.
71 # If these go out tun0 (WARP), Cloudflare's strict NAT drops the returning
72 # hole-punch packets, causing Tailscale to fail P2P and fall back to DERP relays (high latency).
73 # We mark packets originating from port 41642 and force them out the main table (eth0).
74 #
75 # LAN P2P is controlled by two sources (in priority order):
76 # 1. /app/.lan_p2p_detected written by watcher.sh on macOS on every network change
77 #    using system_profiler SPAirPortDataType (802.1x detection) + AP isolation probe. Overrides .env.
78 # 2. TAILSCALE_ALLOW_LAN_P2P env var (from .env) static fallback.
79 # When disabled, we explicitly DROP local RFC1918-destined Tailscale UDP so the
80 # phone cleanly falls back to DERP rather than getting poisoned by macOS gvproxy SNAT.
81 add_tailscale_p2p_bypass() {
82   # Dynamic override from watcher.sh (network auto-detection) takes priority over .env
83   local allow_p2p="${TAILSCALE_ALLOW_LAN_P2P:-false}"
84   if [ -f "/app/.lan_p2p_detected" ]; then
85     allow_p2p=$(cat "/app/.lan_p2p_detected" 2>/dev/null | tr -d '[:space:]')
86   fi
87
88   # Always allow P2P directly to the Docker host gateway (the Mac running this container).
89   # The gateway container is always co-located with the host Mac on the same machine,
90   # so direct Tailscale P2P via the Docker bridge is always safe and avoids DERP overhead.
91   # This RETURN rule must be inserted BEFORE the general DROP rules so it takes priority.
92   local default_gw
93   default_gw=$(ip route show default | awk '{print $3}')
94   if [ -n "$default_gw" ]; then
95     if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null; then
96       iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null || true
97     fi
98   fi
99
100 # Also explicitly whitelist all physical IPs of the host Mac (e.g. 172.17.52.223).
101 # The Tailscale control plane registers the Mac using its physical IP, so the
102 # container will attempt P2P handshakes using that physical IP instead of the
103 # Docker bridge IP. If this physical IP falls within 172.16.0.0/12, it would be dropped.

```

```

104 if [ -f "/app/.host_ips" ]; then
105     while IFS= read -r host_ip; do
106         [ -z "$host_ip" ] && continue
107         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null; then
108             iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null || true
109         fi
110     done < "/app/.host_ips"
111 fi
112
113 if [ "${allow_p2p}" != "true" ]; then
114     # Drop Tailscale UDP packets destined for local subnets to prevent macOS gvproxy SNAT
115     # endpoint poisoning. When disabled (campus/enterprise WPA2-Enterprise with AP isolation),
116     # dropping these forces the S24 to use DERP relays reliably.
117     for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
118         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; then
119             iptables -t mangle -A OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null || true
120         fi
121     done
122 else
123     # Remove DROP rules if they exist (switching from restricted to allowed)
124     for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
125         while iptables -t mangle -D OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; do ;; done
126     done
127 fi
128
129 # Bypass STUN and P2P packets to public IPs (DERP relays and remote peers) out eth0
130 if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null; then
131     iptables -t mangle -A OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null || true
132 fi
133 if ! ip rule show | grep -q 'fwmark 0x8888'; then
134     ip rule add fwmark 0x8888 lookup 254 pref 98 2>/dev/null || true
135 fi
136 }
137
138 add_tailscale_p2p_bypass
139
140 add_onion_transparent_proxy() {
141     for ipt in iptables iptables-legacy; do
142         command -v "$ipt" >/dev/null 2>&1 || continue
143         if ! $ipt -t nat -C PREROUTING -d 198.18.0.0/15 -p tcp -j REDIRECT --to-ports 9040 2>/dev/null; then
144             $ipt -t nat -I PREROUTING -d 198.18.0.0/15 -p tcp -j REDIRECT --to-ports 9040 2>/dev/null || true
145         fi
146         if ! $ipt -C FORWARD -d 198.18.0.0/15 -j ACCEPT 2>/dev/null; then
147             $ipt -I FORWARD -d 198.18.0.0/15 -j ACCEPT 2>/dev/null || true
148         fi
149     done
150 }
151
152 add_tailscale_p2p_bypass
153 add_onion_transparent_proxy
154
155 create_log_drop_chain() {
156     local ipt="$1"
157     local chain="$2"
158     local prefix="$3"
159
160     if ! $ipt -N "$chain" 2>/dev/null; then
161         $ipt -F "$chain" 2>/dev/null || true
162     fi
163     $ipt -A "$chain" -j LOG --log-prefix "$prefix"
164     $ipt -A "$chain" -j DROP 2>/dev/null || true
165 }
166
167 add_country_block() {
168     # To add a country to the blocklist, simply define the 'BLOCKED_COUNTRIES' variable in .env with 2-letter ISO
169     # country codes.
170     # You can find the country code by looking at the filename on http://www.ipdeny.com/ipblocks/data/countries/
171     # For example, to block China (cn) and Russia (ru), set: BLOCKED_COUNTRIES="cn ru" in your .env file
172     for zone in $BLOCKED_COUNTRIES; do
173         set_name="block_${zone}"
174         zone_upper=$(echo "$zone" | tr 'a-z' 'A-Z')
175         chain_name="LOG_DROP_${zone_upper}"
176         log_prefix="IP_BLOCK_${zone_upper}: "
177
178         # Create the ipset if it doesn't exist
179         if ! ipset list "$set_name" >/dev/null 2>&1; then
180             ipset create "$set_name" hash:net 2>/dev/null || true
181             echo "routing-fix: Downloading ${zone_upper} IPs for blocklist..."
182             curl -sS "http://www.ipdeny.com/ipblocks/data/countries/${zone}.zone" | grep -v '^#' | while read -r

```

```

183     done
184 fi
185
186 # Apply LOG_DROP rule to both backends
187 for ipt in iptables iptables-legacy; do
188     command -v "$ipt" >/dev/null 2>&1 || continue
189
190     create_log_drop_chain "$ipt" "$chain_name" "$log_prefix"
191
192     # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
193     while $ipt -D FORWARD -m set --match-set "$set_name" dst -j DROP 2>/dev/null; do ;; done
194
195     if ! $ipt -C FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null; then
196         $ipt -I FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null || true
197     fi
198 done
199 done
200 }
201
202 add_ip_blocklist() {
203     # Only reload when the compiled file has actually changed (mtime check).
204     # This prevents a pointless ipset restore on every 30-second loop tick.
205     if [ ! -f "$IP_BLOCKLIST_FILE" ]; then
206         return 0
207     fi
208
209     CURRENT_MTIME=$(stat -c %Y "$IP_BLOCKLIST_FILE" 2>/dev/null || echo "0")
210     if [ "$CURRENT_MTIME" != "$LAST_IP_MTIME" ]; then
211         echo "routing-fix: IP blocklist changed, reloading..."
212         # ipset restore handles the atomic swap internally:
213         # create_new populate swap with live destroy_new
214         if ipset restore < "$IP_BLOCKLIST_FILE" 2>/dev/null; then
215             LAST_IP_MTIME="$CURRENT_MTIME"
216             echo "routing-fix: IP blocklist loaded ($(ipset list block_malicious | grep -c '[0-9]') entries)."
217         else
218             echo "routing-fix: Warning: ipset restore failed. Retrying next cycle."
219             return 1
220         fi
221     fi
222
223     # Apply FORWARD rules in BOTH iptables backends.
224     for ipt in iptables iptables-legacy; do
225         command -v "$ipt" >/dev/null 2>&1 || continue
226
227         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_DST "IP_BLOCK_MALICIOUS_DST: "
228         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_SRC "IP_BLOCK_MALICIOUS_SRC: "
229
230         # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
231         while $ipt -D FORWARD -m set --match-set block_malicious dst -j DROP 2>/dev/null; do ;; done
232         while $ipt -D FORWARD -m set --match-set block_malicious src -j DROP 2>/dev/null; do ;; done
233
234         if ! $ipt -C FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null; then
235             $ipt -I FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null || true
236         fi
237
238         if ! $ipt -C FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null; then
239             $ipt -I FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null || true
240         fi
241     done
242 }
243
244 # Start background block logger (DNS + IP)
245 if ! pgrep -f "nullexit-logger" >/dev/null; then
246     echo "routing-fix: Starting background block logger..."
247     nullexit-logger &
248 fi
249
250 add_country_block
251 add_ip_blocklist
252
253 echo 'routing-fix: Routes applied.'
254
255 UNHEALTHY_COUNT=0
256
257 # Re-assert loop (every 30 seconds)
258 while true; do
259     # Table 200 routes
260     ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
261
262     # Tunnel health check
263     if ! curl -m 3 -sS --interface tun0 https://cloudflare.com/cdn-cgi/trace >/dev/null 2>&1; then

```

```

264 UNHEALTHY_COUNT=$((UNHEALTHY_COUNT + 1))
265 else
266 UNHEALTHY_COUNT=0
267 if [ -f "/app/TUNNEL_FAILED_CLOSED.marker" ]; then
268 rm -f "/app/TUNNEL_FAILED_CLOSED.marker"
269 fi
270 fi
271
272 if [ "$UNHEALTHY_COUNT" -ge 3 ]; then
273 echo "routing-fix: Tunnel health check failed $UNHEALTHY_COUNT times. Failing closed."
274 ip route del default dev tun0 table 200 2>/dev/null || true
275 ip route replace blackhole default table 200 2>/dev/null || true
276 touch "/app/TUNNEL_FAILED_CLOSED.marker"
277 else
278 ip route replace default dev tun0 table 200 2>/dev/null || true
279 fi
280
281 # Main table: keep Docker subnet route
282 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
283
284 # CGNAT ip rule for Tailscale
285 if ! ip rule show | grep -q '100.64.0.0/10'; then
286 ip rule add to 100.64.0.0/10 lookup 52 pref 99 2>/dev/null || true
287 echo 'routing-fix: CGNAT rule missing, re-injected'
288 fi
289
290 # FORWARD RELATED, ESTABLISHED: Allow return traffic in both iptables
291 # backends. Gluetun may reset the nftables ruleset on VPN reconnect,
292 # and Tailscale may reset the legacy ruleset on auth refresh.
293 add_fwd_related_established
294
295 # Ensure P2P packets bypass WARP to maintain low latency
296 add_tailscale_p2p_bypass
297
298 # Ensure transparent proxying for .onion is active
299 add_onion_transparent_proxy
300
301 # Enforce country blocklist
302 add_country_block
303
304 # Enforce IP blocklist (reloads automatically when file changes)
305 add_ip_blocklist
306
307 sleep 30 &
308 wait $! || true
309 done

```

41 scripts/rule-compiler/go.mod

```

1 module nullexit/rule-compiler
2
3 go 1.22

```

42 scripts/rule-compiler/main.go

```

1 package main
2
3 import (
4     "bufio"
5     "bytes"
6     "crypto/md5"
7     "encoding/base64"
8     "encoding/hex"
9     "fmt"
10    "io"
11    "net/http"
12    "net/netip"
13    "os"
14    "os/exec"
15    "path/filepath"
16    "regexp"
17    "sort"

```

```

18  "strings"
19  "sync"
20  "time"
21 )
22
23 const (
24     BlackListPath = "black_list.txt"
25     WhiteListPath = "white_list.txt"
26     OutputPath    = "adguard/work/userfilters/compiled_rules.txt"
27     IpOutputPath  = "adguard/work/userfilters/ip_blocklist.ipset"
28     IpCacheDir    = "adguard/work/userfilters/cache/ip"
29     CacheDir      = "adguard/work/userfilters/cache"
30 )
31
32 var CoreLists = []string{
33     "https://raw.githubusercontent.com/anudeepND/blacklist/master/adservers.txt",
34     "https://raw.githubusercontent.com/anudeepND/blacklist/master/facebook.txt",
35     "https://raw.githubusercontent.com/crazy-max/WindowsSpyBlocker/master/data/hosts/spy.txt",
36     "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.samsung.txt",
37     "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.apple.txt",
38     "https://abp.oisd.nl/basic/",
39     "https://adaway.org/hosts.txt",
40     "https://pgl.yoyo.org/adservers/serverlist.php?hostformat=adblockplus&showintro=1&mimetype=plaintext",
41     "https://raw.githubusercontent.com/nextdns/cname-cloaking-blocklist/master/domains",
42 }
43
44 var MediumAdditions = []string{
45     "https://raw.githubusercontent.com/lightswitch05/hosts/master/docs/lists/facebook-extended.txt",
46     "https://someonewhocares.org/hosts/zero/hosts",
47     "https://urlhaus.abuse.ch/downloads/hostfile/",
48 }
49
50 var HeavyAdditions = []string{
51     "https://raw.githubusercontent.com/StevenBlack/hosts/master/hosts",
52     "https://raw.githubusercontent.com/Perflyst/PiHoleBlocklist/master/SmartTV-AGH.txt",
53     "https://raw.githubusercontent.com/DandelionSprout/adfilt/master/GameConsoleAdblockList.txt",
54 }
55
56 var IpBlockLists = []string{
57     "https://feodotracker.abuse.ch/downloads/ipblocklist.txt",
58     "https://www.spamhaus.org/drop/drop.txt",
59     "https://rules.emergingthreats.net/fwrules/emerging-Block-IPs.txt",
60     "https://cinsscore.com/list/ci-badguys.txt",
61 }
62
63 func getProfiles() map[string][]string {
64     profiles := make(map[string][]string)
65
66     light := append([]string(nil), CoreLists...)
67     light = append(light, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/light.txt")
68     profiles["light"] = light
69
70     medium := append([]string(nil), CoreLists...)
71     medium = append(medium, MediumAdditions...)
72     medium = append(medium, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/multi.txt")
73     profiles["medium"] = medium
74
75     heavy := append([]string(nil), CoreLists...)
76     heavy = append(heavy, MediumAdditions...)
77     heavy = append(heavy, HeavyAdditions...)
78     heavy = append(heavy, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/pro.txt")
79     profiles["heavy"] = heavy
80
81     return profiles
82 }
83
84 func loadEnvProfile() string {
85     profile := "medium"
86
87     if bytesData, err := os.ReadFile(".env"); err == nil {
88         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
89         for scanner.Scan() {
90             line := strings.TrimSpace(scanner.Text())
91             if line != "" && !strings.HasPrefix(line, "#") {
92                 parts := strings.SplitN(line, "=", 2)
93                 if len(parts) == 2 {
94                     key := strings.TrimSpace(parts[0])
95                     val := strings.TrimSpace(parts[1])
96                     if key == "GATEWAY_RULE_PROFILE" {
97                         val = strings.Trim(val, "\"'")
98                         profile = strings.ToLower(val)

```

```

99     }
100   }
101 }
102 }
103 } else if !os.IsNotExist(err) {
104   fmt.Printf("Warning: Failed to read .env file for profile configuration (%v). Using default.\n", err)
105 }
106
107 if envVal := os.Getenv("GATEWAY_RULE_PROFILE"); envVal != "" {
108   profile = strings.ToLower(envVal)
109 }
110
111 profiles := getProfiles()
112 if _, exists := profiles[profile]; !exists {
113   fmt.Printf("Warning: Profile '%s' is invalid. Falling back to 'medium'.\n", profile)
114   profile = "medium"
115 }
116 return profile
117 }
118
119 func loadDomains(filepath string) map[string]bool {
120   domains := make(map[string]bool)
121   if bytesData, err := os.ReadFile(filepath); err == nil {
122     scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
123     for scanner.Scan() {
124       line := strings.TrimSpace(scanner.Text())
125       if line != "" && !strings.HasPrefix(line, "#") {
126         domains[strings.ToLower(line)] = true
127       }
128     }
129   }
130   return domains
131 }
132
133 func parseDomainsFromContent(content string) map[string]bool {
134   domains := make(map[string]bool)
135   scanner := bufio.NewScanner(strings.NewReader(content))
136   ipRegex := regexp.MustCompile(`^\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}$`)
137   agRegex := regexp.MustCompile(`^\|([a-z0-9.-]+)\|^$`)
138   domainRegex := regexp.MustCompile(`^[a-z0-9.-]+$`)
139
140   for scanner.Scan() {
141     line := strings.TrimSpace(scanner.Text())
142     if line == "" || strings.HasPrefix(line, "!") || strings.HasPrefix(line, "[") {
143       continue
144     }
145     parts := strings.SplitN(line, "#", 2)
146     line = strings.TrimSpace(parts[0])
147     if line == "" {
148       continue
149     }
150
151     fields := strings.Fields(line)
152     var rule string
153     if len(fields) >= 2 {
154       if ipRegex.MatchString(fields[0]) {
155         rule = fields[1]
156       } else {
157         rule = fields[0]
158       }
159     } else {
160       rule = fields[0]
161     }
162
163     rule = strings.TrimSpace(strings.ToLower(rule))
164
165     var domain string
166     if m := agRegex.FindStringSubmatch(rule); m != nil {
167       domain = m[1]
168     } else if domainRegex.MatchString(rule) {
169       domain = rule
170     } else {
171       domain = rule
172     }
173
174     if domain != "" && domain != "localhost" && domain != "0.0.0.0" && domain != "127.0.0.1" && domain != "
175       broadcasthost" {
176       domains[domain] = true
177     }
178   }
179   return domains

```

```

179 }
180
181 // WARNING: Parsing AdGuardHome.yaml using regex instead of a proper YAML parser
182 // is brittle and assumes the exact format currently emitted by AdGuard.
183 // If an AdGuard version bump updates the configuration schema format,
184 // this parser will fail silently or loudly and should be refactored to use gopkg.in/yaml.v3.
185 func getAdguardNativeLists() []string {
186     yamlPath := "adguard/conf/AdGuardHome.yaml"
187     var enabledUrls []string
188     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
189         return enabledUrls
190     }
191
192     contentBytes, err := os.ReadFile(yamlPath)
193     if err != nil {
194         fmt.Printf("Warning: Failed to parse AdGuardHome.yaml for native lists (%v)\n", err)
195         return enabledUrls
196     }
197     content := string(contentBytes)
198
199     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*)"?(?:[a-zA-Z_]+:|\z)`
200     match := filtersRegex.FindStringSubmatch(content)
201     if len(match) > 1 {
202         filtersText := match[1]
203         blocks := strings.Split(filtersText, "\n - ")
204         urlRegex := regexp.MustCompile(`url:\s*(\S+)`)
205
206         for _, block := range blocks {
207             if strings.TrimSpace(block) == "" {
208                 continue
209             }
210             if strings.Contains(block, "enabled: true") {
211                 urlMatch := urlRegex.FindStringSubmatch(block)
212                 if len(urlMatch) > 1 {
213                     url := urlMatch[1]
214                     if !strings.Contains(url, "compiled_rules.txt") {
215                         enabledUrls = append(enabledUrls, url)
216                     }
217                 }
218             }
219         }
220     }
221     return enabledUrls
222 }
223
224 func getAdguardCompiledRulesCachePath() string {
225     yamlPath := "adguard/conf/AdGuardHome.yaml"
226     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
227         return ""
228     }
229
230     contentBytes, err := os.ReadFile(yamlPath)
231     if err != nil {
232         fmt.Printf("Warning: Failed to resolve compiled_rules.txt cache path (%v)\n", err)
233         return ""
234     }
235     content := string(contentBytes)
236
237     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*)"?(?:[a-zA-Z_]+:|\z)`
238     match := filtersRegex.FindStringSubmatch(content)
239     if len(match) > 1 {
240         blocks := strings.Split(match[1], "\n - ")
241         idRegex := regexp.MustCompile(`id:\s*(\d+)`)
242         for _, block := range blocks {
243             if !strings.Contains(block, "compiled_rules.txt") {
244                 continue
245             }
246             idMatch := idRegex.FindStringSubmatch(block)
247             if len(idMatch) > 1 {
248                 return fmt.Sprintf("adguard/work/data/filters/%s.txt", idMatch[1])
249             }
250         }
251     }
252     return ""
253 }
254
255 func handleFetchError(urlStr string, cacheFile string, err error) map[string]bool {
256     if _, statErr := os.Stat(cacheFile); statErr == nil {
257         fmt.Printf("-> Warning: Failed to fetch %s (%v). Falling back to expired local cache.\n", urlStr, err)
258         content, readErr := os.ReadFile(cacheFile)
259         if readErr == nil {

```

```

260     domains := parseDomainsFromContent(string(content))
261     fmt.Printf(" -> Loaded from expired cache (%d domains).\n", len(domains))
262     return domains
263 }
264 fmt.Printf(" -> Warning: Failed to read expired cache (%v). Using local lists only.\n", readErr)
265 } else {
266     fmt.Printf(" -> Warning: Failed to fetch %s (%v). Using local lists only for this source.\n", urlStr, err)
267 }
268 return make(map[string]bool)
269 }
270
271 func fetchRemoteDomains(urlStr string) map[string]bool {
272     os.MkdirAll(CacheDir, 0755)
273
274     hash := md5.Sum([]byte(urlStr))
275     urlHash := hex.EncodeToString(hash[:])
276     cacheFile := filepath.Join(CacheDir, fmt.Sprintf("%s.txt", urlHash))
277
278     if stat, err := os.Stat(cacheFile); err == nil {
279         fileAge := time.Since(stat.ModTime()).Seconds()
280         if fileAge < 86400 {
281             fmt.Printf("Loading remote blacklist from cache: %s\n", urlStr)
282             content, err := os.ReadFile(cacheFile)
283             if err == nil {
284                 domains := parseDomainsFromContent(string(content))
285                 fmt.Printf(" -> Loaded from cache (copy is %.1f hours old, %d domains).\n", fileAge/3600.0, len(domains))
286             }
287             return domains
288         }
289         fmt.Printf(" -> Warning: Failed to read cache file %s (%v). Re-fetching...\n", cacheFile, err)
290     }
291
292     fmt.Printf("Fetching remote blacklist from: %s ...\n", urlStr)
293
294     req, err := http.NewRequest("GET", urlStr, nil)
295     if err != nil {
296         return handleFetchError(urlStr, cacheFile, err)
297     }
298     req.Header.Set("User-Agent", "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)")
299
300     client := &http.Client{Timeout: 15 * time.Second}
301     resp, err := client.Do(req)
302     if err != nil {
303         return handleFetchError(urlStr, cacheFile, err)
304     }
305     defer resp.Body.Close()
306
307     if resp.StatusCode != 200 {
308         return handleFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
309     }
310
311     bodyBytes, err := io.ReadAll(resp.Body)
312     if err != nil {
313         return handleFetchError(urlStr, cacheFile, err)
314     }
315
316     content := string(bodyBytes)
317     domains := parseDomainsFromContent(content)
318
319     if len(domains) < 10 {
320         return handleFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only found %d domains. Possible 404 or bad URL", len(domains)))
321     }
322
323     err = os.WriteFile(cacheFile, bodyBytes, 0644)
324     if err != nil {
325         fmt.Printf(" -> Warning: Failed to save cache file (%v)\n", err)
326     }
327
328     fmt.Printf(" -> Successfully fetched and cached %d domains.\n", len(domains))
329     return domains
330 }
331
332 func optimizeSubdomains(domains map[string]bool, listName string) map[string]bool {
333     fmt.Printf("Optimizing %s by removing redundant subdomains...\n", listName)
334     rawCount := len(domains)
335     if rawCount == 0 {
336         return make(map[string]bool)
337     }
338

```

```

339 optimized := make(map[string]bool)
340 for domain := range domains {
341     if strings.ContainsAny(domain, "|^@$/") {
342         optimized[domain] = true
343         continue
344     }
345
346     parts := strings.Split(domain, ".")
347     hasParent := false
348
349     for i := 1; i < len(parts)-1; i++ {
350         parent := strings.Join(parts[i:], ".")
351         if domains[parent] {
352             hasParent = true
353             break
354         }
355     }
356
357     if !hasParent {
358         optimized[domain] = true
359     }
360 }
361
362 saved := rawCount - len(optimized)
363 reduction := float64(0)
364 if rawCount > 0 {
365     reduction = (float64(saved) / float64(rawCount)) * 100
366 }
367 fmt.Printf(" -> Reduced %s from %d to %d domains (-%d / %.1f%% reduction).\n", listName, rawCount, len(
    optimized), saved, reduction)
368 return optimized
369 }
370
371 func parseIpsFromContent(content string) map[string]bool {
372     ips := make(map[string]bool)
373     scanner := bufio.NewScanner(strings.NewReader(content))
374
375     for scanner.Scan() {
376         line := strings.TrimSpace(scanner.Text())
377         if line == "" || strings.HasPrefix(line, "#") || strings.HasPrefix(line, ";") {
378             continue
379         }
380
381         fields := strings.Fields(line)
382         if len(fields) > 0 {
383             token := strings.TrimRight(fields[0], ";,")
384             if _, err := netip.ParsePrefix(token); err == nil {
385                 ips[token] = true
386             } else if _, err := netip.ParseAddr(token); err == nil {
387                 ips[token] = true
388             }
389         }
390     }
391     return ips
392 }
393
394 func handleIpFetchError(urlStr string, cacheFile string, err error) map[string]bool {
395     if _, statErr := os.Stat(cacheFile); statErr == nil {
396         fmt.Printf(" -> Warning: fetch failed (%v). Falling back to stale cache.\n", err)
397         content, readErr := os.ReadFile(cacheFile)
398         if readErr == nil {
399             ips := parseIpsFromContent(string(content))
400             fmt.Printf(" -> %d IPs loaded from stale cache.\n", len(ips))
401             return ips
402         }
403         fmt.Printf(" -> Warning: stale cache also failed (%v).\n", readErr)
404     } else {
405         fmt.Printf(" -> Warning: %s failed (%v). No cache available. Skipping.\n", urlStr, err)
406     }
407     return make(map[string]bool)
408 }
409
410 func fetchRemoteIps(urlStr string) map[string]bool {
411     os.MkdirAll(IpCacheDir, 0755)
412
413     hash := md5.Sum([]byte(urlStr))
414     urlHash := hex.EncodeToString(hash[:])
415     cacheFile := filepath.Join(IpCacheDir, fmt.Sprintf("%s.txt", urlHash))
416
417     if stat, err := os.Stat(cacheFile); err == nil {
418         fileAge := time.Since(stat.ModTime()).Seconds()

```

```

419     if fileAge < 86400 {
420         fmt.Printf("Loading IP feed from cache: %s\n", urlStr)
421         content, err := os.ReadFile(cacheFile)
422         if err == nil {
423             ips := parseIpsFromContent(string(content))
424             fmt.Printf(" -> %d IPs (cache is %.1fh old).\n", len(ips), fileAge/3600.0)
425             return ips
426         }
427         fmt.Printf(" -> Warning: cache read failed (%v). Re-fetching...\n", err)
428     }
429 }
430
431 fmt.Printf("Fetching IP feed: %s ...\n", urlStr)
432 req, err := http.NewRequest("GET", urlStr, nil)
433 if err != nil {
434     return handleIpFetchError(urlStr, cacheFile, err)
435 }
436 req.Header.Set("User-Agent", "Mozilla/5.0")
437
438 client := &http.Client{Timeout: 15 * time.Second}
439 resp, err := client.Do(req)
440 if err != nil {
441     return handleIpFetchError(urlStr, cacheFile, err)
442 }
443 defer resp.Body.Close()
444
445 if resp.StatusCode != 200 {
446     return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
447 }
448
449 bodyBytes, err := io.ReadAll(resp.Body)
450 if err != nil {
451     return handleIpFetchError(urlStr, cacheFile, err)
452 }
453
454 content := string(bodyBytes)
455 ips := parseIpsFromContent(content)
456 if len(ips) < 1 {
457     return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only %d IPs found. Possible
458     404", len(ips)))
459 }
460
461 err = os.WriteFile(cacheFile, bodyBytes, 0644)
462 if err != nil {
463     fmt.Printf(" -> Warning: cache write failed (%v)\n", err)
464 }
465
466 fmt.Printf(" -> %d IPs fetched and cached.\n", len(ips))
467 return ips
468 }
469
470 func compileIpBlocklist() int {
471     fmt.Println("\n IP Blocklist Compilation ")
472
473     allIps := make(map[string]bool)
474     var mu sync.Mutex
475     var wg sync.WaitGroup
476
477     maxThreads := 8
478     if len(IpBlockLists) < 8 {
479         maxThreads = len(IpBlockLists)
480     }
481     semaphore := make(chan struct{}, maxThreads)
482
483     for _, urlStr := range IpBlockLists {
484         wg.Add(1)
485         go func(u string) {
486             defer wg.Done()
487             semaphore <- struct{}{}
488             defer func() { <-semaphore }()
489
490             ips := fetchRemoteIps(u)
491             mu.Lock()
492             for ip := range ips {
493                 allIps[ip] = true
494             }
495             mu.Unlock()
496         }(urlStr)
497     }
498     wg.Wait()

```

```

499     fmt.Printf("Total unique entries before filtering: %d\n", len(allIps))
500
501     reservedStr := []string{
502         "10.0.0.0/8",
503         "172.16.0.0/12",
504         "192.168.0.0/16",
505         "127.0.0.0/8",
506         "169.254.0.0/16",
507         "100.64.0.0/10",
508         "0.0.0.0/8",
509     }
510     var reserved []netip.Prefix
511     for _, s := range reservedStr {
512         p, _ := netip.ParsePrefix(s)
513         reserved = append(reserved, p)
514     }
515
516     cleanIps := make(map[string]bool)
517     for entry := range allIps {
518         var prefix netip.Prefix
519         p, err := netip.ParsePrefix(entry)
520         if err != nil {
521             addr, err2 := netip.ParseAddr(entry)
522             if err2 != nil {
523                 continue
524             }
525             prefix = netip.PrefixFrom(addr, addr.BitLen())
526         } else {
527             prefix = p
528         }
529
530         overlaps := false
531         for _, r := range reserved {
532             if r.Overlaps(prefix) {
533                 overlaps = true
534                 break
535             }
536         }
537         if !overlaps {
538             cleanIps[entry] = true
539         }
540     }
541
542     removed := len(allIps) - len(cleanIps)
543     if removed > 0 {
544         fmt.Printf(" -> Removed %d private/reserved entries.\n", removed)
545     }
546     fmt.Printf(" -> Final IP blocklist: %d entries.\n", len(cleanIps))
547
548     os.MkdirAll(filepath.Dir(IpOutputPath), 0755)
549
550     f, err := os.Create(IpOutputPath)
551     if err != nil {
552         fmt.Printf("Error writing IP output file: %v\n", err)
553         return len(cleanIps)
554     }
555     defer f.Close()
556
557     f.WriteString("# nullexit Compiled IP Blocklist\n")
558     f.WriteString("# Sources: Feodo Tracker, Spamhaus DROP/EDROP, Emerging Threats, CINS\n")
559     f.WriteString(fmt.Sprintf("# Entries: %d\n", len(cleanIps)))
560
561     f.WriteString("create block_malicious_new hash:net maxelem 200000 -exist\n")
562
563     var sortedIps []string
564     for ip := range cleanIps {
565         sortedIps = append(sortedIps, ip)
566     }
567     sort.Strings(sortedIps)
568
569     for _, ip := range sortedIps {
570         f.WriteString(fmt.Sprintf("add block_malicious_new %s -exist\n", ip))
571     }
572
573     f.WriteString("create block_malicious hash:net maxelem 200000 -exist\n")
574     f.WriteString("swap block_malicious block_malicious_new\n")
575     f.WriteString("destroy block_malicious_new\n")
576
577     fmt.Printf("IP blocklist written to %s\n", IpOutputPath)
578     return len(cleanIps)
579 }

```

```

580
581 func main() {
582     profile := loadEnvProfile()
583     fmt.Printf("Active memory profile: '%s'\n", strings.ToUpper(profile))
584
585     localBlackList := loadDomains(BlackListPath)
586     whiteList := loadDomains(WhiteListPath)
587
588     blackList := make(map[string]bool)
589     for k := range localBlackList {
590         blackList[k] = true
591     }
592
593     profiles := getProfiles()
594     urlsToFetch := profiles[profile]
595
596     fmt.Printf("\nStarting concurrent downloads for %d lists...\n", len(urlsToFetch))
597     startTime := time.Now()
598
599     var mu sync.Mutex
600     var wg sync.WaitGroup
601
602     maxThreads := 16
603     if len(urlsToFetch) < 16 {
604         maxThreads = len(urlsToFetch)
605     }
606     semaphore := make(chan struct{}, maxThreads)
607
608     for _, urlStr := range urlsToFetch {
609         wg.Add(1)
610         go func(u string) {
611             defer wg.Done()
612             semaphore <- struct{}{}
613             defer func() { <-semaphore }()
614
615             domains := fetchRemoteDomains(u)
616             mu.Lock()
617             for d := range domains {
618                 blackList[d] = true
619             }
620             mu.Unlock()
621         }(urlStr)
622     }
623     wg.Wait()
624
625     fmt.Printf("Finished concurrent downloads in %.2f seconds using %d threads.\n", time.Since(startTime).Seconds(), maxThreads)
626
627     adguardNativeUrls := getAdguardNativeLists()
628     var adguardNativeDomains map[string]bool
629     if len(adguardNativeUrls) > 0 {
630         fmt.Printf("\nFetching %d AdGuard native list(s) for deduplication...\n", len(adguardNativeUrls))
631         adguardNativeDomains = make(map[string]bool)
632
633         maxNativeThreads := 4
634         if len(adguardNativeUrls) < 4 {
635             maxNativeThreads = len(adguardNativeUrls)
636         }
637         nativeSemaphore := make(chan struct{}, maxNativeThreads)
638
639         for _, urlStr := range adguardNativeUrls {
640             wg.Add(1)
641             go func(u string) {
642                 defer wg.Done()
643                 nativeSemaphore <- struct{}{}
644                 defer func() { <-nativeSemaphore }()
645
646                 domains := fetchRemoteDomains(u)
647                 mu.Lock()
648                 for d := range domains {
649                     adguardNativeDomains[d] = true
650                 }
651                 mu.Unlock()
652             }(urlStr)
653         }
654         wg.Wait()
655
656         if len(adguardNativeDomains) > 0 {
657             fmt.Println("Deduplicating compiled rules against AdGuard native lists...")
658             originalSize := len(blackList)
659             for d := range adguardNativeDomains {

```

```

660     delete(blackList, d)
661 }
662 fmt.Printf(" -> Removed %d redundant rules already covered by AdGuard.\n", originalSize-len(blackList))
663 }
664 }
665
666 var contradictions []string
667 for d := range whiteList {
668     if blackList[d] {
669         contradictions = append(contradictions, d)
670     }
671 }
672
673 if len(contradictions) > 0 {
674     fmt.Printf("Detected %d contradictions. Whitelist taking priority.\n", len(contradictions))
675     sort.Strings(contradictions)
676     for _, domain := range contradictions {
677         if localBlackList[domain] {
678             fmt.Printf(" -> Removed local blacklist domain: %s\n", domain)
679         }
680         delete(blackList, domain)
681     }
682 }
683
684 blackList = optimizeSubdomains(blackList, "blacklist")
685 whiteList = optimizeSubdomains(whiteList, "whitelist")
686
687 os.MkdirAll(filepath.Dir(OutputPath), 0755)
688 outFile, err := os.Create(OutputPath)
689 if err != nil {
690     fmt.Printf("Error creating output file: %v\n", err)
691     return
692 }
693 defer outFile.Close()
694
695 outFile.WriteString("! Custom Compiled Rules (Auto-Generated)\n")
696 outFile.WriteString(fmt.Sprintf("! Memory Profile: %s\n", strings.ToUpper(profile)))
697 outFile.WriteString(fmt.Sprintf("! Total Block Rules: %d\n", len(blackList)))
698 outFile.WriteString(fmt.Sprintf("! Native AdGuard Rules: %d\n", len(adguardNativeDomains)))
699 outFile.WriteString(fmt.Sprintf("! Total Whitelist Rules: %d\n", len(whiteList)))
700
701 outFile.WriteString("! --- Blacklist Rules ---\n")
702
703 var sortedBlacklist []string
704 for d := range blackList {
705     sortedBlacklist = append(sortedBlacklist, d)
706 }
707 sort.Strings(sortedBlacklist)
708
709 for _, domain := range sortedBlacklist {
710     if strings.Contains(domain, "$") {
711         parts := strings.SplitN(domain, "$", 2)
712         dom, mod := parts[0], parts[1]
713         if strings.HasPrefix(dom, "|") {
714             if strings.HasSuffix(dom, "~") {
715                 outFile.WriteString(fmt.Sprintf("%s%s\n", dom, mod))
716             } else {
717                 outFile.WriteString(fmt.Sprintf("%s~%s\n", dom, mod))
718             }
719         } else {
720             outFile.WriteString(fmt.Sprintf("||%s~%s\n", dom, mod))
721         }
722     } else if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") || strings.HasSuffix(domain, "~") {
723         outFile.WriteString(fmt.Sprintf("%s\n", domain))
724     } else {
725         outFile.WriteString(fmt.Sprintf("||%s~\n", domain))
726     }
727 }
728
729 outFile.WriteString("\n! --- Whitelist Rules ---\n")
730 var sortedWhitelist []string
731 for d := range whiteList {
732     sortedWhitelist = append(sortedWhitelist, d)
733 }
734 sort.Strings(sortedWhitelist)
735
736 for _, domain := range sortedWhitelist {
737     if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") || strings.HasSuffix(domain, "~") || strings.HasPrefix(domain, "@@") {
738         rule := domain

```

```

739     if !strings.HasPrefix(domain, "@@") {
740         rule = "@@" + domain
741     }
742     outFile.WriteString(fmt.Sprintf("%s\n", rule))
743 } else {
744     outFile.WriteString(fmt.Sprintf("@@||%s^\n", domain))
745 }
746 }
747
748 fmt.Printf("\nSuccessfully compiled %d block rules and %d allow rules to %s\n", len(blackList), len(whiteList)
749 ), OutputPath)
750
751 cachedFilterPath := getAdguardCompiledRulesCachePath()
752 if cachedFilterPath != "" {
753     input, err := os.ReadFile(OutputPath)
754     if err == nil {
755         err = os.WriteFile(cachedFilterPath, input, 0644)
756         if err == nil {
757             fmt.Printf("Updated AdGuard filter cache: %s\n", cachedFilterPath)
758         } else {
759             fmt.Printf("Warning: Could not update AdGuard filter cache %s (%v)\n", cachedFilterPath, err)
760         }
761     } else {
762         fmt.Printf("Warning: Could not read OutputPath to copy to cache (%v)\n", err)
763     }
764 } else {
765     fmt.Println("Warning: Could not determine compiled_rules.txt cache path from AdGuardHome.yaml; skipping
766     cache update.")
767 }
768
769 adguardReachable := false
770
771 dockerComposeYamlExists := false
772 if _, err := os.Stat("docker-compose.yml"); err == nil {
773     dockerComposeYamlExists = true
774 }
775
776 dockerPath, err := exec.LookPath("docker")
777 if err == nil && dockerComposeYamlExists {
778     cmd := exec.Command(dockerPath, "compose", "ps", "--status", "running", "-q", "adguardhome")
779     out, err := cmd.Output()
780     if err == nil && strings.TrimSpace(string(out)) != "" {
781         fmt.Println("Force-recreating AdGuard Home container to drop persisted filter state...")
782         upCmd := exec.Command(dockerPath, "compose", "up", "-d", "--force-recreate", "--no-deps", "adguardhome")
783         err := upCmd.Run()
784         if err == nil {
785             fmt.Println("AdGuard Home force-recreated successfully.")
786             time.Sleep(8 * time.Second)
787             adguardReachable = true
788         } else {
789             fmt.Println("Failed to restart AdGuard Home. Is it running?")
790         }
791     }
792 }
793
794 if adguardReachable {
795     agPass := "nullexit"
796     if bytesData, err := os.ReadFile(".env"); err == nil {
797         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
798         for scanner.Scan() {
799             line := scanner.Text()
800             if strings.HasPrefix(line, "ADGUARD_PASSWORD=") {
801                 parts := strings.SplitN(line, "=", 2)
802                 if len(parts) == 2 {
803                     agPass = strings.Trim(strings.TrimSpace(parts[1]), "\"'")
804                 }
805             }
806         }
807     }
808
809     creds := base64.StdEncoding.EncodeToString([]byte(fmt.Sprintf("admin:%s", agPass)))
810     req, err := http.NewRequest("POST", "http://127.0.0.1:3000/control/filtering/refresh", bytes.NewBuffer([]
811     byte("{}")))
812     if err == nil {
813         req.Header.Set("Authorization", fmt.Sprintf("Basic %s", creds))
814         req.Header.Set("Content-Type", "application/json")
815         client := &http.Client{Timeout: 15 * time.Second}
816         resp, err := client.Do(req)
817         if err == nil {
818             fmt.Printf("Triggered AdGuard filter refresh via REST API (HTTP=%d).\n", resp.StatusCode)
819             resp.Body.Close()

```

```

817     } else {
818         fmt.Printf("Warning: AdGuard filter refresh API call failed (%v). New whitelist will not load until
next refresh tick (~24h) or manual refresh.\n", err)
819     }
820 }
821 }
822
823 compileIpBlocklist()
824 }

```

43 scripts/setup-common.sh

```

1  #!/bin/bash
2  # 1. Docker
3  step "Checking Docker"
4
5  if ! command -v docker >> output.log 2>&1; then
6      echo ""
7      if [[ "$OSTYPE" == "darwin"* ]]; then
8          echo " Docker is not installed."
9          # Note: Homebrew aggressively rolls forward and discourages version pinning.
10         # This setup is confirmed stable on Colima v0.10.3 and Docker v29.6.1.
11         read -rp " Would you like to automatically install Colima & Docker via Homebrew? [y/N]: "
12         INSTALL_DOCKER
13         if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
14             step "Installing Colima and Docker..."
15             brew install colima docker docker-compose
16             step "Starting Colima VM..."
17             colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged
18         else
19             echo ""
20             echo " Please install Docker manually. Two options:"
21             echo " A) Colima (Recommended): brew install colima docker docker-compose && colima start --memory
0.6 --vm-type vz --network-address --network-mode bridged"
22             echo " B) Docker Desktop: https://www.docker.com/products/docker-desktop/"
23             die "Install Docker, then re-run this script."
24         fi
25     else
26         echo " Docker is not installed."
27         # Note: The Docker convenience script always fetches the latest stable release.
28         # This setup is confirmed stable on Docker Engine v29.6.1.
29         read -rp " Would you like to automatically install Docker? (Requires sudo) [y/N]: " INSTALL_DOCKER
30         if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
31             step "Installing Docker..."
32             curl -fsSL https://get.docker.com | sh
33             sudo usermod -aG docker "$USER"
34             echo -e "\n\033[0;32m[OK] Docker installed successfully!\033[0m"
35             echo -e "\033[0;33m[!] CRITICAL: You must log out of your SSH session and log back in for Docker
permissions to apply.\033[0m"
36             die "Please log out, log back in, and re-run setup.sh."
37         else
38             echo " Please install Docker manually:"
39             echo " curl -fsSL https://get.docker.com | sh"
40             echo " sudo usermod -aG docker \"$USER && newgrp docker"
41             die "Install Docker, then re-run this script."
42         fi
43     fi
44 fi
45
46 if ! docker info >> output.log 2>&1; then
47     echo ""
48     if [[ "$OSTYPE" == "darwin"* ]]; then
49         echo " Docker is installed but not running."
50         echo " If using Colima: colima start --memory 0.6 --vm-type vz --network-address --network-mode
bridged"
51         echo " If using Docker Desktop: open the app."
52     else
53         echo " Docker is installed but not running."
54         echo " Run: sudo systemctl start docker"
55     fi
56     die "Start Docker, then re-run this script."
57 fi
58
59 if ! docker compose version >> output.log 2>&1; then
60     die "Docker Compose v2 not found. Update Docker or install the compose plugin:\n https://docs.docker.com/
compose/install/"

```

```

60 fi
61
62 ok "Docker $(docker --version | grep -oE '[0-9.]+' | head -1) is running."
63
64 # 1.5 Python 3
65 step "Checking Python 3"
66
67 if ! command -v python3 >> output.log 2>&1; then
68     echo ""
69     if [[ "$OSTYPE" == "darwin"* ]]; then
70         echo " Python 3 is not installed."
71         echo " macOS includes it via Xcode Command Line Tools, or you can install it via Homebrew:"
72         echo "         brew install python3"
73         die "Install Python 3, then re-run this script."
74     else
75         echo " Python 3 is not installed."
76         echo " Please install it using your system's package manager. For example:"
77         echo "     Debian/Ubuntu: sudo apt install python3"
78         echo "     Fedora:         sudo dnf install python3"
79         echo "     Arch:           sudo pacman -S python"
80         die "Install Python 3, then re-run this script."
81     fi
82 fi
83
84 # Note: This setup is confirmed stable on Python 3.9.6 (macOS default).
85 PYTHON_VER=$(python3 --version 2>&1 | head -n 1)
86 ok "$PYTHON_VER is installed."
87
88 # 2. Tailscale (host)
89 step "Checking Tailscale (host)"
90 # The host needs its own Tailscale installation separate from the Docker
91 # container so Toggle-Gateway can run 'tailscale down' before stopping
92 # containers and 'tailscale up' after starting them. Without this, the
93 # exit-node deadlock that setup.sh is designed to prevent can still occur.
94
95 # Resolve the tailscale binary: CLI (brew/package) or bundled macOS app
96 TAILSCALE_BIN=""
97 if command -v tailscale >> output.log 2>&1; then
98     TAILSCALE_BIN="tailscale"
99 elif [[ -x "/Applications/Tailscale.app/Contents/MacOS/Tailscale" ]]; then
100     TAILSCALE_BIN="/Applications/Tailscale.app/Contents/MacOS/Tailscale"
101 fi
102
103 RECOMMENDED_TS_VERSION="1.98.5"
104
105 if [[ -z "$TAILSCALE_BIN" ]]; then
106     warn "Tailscale not found on host installing..."
107
108     if [[ "$OSTYPE" == "darwin"* ]]; then
109         if command -v brew >> output.log 2>&1; then
110             step "Installing Tailscale CLI formula via Homebrew (standalone, no .app GUI)..."
111             # Note: We install Tailscale via Homebrew, targeting the stable version (recommended: 1.98.5)
112             brew install tailscale -q
113             TAILSCALE_BIN="tailscale"
114
115             step "Starting tailscaled as a per-user LaunchAgent (auto-starts at login)..."
116             if brew services start tailscale 2>&1 | grep -qE 'Successfully|started'; then
117                 ok "tailscaled LaunchAgent registered."
118             else
119                 warn "Could not auto-start tailscaled. Run 'brew services start tailscale' manually."
120                 warn "For a system-wide LaunchDaemon that runs before login, run instead:"
121                 warn "         sudo brew services start tailscale"
122             fi
123         else
124             echo ""
125             die "Homebrew not found. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION}):
126             https://tailscale.com/download/mac\n Then re-run this script."
127         fi
128     elif [[ "$OSTYPE" == "linux-gnu"* ]]; then
129         curl -fsSL https://tailscale.com/install.sh | sh
130         sudo systemctl enable --now tailscaled 2>> output.log || true
131         TAILSCALE_BIN="tailscale"
132     else
133         die "Unsupported OS. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION}):
134         https://tailscale.com/download\n Then re-run this script."
135     fi
136
137     TAILSCALE_VER=$(("${TAILSCALE_BIN}" version 2>> output.log | head -n 1)
138     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then

```

```

139     warn "Tailscale installed, but version ($TAILSCALE_VER) differs from recommended version (
140     $RECOMMENDED_TS_VERSION)."
141 else
142     ok "Tailscale installed (version $TAILSCALE_VER)."
143 fi
144 else
145     TAILSCALE_VER=$((${TAILSCALE_BIN} version 2>> output.log | head -n 1)
146     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
147         warn "Tailscale is already installed ($TAILSCALE_VER), but recommended version is
148         $RECOMMENDED_TS_VERSION for host/container consistency."
149     else
150         ok "Tailscale is already installed at recommended version ($TAILSCALE_VER)."
151     fi
152 fi
153 # Authenticate the host machine if it isn't already connected.
154 # This is an interactive step it opens a browser login URL.
155 # We do it now, before containers start, so Toggle-Gateway can
156 # safely run 'tailscale down/up' from day one.
157 if ! ${TAILSCALE_BIN} status >> output.log 2>&1; then
158     echo ""
159     echo " Your host machine needs to join your Tailscale network."
160     echo " A browser window or login URL will appear authenticate, then"
161     echo " return here. The script will continue automatically."
162     echo ""
163     if [[ "$OSTYPE" == "linux-gnu*" ]]; then
164         sudo ${TAILSCALE_BIN} up
165     else
166         ${TAILSCALE_BIN} up
167     fi
168     ok "Host machine connected to Tailscale."
169 else
170     ok "Host machine is already connected to Tailscale."
171 fi
172 # 3. wgcf
173 step "Checking wgcf (Cloudflare WARP key tool)"
174
175 if ! command -v wgcf >> output.log 2>&1; then
176     warn "wgcf not found installing..."
177
178     if [[ "$OSTYPE" == "darwin*" ]]; then
179         command -v brew >> output.log 2>&1 || die "Homebrew not found.\nInstall wgcf manually: https://github.
180         com/ViRb3/wgcf/releases"
181         brew install wgcf -q
182     elif [[ "$OSTYPE" == "linux-gnu*" ]]; then
183         ARCH=$(uname -m)
184         case $ARCH in
185             x86_64) WGCF_ARCH="amd64" ;;
186             aarch64) WGCF_ARCH="arm64" ;;
187             armv7l) WGCF_ARCH="armv7" ;;
188             *) die "Unsupported architecture: $ARCH.\nInstall wgcf manually: https://github.com/ViRb3/wgcf/
189             releases" ;;
190         esac
191
192         echo " Fetching latest release..."
193         WGCF_VERSION=$(curl -sf https://api.github.com/repos/ViRb3/wgcf/releases/latest \
194         | grep "tag_name" | cut -d'"' -f4)
195         [[ -z "$WGCF_VERSION" ]] && die "Could not fetch wgcf version. Check your internet connection."
196
197         sudo curl -sfl \
198         "https://github.com/ViRb3/wgcf/releases/download/${WGCF_VERSION}/wgcf_${WGCF_VERSION#v}_linux_${
199         WGCF_ARCH}" \
200         -o /usr/local/bin/wgcf
201         sudo chmod +x /usr/local/bin/wgcf
202     else
203         die "Unsupported OS. Install wgcf manually: https://github.com/ViRb3/wgcf/releases"
204     fi
205 fi
206 ok "wgcf is ready."
207 # 4. WARP profile
208 step "Generating Cloudflare WARP profile"
209
210 if [[ -f "wgcf-profile.conf" ]]; then
211     warn "wgcf-profile.conf already exists skipping key generation."
212 else
213     wgcf register --accept-tos 2>> output.log || die "wgcf register failed. Check your internet connection."
214     wgcf generate 2>> output.log || die "wgcf generate failed."

```

```

215     ok "WARP profile generated."
216 fi
217
218 # Parse keys (IPv4 address only gluetun doesn't need the IPv6 one)
219 PRIVATE_KEY=$(awk '/PrivateKey/{print $3}' wgcf-profile.conf)
220 PUBLIC_KEY=$(awk '/PublicKey/{print $3}' wgcf-profile.conf)
221 ADDRESSES=$(awk '/Address/{print $3}' wgcf-profile.conf | grep -v ':' | head -1)
222
223 [[ -z "$PRIVATE_KEY" ]] && die "Could not parse PrivateKey from wgcf-profile.conf"
224 [[ -z "$PUBLIC_KEY" ]] && die "Could not parse PublicKey from wgcf-profile.conf"
225 [[ -z "$ADDRESSES" ]] && die "Could not parse IPv4 Address from wgcf-profile.conf"
226
227 ok "WARP keys extracted."
228
229 # 5. Tailscale auth key
230 step "Tailscale authentication"
231
232 TS_AUTHKEY=""
233 if [[ -f ".env" ]]; then
234     EXISTING_TS_KEY=$(read_env_var TS_AUTHKEY)
235     if [[ -n "$EXISTING_TS_KEY" ]]; then
236         TS_AUTHKEY="$EXISTING_TS_KEY"
237         warn "Found existing TS_AUTHKEY in .env. Skipping prompt."
238     fi
239 fi
240
241 if [[ -z "$TS_AUTHKEY" ]]; then
242     echo ""
243     echo "Generate a key at: https://login.tailscale.com/admin/settings/keys"
244     echo "'Generate auth key' enable Reusable if you plan to re-run setup"
245     echo ""
246     read -rp "Paste your Tailscale auth key: " TS_AUTHKEY
247     echo ""
248 fi
249
250 [[ -z "$TS_AUTHKEY" ]] && die "Tailscale auth key is required."
251 [[ "$TS_AUTHKEY" != tskey-auth-* ]] && warn "Key doesn't look like a Tailscale auth key proceeding anyway."
252 ok "Auth key accepted."
253
254 # 6. AdGuard Home admin password
255 step "AdGuard Home admin account"
256
257 # Hardcoded default credentials to prevent lockout
258 # Username: admin
259 # Password: nullexit
260 if [[ -f ".env" ]]; then
261     EXISTING_AG_PASS=$(read_env_var ADGUARD_PASSWORD)
262     if [[ -n "$EXISTING_AG_PASS" ]]; then
263         ADGUARD_PASSWORD="$EXISTING_AG_PASS"
264         ADGUARD_PWD_ESC="$EXISTING_AG_PASS"
265         ok "Found existing ADGUARD_PASSWORD in .env."
266     else
267         ADGUARD_PASSWORD="nullexit"
268         ADGUARD_PWD_ESC="nullexit"
269         ok "Default password set to 'nullexit'."
270     fi
271 else
272     ADGUARD_PASSWORD="nullexit"
273     ADGUARD_PWD_ESC="nullexit"
274     ok "Default password set to 'nullexit'."
275 fi
276
277
278 # 7. Write .env
279 step "Writing .env"
280
281 if [[ -f ".env" ]]; then
282     read -rp ".env already exists. Overwrite? [y/N]: " OVERWRITE
283     if [[ "$OVERWRITE" != "y" && "$OVERWRITE" != "Y" ]]; then
284         warn "Keeping existing .env."
285     else
286         WRITE_ENV=1
287     fi
288 else
289     WRITE_ENV=1
290 fi
291
292 if [[ "${WRITE_ENV:-0}" == "1" ]]; then
293     # Preserve existing configuration flags if .env already exists
294     EXISTING_PROFILE="medium"
295     EXISTING_MSS="1120"

```

```

296 EXISTING_HIJACK="true"
297 EXISTING_EXIT="true"
298 EXISTING_THRESH="6"
299 EXISTING_KILL="false"
300 EXISTING_BLOCKED="kp il"
301 EXISTING_HOST_MTU="1200"
302
303 if [[ -f ".env" ]]; then
304     [[ -n "$(read_env_var GATEWAY_RULE_PROFILE)" ]] && EXISTING_PROFILE=$(read_env_var GATEWAY_RULE_PROFILE
305 )
306     [[ -n "$(read_env_var GATEWAY_MSS)" ]] && EXISTING_MSS=$(read_env_var GATEWAY_MSS)
307     [[ -n "$(read_env_var GATEWAY_HIJACK_HOST)" ]] && EXISTING_HIJACK=$(read_env_var GATEWAY_HIJACK_HOST)
308     [[ -n "$(read_env_var GATEWAY_USE_EXIT_NODE)" ]] && EXISTING_EXIT=$(read_env_var GATEWAY_USE_EXIT_NODE)
309     [[ -n "$(read_env_var WARP_FAIL_THRESHOLD)" ]] && EXISTING_THRESH=$(read_env_var WARP_FAIL_THRESHOLD)
310     [[ -n "$(read_env_var KILL_SWITCH)" ]] && EXISTING_KILL=$(read_env_var KILL_SWITCH)
311     [[ -n "$(read_env_var BLOCKED_COUNTRIES)" ]] && EXISTING_BLOCKED=$(read_env_var BLOCKED_COUNTRIES)
312     [[ -n "$(read_env_var HOST_MTU)" ]] && EXISTING_HOST_MTU=$(read_env_var HOST_MTU)
313 fi
314
315 cat > .env <<EOF
316 WIREGUARD_PRIVATE_KEY=${PRIVATE_KEY}
317 WIREGUARD_PUBLIC_KEY=${PUBLIC_KEY}
318 WIREGUARD_ADDRESSES=${ADDRESSES}
319 TS_AUTHKEY=${TS_AUTHKEY}
320 GATEWAY_RULE_PROFILE=${EXISTING_PROFILE}
321 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
322 # have a healthy path.
323 GATEWAY_MSS=${EXISTING_MSS}
324 # Lower the host Tailscale MTU to 1200 to prevent fragmentation when passing through the 1280-byte WARP tunnel.
325 HOST_MTU=${EXISTING_HOST_MTU}
326 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
327 # internet is NOT hijacked).
328 GATEWAY_HIJACK_HOST=${EXISTING_HIJACK}
329 GATEWAY_USE_EXIT_NODE=${EXISTING_EXIT}
330 WARP_FAIL_THRESHOLD=${EXISTING_THRESH}
331 KILL_SWITCH=${EXISTING_KILL}
332 ADGUARD_PASSWORD=${ADGUARD_PASSWORD}
333 BLOCKED_COUNTRIES="${EXISTING_BLOCKED}"
334 EOF
335 ok ".env written."
336 fi
337
338 # 8. Prepare directories
339 step "Preparing AdGuard directories"
340
341 mkdir -p adguard/conf adguard/work/userfilters
342
343 # Remove empty AdGuardHome.yaml that would cause a crash loop (same logic as Toggle-Gateway scripts)
344 if [[ -f "adguard/conf/AdGuardHome.yaml" ]] && ! -s "adguard/conf/AdGuardHome.yaml" ]; then
345     rm "adguard/conf/AdGuardHome.yaml"
346     warn "Removed empty AdGuardHome.yaml to prevent crash loop."
347 fi
348
349 ok "Directories ready."
350
351 # 9. Compile DNS filter rules
352 step "Compiling DNS filter rules (this may take a minute)"
353
354 # 10. Start containers
355 step "Starting containers"
356 # Note: tailscale depends on adguardhome being healthy (port 5335 up),
357 # so Docker Compose will hold it back automatically until the wizard is done.
358 docker compose up -d
359 ok "Containers starting."
360
361 # 11. Wait for AdGuard Home setup wizard
362 step "Waiting for AdGuard Home setup wizard"
363 echo " (up to 60 seconds...)"
364
365 MAX=60; ELAPSED=0
366 until docker exec routing-fix curl -sf http://127.0.0.1:3000 >> output.log 2>&1; do
367     sleep 2; ELAPSED=$((ELAPSED + 2))
368     [[ $ELAPSED -ge $MAX ]] && die "AdGuard Home didn't come up.\nDebug with: docker compose logs adguardhome"
369 done
370
371 ok "AdGuard Home is up."
372
373 # 12. Configure AdGuard Home via API
374 step "Configuring AdGuard Home"

```

```

374 # Run all API calls in a single docker exec sh -c so the session cookie file
375 # is shared across calls without leaving the container.
376 docker exec routing-fix sh -c "
377     set -e
378
379     # Step 1: Complete the setup wizard (sets admin creds + DNS port to 5335)
380     curl -sf -X POST http://127.0.0.1:3000/control/install/configure \
381         -H 'Content-Type: application/json' \
382         -d '{"web":{"ip":"0.0.0.0"},"port":3000},"dns":{"ip":"0.0.0.0"},"port":5335},"username":"admin","password":"'${ADGUARD_PWD_ESC}'" \
383         >/dev/null || true
384
385     # Step 2: Wait for AdGuard to restart after wizard
386     sleep 4
387     WAITED=0
388     until curl -sf http://127.0.0.1:3000 >/dev/null 2>&1; do
389         sleep 2; WAITED=$((WAITED + 2))
390         [ $WAITED -ge 30 ] && { echo 'AdGuard did not restart in time'; exit 1; }
391     done
392
393     # Step 3: Login to get session cookie
394     curl -sf -X POST http://127.0.0.1:3000/control/login \
395         -H 'Content-Type: application/json' \
396         -d '{"name":"admin","password":"'${ADGUARD_PWD_ESC}'" \
397         -c /tmp/agh.cookie >/dev/null
398
399     # Step 4: Point upstream DNS back through the WARP tunnel
400     curl -sf -X POST http://127.0.0.1:3000/control/dns_config \
401         -H 'Content-Type: application/json' \
402         -b /tmp/agh.cookie \
403         -d '{"upstream_dns":["127.0.0.1:53","[/onion/]127.0.0.1:5353"],"bootstrap_dns":["1.1.1.1","8.8.8.8"],"upstream_mode":"load_balance"}' \
404         >/dev/null
405
406     # Step 5: Register the compiled rules file as a filter list
407     curl -sf -X POST http://127.0.0.1:3000/control/filtering/add_url \
408         -H 'Content-Type: application/json' \
409         -b /tmp/agh.cookie \
410         -d '{"name":"Custom Compiled Rules","url":"/opt/adguardhome/work/userfilters/compiled_rules.txt"}' \
411         >/dev/null || true
412
413     # Step 6: Force a filter refresh to load the rules immediately
414     curl -sf -X POST http://127.0.0.1:3000/control/filtering/refresh \
415         -H 'Content-Type: application/json' \
416         -b /tmp/agh.cookie \
417         -d '{"whitelist":false}' \
418         >/dev/null || true
419
420     rm -f /tmp/agh.cookie
421 "
422
423 ok "AdGuard Home configured."
424
425 # 13. Wait for Tailscale
426 step "Waiting for Tailscale to authenticate"
427 echo " (Tailscale only starts once AdGuard's healthcheck passes up to 90s)"
428
429 MAX=90; ELAPSED=0; TS_IP=""
430 until [[ -n "$TS_IP" ]]; do
431     TS_IP=$(docker exec tailscale tailscale ip -4 2>> output.log || true)
432     if [[ -z "$TS_IP" ]]; then
433         sleep 3; ELAPSED=$((ELAPSED + 3))
434         if [[ $ELAPSED -ge $MAX ]]; then
435             warn "Tailscale hasn't authenticated yet."
436             warn "Once it does, re-run the toggle script it will resolve the IP dynamically."
437             break
438         fi
439     fi
440 done
441
442 if [[ -n "$TS_IP" ]]; then
443     ok "Tailscale IP: ${TS_IP} (resolved dynamically by the toggle script on each startup)"
444 fi

```

44 scripts/sweep.sh

```

1 #!/bin/bash
2 # scripts/sweep.sh  nullexit gateway health sweep
3 #
4 # Automates the 5-check post-restart health sweep documented in sweep.md 1.
5 # It answers one question: "after a toggle/restart/wake, is the gateway
6 # actually healthy end-to-end no leak, direct P2P, DNS filtering live,
7 # kill-switch armed?" It is READ-ONLY: it never mutates routing, PF, DNS,
8 # or containers, so it is safe to run against a live gateway at any time
9 # (including from a remote session see the caveat in agent.md L16).
10 #
11 # The 5 checks (mirroring sweep.md 1):
12 #   1. Containers      warp, adguardhome, tailscale, tor, routing-fix up/healthy
13 #   2. Double-tunnel/leak  container WARP exit IP == host exit IP, both warp=on
14 #   3. Hostcontainer path  Tailscale exit-node is a DIRECT P2P conn, not DERP relay
15 #   4. AdGuard filtering  compiled rule counts present + live DNS interception
16 #   5. PF kill-switch     pfctl Enabled + anchor com.apple/nullexit carries rules
17 #
18 # A timestamped report is written to the project root (one file per run, so
19 # runs can be diffed). stdout carries a coloured summary + a final PASS/FAIL
20 # verdict; exit code is 0 when every check passes, 1 otherwise (so it can gate
21 # CI / launchd / a post-restart hook).
22 #
23 # Usage:
24 #   bash scripts/sweep.sh           # run the sweep, write report, print verdict
25 #   bash scripts/sweep.sh --quiet   # only the final verdict line + non-zero on FAIL
26 #   bash scripts/sweep.sh --help    # this message
27
28 set -uo pipefail
29 # NOTE: errexit is intentionally OFF. Several probes are EXPECTED to fail on an
30 # unhealthy gateway and that failure IS the diagnostic signal we record each
31 # exit status rather than letting one abort the whole sweep.
32
33 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
34 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
35 TIMESTAMP="$(date -u +%Y%m%dT%H%M%S)"
36 cd "$PROJECT_ROOT" || { echo "[FATAL] cannot cd to $PROJECT_ROOT" >&2; exit 1; }
37 source "$SCRIPT_DIR/common.sh"
38 LOG_FILE="$PROJECT_ROOT/output.log"
39 REPORT_FILE="$PROJECT_ROOT/sweep-$TIMESTAMP.txt"
40
41 # Argument parsing
42 QUIET=false
43 case "${1:-}" in
44   --quiet|-q) QUIET=true ;;
45   --help|-h)  sed -n '2,26p' "$0"; exit 0 ;;
46   *)          : ;;
47   *)          echo "Unknown argument: $1" >&2
48               echo "Run with --help for usage." >&2
49               exit 2 ;;
50 esac
51
52 # Colours (auto-disabled when stdout is not a tty, e.g. piped)
53 if [ -t 1 ]; then
54   RED='\033[0;31m'; GREEN='\033[0;32m'; YELLOW='\033[1;33m'
55   BOLD='\033[1m'; NC='\033[0m'
56 else
57   RED=''; GREEN=''; YELLOW=''; BOLD=''; NC=''
58 fi
59
60 # Homebrew on PATH (same as toggle.sh / recover.sh / diagnose-host-leak)
61 export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"
62
63 # Output helpers write to BOTH stdout and the report file
64 exec 3>> "$REPORT_FILE" || {
65   printf '\n[FATAL] cannot open %s for write\n' "$REPORT_FILE" >&2
66   exit 1
67 }
68
69 emit() { [ "$QUIET" = true ] || printf '%b\n' "$1"; printf '%b\n' "$1" | sed 's/\x1b\[0-9;]*m//g' >&3; }
70 section() { emit ""; emit "${BOLD} $1 ${NC}"; }
71 line() { emit " $1"; }
72 ok() { line "${GREEN}${NC} $1"; }
73 warn() { line "${YELLOW}${NC} $1"; }
74 fail() { line "${RED}${NC} $1"; }
75
76 # common.sh redefines run_with_timeout; it is a pure-bash macOS-safe timeout
77 # (GNU `timeout` is absent on stock macOS). We reuse the sourced version.
78
79 # Result accounting
80 # Each check appends "PASS"/"WARN"/"FAIL" so the verdict is a pure roll-up.
81 declare -a RESULTS=()

```

```

81 record() { RESULTS+=("$1|$2"); } # status|label
82
83 # Header
84 if [ "$QUIET" != true ]; then
85     emit ""
86     emit ""
87     emit " n u l l e x i t   h e a l t h   s w e e p "
88     emit ""
89     emit " Timestamp (UTC): $TIMESTAMP"
90     emit " Project root:    $PROJECT_ROOT"
91     emit " Report file:     $REPORT_FILE"
92     emit ""
93 fi
94
95 GATEWAY_TS_IP="$(read_adguard_ip)" # gateway Tailscale IP from .gateway_ip
96
97 #
98 section "1/5 Containers"
99 #
100 # All five services must be running; warp + adguardhome additionally expose a
101 # healthcheck and must be 'healthy'. `docker compose ps` is the source of truth.
102 EXPECTED_SERVICES="warp adguardhome tailscale tor routing-fix"
103 if ! command -v docker >/dev/null 2>&1; then
104     fail "docker CLI not on PATH cannot inspect containers"
105     record FAIL "containers"
106 else
107     PS_OUT=$(run_with_timeout 20 docker compose ps 2>>"$LOG_FILE")
108     missing=""; unhealthy=""
109     for svc in $EXPECTED_SERVICES; do
110         row=$(printf '%s\n' "$PS_OUT" | awk -v s="$svc" '$1==s || $0 ~ ("[:space:]"s"[:space:]")')
111         if [ -z "$row" ] || ! printf '%s' "$row" | grep -qiE 'up|running'; then
112             missing="$missing $svc"
113         elif printf '%s' "$row" | grep -qi 'unhealthy'; then
114             unhealthy="$unhealthy $svc"
115         fi
116     done
117     if [ -z "$missing" ] && [ -z "$unhealthy" ]; then
118         ok "all 5 services up (warp/adguardhome healthy, tailscale/tor/routing-fix running)"
119         record PASS "containers"
120     else
121         [ -n "$missing" ] && fail "not up/running:$missing"
122         [ -n "$unhealthy" ] && fail "reporting unhealthy:$unhealthy"
123         record FAIL "containers"
124     fi
125     line " docker compose ps "
126     printf '%s\n' "$PS_OUT" | awk '{print "    "$0}' >&3
127     [ "$QUIET" = true ] || printf '%s\n' "$PS_OUT" | awk '{print "    "$0}'
128 fi
129 #
130 #
131 section "2/5 Double-tunnel / IP-leak"
132 #
133 # The container's WARP egress and the host's egress must be the SAME public IP
134 # with warp=on for both. Equal IPs all host traffic is exiting through the
135 # same WARP tunnel the container uses (no split / raw-ISP leak).
136 # The warp container (gluetun) ships wget, not curl fetch the trace from
137 # inside its netns with whichever client is present.
138 CTR_TRACE=$(run_with_timeout 12 docker compose exec -T warp sh -c \
139     'curl -s --max-time 8 https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null \
140     || wget -qO- --timeout=8 https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null' \
141     2>>"$LOG_FILE")
142 HOST_TRACE=$(run_with_timeout 12 curl -s --max-time 8 \
143     https://www.cloudflare.com/cdn-cgi/trace 2>>"$LOG_FILE")
144 CTR_IP=$(printf '%s' "$CTR_TRACE" | awk -F= '/^ip=/ {print $2;exit}')
145 CTR_WARP=$(printf '%s' "$CTR_TRACE" | awk -F= '/^warp=/ {print $2;exit}')
146 HOST_IP=$(printf '%s' "$HOST_TRACE" | awk -F= '/^ip=/ {print $2;exit}')
147 HOST_WARP=$(printf '%s' "$HOST_TRACE" | awk -F= '/^warp=/ {print $2;exit}')
148 line "container exit: ip=${CTR_IP:-?} warp=${CTR_WARP:-?}"
149 line "host exit: ip=${HOST_IP:-?} warp=${HOST_WARP:-?}"
150 if [ -z "$CTR_IP" ] || [ -z "$HOST_IP" ]; then
151     fail "could not obtain both egress IPs (container or host trace failed)"
152     record FAIL "leak"
153 elif [ "$CTR_IP" = "$HOST_IP" ] && [ "$HOST_WARP" = "on" ]; then
154     ok "no leak host egress == container egress ($HOST_IP), warp=on"
155     record PASS "leak"
156 elif [ "$HOST_WARP" != "on" ]; then
157     fail "host warp=$HOST_WARP host traffic is NOT going through WARP (LEAK)"
158     record FAIL "leak"
159 else
160     fail "egress mismatch: host $HOST_IP != container $CTR_IP split/leak"
161     record FAIL "leak"

```

```

162 fi
163
164 #
165 section "3/5 Host container path (direct P2P, not DERP relay)"
166 #
167 # `tailscale status` annotates each peer with 'direct <ip:port>' or
168 # 'relay "<derp>". The gateway peer should be DIRECT a DERP relay hop adds
169 # latency and signals NAT traversal failure between host and gateway.
170 if ! command -v tailscale >/dev/null 2>&1; then
171     fail "tailscale CLI not on PATH"
172     record FAIL "p2p"
173 else
174     TS_OUT=$(run_with_timeout 6 tailscale status 2>>"$LOG_FILE")
175     PEER_LINE=$(printf '%s\n' "$TS_OUT" | grep -E "(^|[:space:])$GATEWAY_TS_IP([[:space:]]|$)" | head -1)
176     if [ -z "$PEER_LINE" ]; then
177         warn "gateway $GATEWAY_TS_IP not found in host peer list"
178         line "$(printf '%s\n' "$TS_OUT" | head -5 | awk '{print "    "$0}')"
179         record WARN "p2p"
180     elif printf '%s' "$PEER_LINE" | grep -q 'direct '; then
181         conn=$(printf '%s' "$PEER_LINE" | grep -oE 'direct [0-9.]+:[0-9]+')
182         ok "direct P2P to gateway $conn (no DERP relay)"
183         record PASS "p2p"
184     elif printf '%s' "$PEER_LINE" | grep -q 'relay '; then
185         relay=$(printf '%s' "$PEER_LINE" | grep -oE 'relay "[^"]+')
186         warn "gateway reached via DERP $relay (not direct) NAT traversal degraded"
187         record WARN "p2p"
188     else
189         warn "gateway peer state indeterminate (idle/no active conn)"
190         line "    $PEER_LINE"
191         record WARN "p2p"
192     fi
193 fi
194
195 #
196 section "4/5 AdGuard filtering"
197 #
198 # Two signals: (a) the compiled ruleset exists with a non-trivial block count,
199 # (b) the gateway resolver actually answers a query (live interception).
200 COMPILED="adguard/work/userfilters/compiled_rules.txt"
201 if [ -f "$COMPILED" ]; then
202     TOTAL=$(grep -m1 '! Total Block Rules:' "$COMPILED" | awk -F': ' '{print $2}' | tr -d '[:space:]')
203     NATIVE=$(grep -m1 '! Native AdGuard Rules:' "$COMPILED" | awk -F': ' '{print $2}' | tr -d '[:space:]')
204     if [ -n "$TOTAL" ] && [ "$TOTAL" -gt 0 ] 2>/dev/null; then
205         ok "compiled ruleset present Total Block Rules: $TOTAL / Native AdGuard: ${NATIVE:-?}"
206         RULES_OK=true
207     else
208         warn "compiled_rules.txt present but block-rule count unreadable"
209         RULES_OK=false
210     fi
211 else
212     warn "compiled_rules.txt missing (rule-compile may not have run)"
213     RULES_OK=false
214 fi
215 # Live interception: the gateway resolver must answer (dig via .gateway_ip).
216 DNS_OK=false
217 if command -v dig >/dev/null 2>&1 && [ -n "$GATEWAY_TS_IP" ]; then
218     DIG_OUT=$(run_with_timeout 8 dig +time=3 +tries=1 @"$GATEWAY_TS_IP" google.com A 2>>"$LOG_FILE")
219     if printf '%s' "$DIG_OUT" | grep -qE 'ANSWER SECTION|status: NOERROR'; then
220         ok "live DNS interception gateway $GATEWAY_TS_IP resolved google.com"
221         DNS_OK=true
222     else
223         warn "gateway $GATEWAY_TS_IP did not answer a test query"
224     fi
225 else
226     warn "skipped live-resolution probe (dig missing or gateway IP unknown)"
227 fi
228 if [ "$RULES_OK" = true ] && [ "$DNS_OK" = true ]; then
229     record PASS "adguard"
230 elif [ "$RULES_OK" = true ] || [ "$DNS_OK" = true ]; then
231     record WARN "adguard"
232 else
233     record FAIL "adguard"
234 fi
235
236 #
237 section "5/5 PF kill-switch"
238 #
239 # PF must be Enabled AND the com.apple/nullexit anchor must carry rules.
240 # An empty/absent anchor means the fail-closed lane is not actually armed.
241 PF_STATUS=$(run_with_timeout 6 sudo -n pfctl -s info 2>>"$LOG_FILE" | awk '/^Status:/{print $2;exit}')
242 ANCHOR_RULES=$(run_with_timeout 6 sudo -n pfctl -a com.apple/nullexit -sr 2>>"$LOG_FILE")

```

```

243 if [ "$PF_STATUS" = "Enabled" ] && [ -n "$ANCHOR_RULES" ]; then
244     RULE_N=$(printf '%s\n' "$ANCHOR_RULES" | grep -cvE '\s*$')
245     ok "PF Enabled + anchor com.apple/nullexit armed ($RULE_N rules)"
246     line "$(printf '%s\n' "$ANCHOR_RULES" | head -4 | awk '{print "    "$0}')"
247     record PASS "pf"
248 elif [ "$PF_STATUS" = "Enabled" ]; then
249     fail "PF Enabled but anchor com.apple/nullexit has NO rules  kill-switch not armed"
250     record FAIL "pf"
251 elif [ -z "$PF_STATUS" ]; then
252     warn "could not read PF status (needs passwordless sudo for 'pfctl -s info')"
253     record WARN "pf"
254 else
255     fail "PF Status: $PF_STATUS  kill-switch disabled"
256     record FAIL "pf"
257 fi
258
259 #
260 section "VERDICT"
261 #
262 pass=0; warnc=0; failc=0
263 for r in "${RESULTS[@]"; do
264     case "${r%[*]}" in PASS) pass=$((pass+1));; WARN) warnc=$((warnc+1));; FAIL) failc=$((failc+1));; esac
265 done
266 for r in "${RESULTS[@]"; do
267     st="${r%[*]}"; lbl="${r###*}"
268     case "$st" in
269         PASS) ok "$lbl" ;;
270         WARN) warn "$lbl" ;;
271         FAIL) fail "$lbl" ;;
272     esac
273 done
274 emit ""
275 if [ "$failc" -eq 0 ] && [ "$warnc" -eq 0 ]; then
276     emit "${GREEN}${BOLD}SWEEP PASS  gateway healthy (${pass}/5 checks green)${NC}"
277     EXIT=0
278 elif [ "$failc" -eq 0 ]; then
279     emit "${YELLOW}${BOLD}SWEEP PASS with warnings  ${pass} pass, ${warnc} warn, 0 fail${NC}"
280     EXIT=0
281 else
282     emit "${RED}${BOLD}SWEEP FAIL  ${pass} pass, ${warnc} warn, ${failc} fail${NC}"
283     EXIT=1
284 fi
285 emit ""
286 [ "$QUIET" = true ] || emit "Full report: $REPORT_FILE"
287 exit "$EXIT"

```

45 scripts/unlock-files.sh

```

1  #!/bin/bash
2  # Resolve project root relative to this script's location, so the script
3  # works regardless of the caller's CWD.
4  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
5  PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
6
7  echo "Unlocking .env..."
8  if [ -f "$PROJECT_ROOT/.env" ]; then
9      sudo cat "$PROJECT_ROOT/.env" > "$PROJECT_ROOT/.env.tmp" && mv "$PROJECT_ROOT/.env.tmp" "$PROJECT_ROOT/.env"
10     echo ".env unlocked"
11 fi
12
13 echo "Unlocking AdGuard cache files..."
14 for f in "$PROJECT_ROOT"/adguard/work/userfilters/compiled_rules.txt \
15     "$PROJECT_ROOT"/adguard/work/userfilters/ip_blocklist.ipset \
16     "$PROJECT_ROOT"/adguard/work/userfilters/cache/*.txt \
17     "$PROJECT_ROOT"/adguard/work/userfilters/cache/ip/*.txt \
18     "$PROJECT_ROOT"/adguard/work/data/filters/*.txt; do
19     if [ -f "$f" ]; then
20         cat "$f" > "$f.tmp" && mv "$f.tmp" "$f"
21         echo "Unlocked $f"
22     fi
23 done
24
25 echo ""
26 echo "All locked files have been successfully bypassed via atomic rename!"

```

46 tor-bridges.txt

47 white_list.txt

```
1 0.client-channel.google.com
2 1drv.com
3 2.android.pool.ntp.org
4 akamaihd.net
5 akamaitechnologies.com
6 akamaized.net
7 amazonaws.com
8 android.clients.google.com
9 api-adservices.apple.com
10 api.ipify.org
11 api.rlje.net
12 app-api.ted.com
13 appleid.apple.com
14 apps.skype.com
15 appsbackup-pa.clients6.google.com
16 appsbackup-pa.googleapis.com
17 appspot-preview.l.google.com
18 apt.sonarr.tv
19 aspnetscdn.com
20 attestation.xboxlive.com
21 ax.phobos.apple.com.edgesuite.net
22 books-analytics-events.apple.com
23 brightcove.net
24 c.s-microsoft.com
25 cdn.cloudflare.net
26 cdn.embedly.com
27 cdn.optimizely.com
28 cdn.vidible.tv
29 cdn2.optimizely.com
30 cdn3.optimizely.com
31 cdnjs.cloudflare.com
32 cert.mgt.xboxlive.com
33 clientconfig.passport.net
34 clients1.google.com
35 clients2.google.com
36 clients3.google.com
37 clients4.google.com
38 clients5.google.com
39 clients6.google.com
40 connectivitycheck.android.com
41 connectivitycheck.gstatic.com
42 continuum.dds.microsoft.com
43 cpms.spop10.ams.plex.bz
44 cpms35.spop10.ams.plex.bz
45 cse.google.com
46 ctldl.windowsupdate.com
47 cws.conviva.com
48 d2c8v52ll5s99u.cloudfront.net
49 d2gatte9o95jao.cloudfront.net
50 dashboard.plex.tv
51 dataplicity.com
52 def-vef.xboxlive.com
53 delivery.vidible.tv
54 dev.virtualearth.net
55 device.auth.xboxlive.com
56 display.ugc.bazaarvoice.com
57 displaycatalog.mp.microsoft.com
58 dl.delivery.mp.microsoft.com
59 dl.dropbox.com
60 dl.dropboxusercontent.com
61 dns.msftncsi.com
62 download.sonarr.tv
63 drift.com
64 driftt.com
65 dynupdate.no-ip.com
66 ecn.dev.virtualearth.net
67 edge.api.brightcove.com
68 eds.xboxlive.com
69 fonts.gstatic.com
70 forums.sonarr.tv
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71 g.live.com
72 geo-prod.do.dsp.mp.microsoft.com
73 geo3.ggpht.com
74 gfws1.geforce.com
75 giphy.com
76 github.com
77 github.io
78 googleapis.com
79 gravatar.com
80 gstatic.com
81 help.ui.xboxlive.com
82 hls.ted.com
83 i.ytimg.com
84 il.ytimg.com
85 iadsdk.apple.com
86 imagesak.secureserver.net
87 img.vidible.tv
88 imgix.net
89 imgs.xkcd.com
90 instantmessaging-pa.googleapis.com
91 intercom.io
92 jquery.com
93 jsdelivr.net
94 keystone.mwbsys.com
95 lastfm-img2.akamaized.net
96 licensing.xboxlive.com
97 live.com
98 livepassdl.conviva.com
99 login.live.com
100 login.microsoftonline.com
101 manifest.googlevideo.com
102 meta-db-worker02.pop.ric.plex.bz
103 meta.plex.bz
104 meta.plex.tv
105 microsoftonline.com
106 msftncsi.com
107 my.plexapp.com
108 nexusrules.officeapps.live.com
109 nine.plugins.plexapp.com
110 no-ip.com
111 node.plexapp.com
112 notes-analytics-events.apple.com
113 notify.xboxlive.com
114 npr-news.streaming.adswizz.com
115 ns1.dropbox.com
116 ns2.dropbox.com
117 o1.email.plex.tv
118 o2.sg0.plex.tv
119 ocsp.apple.com
120 office.com
121 office.net
122 office365.com
123 officeclient.microsoft.com
124 om.cbsi.com
125 onedrive.live.com
126 outlook.live.com
127 outlook.office365.com
128 pings.conviva.com
129 placeholder.it
130 placeholderit.imgix.net
131 players.brightcove.net
132 pricelist.skype.com
133 products.office.com
134 proxy.plex.bz
135 proxy.plex.tv
136 proxy02.pop.ord.plex.bz
137 pubsub.plex.bz
138 pubsub.plex.tv
139 raw.githubusercontent.com
140 redirector.googlevideo.com
141 res.cloudinary.com
142 s.gateway.messenger.live.com
143 s.marketwatch.com
144 s.youtube.com
145 s.ytimg.com
146 s1.wp.com
147 s2.youtube.com
148 s3.amazonaws.com
149 sa.symcb.com
150 secure.avangate.com
151 secure.brightcove.com

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152 secure.surveymonkey.com
153 services.sonarr.tv
154 skyhook.sonarr.tv
155 spclient.wg.spotify.com
156 ssl.p.jwpcdn.com
157 staging.plex.tv
158 status.plex.tv
159 t.co
160 t0.ssl.ak.dynamic.tiles.virtualearth.net
161 t0.ssl.ak.tiles.virtualearth.net
162 tawk.to
163 tedcdn.com
164 themoviedb.com
165 thetvdb.com
166 tinyurl.com
167 title.auth.xboxlive.com
168 title.mgt.xboxlive.com
169 traffic.libsyn.com
170 tvdb2.plex.tv
171 tvthemes.plexapp.com
172 twimg.com
173 ui.skype.com
174 video-stats.l.google.com
175 videos.vidible.tv
176 vidtech.cbsinteractive.com
177 weather-analytics-events.apple.com
178 widget-cdn.rpxnow.com
179 win10.ipv6.microsoft.com
180 wp.com
181 ws.audioscrobbler.com
182 www.dataplicity.com
183 www.googleapis.com
184 www.msftconnecttest.com
185 www.msftncsi.com
186 www.no-ip.com
187 www.youtube-nocookie.com
188 xbox.ipv6.microsoft.com
189 xboxexperiencesprod.experimentation.xboxlive.com
190 xflight.xboxlive.com
191 xkms.xboxlive.com
192 xsts.auth.xboxlive.com
193youtu.be
194 youtube-nocookie.com
195 yt3.ggpht.com
196 zee.cws.conviva.com
197
198 # YouTube family base domains so every subdomain (video stream servers on
199 # *.googlevideo.com, image hosts on *.ytimg.com, profile thumbnails on
200 # *.ggpht.com, etc.) is automatically allowlisted via the AdGuard '||domain^'
201 # wildcard. The specific s.youtube.com / manifest.googlevideo.com entries
202 # above are kept for clarity but will be optimized away by rule-compiler
203 # because their parents are now in this list.
204 youtube.com
205 googlevideo.com
206 ytimg.com
207 ggpht.com
208 gvt1.com
209 gvt2.com
210 gvt3.com
211 gvt1.net
212 gvt2.net
213 gvt3.net
214
215 edge-mqtt.facebook.com
216 gateway.instagram.com
217 i.instagram.com
218
219 facebook.com
220 fb.com
221 fbcdn.net
222 fbsbx.com
223 messenger.com
224 whatsapp.com
225 web.whatsapp.com
226 instagram.com
227 cdninstagram.com
228 static.whatsapp.net
229 media-bos5-1.cdn.whatsapp.net
230 media-yyz1-1.cdn.whatsapp.net
231 webtp.whatsapp.net
232 mmg.whatsapp.net

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